



GE Nuclear Energy

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ABWR Design Control Document

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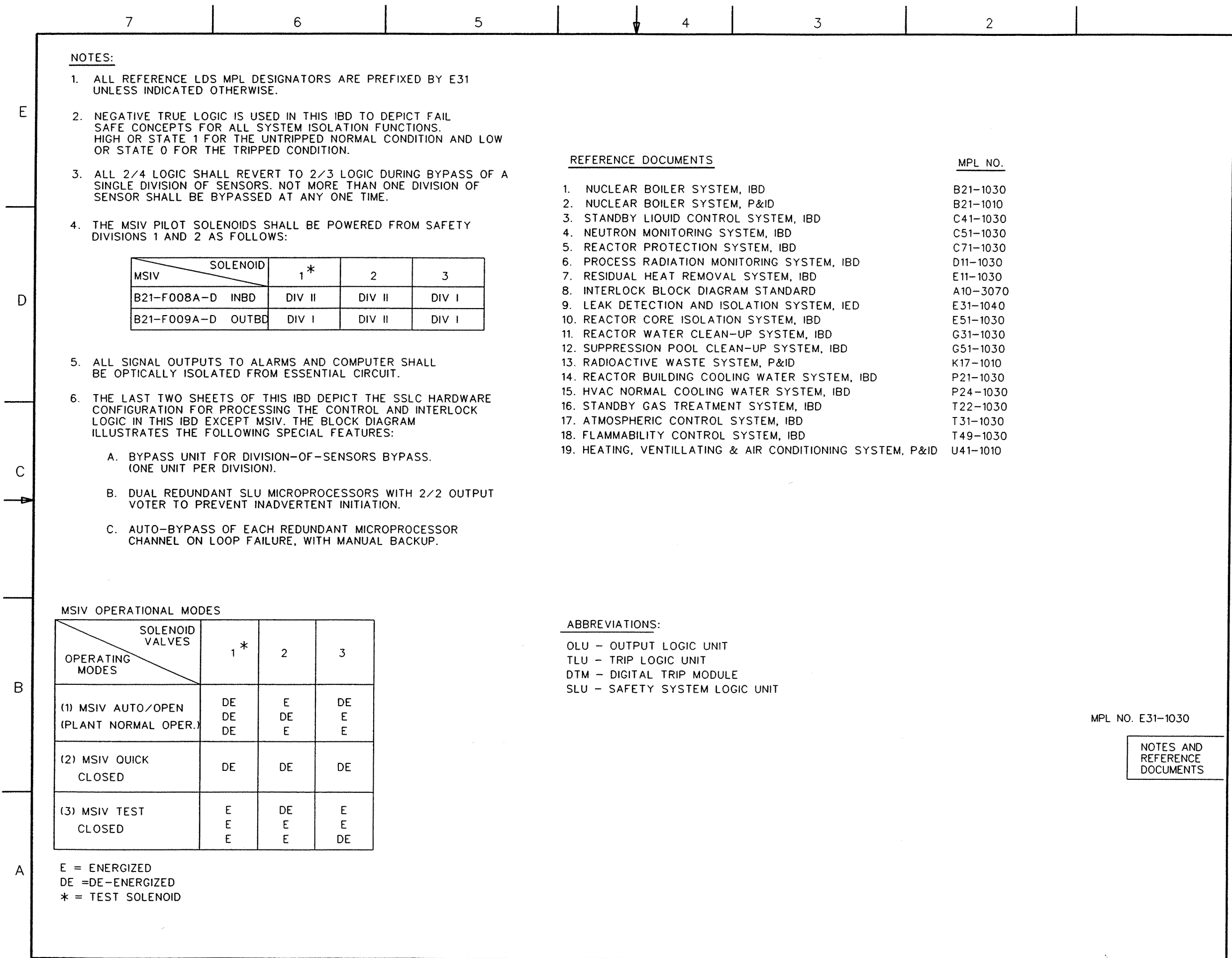
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	19	MSIV AUTO ISOLATION TRIP LOGIC – DIV 3	49	RHR AREA B TEMP TRIP LOGIC		
	20	MSIV VOTING & CHANNEL BYPASS LOGIC – DIV 4	50	RHR AREA C TEMP TRIP LOGIC		
	21	MSIV VOTING & CHANNEL BYPASS LOGIC – DIV 4	51	RHR LOOP A ISOLATION LOGIC		
	22	MSIV VOTING & CHANNEL BYPASS LOGIC – DIV 4	52	RHR LOOP B ISOLATION LOGIC		
B	23	MSIV AUTO ISOLATION TRIP LOGIC – DIV 4	53	RHR LOOP C ISOLATION LOGIC		
	24	MSIV POSITION SENSORS LOGIC	54	RCIC AREA TEMP TRIP LOGIC		
	25	MSL A INBOARD MSIV CONTROL LOGIC	55	RCIC LOW SL PRESSURE TRIP LOGIC		
	26	MSL B INBOARD MSIV CONTROL LOGIC	56	RCIC HIGH SL FLOW TRIP LOGIC		
	27	MSL C INBOARD MSIV CONTROL LOGIC	57	RCIC ISOLATION LOGIC		
	28	MSL D INBOARD MSIV CONTROL LOGIC	58	CUW MASS DIFF FLOW TRIP LOGIC		
	29	MSL A OUTBOARD MSIV CONTROL LOGIC	59	CUW REG HX TEMP TRIP LOGIC		
	30	MSL B OUTBOARD MSIV CONTROL LOGIC	60	CUW NON-REG HX TEMP TRIP LOGIC		
A						
	CONTENTS					

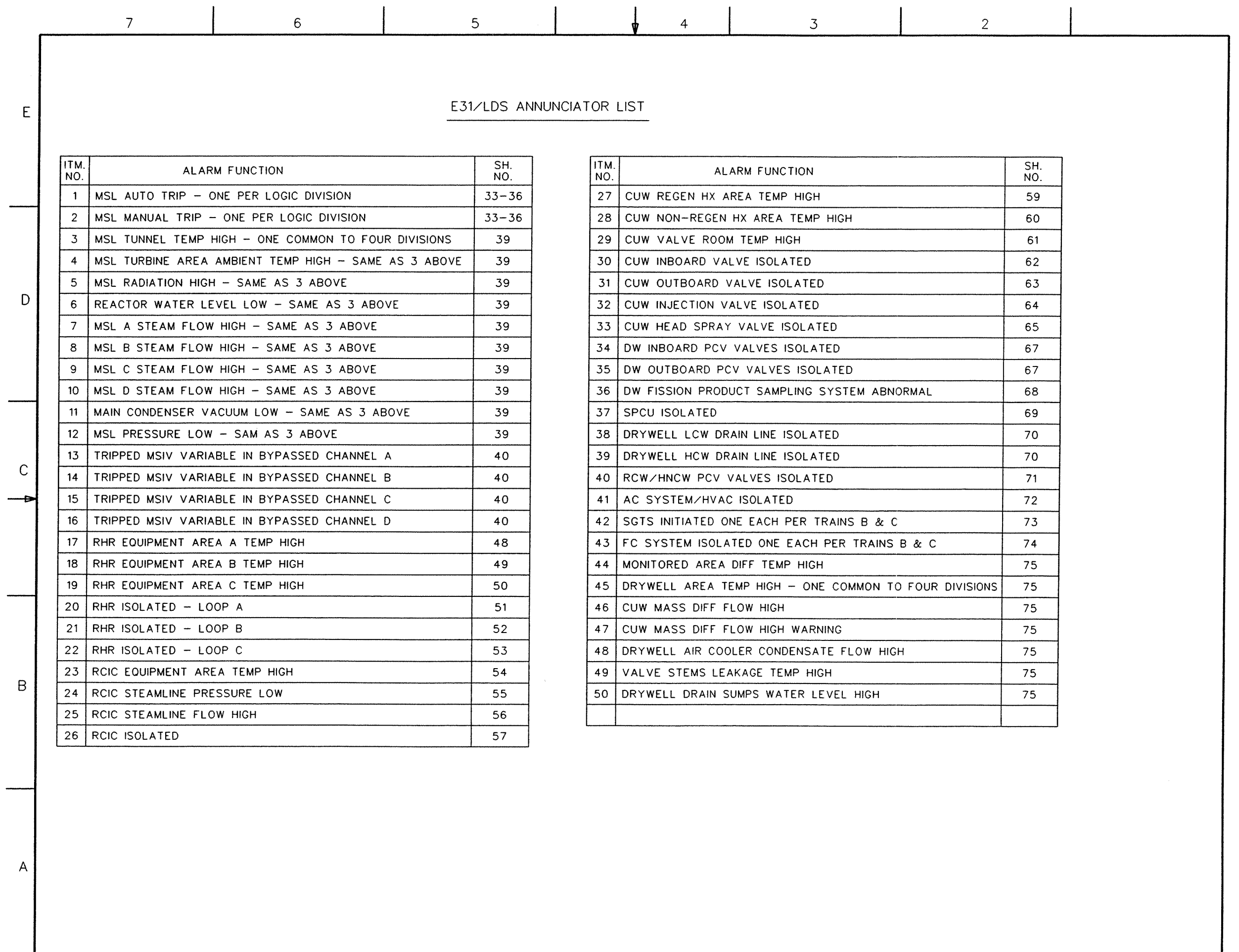


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 3 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.3

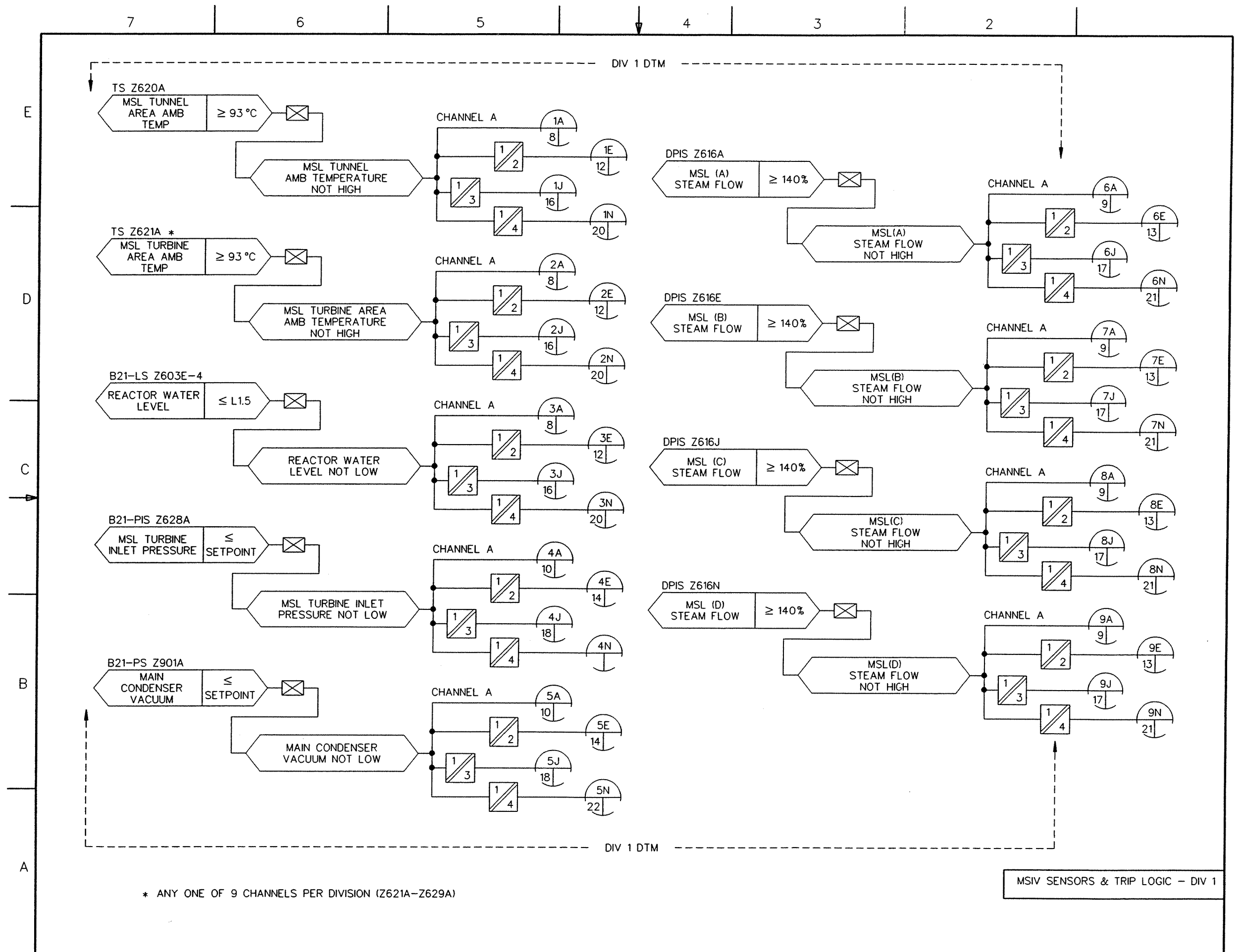


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 4 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.4

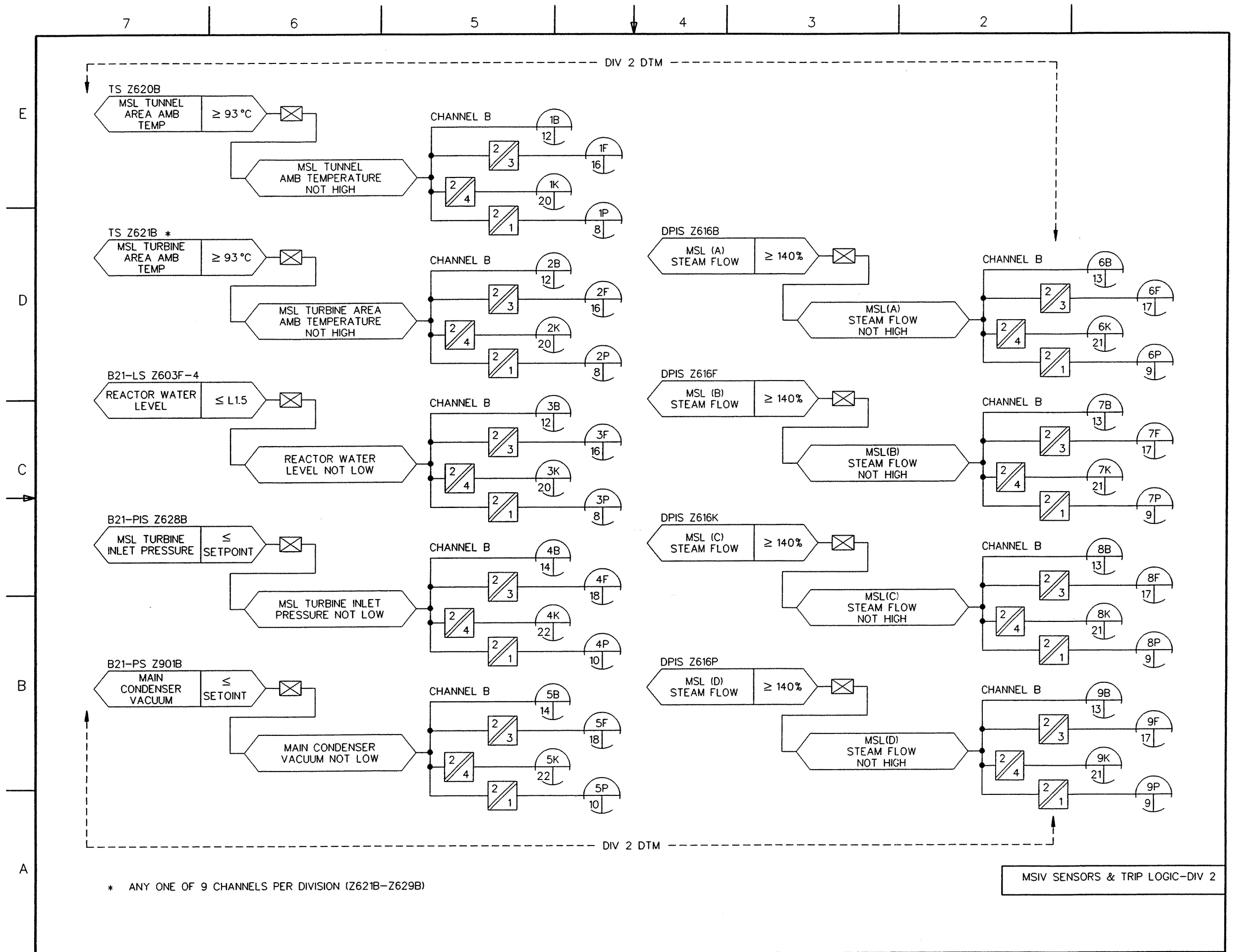


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 5 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.5

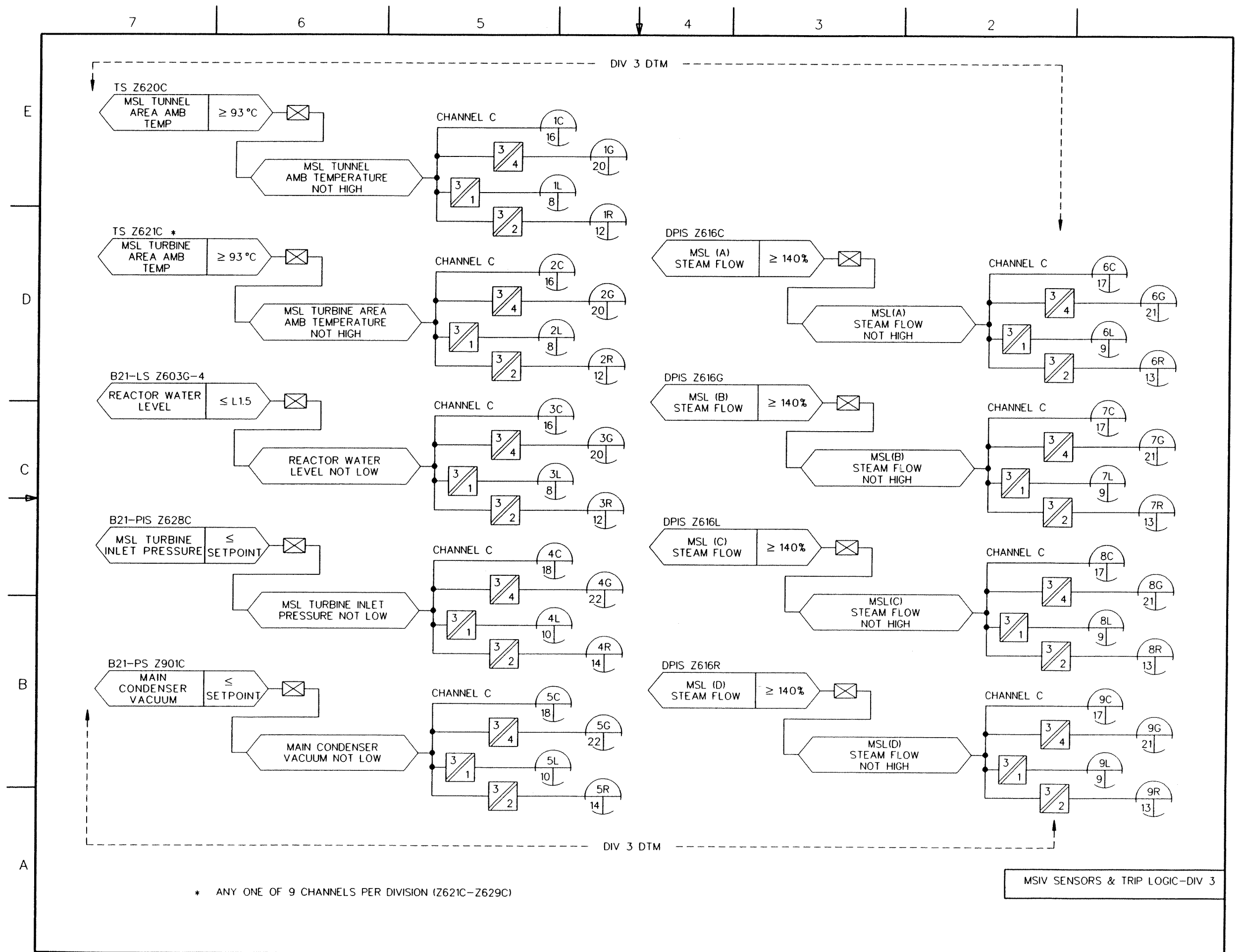


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 6 of 77)

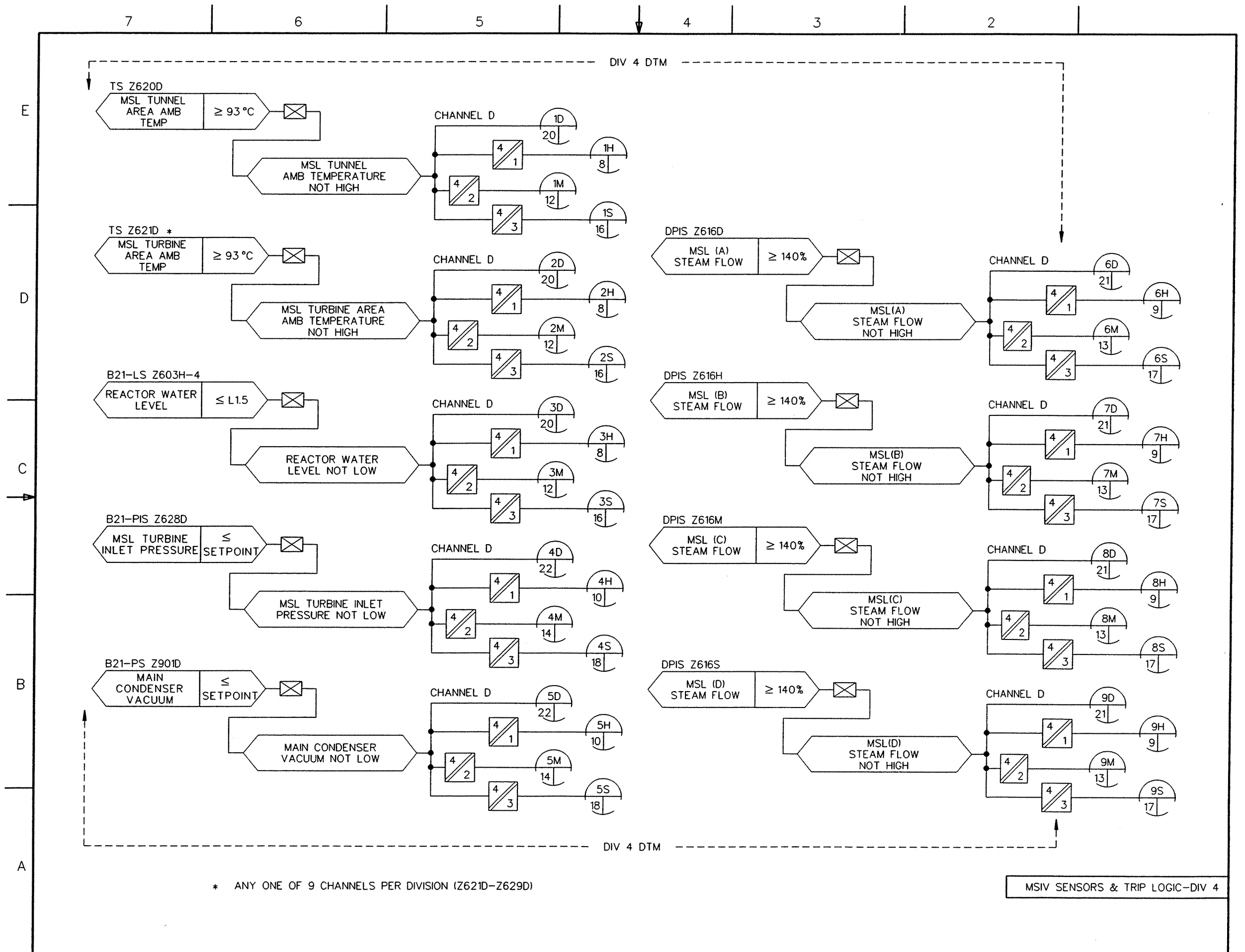


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 7 of 77)

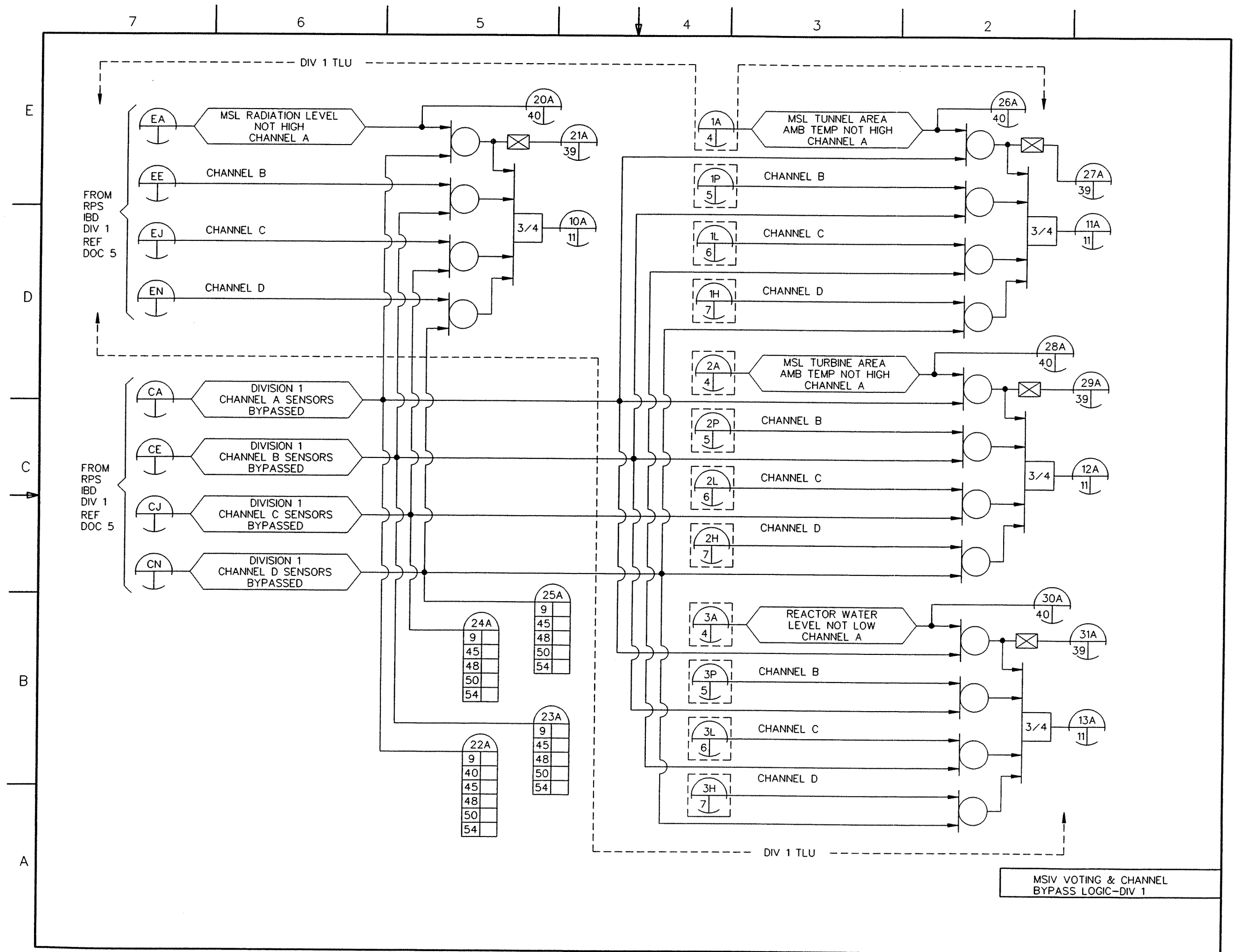


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 8 of 77)

ABWR DCD/Tier 2

Rev. 0

21-228.8

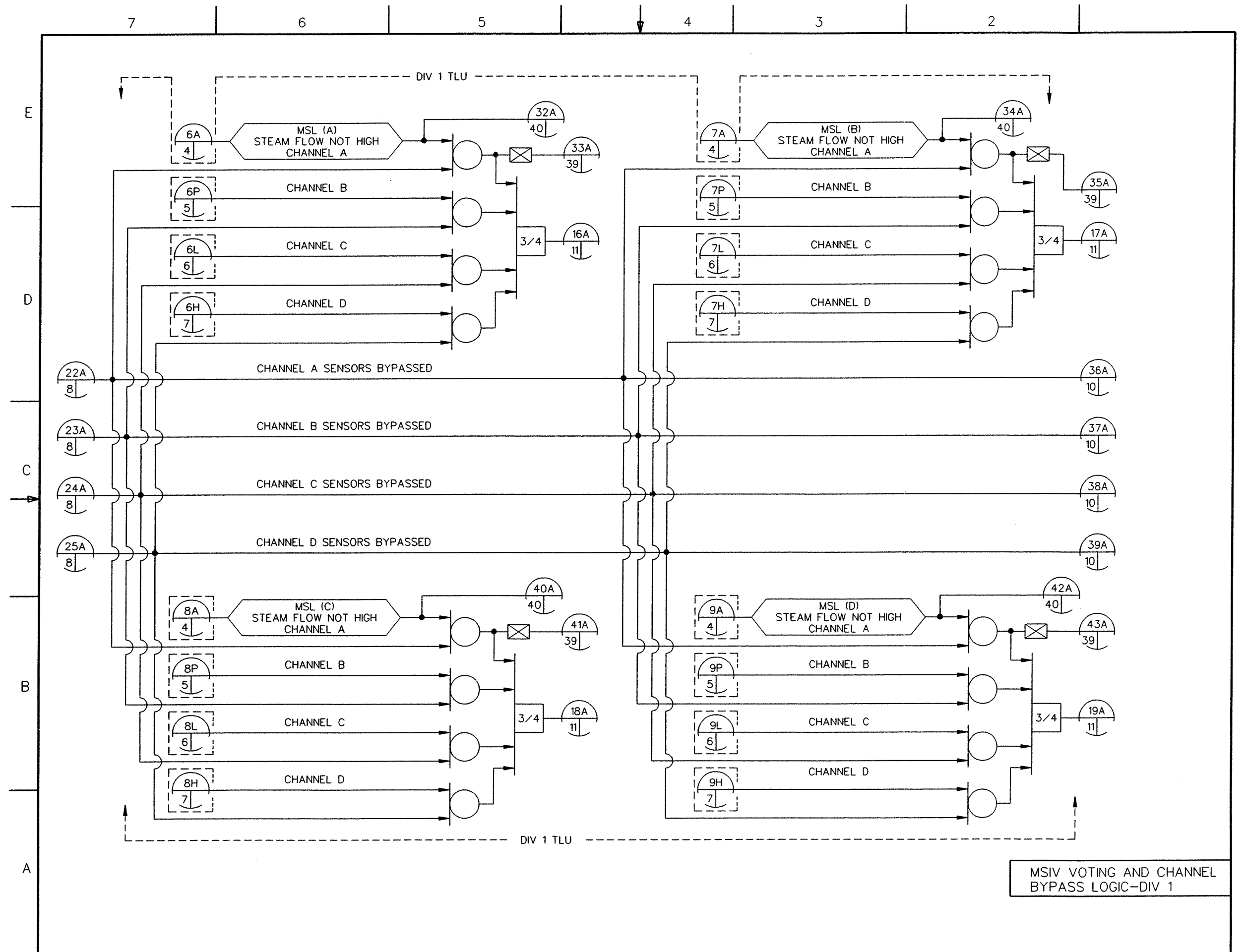


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 9 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.9

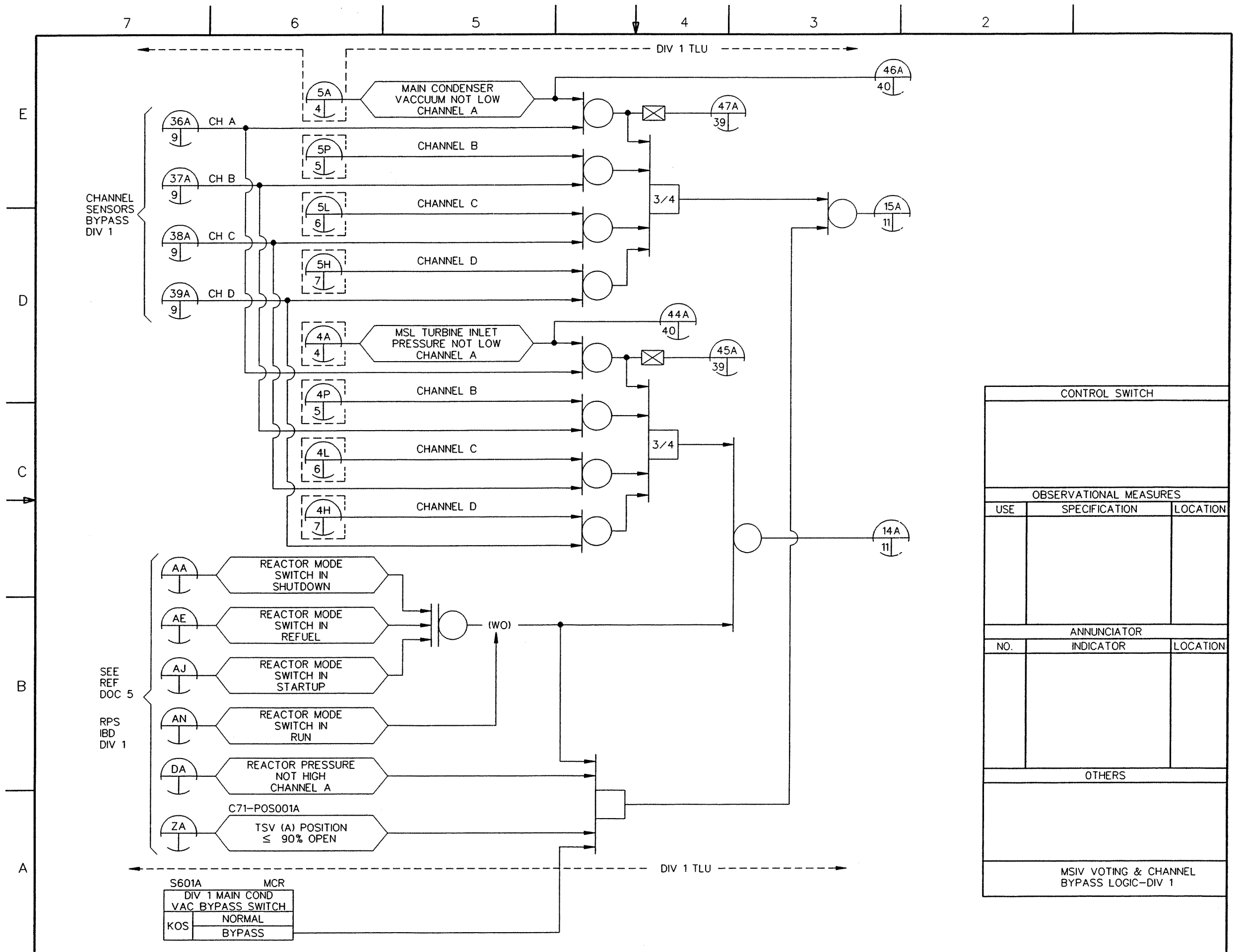


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 10 of 77)

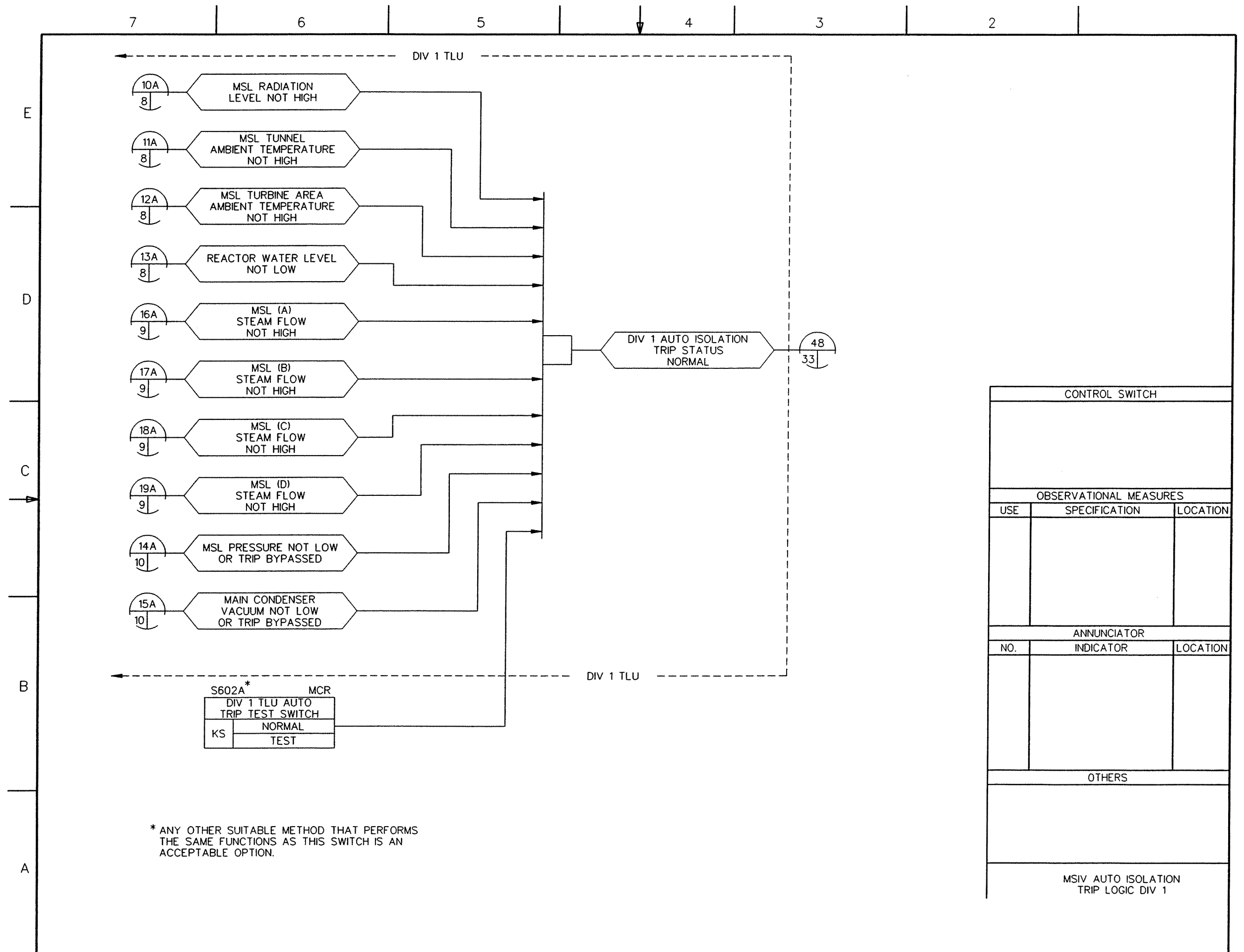


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 11 of 77)

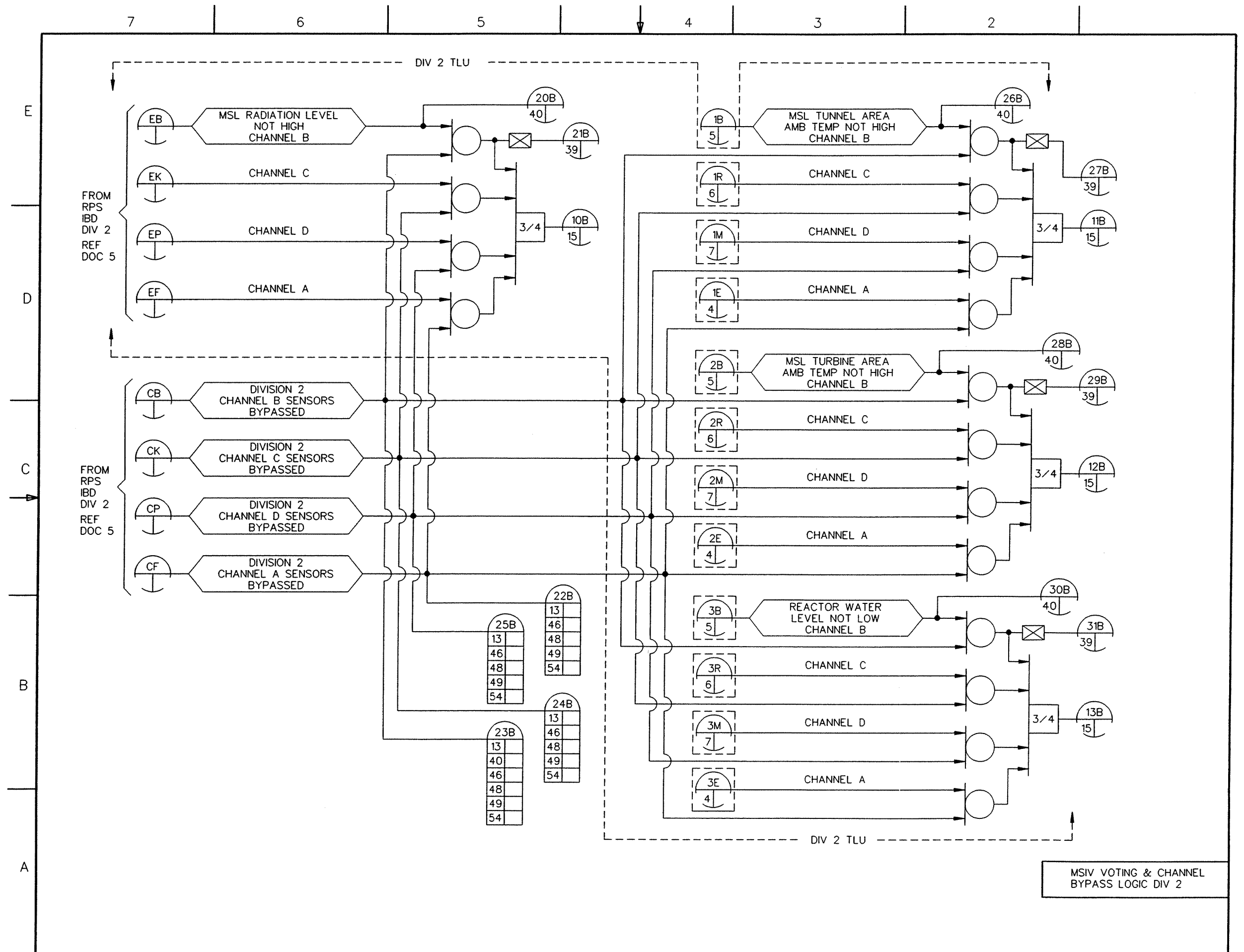


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 12 of 77)

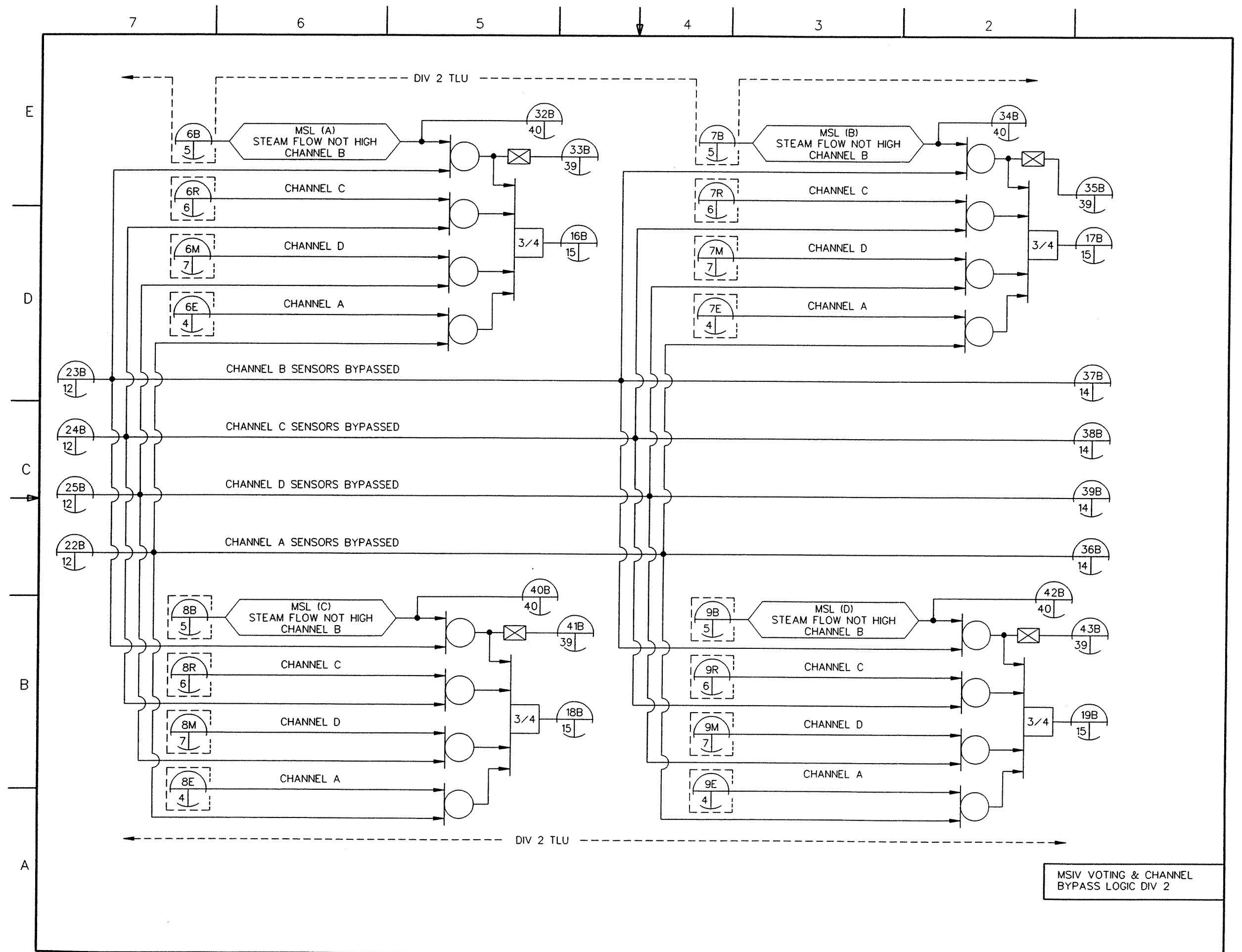


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 13 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.13

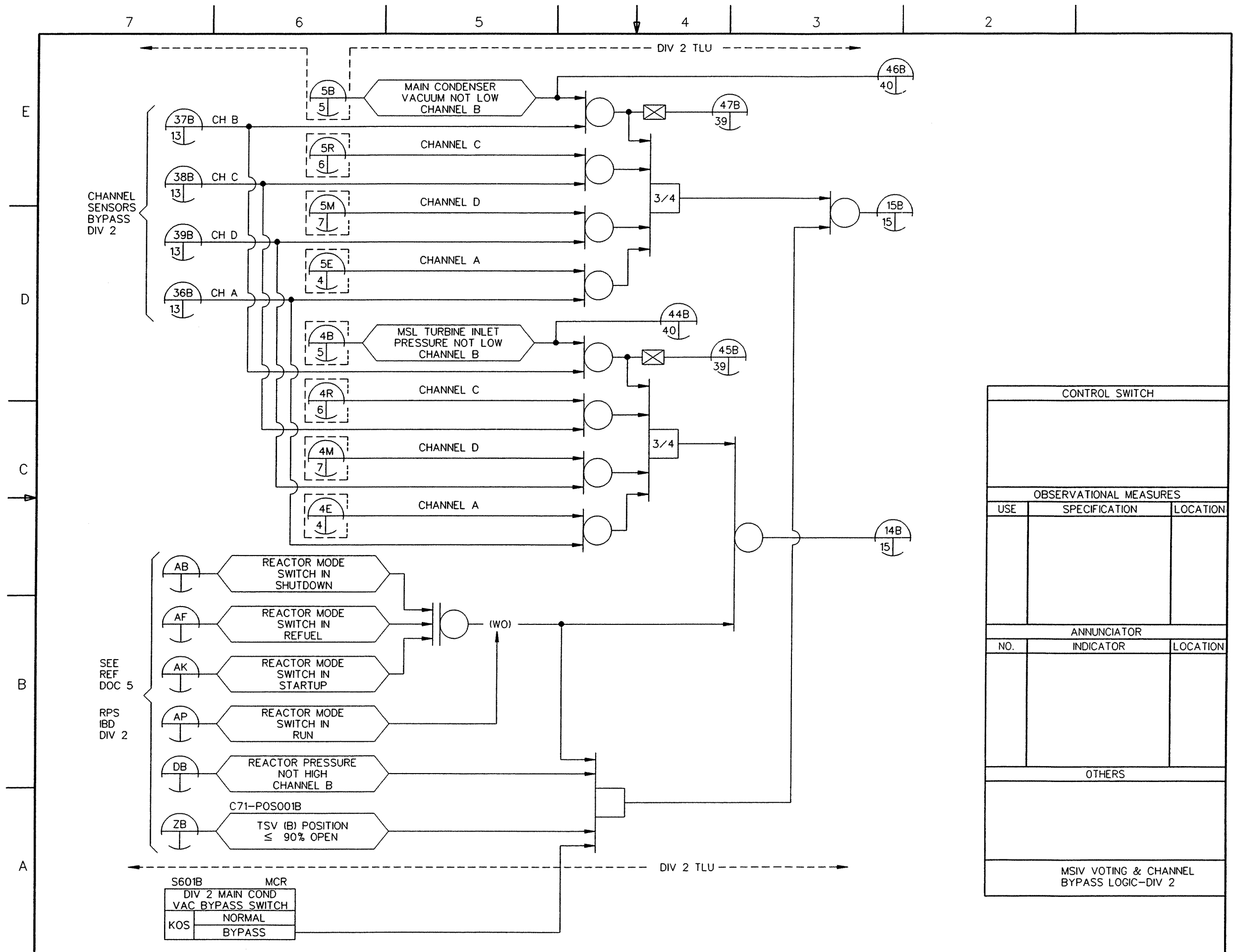


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 14 of 77)

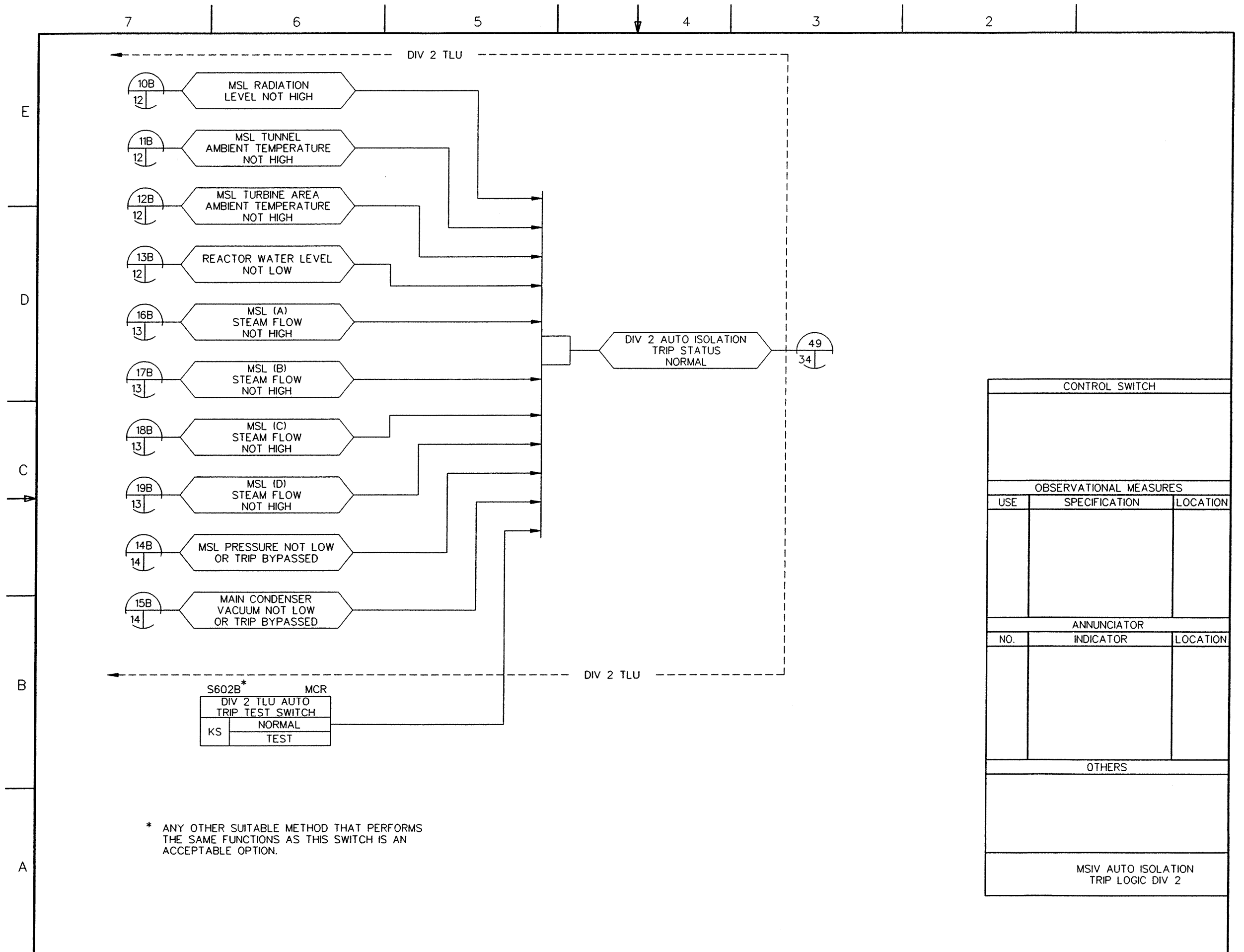


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 15 of 77)

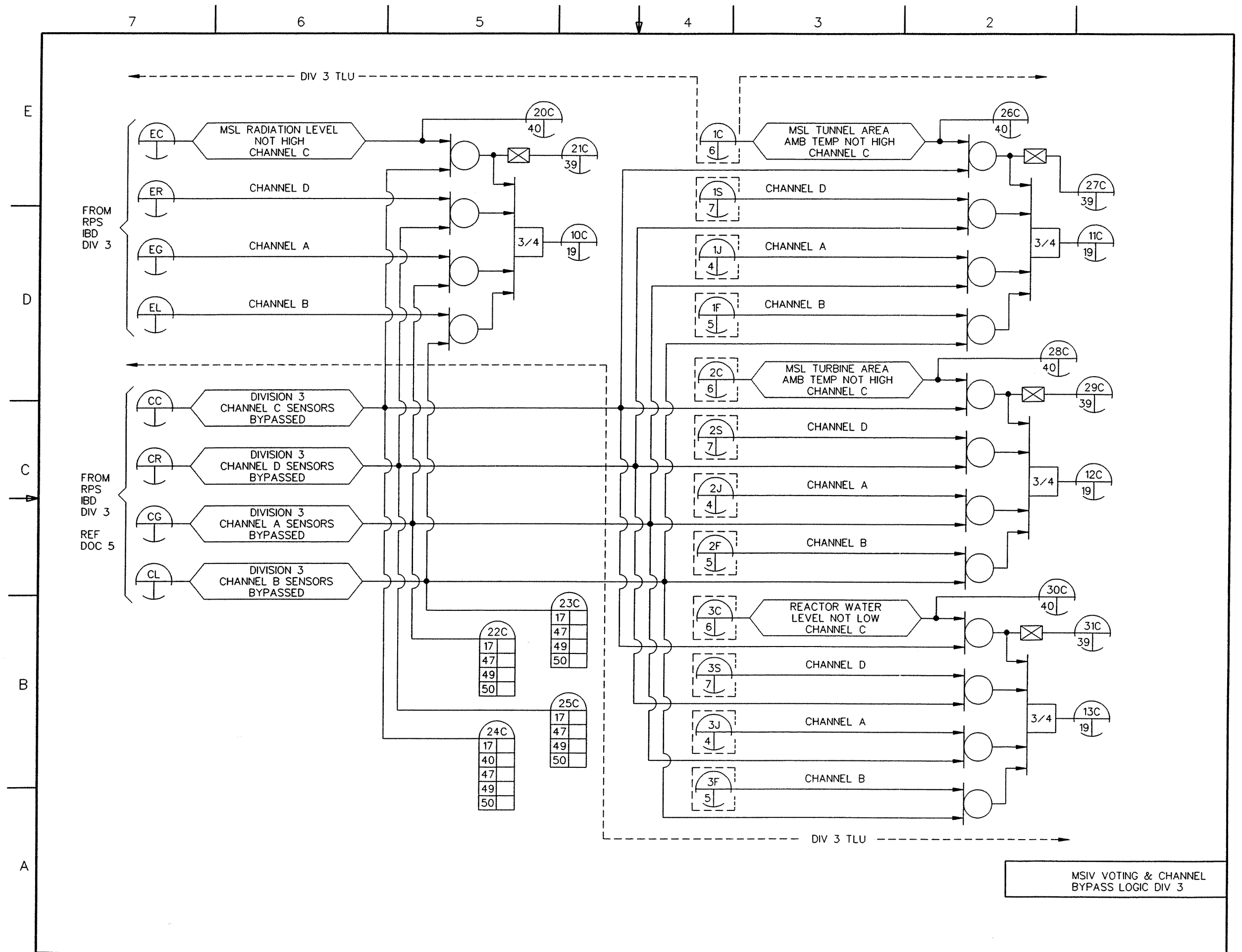


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 16 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.16

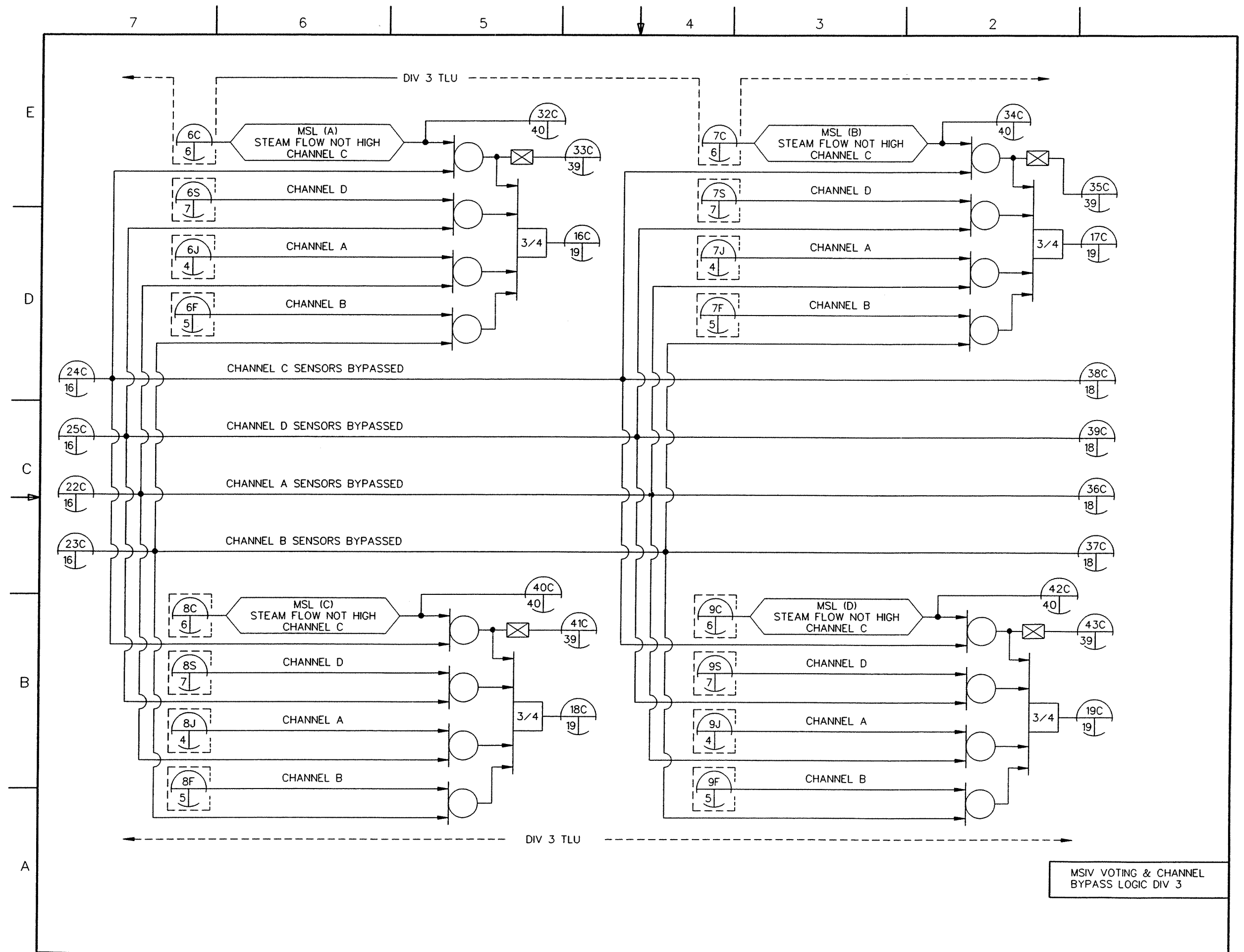


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 17 of 77)

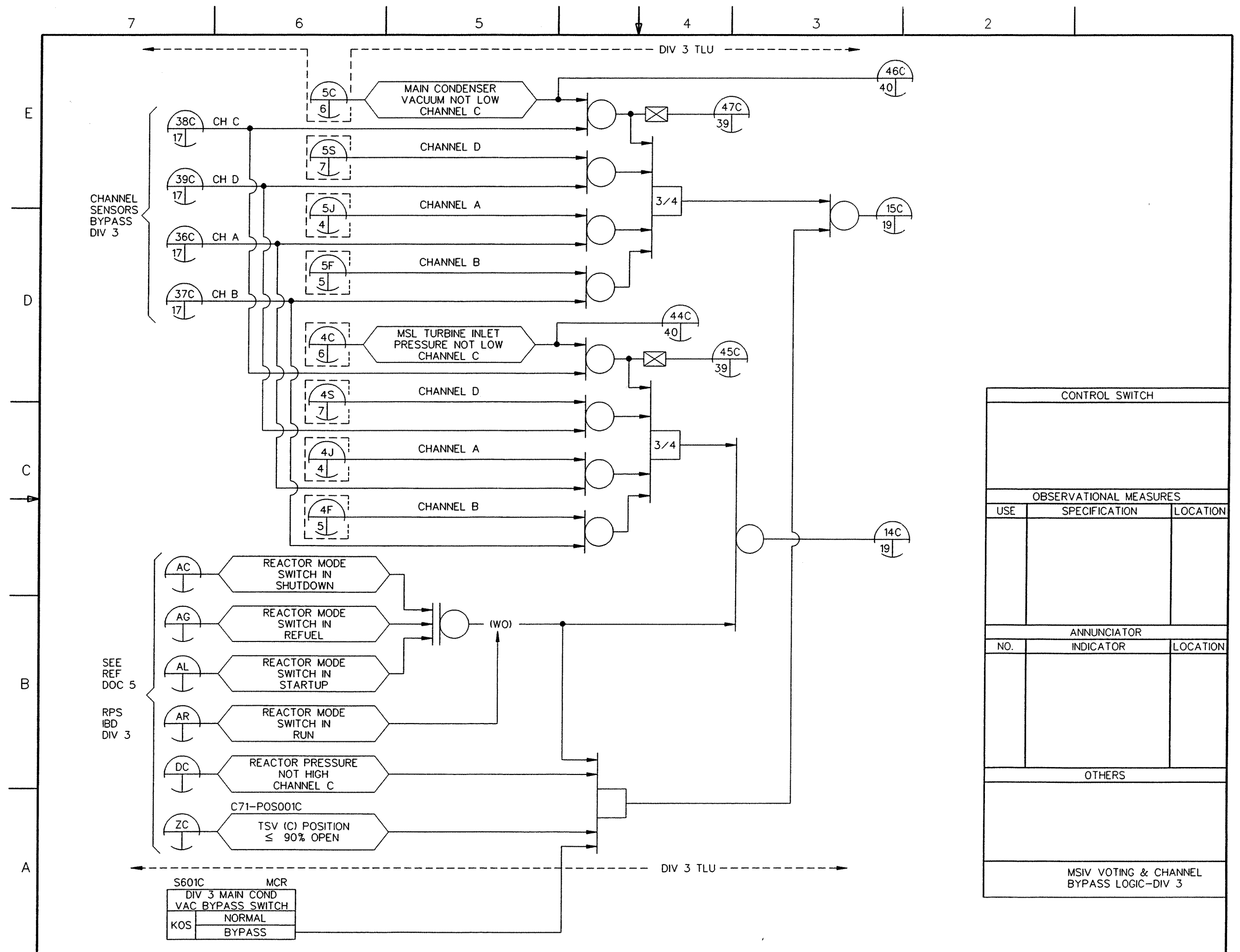


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 18 of 77)

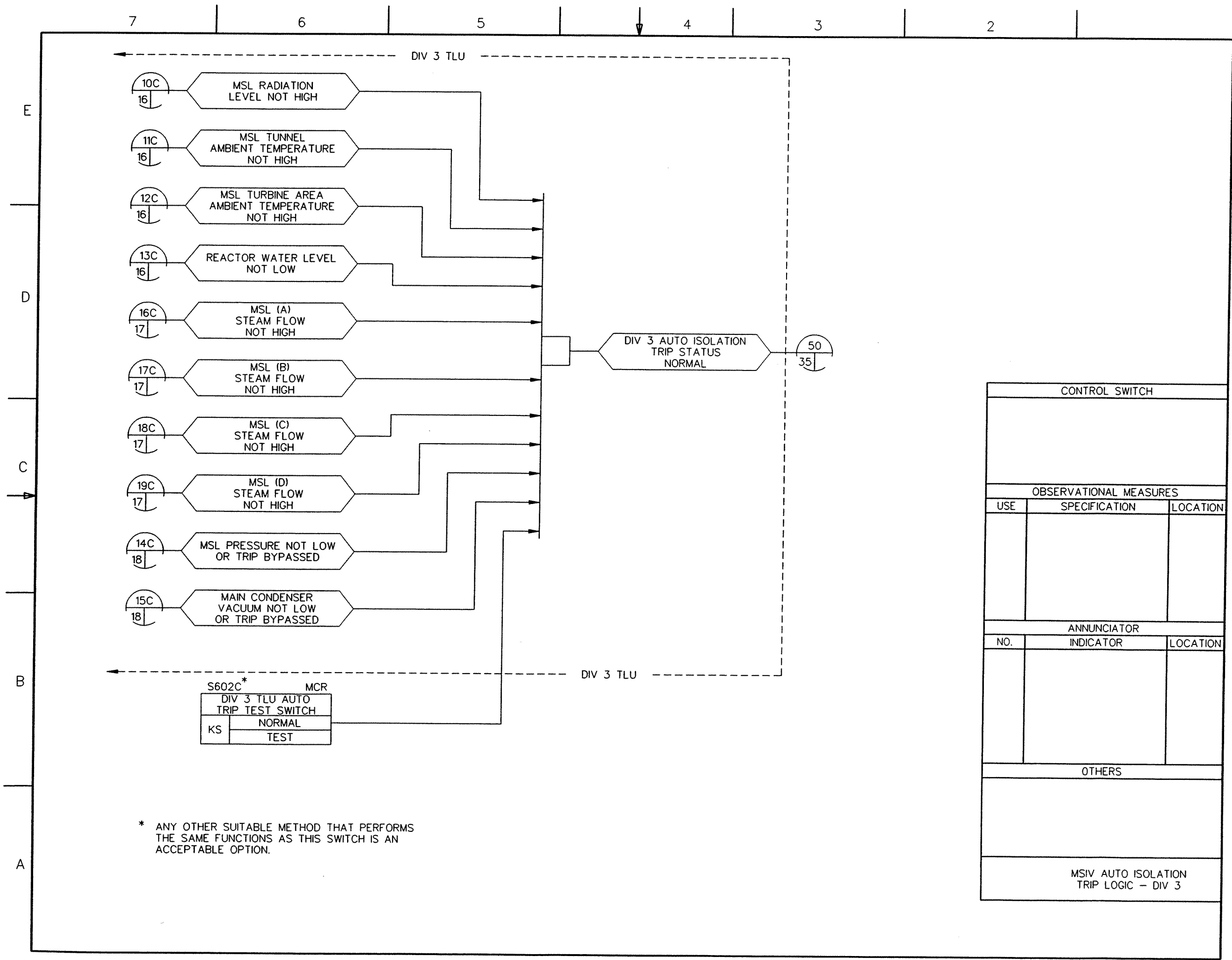


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 19 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.19

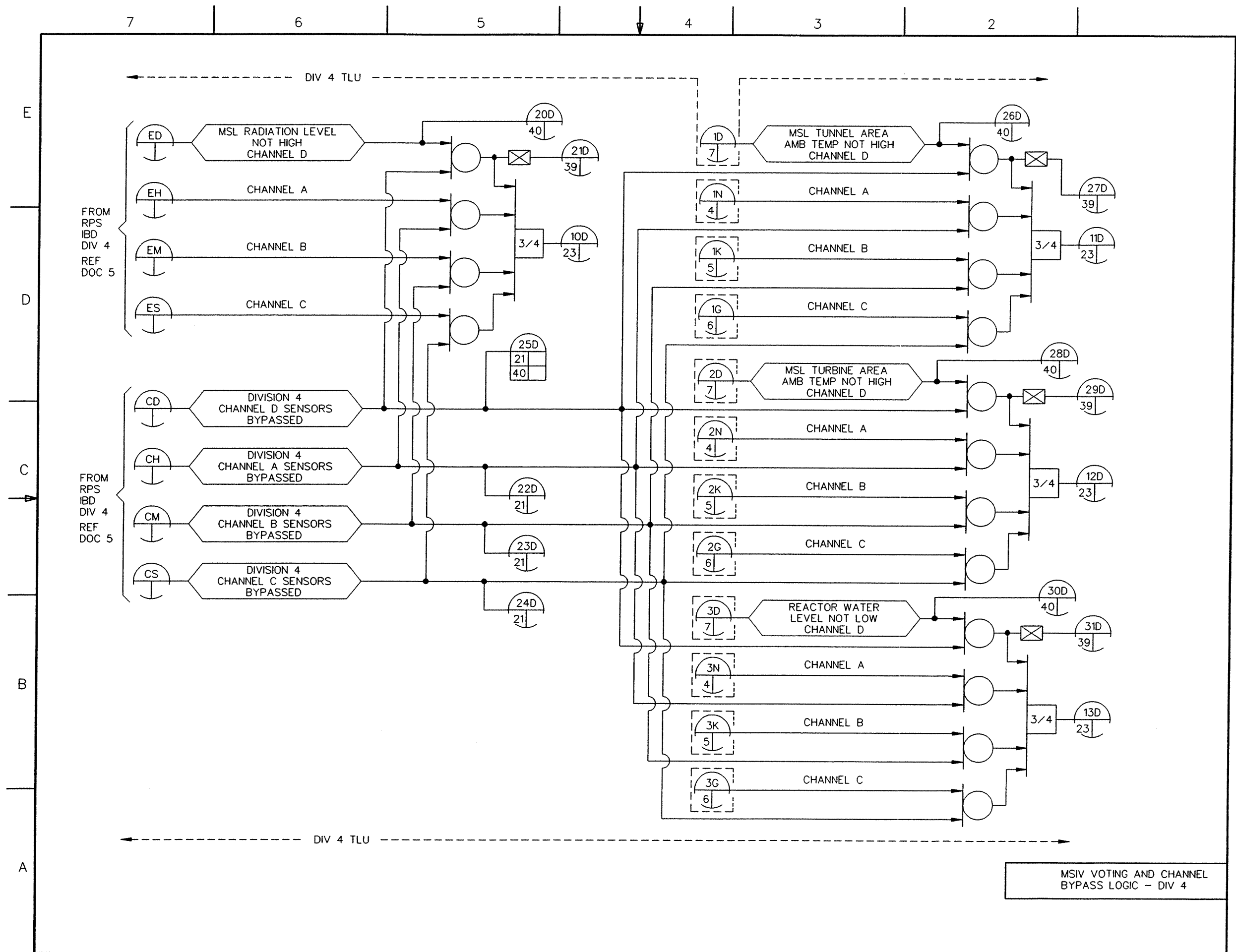


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 20 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.20

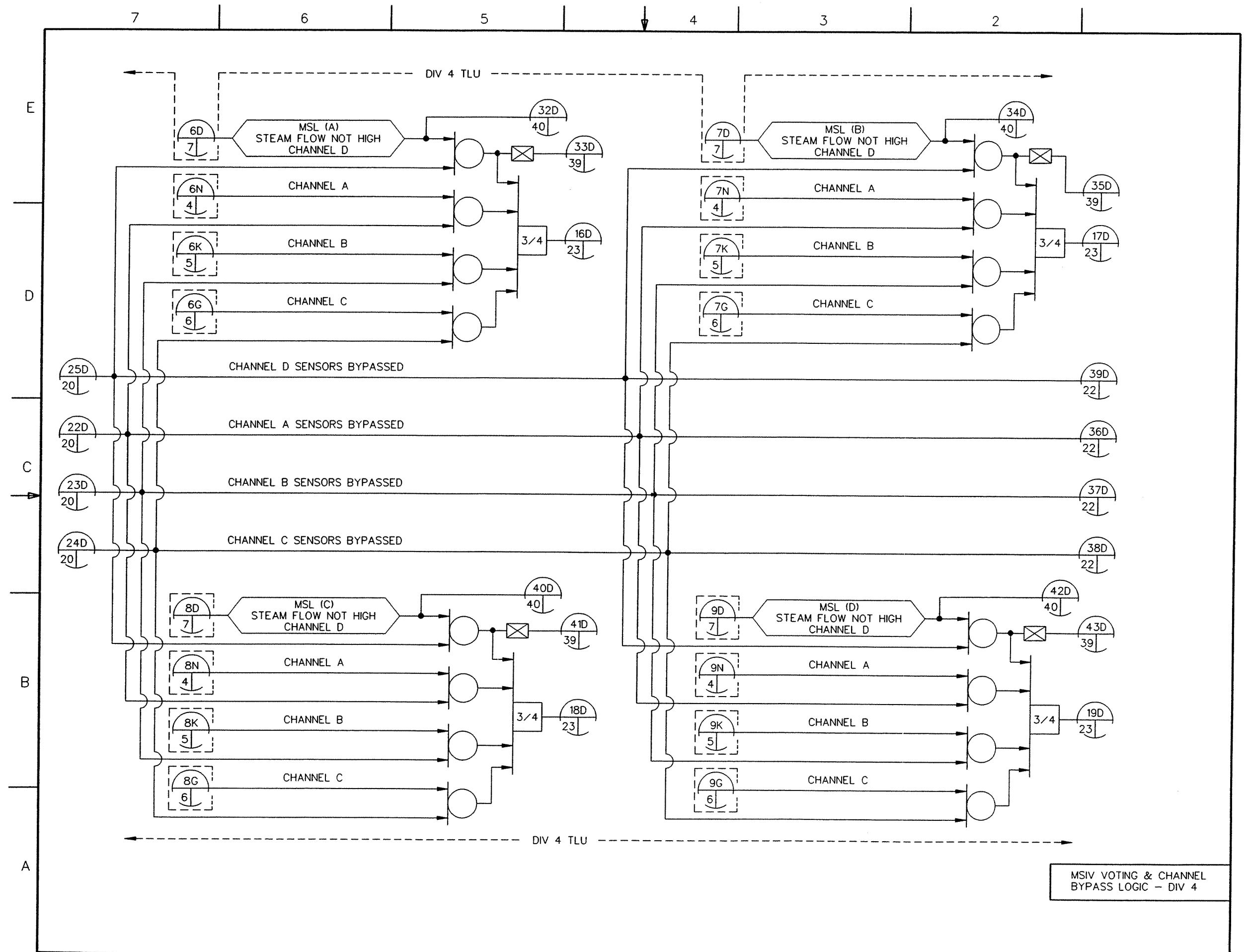


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 21 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.21

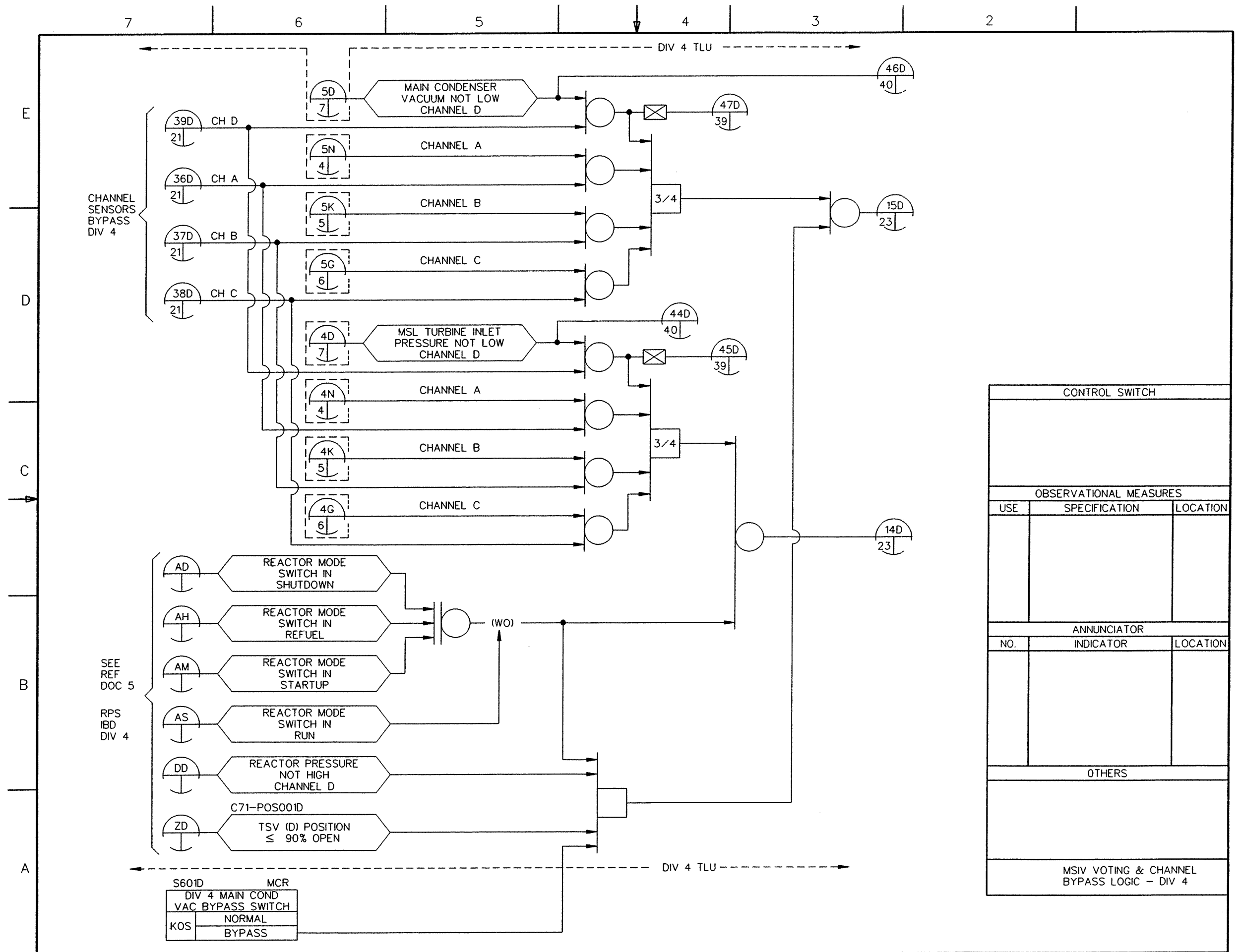


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 22 of 77)

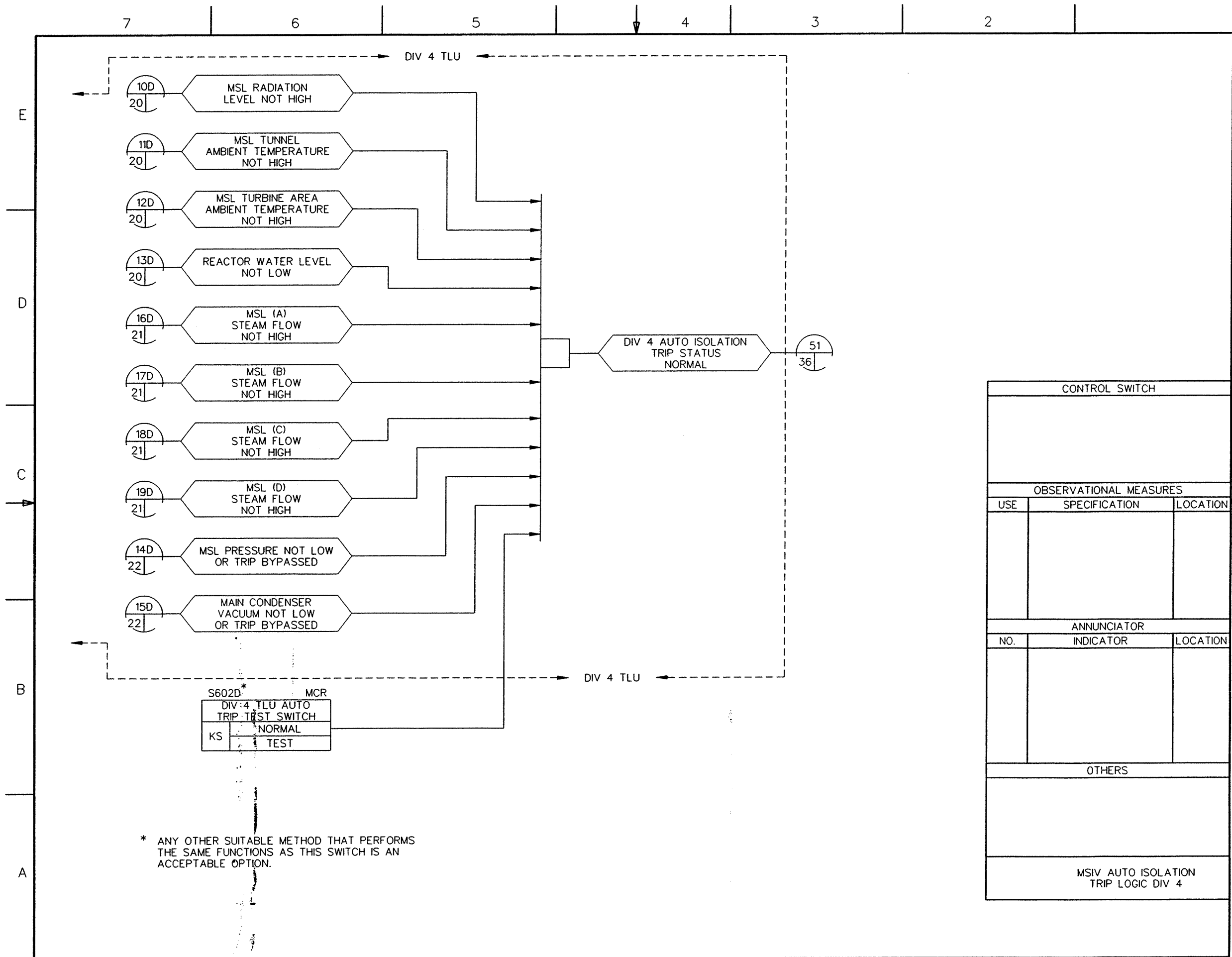


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 23 of 77)

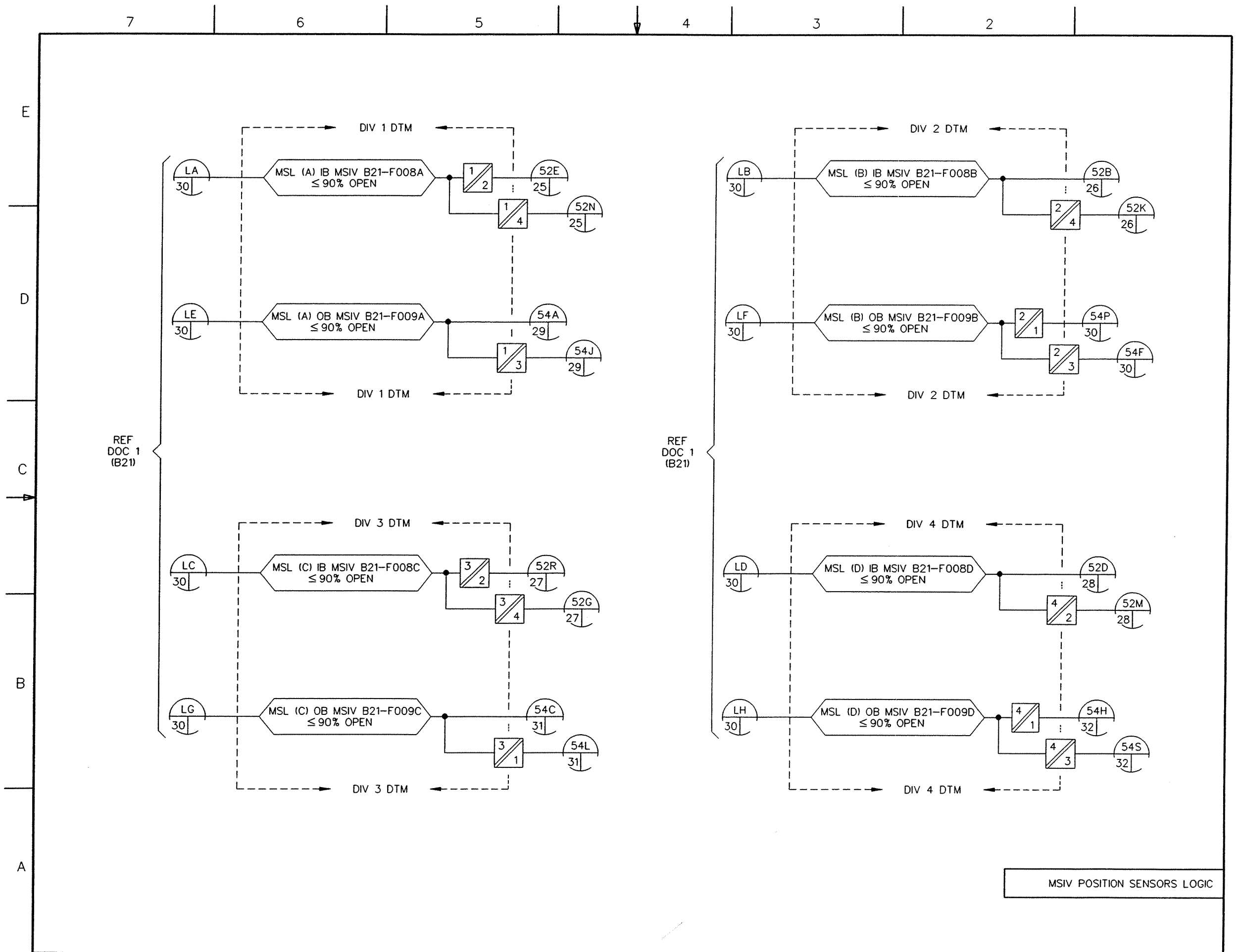
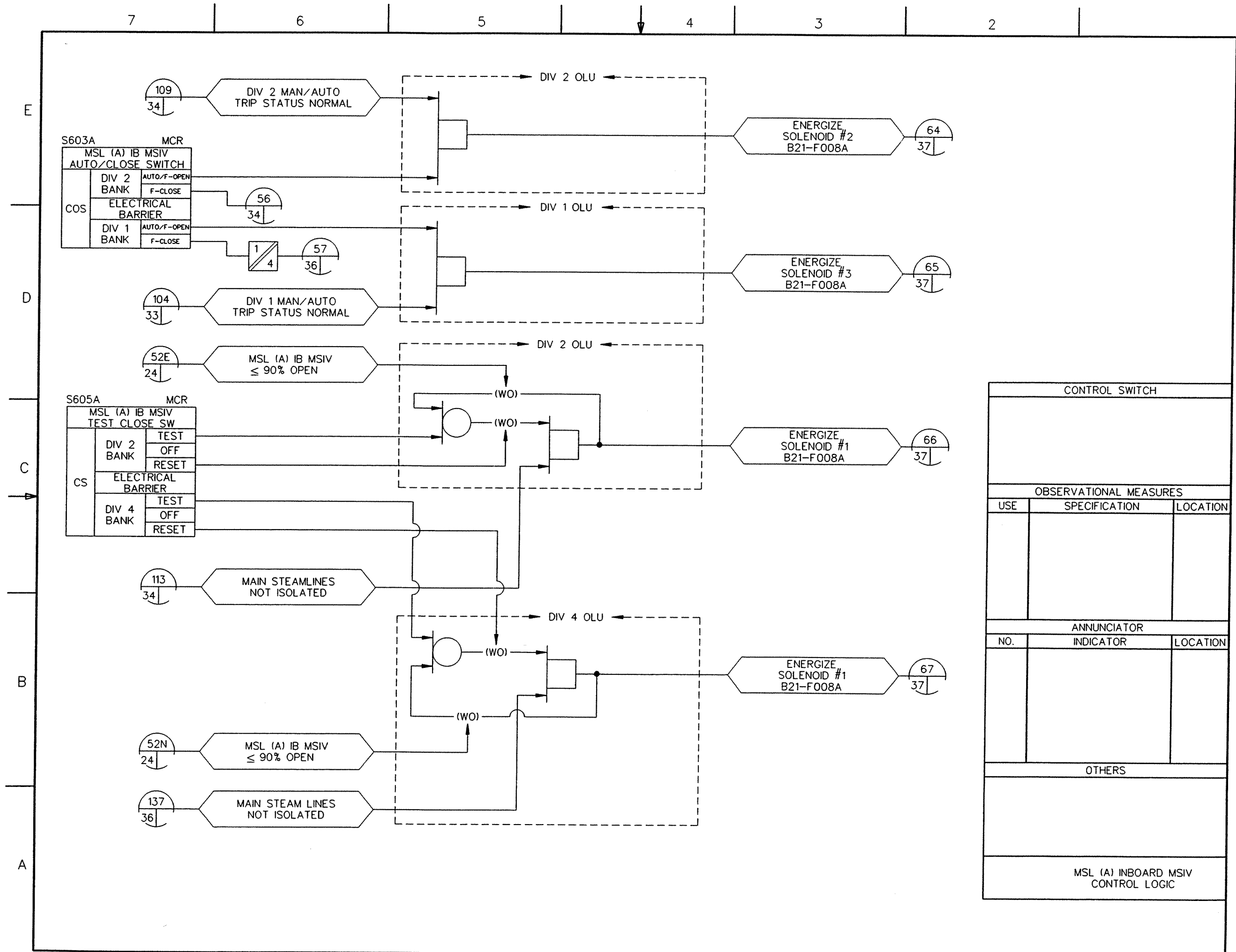
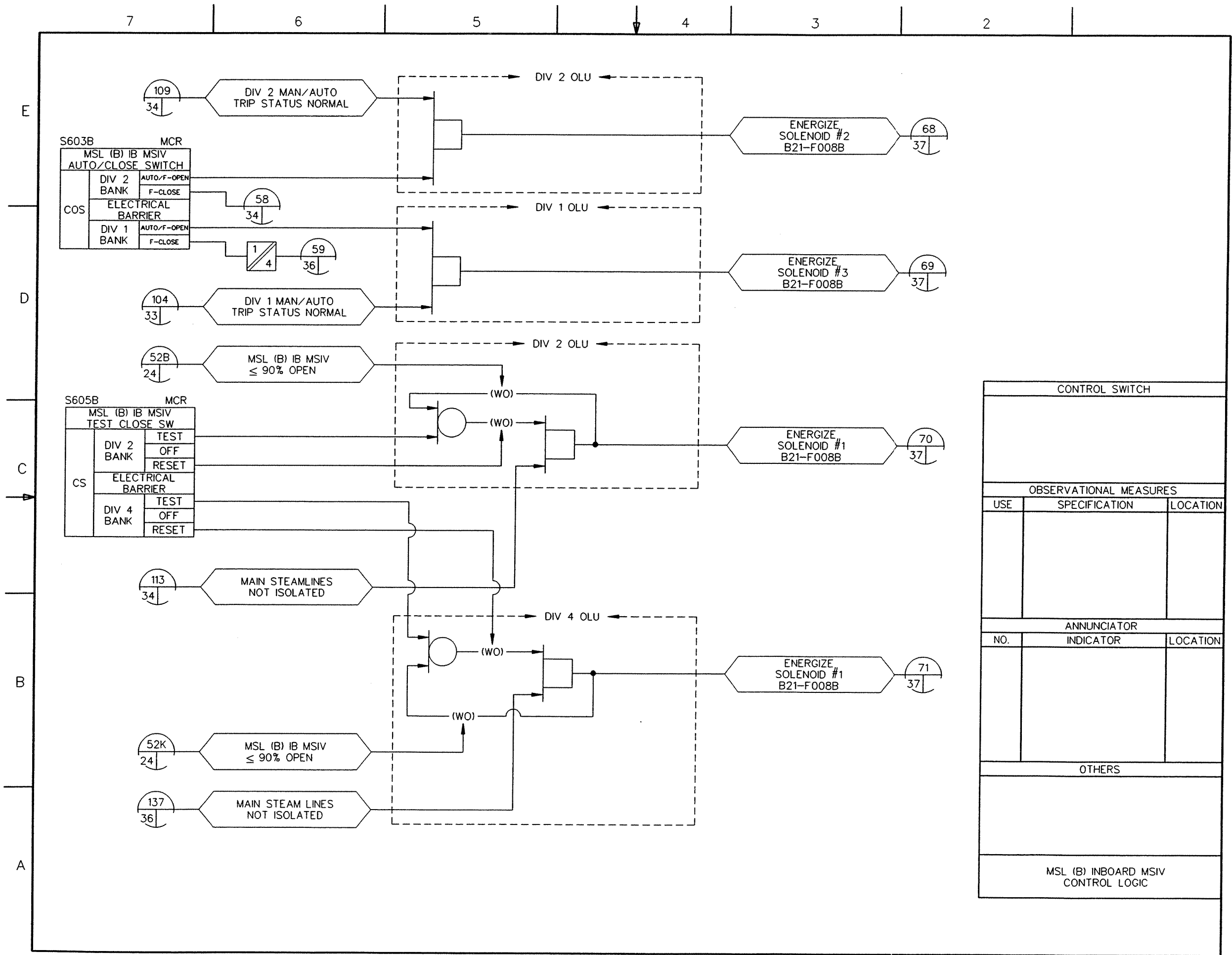


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 24 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.24



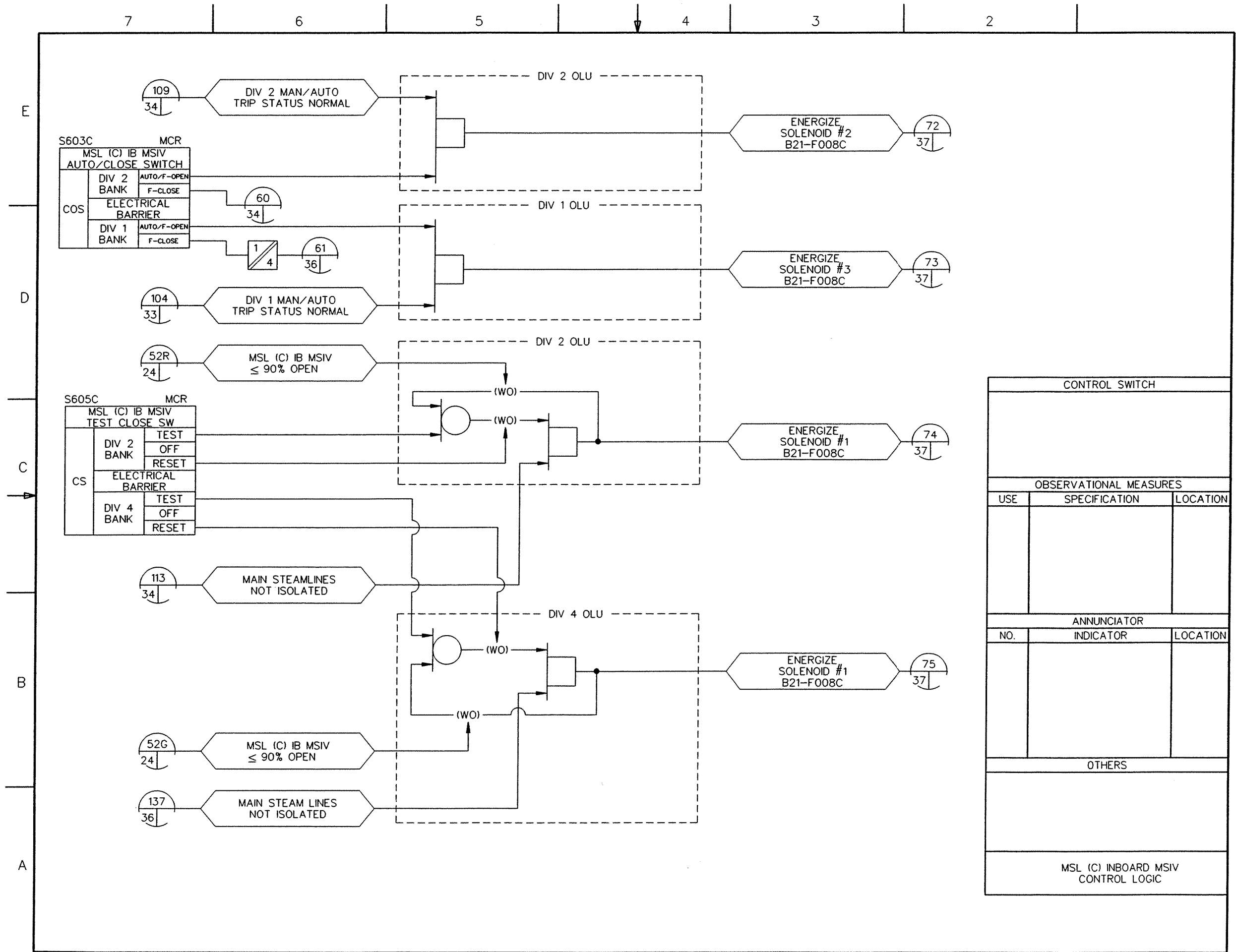
CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
MSL (A) INBOARD MSIV CONTROL LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 25 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.25



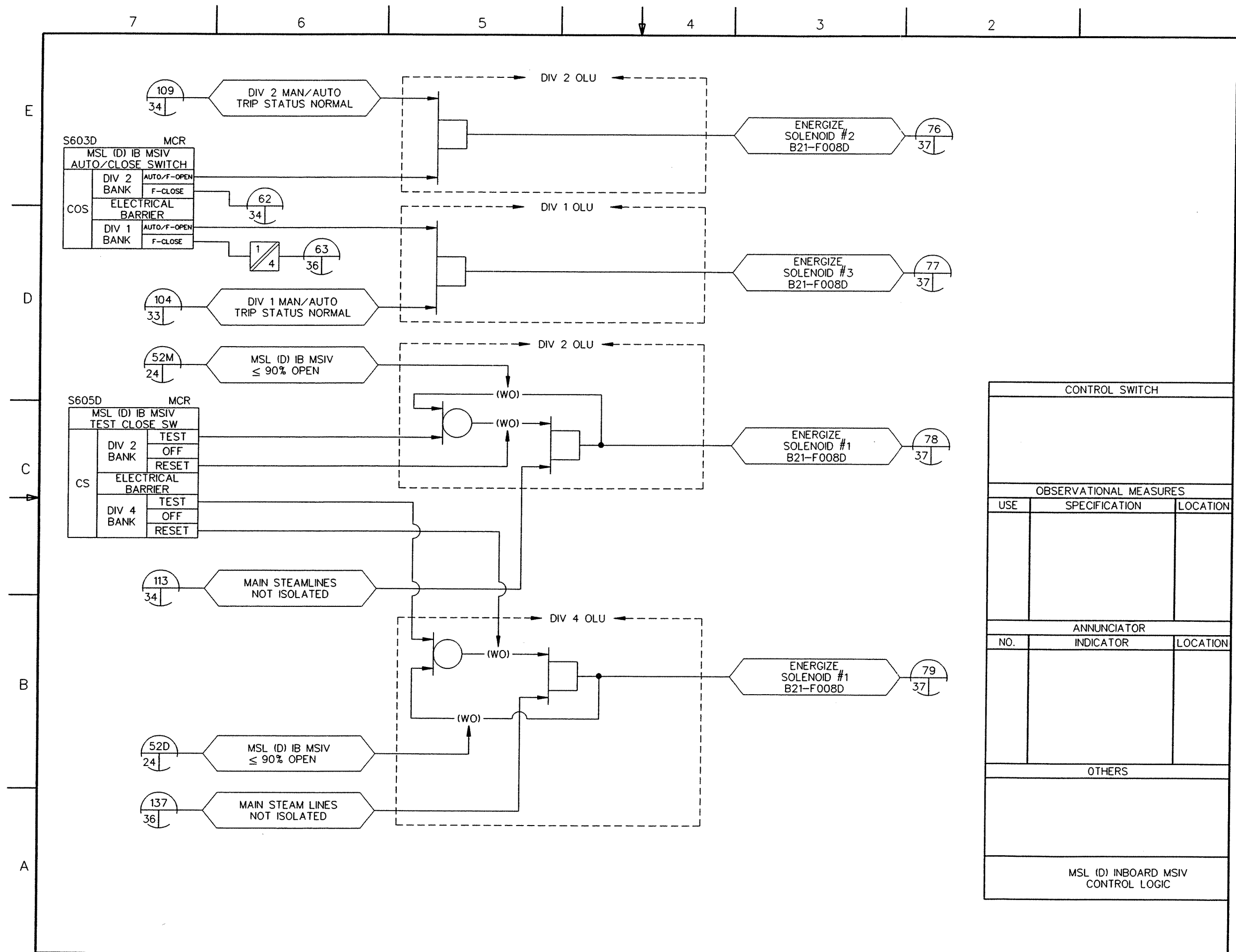
CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
MSL (B) INBOARD MSIV CONTROL LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 26 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.26



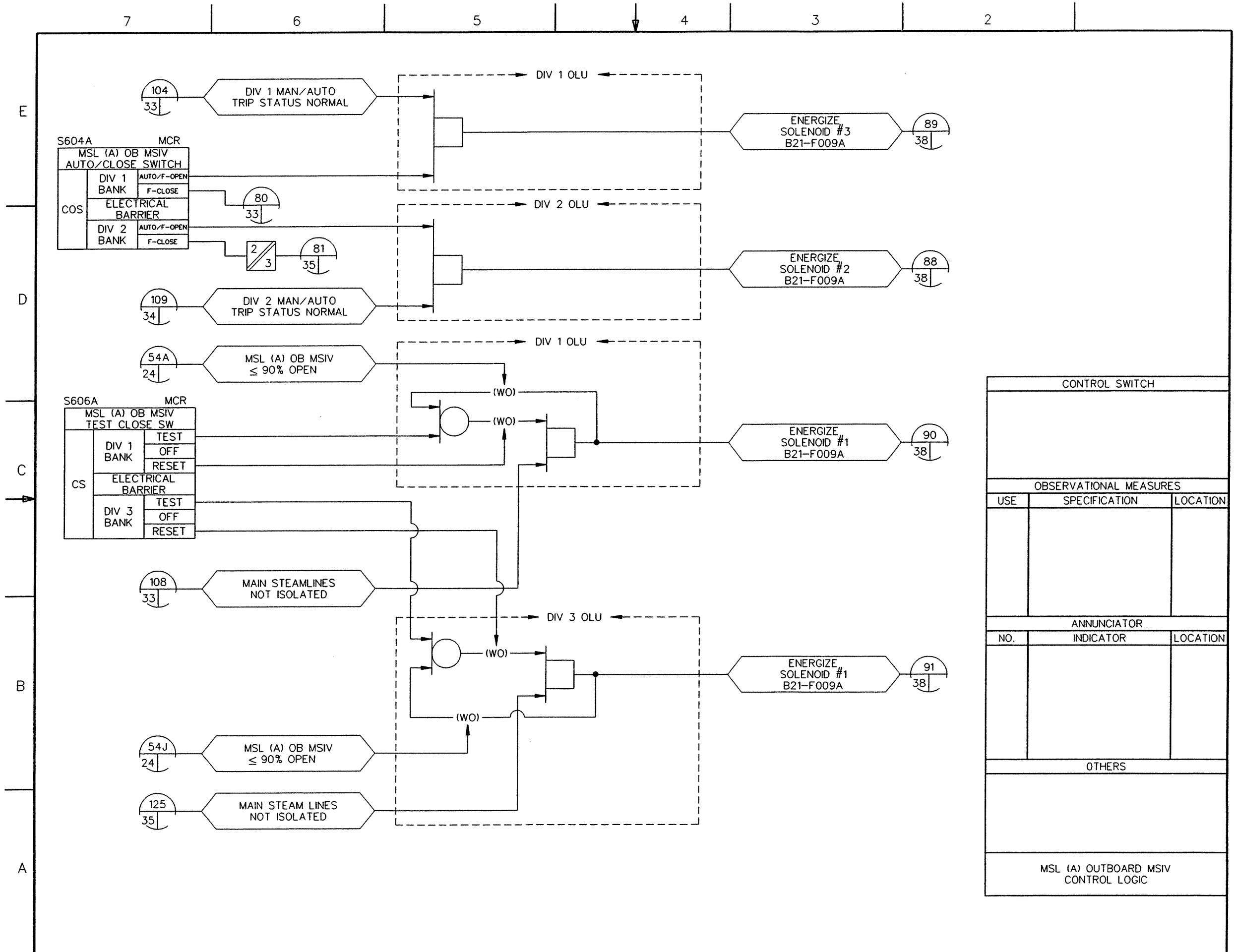
CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
MSL (C) INBOARD MSIV CONTROL LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 27 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.27



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
MSL (D) INBOARD MSIV CONTROL LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 28 of 77)



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
MSL (A) OUTBOARD MSIV CONTROL LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 29 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.29

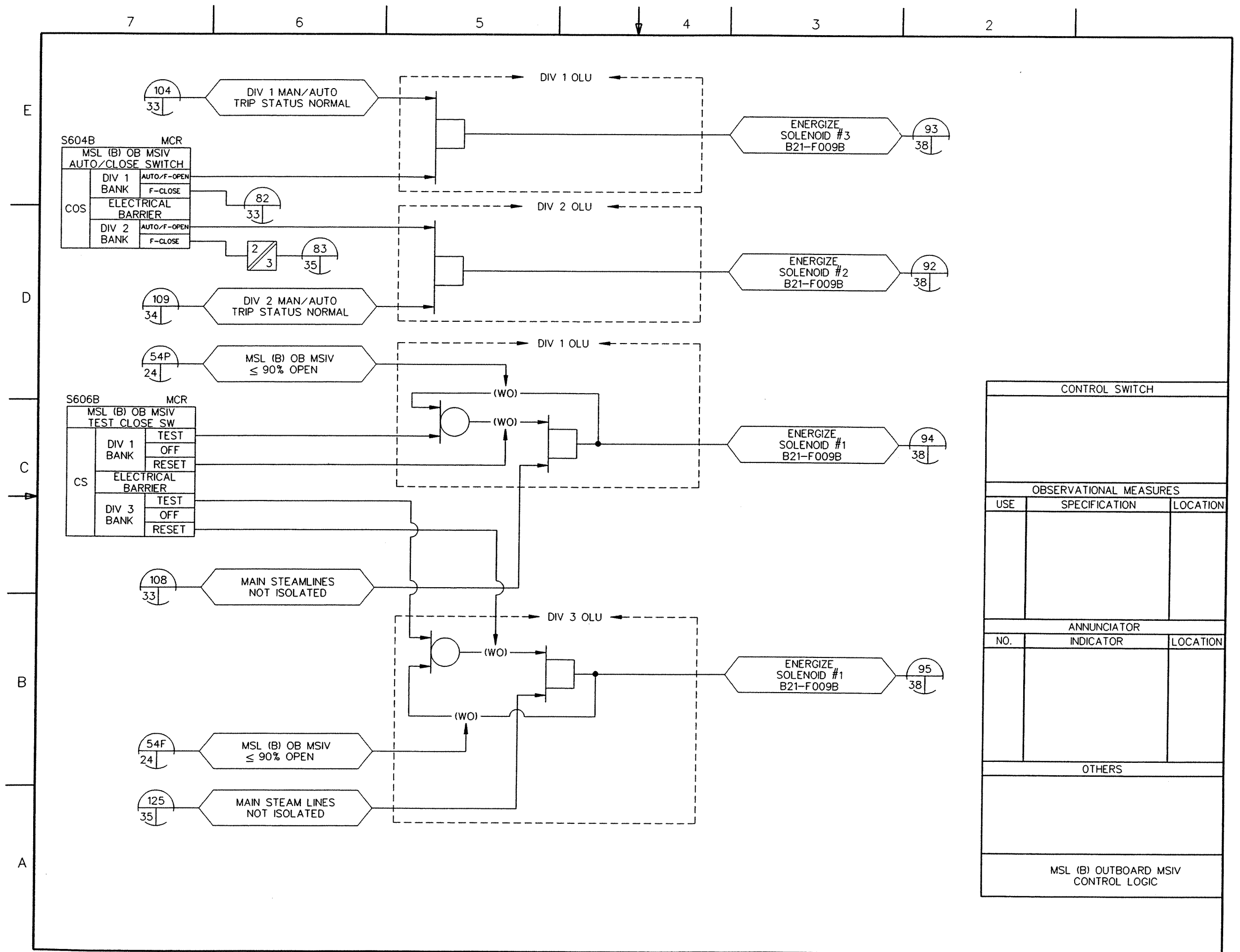
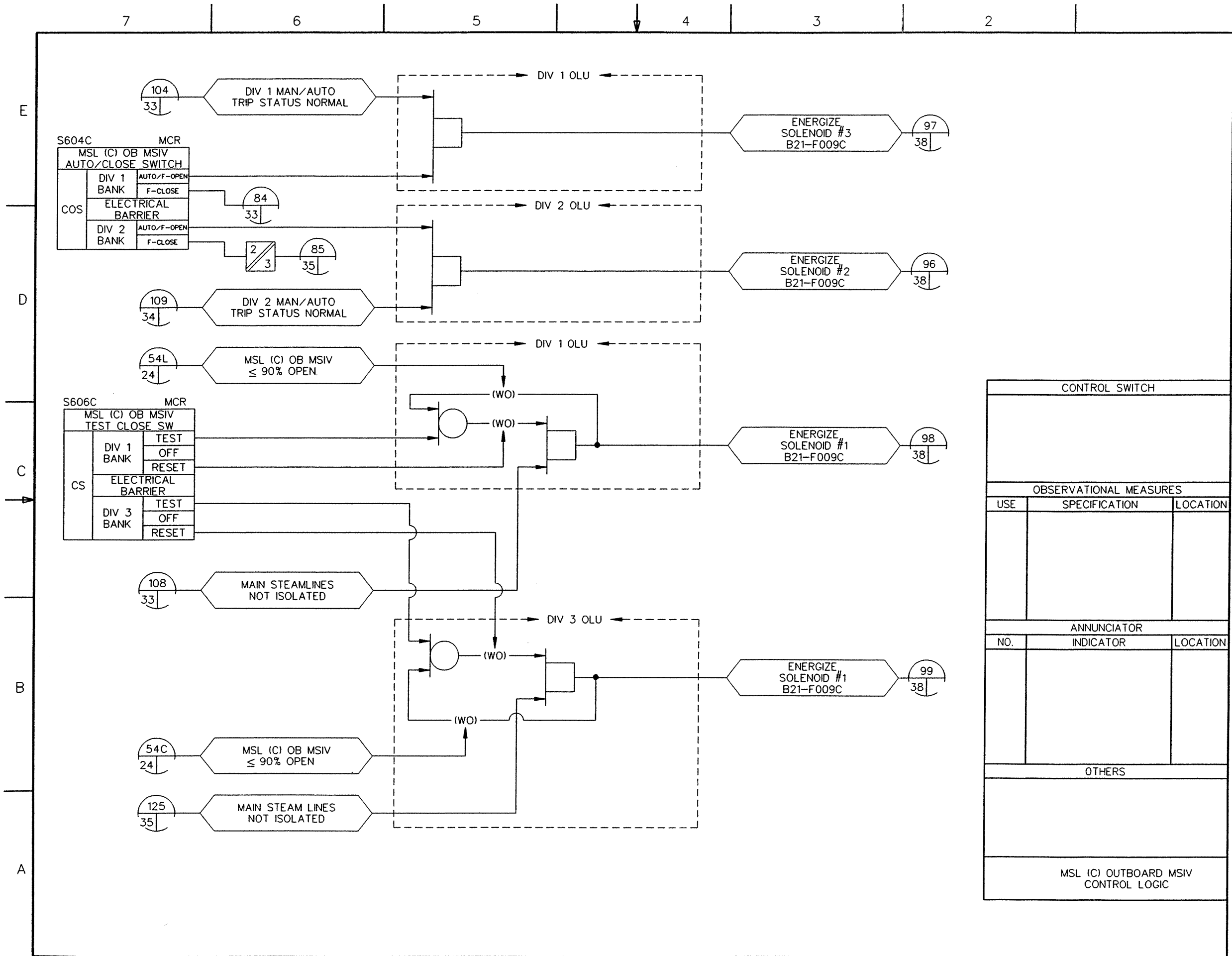


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 30 of 77)



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
MSL (C) OUTBOARD MSIV CONTROL LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 31 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.31

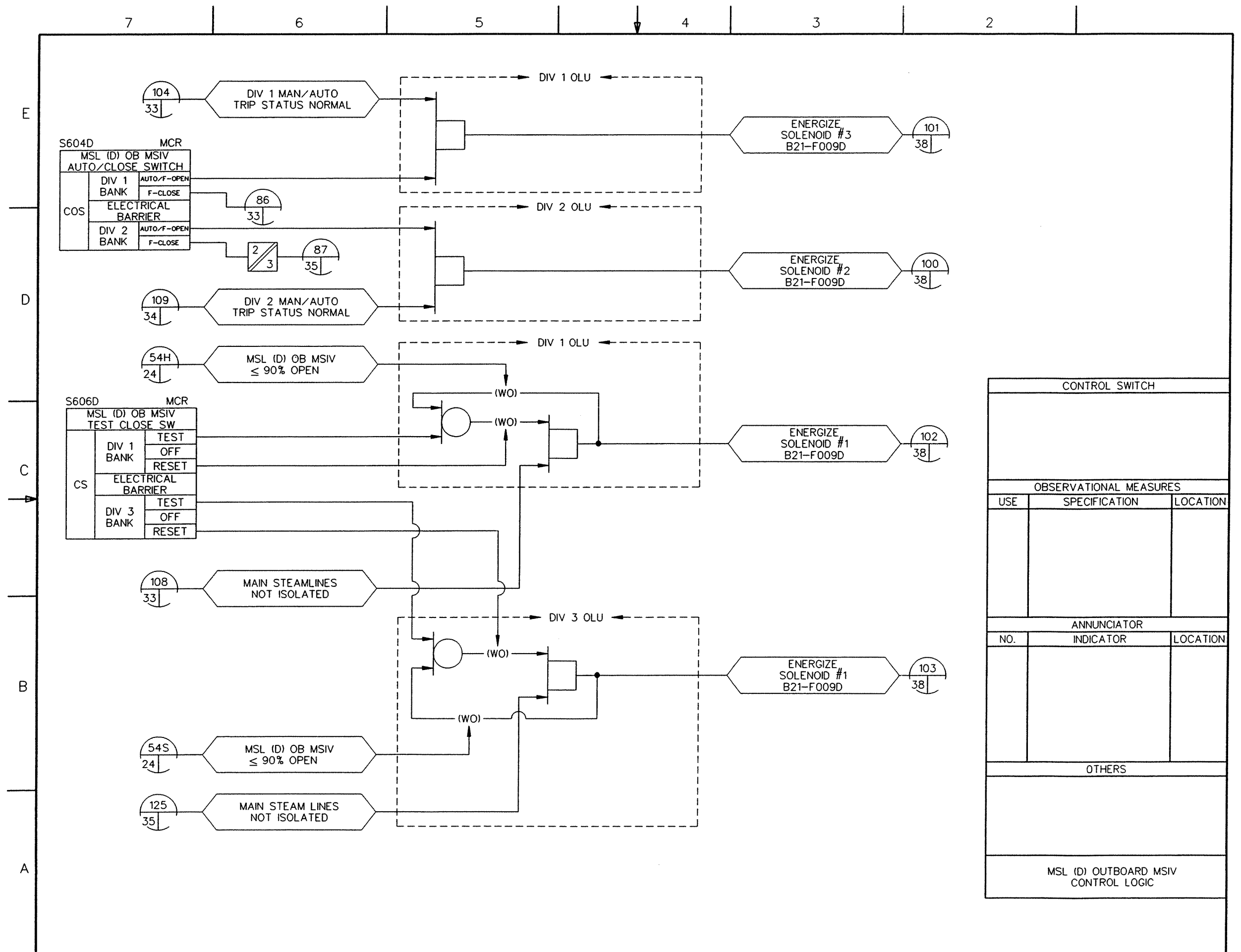


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 32 of 77)

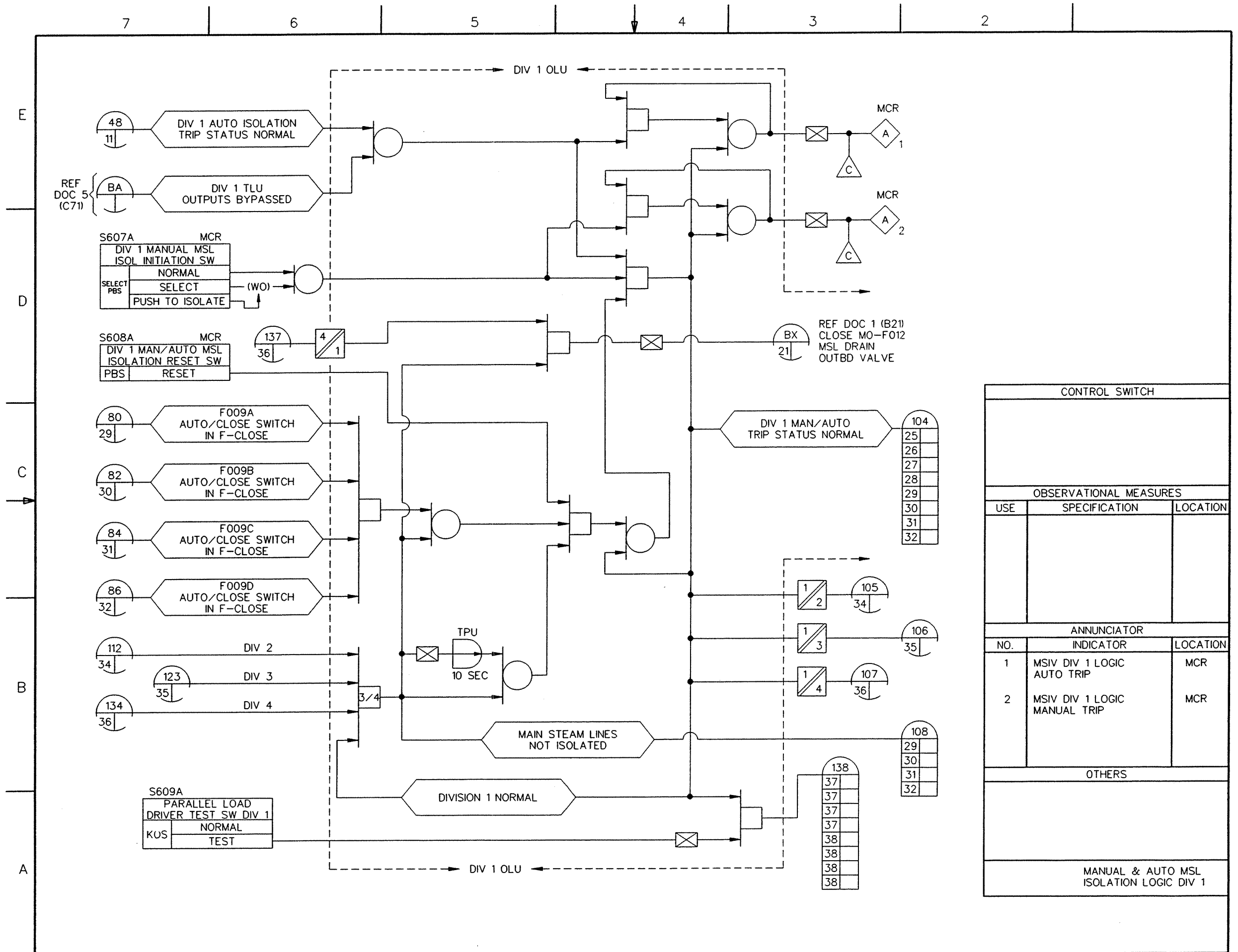


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 33 of 77)

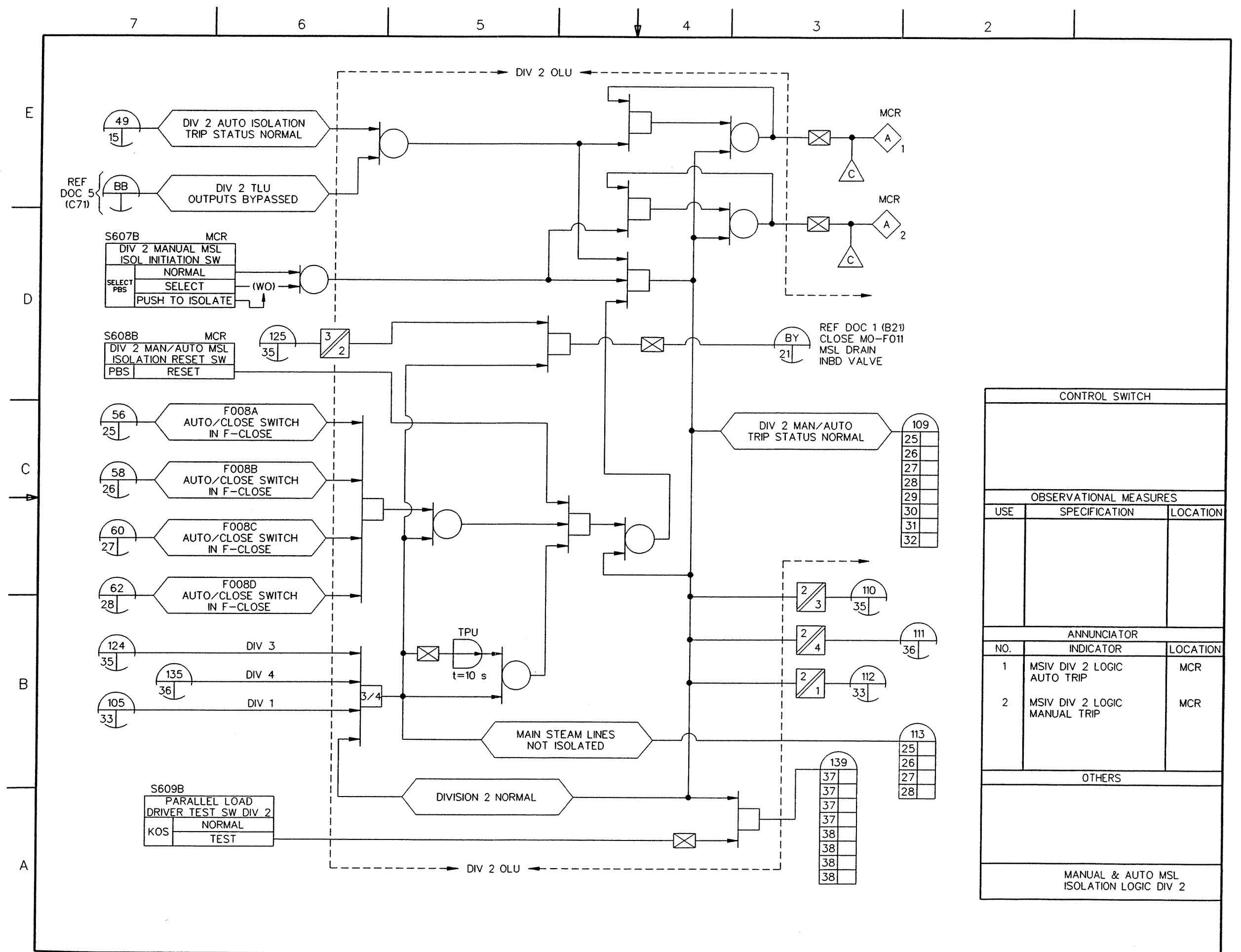


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 34 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.34

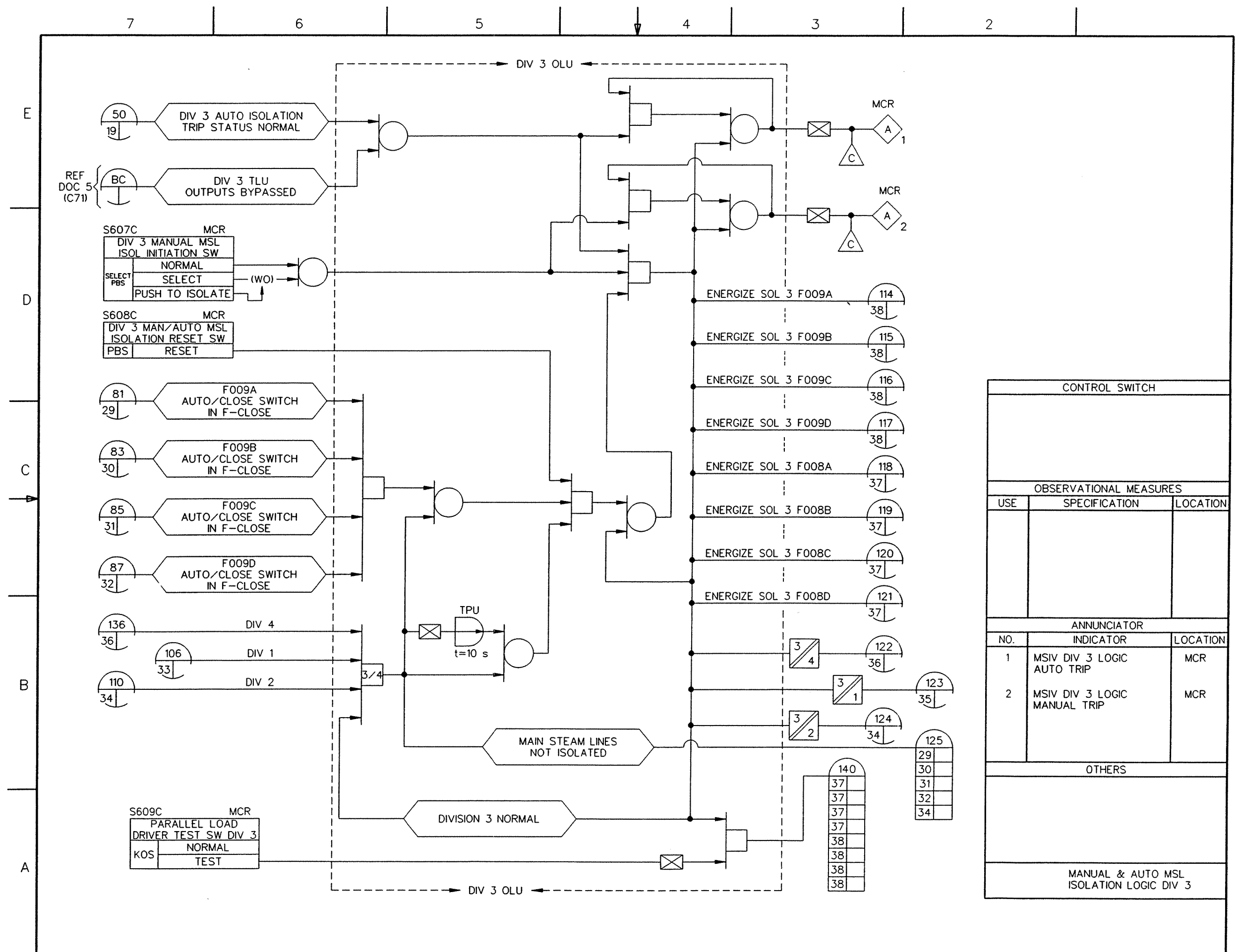


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 35 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.35

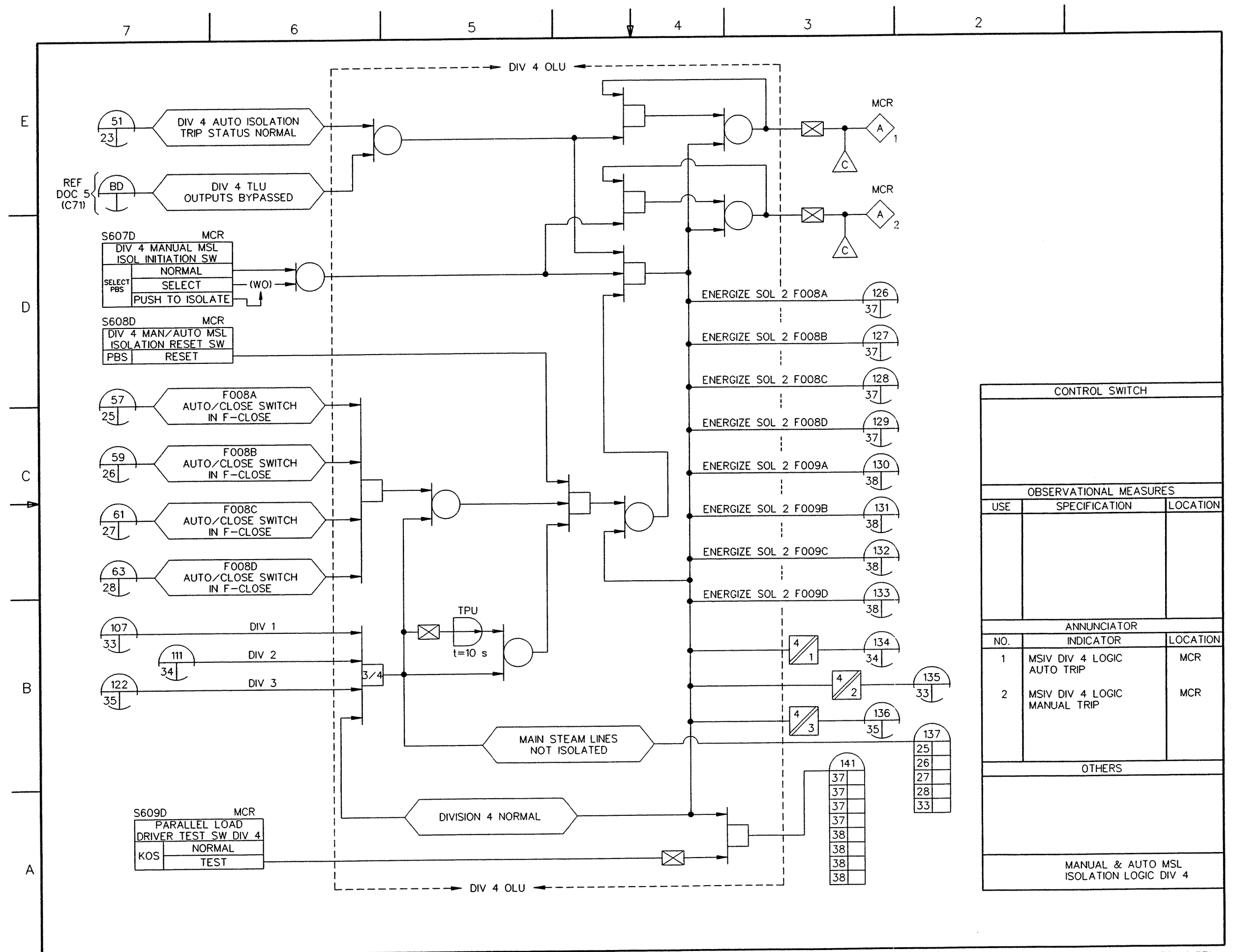


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 36 of 77)
 ABWR DCD/Tier 2
 Rev. 0
 21-228.36

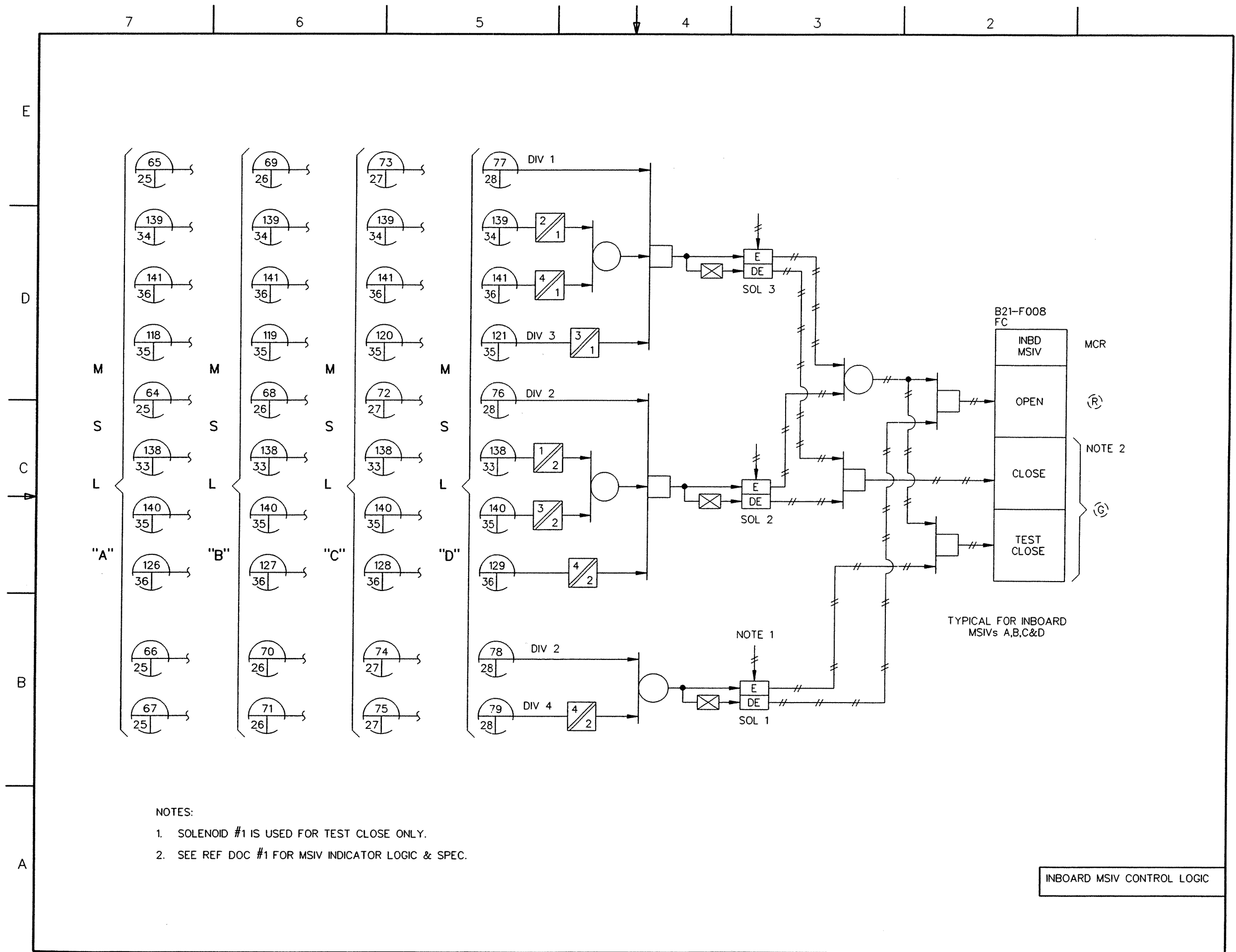


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 37 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.37

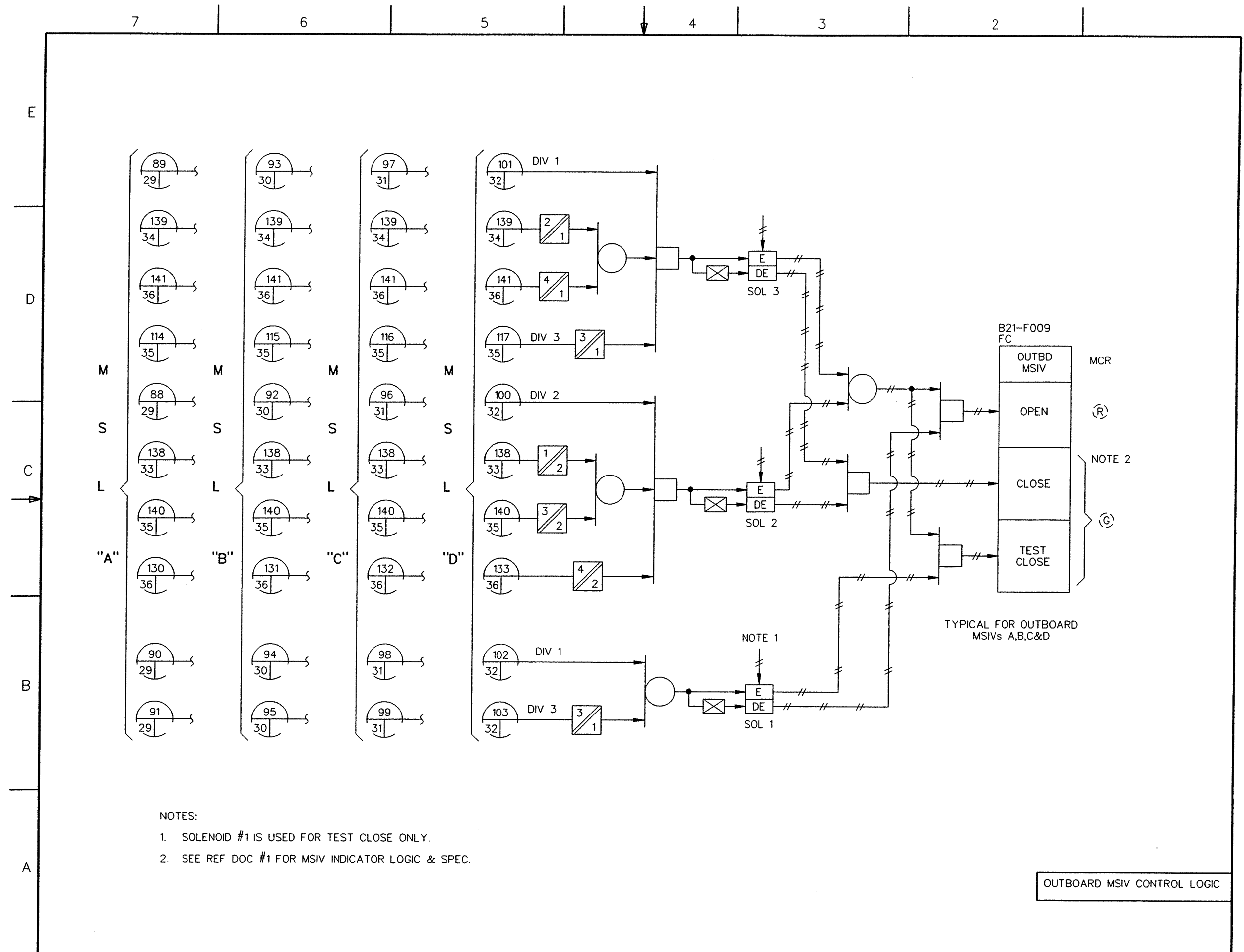


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 38 of 77)

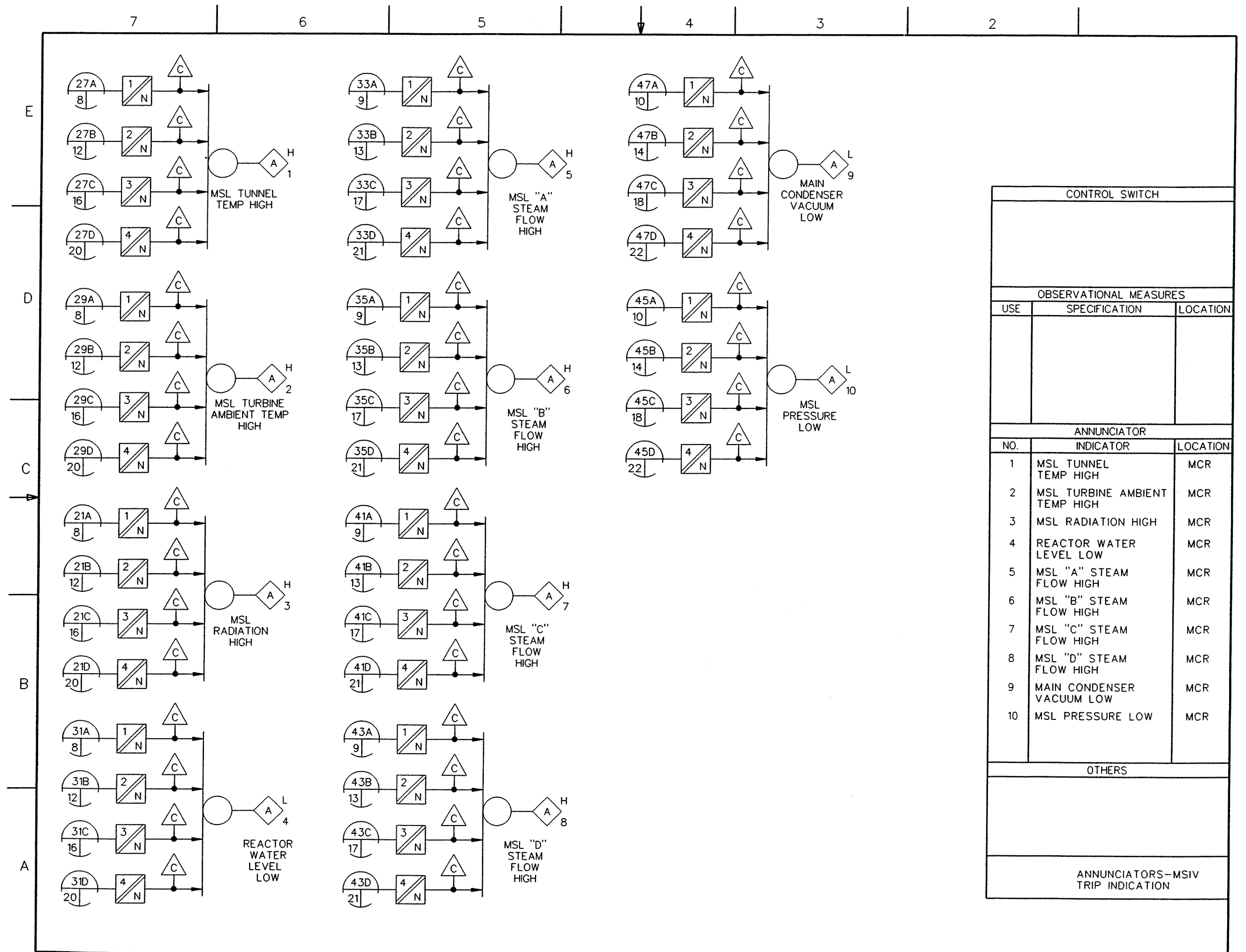
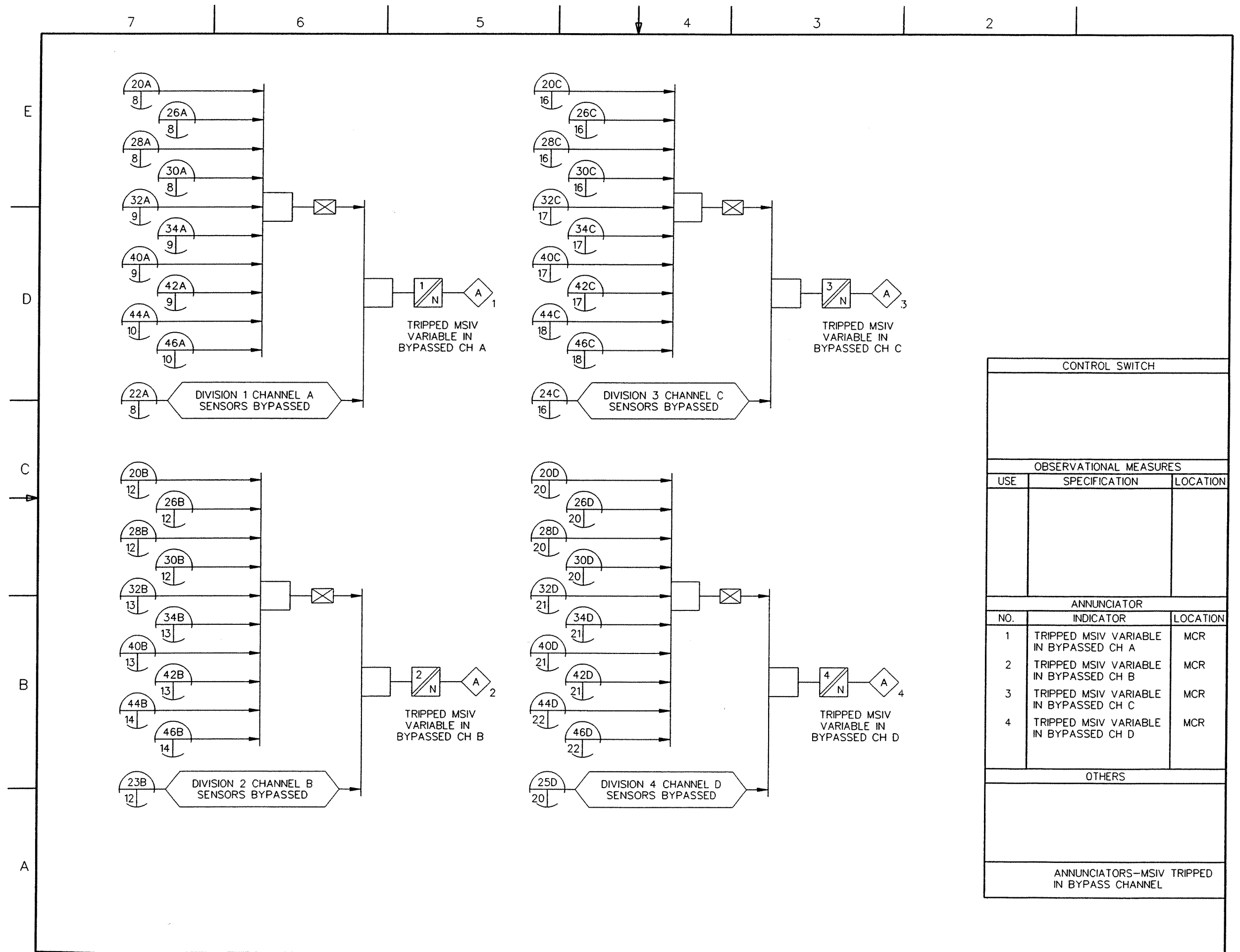


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 39 of 77)



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	TRIPPED MSIV VARIABLE IN BYPASSED CH A	MCR
2	TRIPPED MSIV VARIABLE IN BYPASSED CH B	MCR
3	TRIPPED MSIV VARIABLE IN BYPASSED CH C	MCR
4	TRIPPED MSIV VARIABLE IN BYPASSED CH D	MCR
OTHERS		
ANNUNCIATORS-MSIV TRIPPED IN BYPASS CHANNEL		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 40 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.40

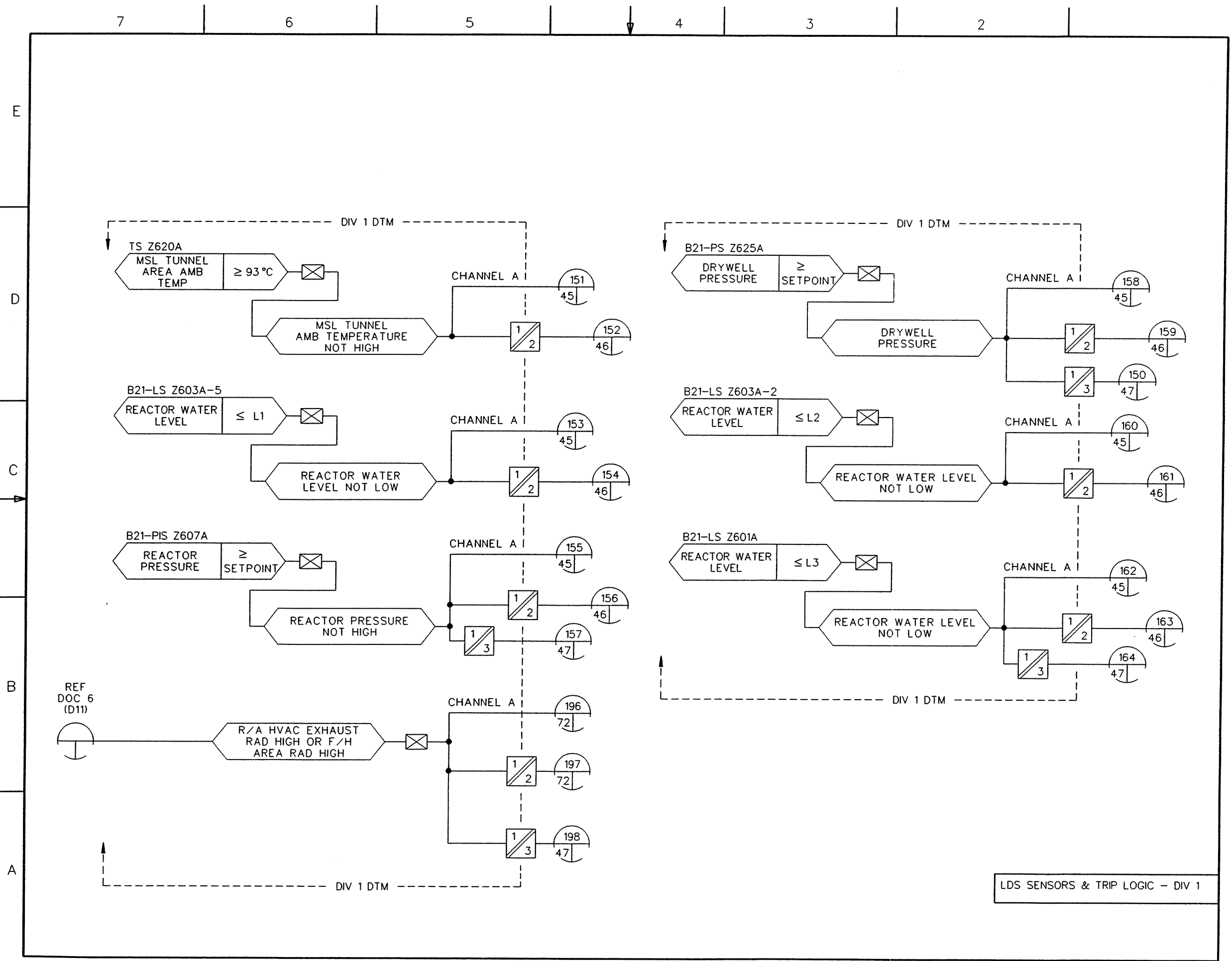


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 41 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.41

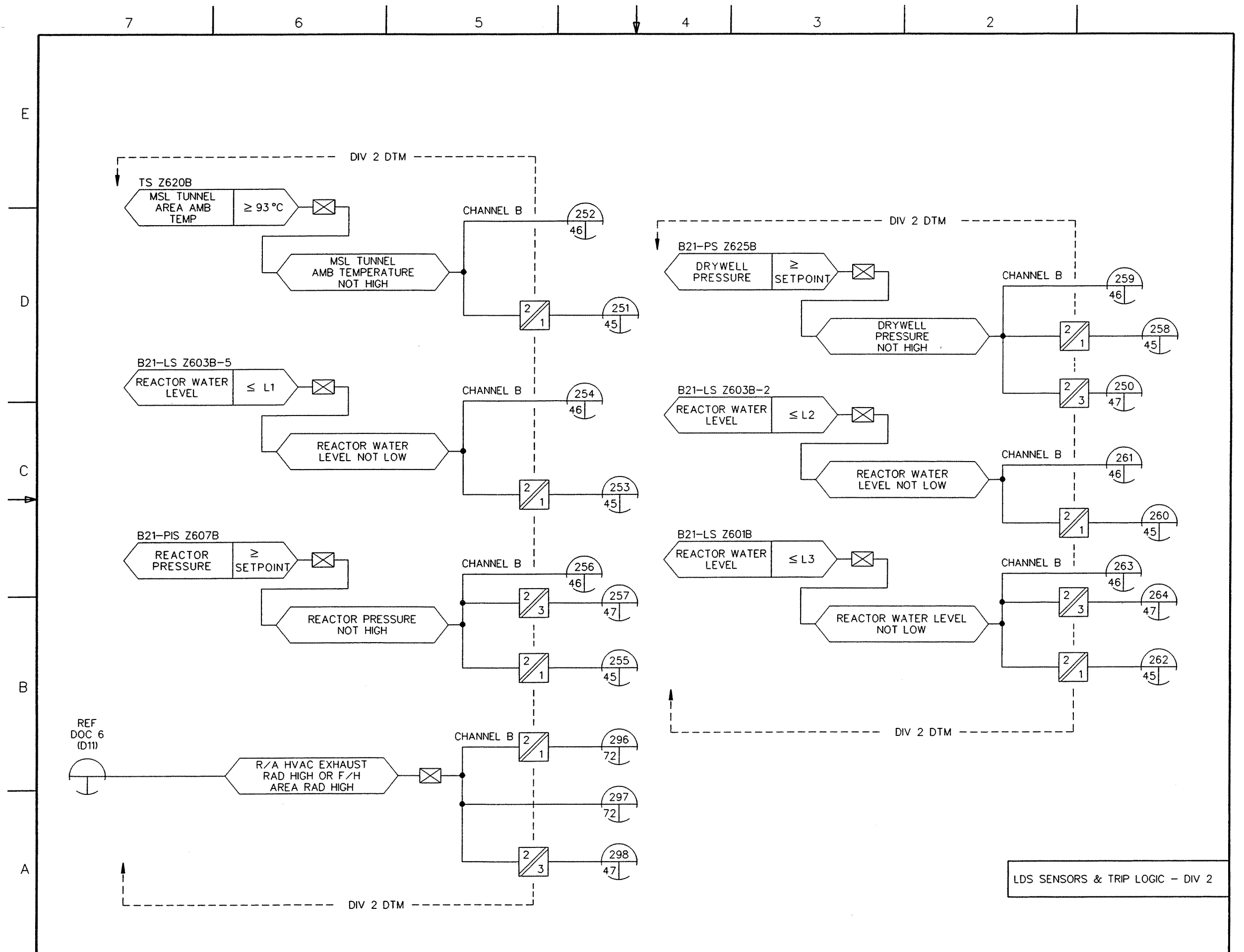


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 42 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.42

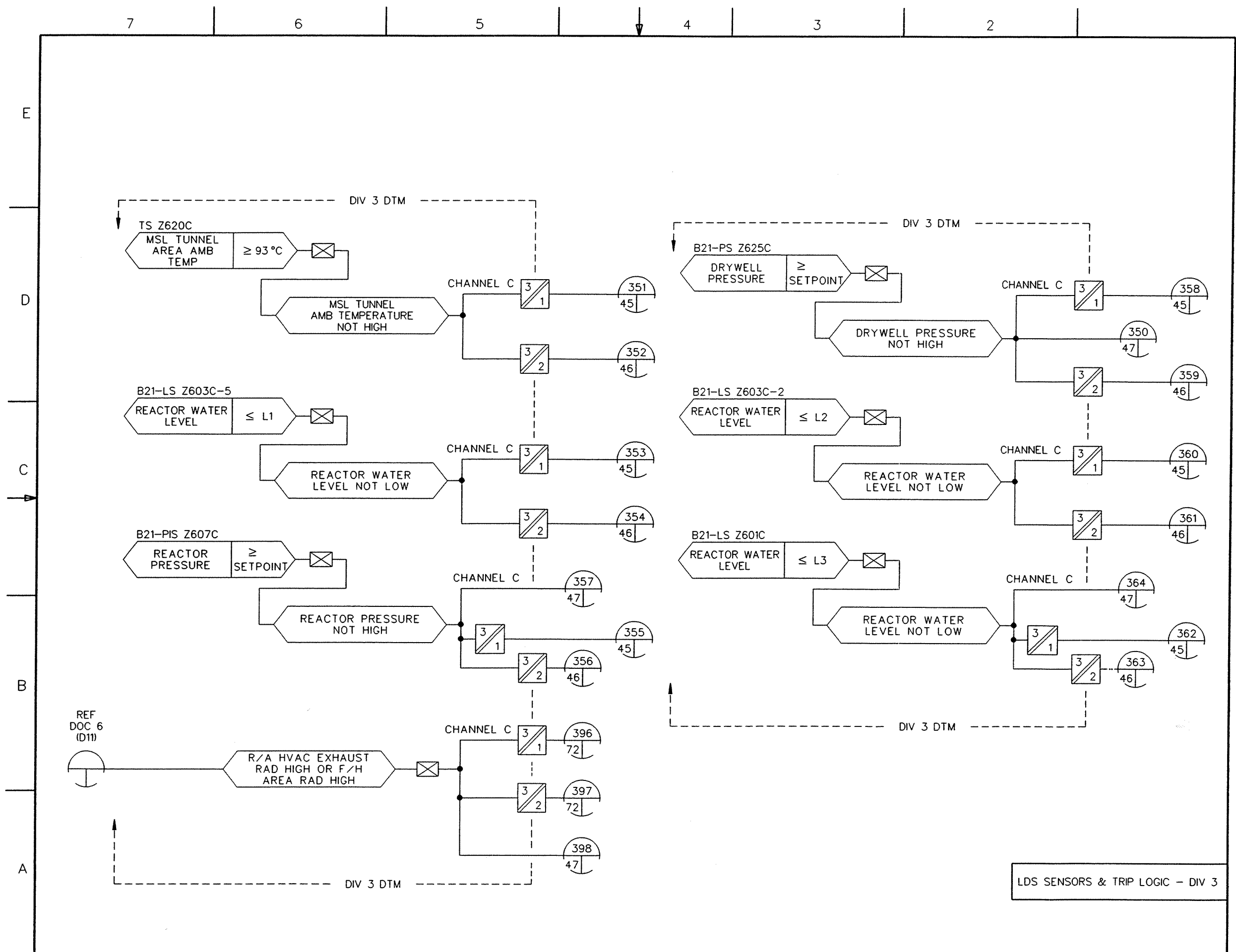


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 43 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.43

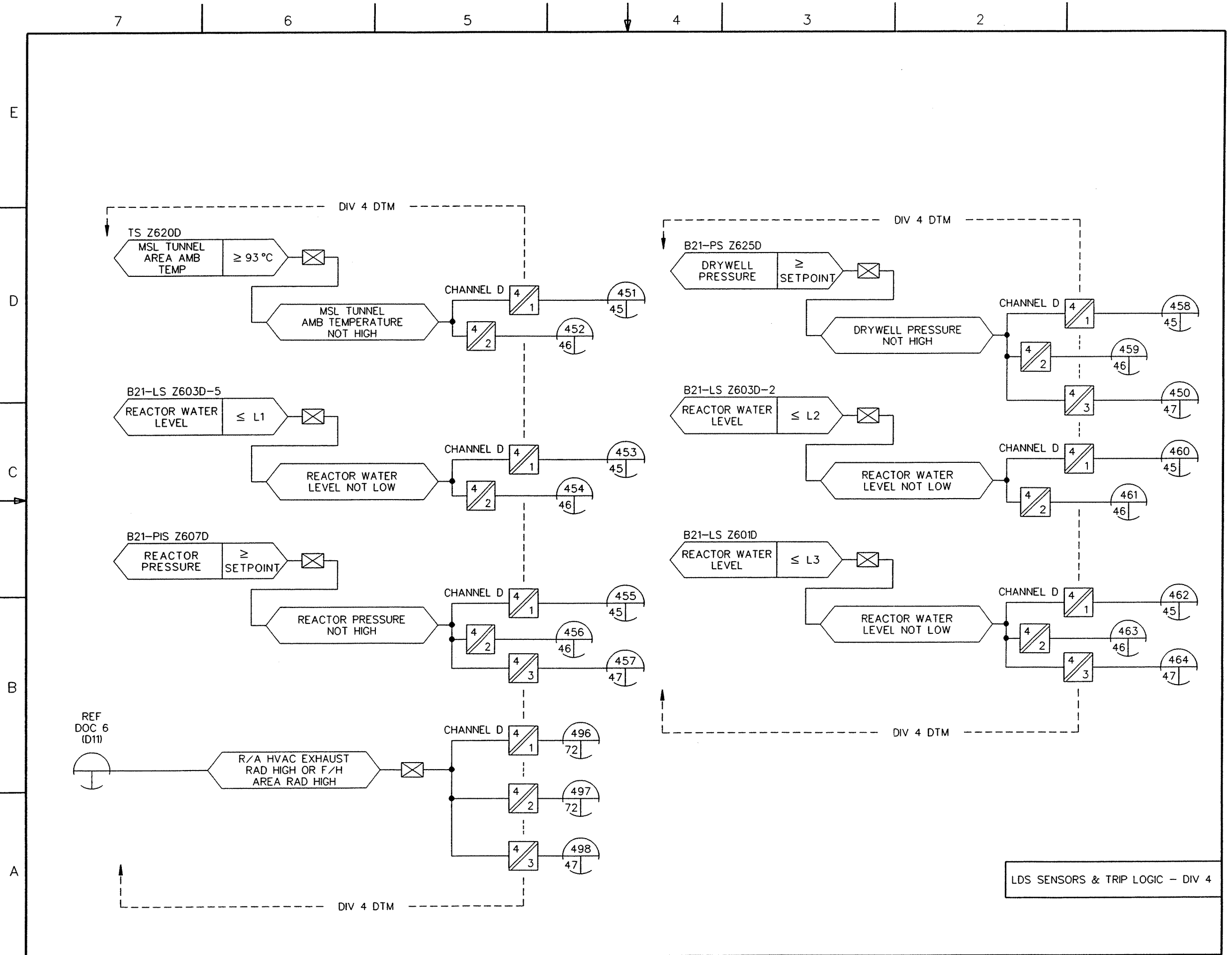


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 44 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.44

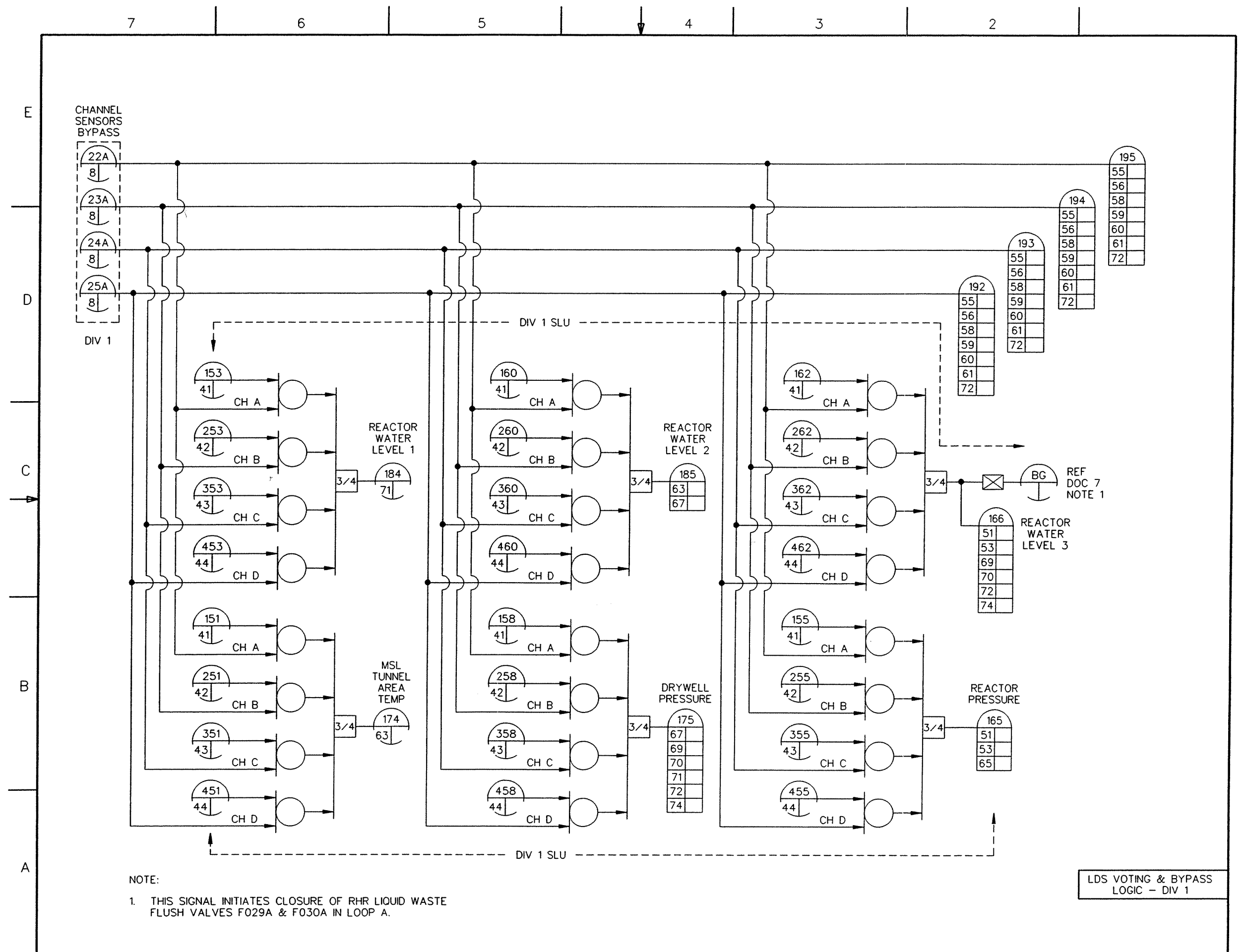


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 45 of 77)

ABWR DCD/Tier 2

Rev. 0

21-228.45

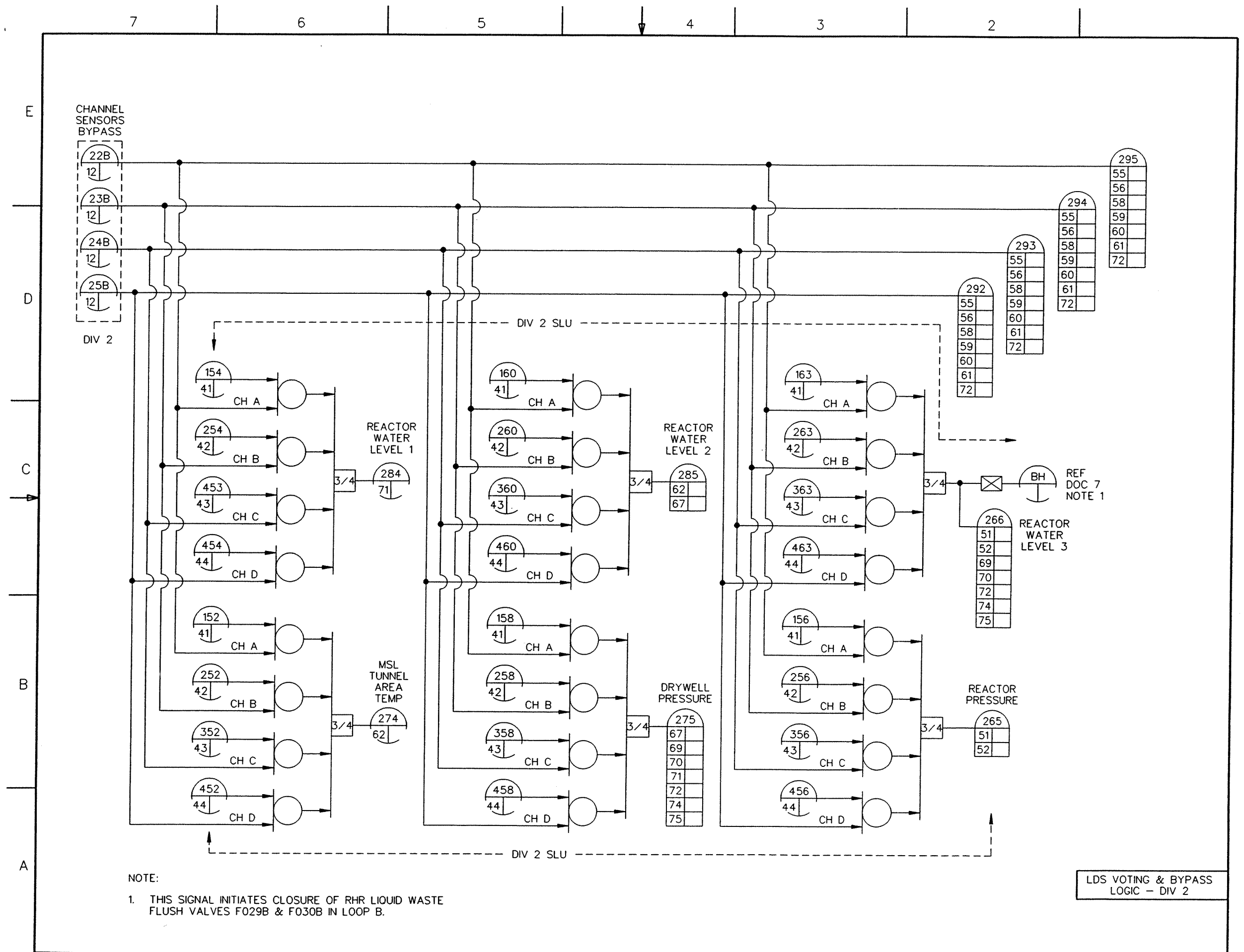


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 46 of 77)

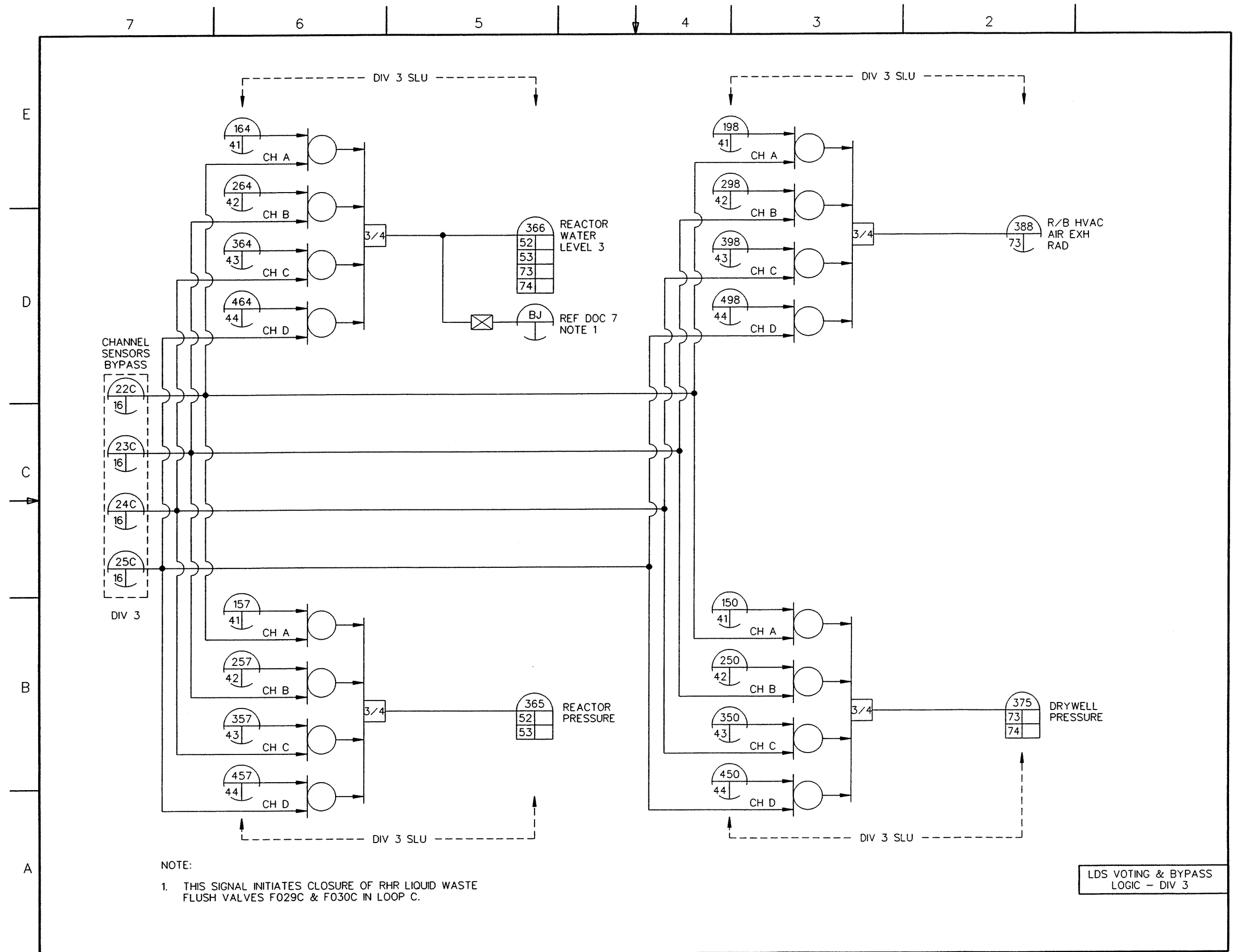


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 47 of 77)

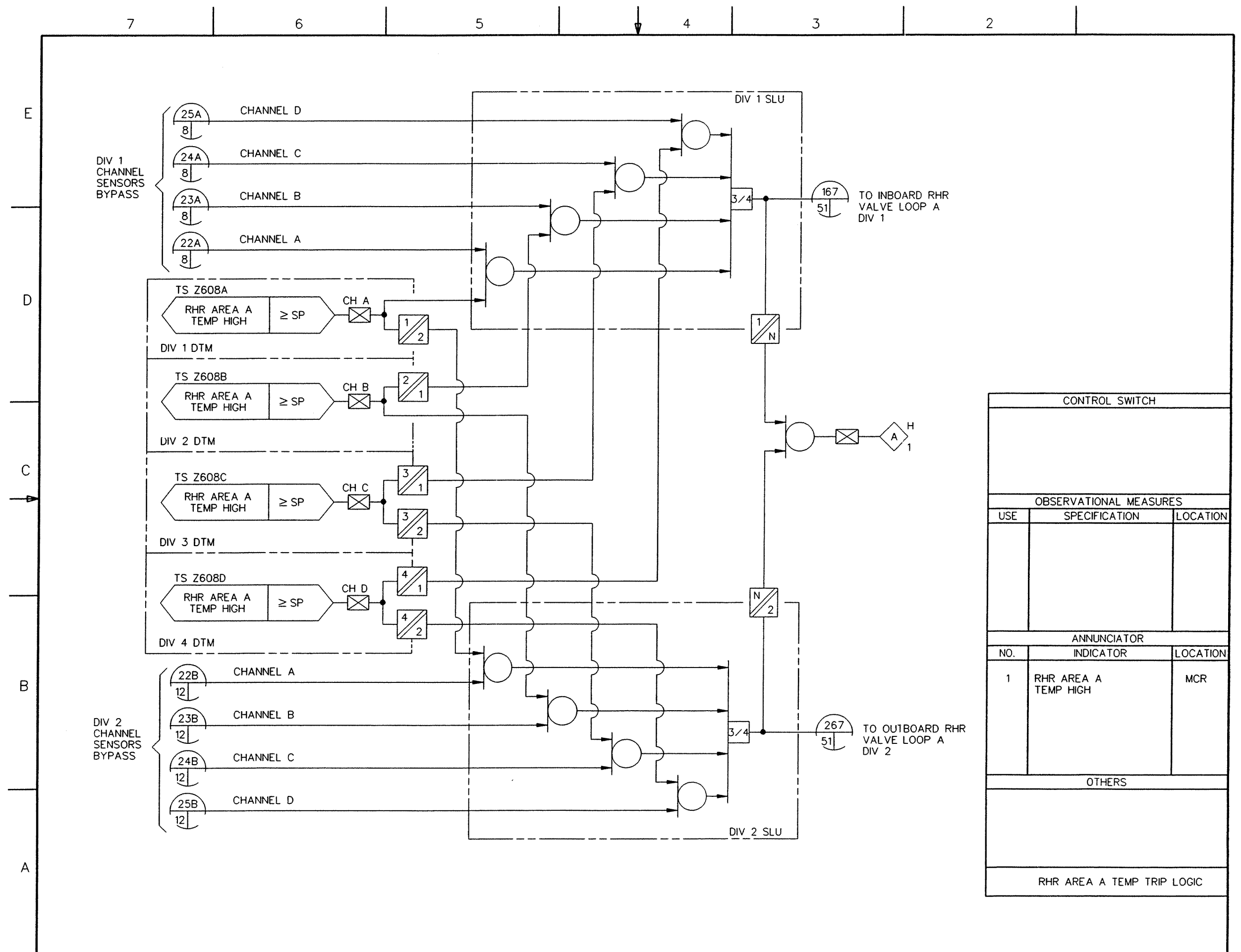


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 48 of 77)

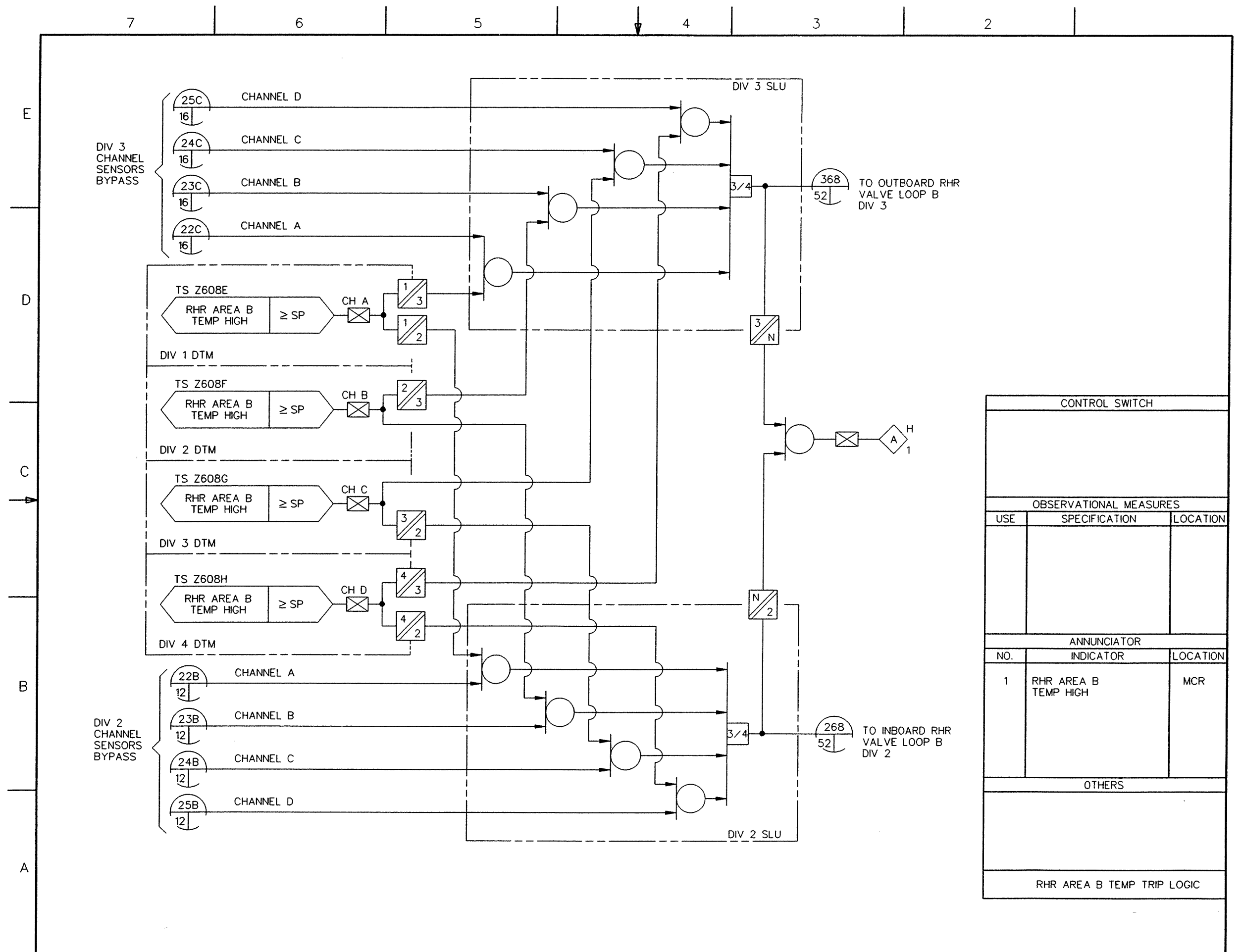
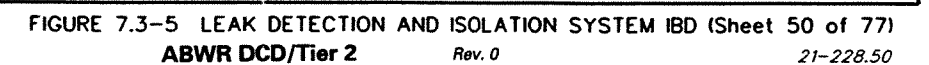


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 49 of 77)



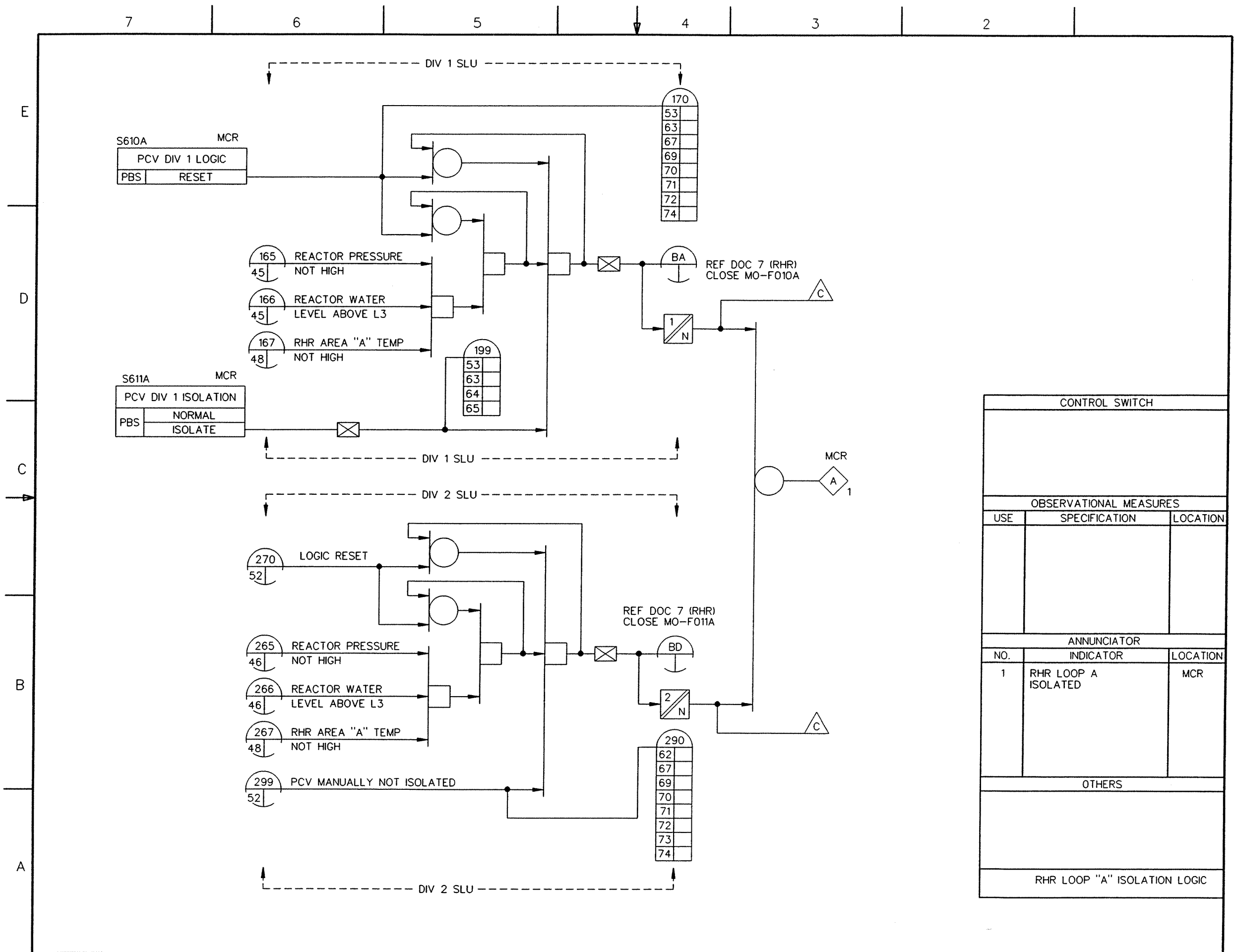


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 51 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.51

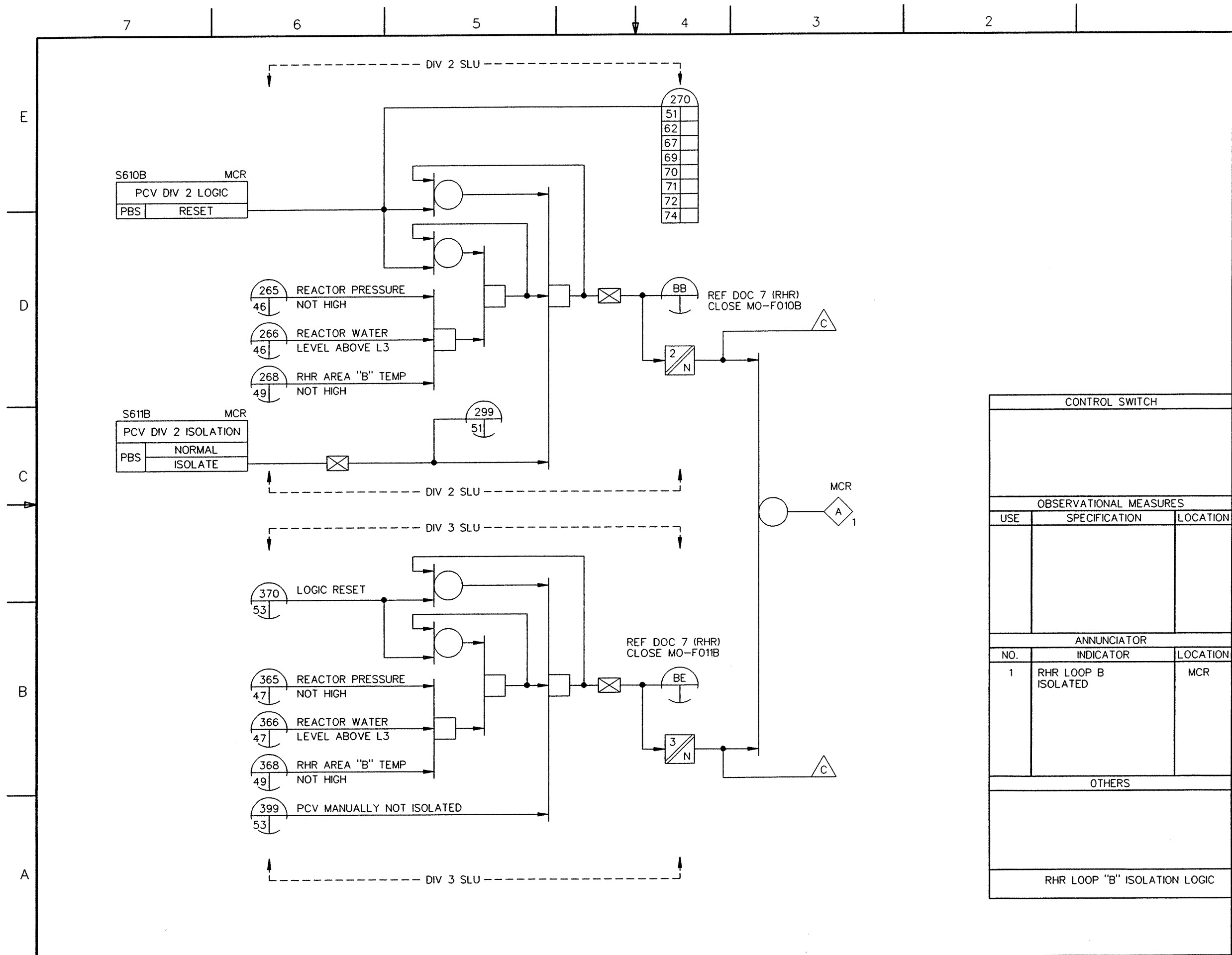


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 52 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.52

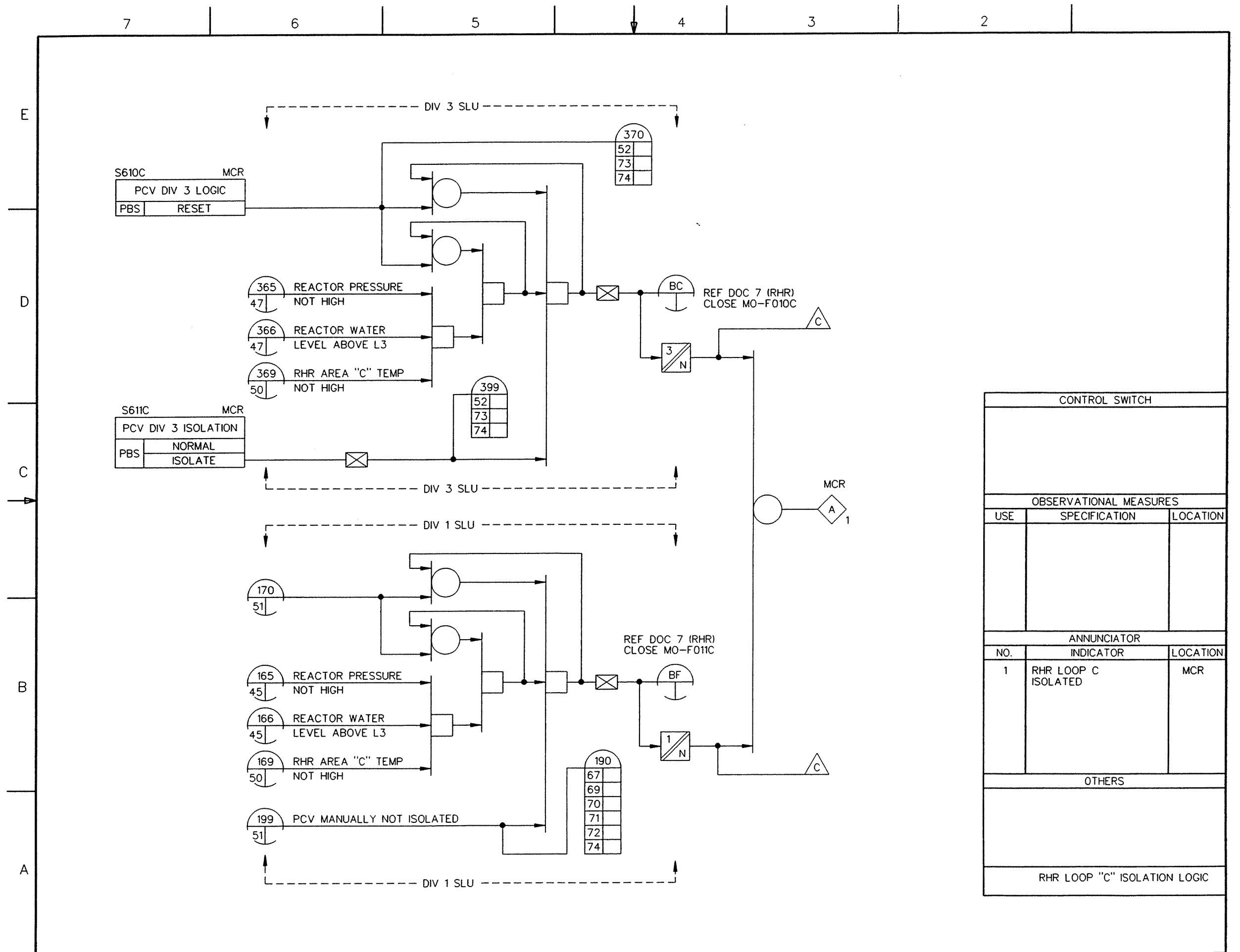


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 53 of 77)

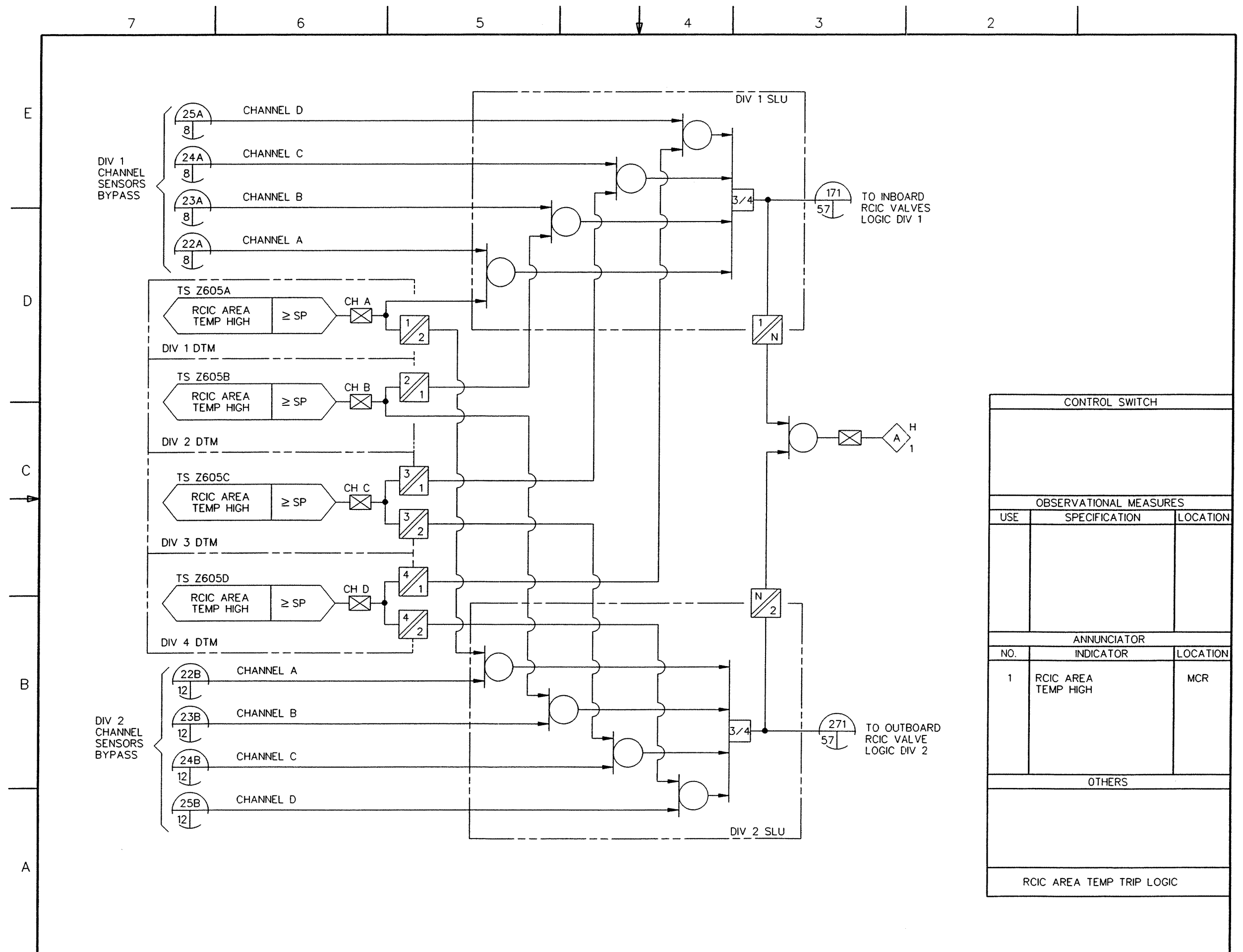
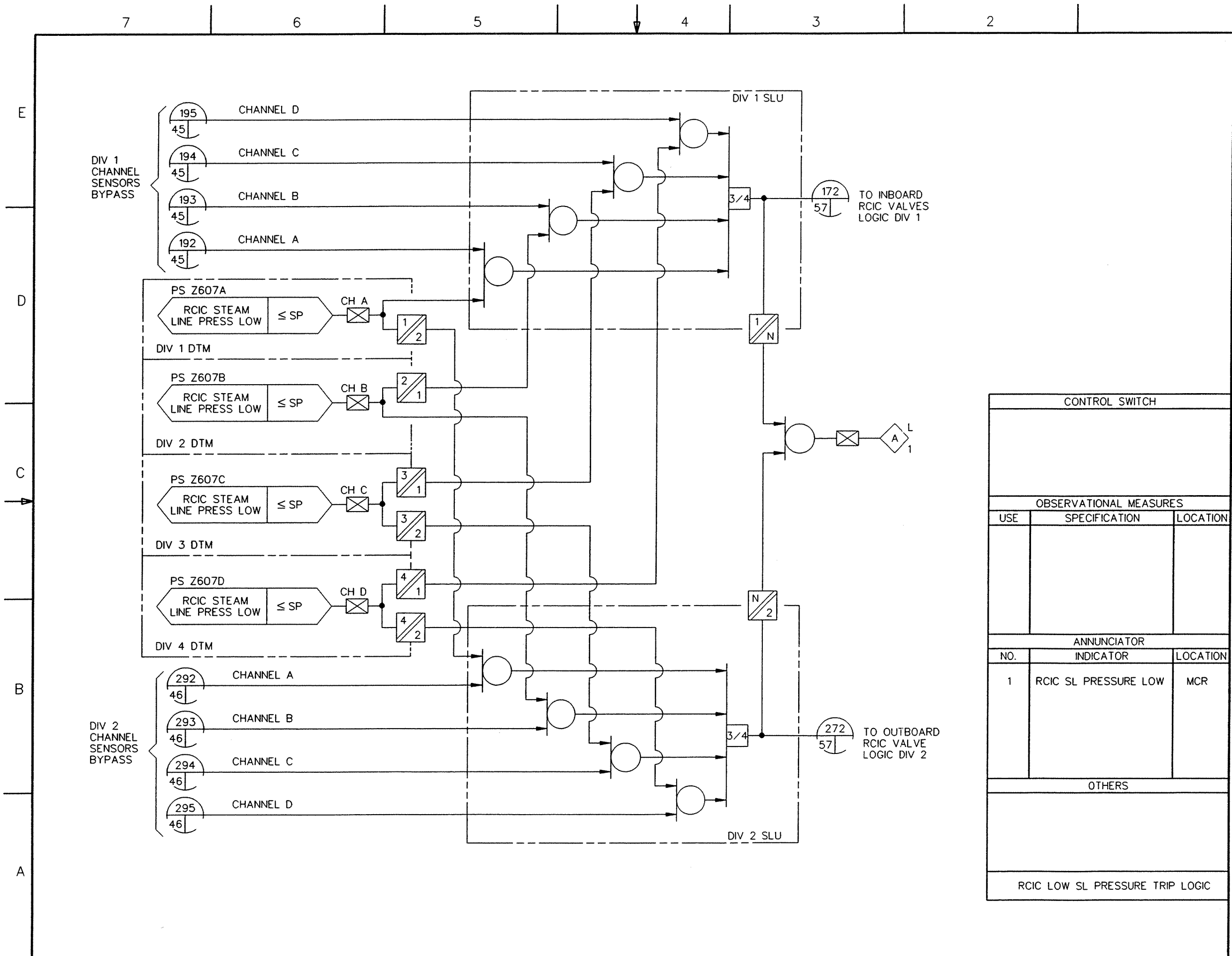


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 54 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.54



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	RCIC SL PRESSURE LOW	MCR
OTHERS		
RCIC LOW SL PRESSURE TRIP LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 55 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.55

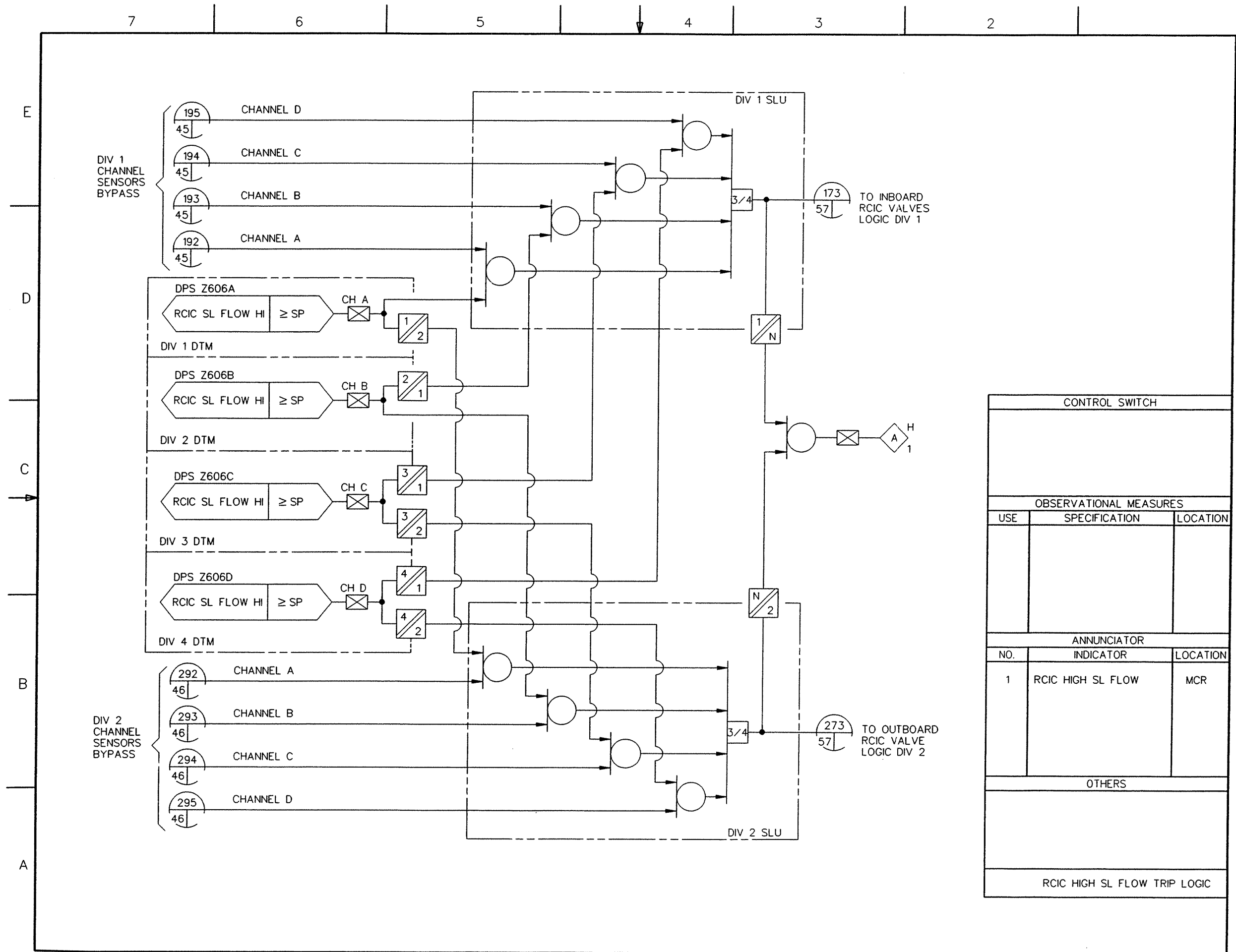


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 56 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.56

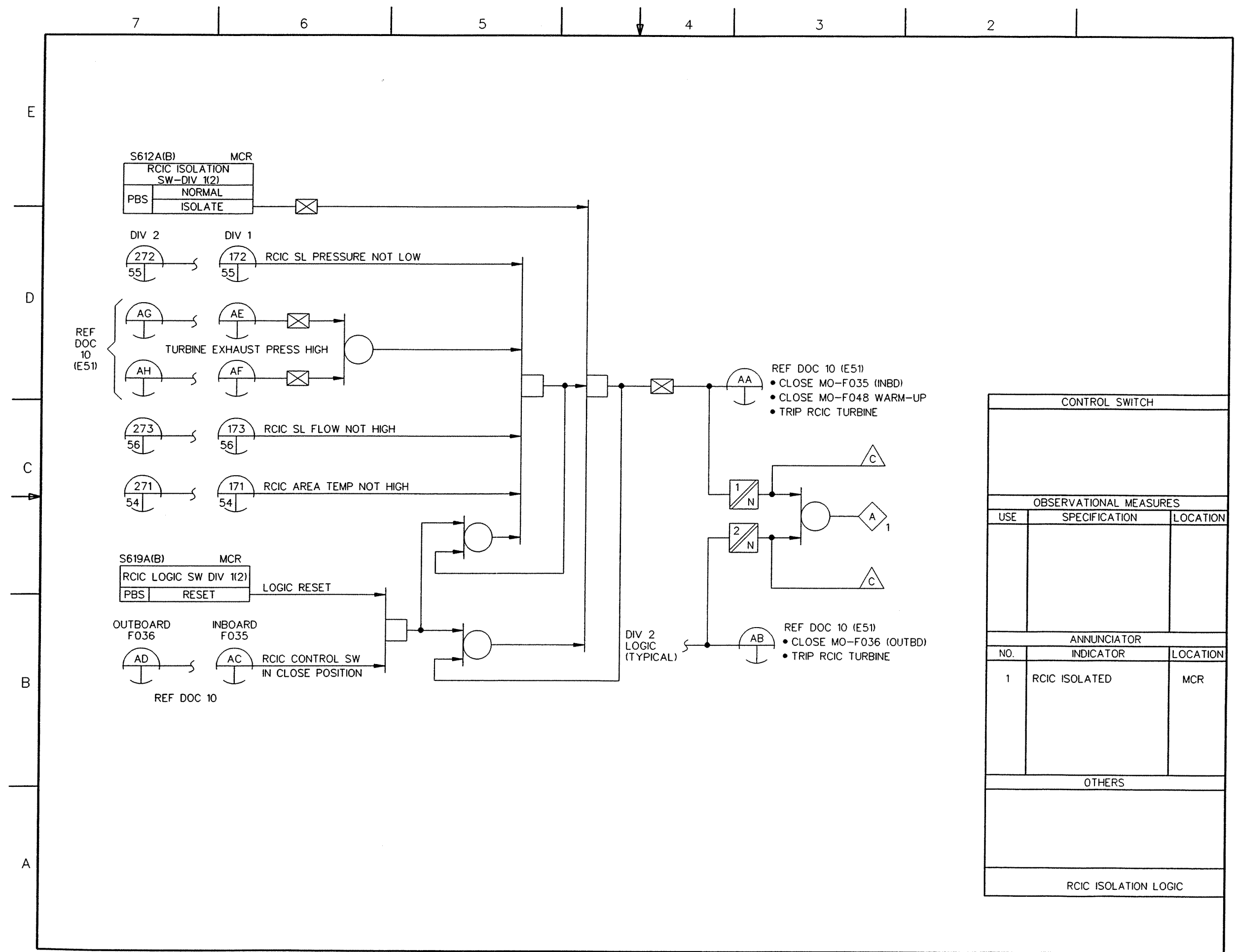
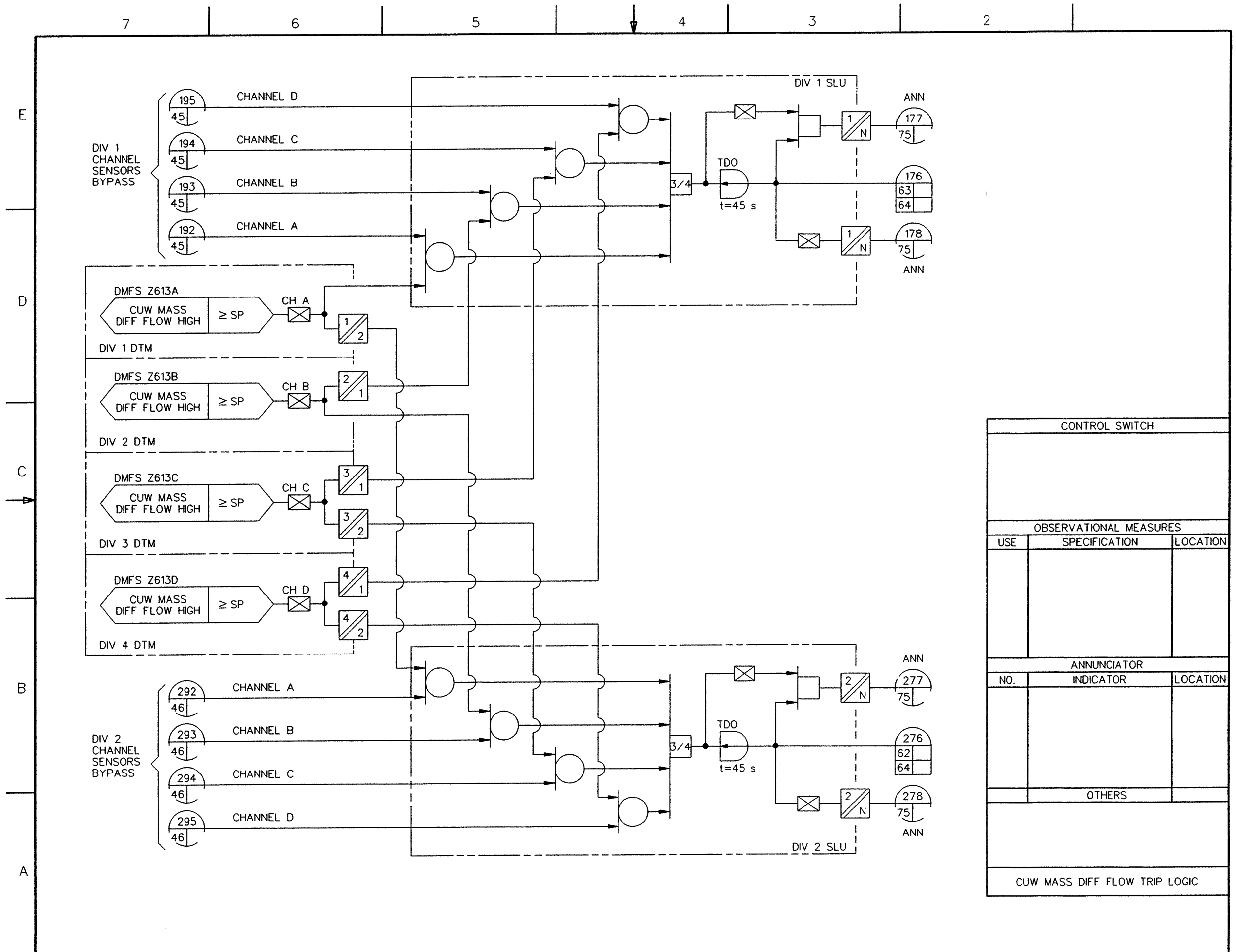


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 57 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.57



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
CUW MASS DIFF FLOW TRIP LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 58 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.58

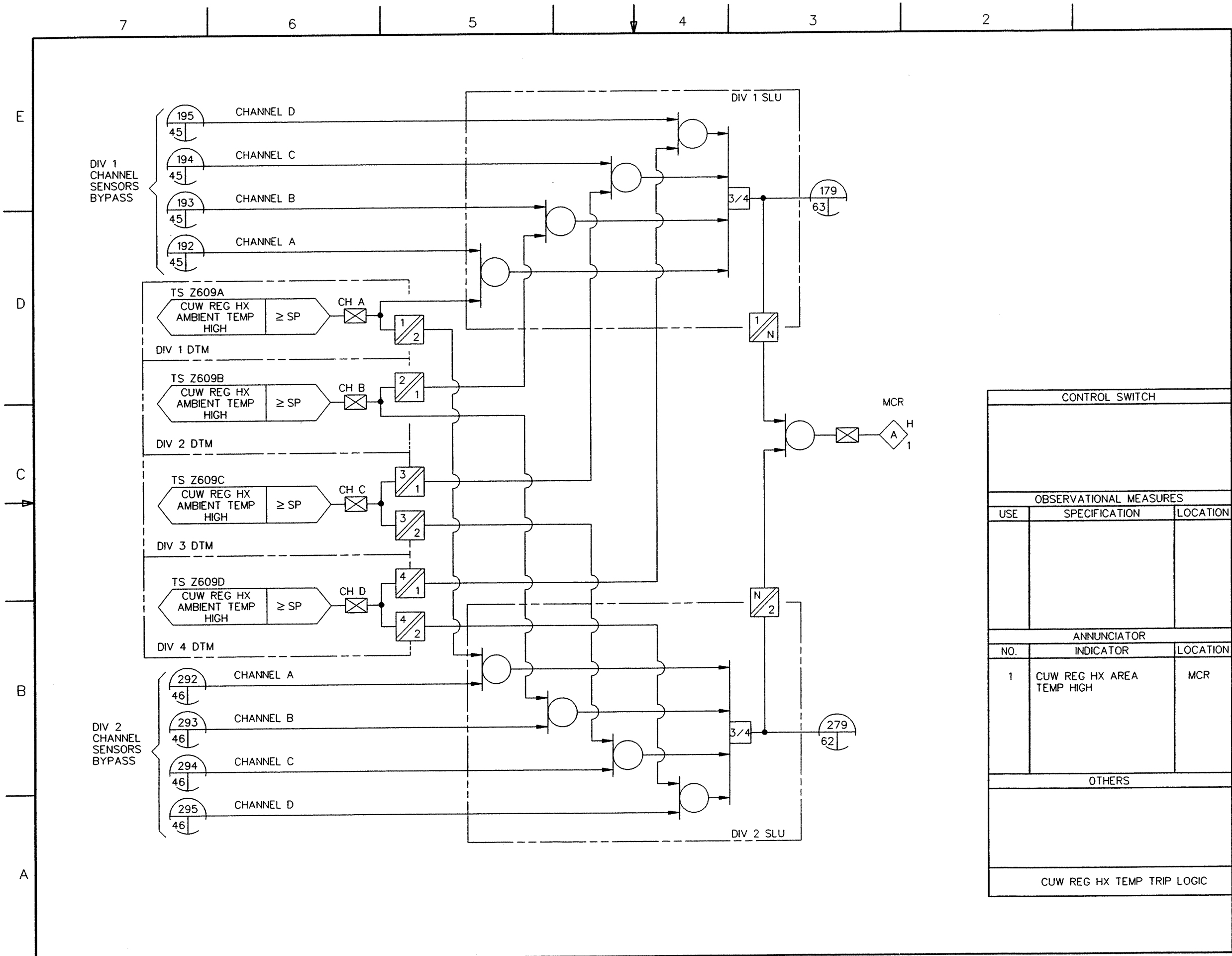


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 59 of 77)

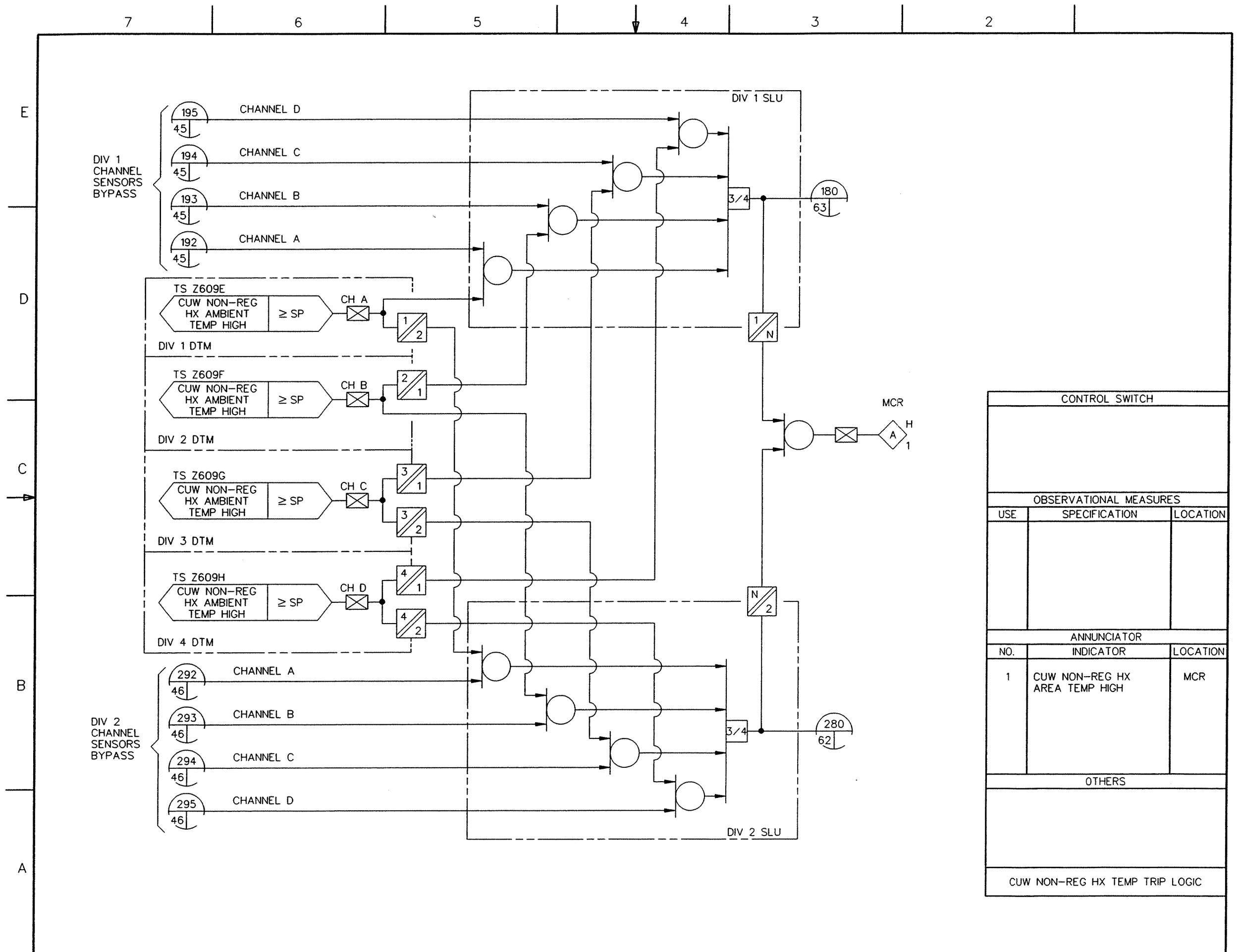


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 60 of 77)

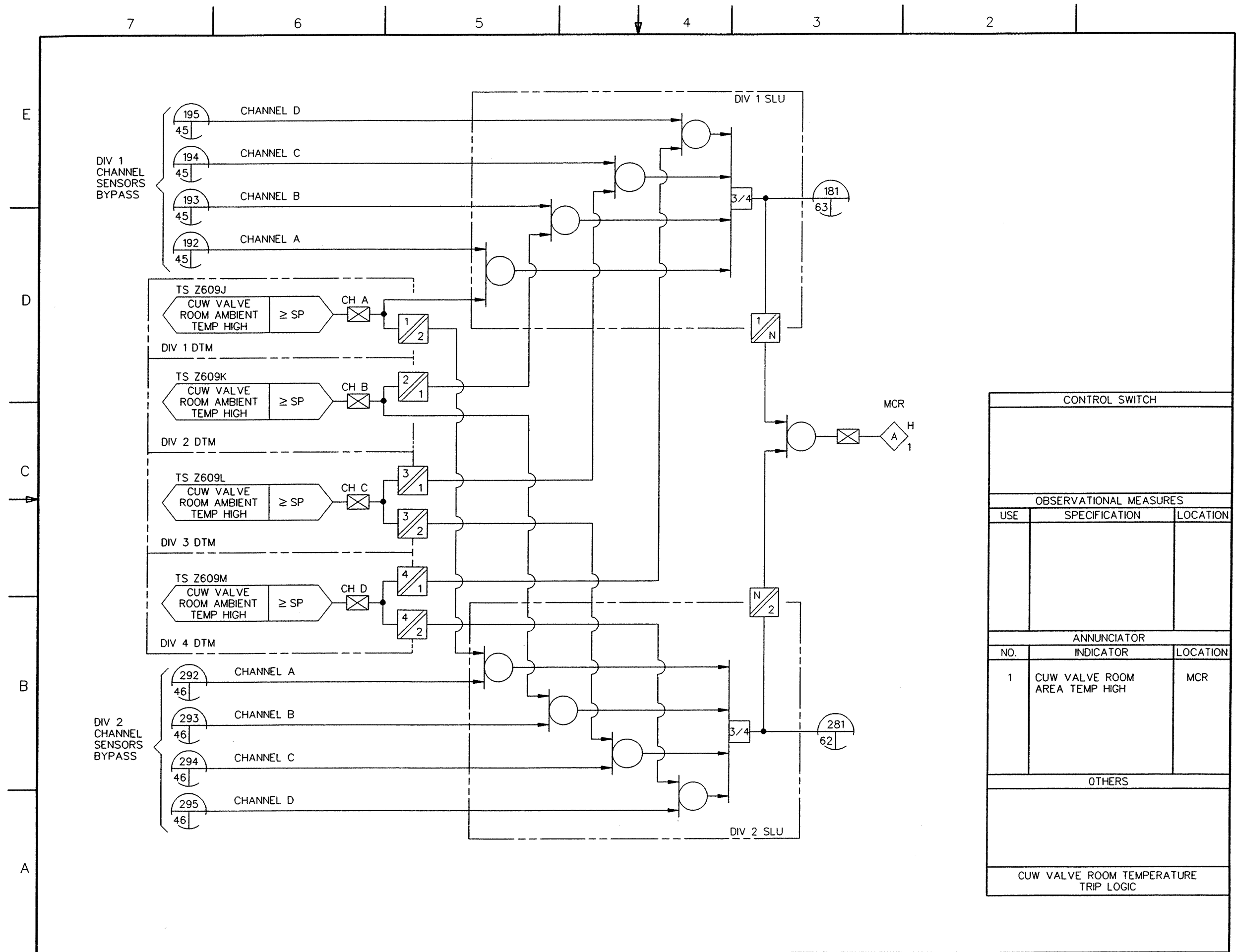
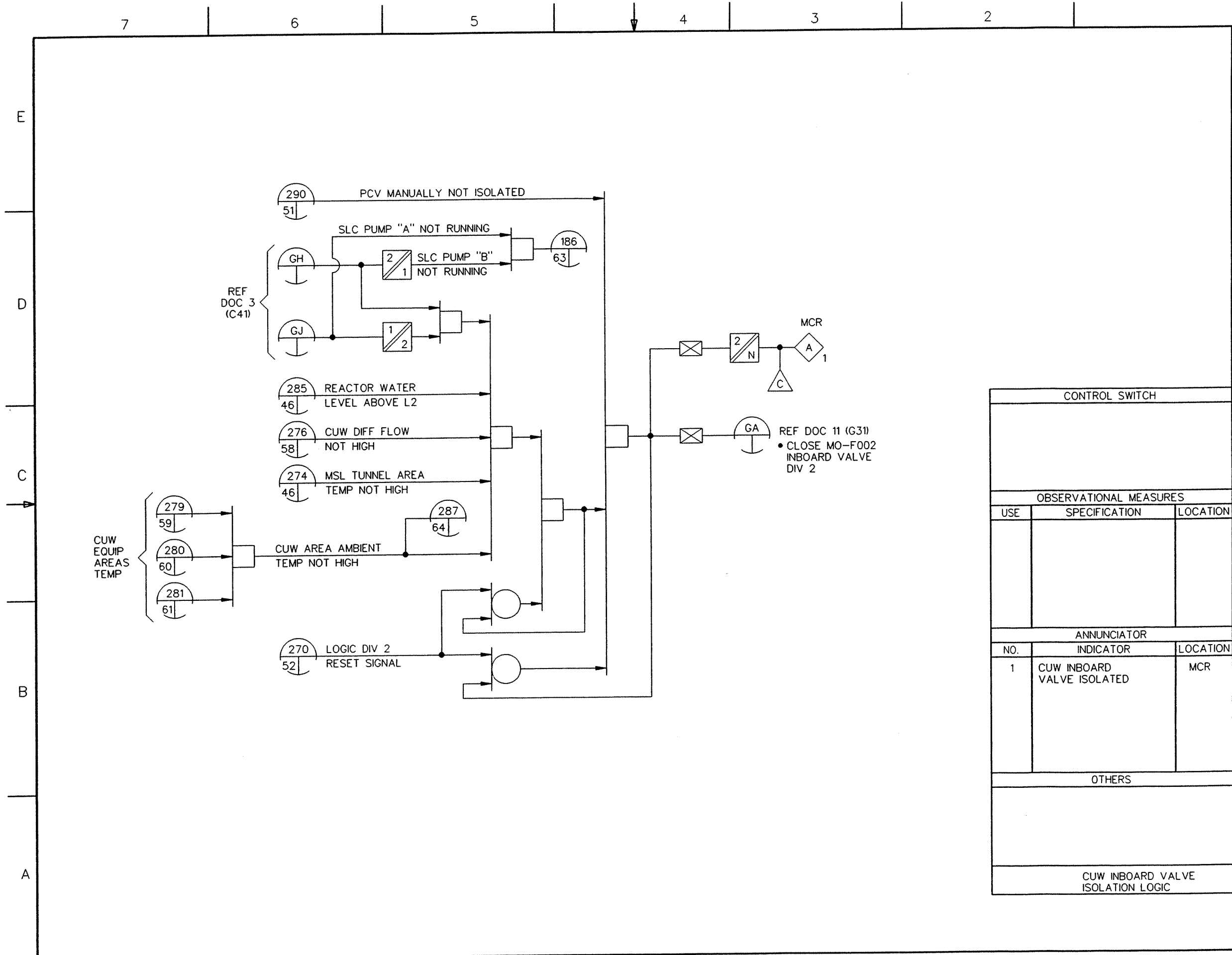
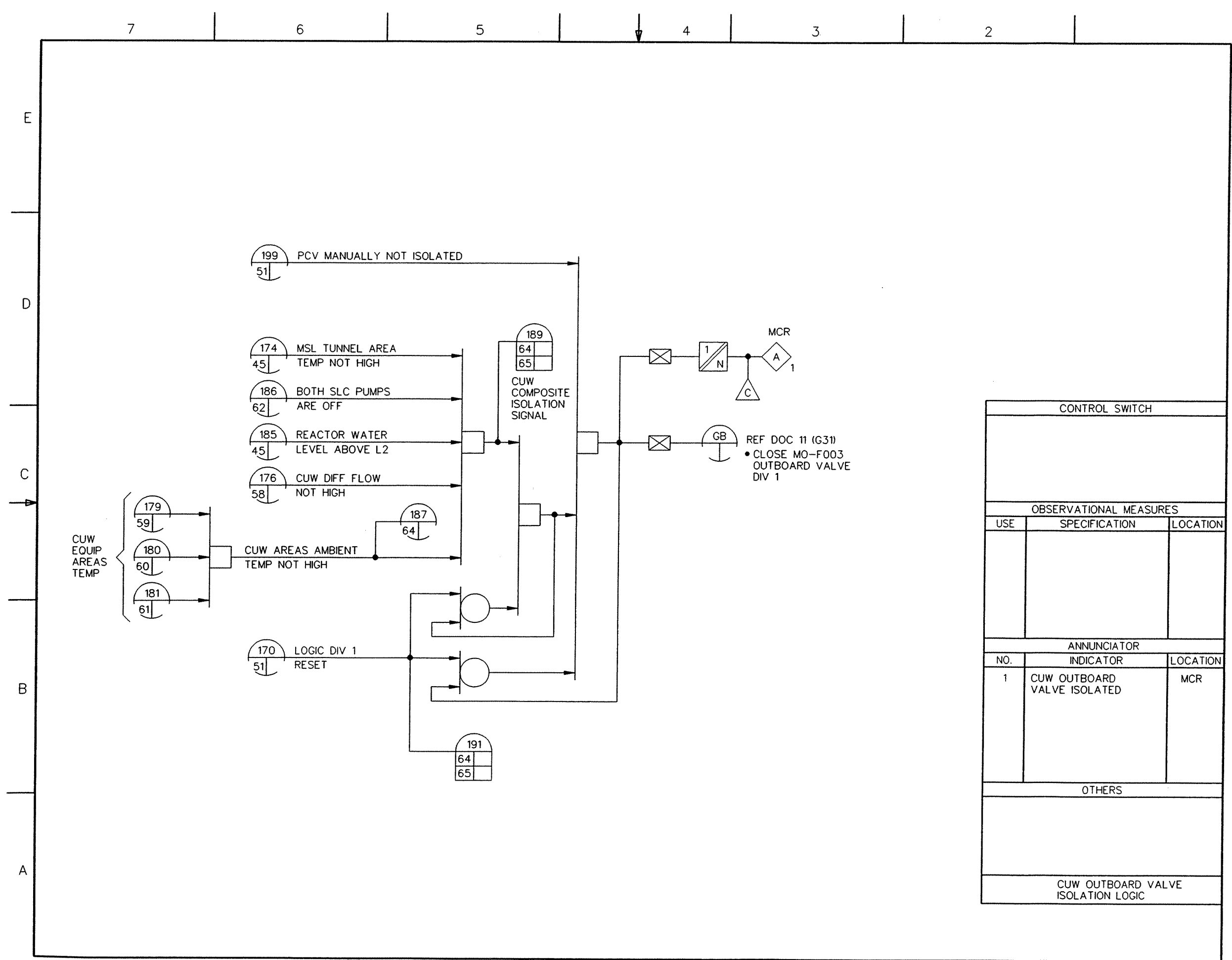


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 61 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.61



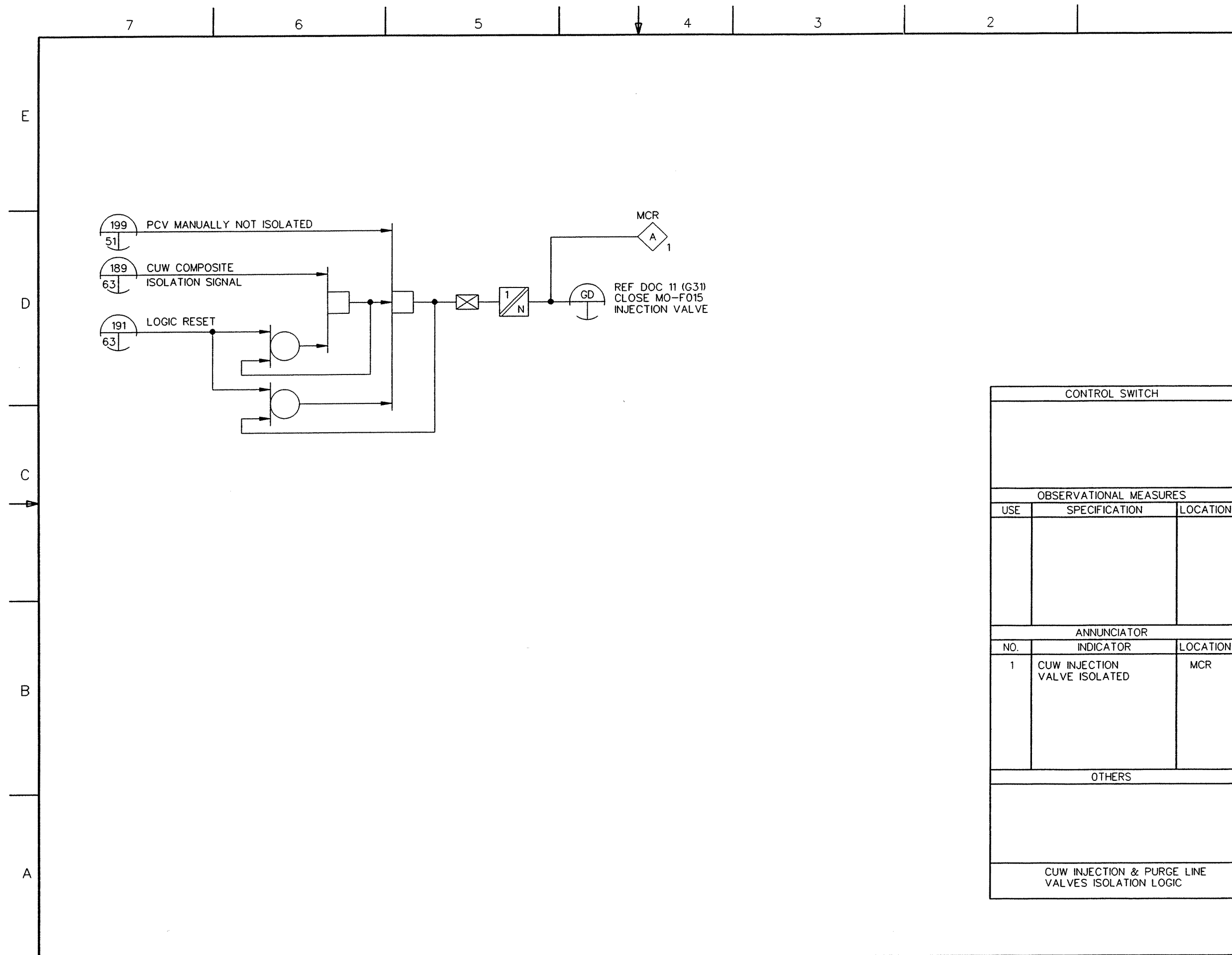
CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	CUW INBOARD VALVE ISOLATED	MCR
OTHERS		
CUW INBOARD VALVE ISOLATION LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 62 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.62



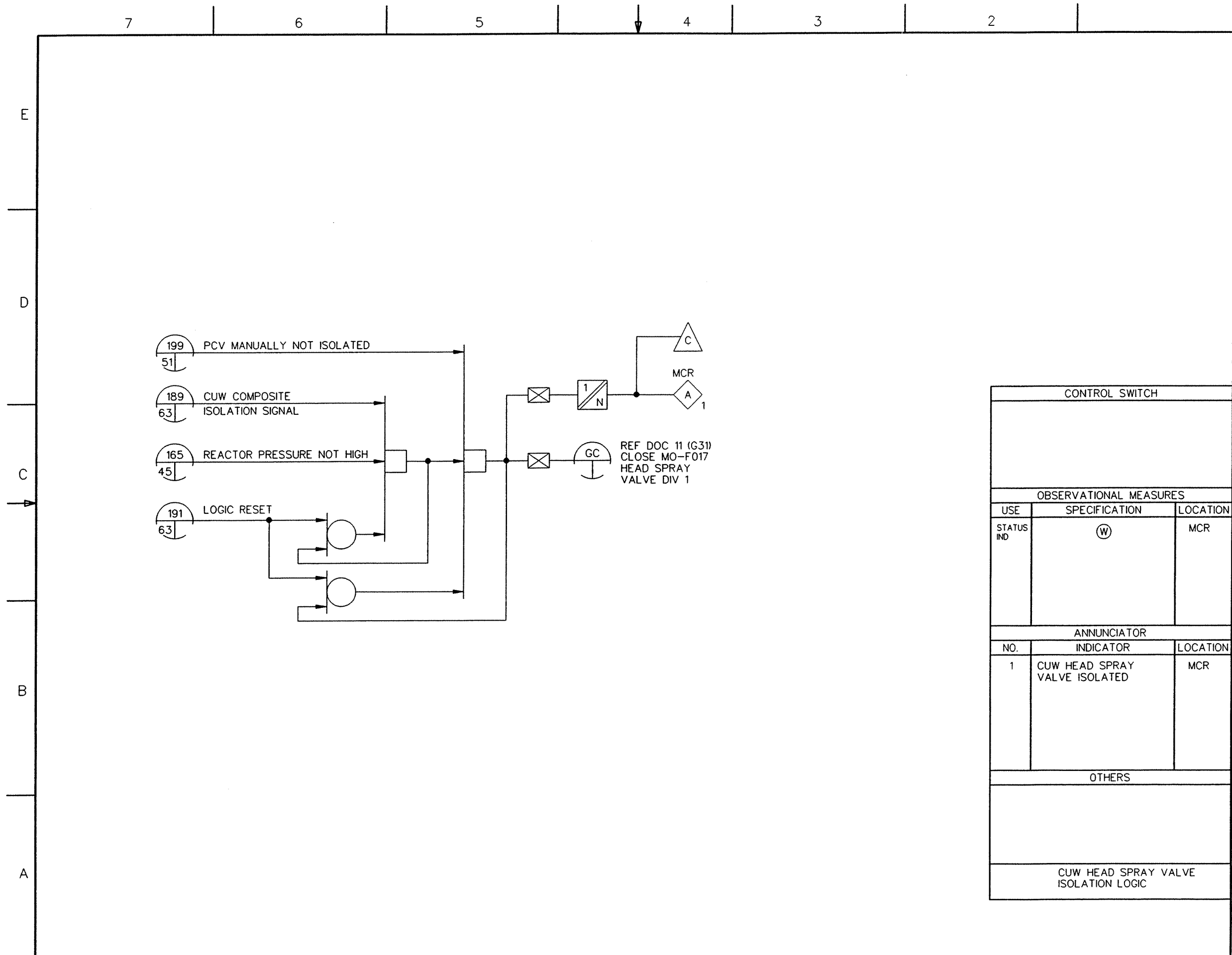
CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	CUW OUTBOARD VALVE ISOLATED	MCR
OTHERS		
CUW OUTBOARD VALVE ISOLATION LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 63 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.63



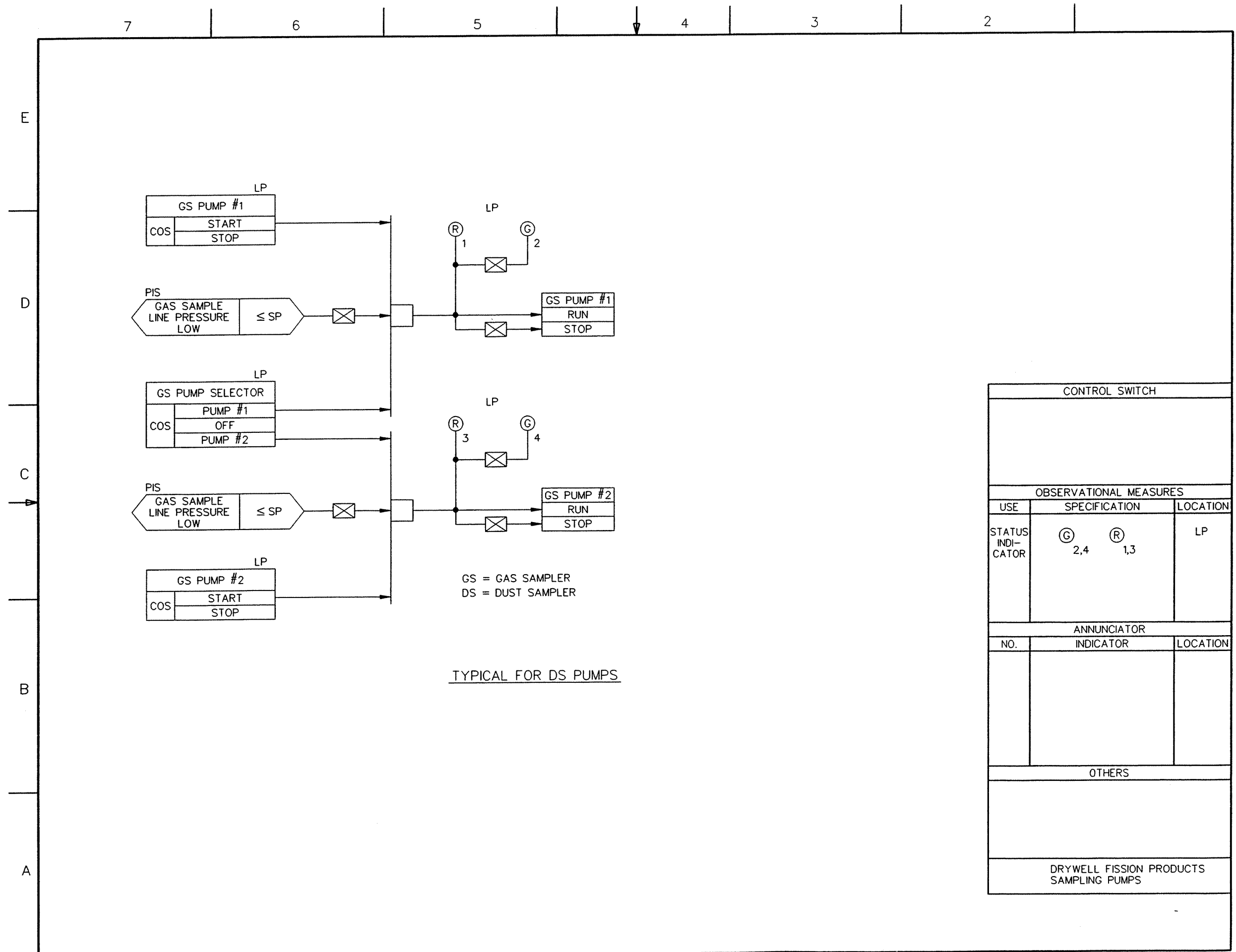
CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	CUW INJECTION VALVE ISOLATED	MCR
OTHERS		
CUW INJECTION & PURGE LINE VALVES ISOLATION LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 64 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.64



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
STATUS IND	(W)	MCR
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	CUW HEAD SPRAY VALVE ISOLATED	MCR
OTHERS		
CUW HEAD SPRAY VALVE ISOLATION LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 65 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.65



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
STATUS INDI- CATOR	Ⓒ 2,4 Ⓓ 1,3	LP
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
DRYWELL FISSION PRODUCTS SAMPLING PUMPS		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 66 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.66

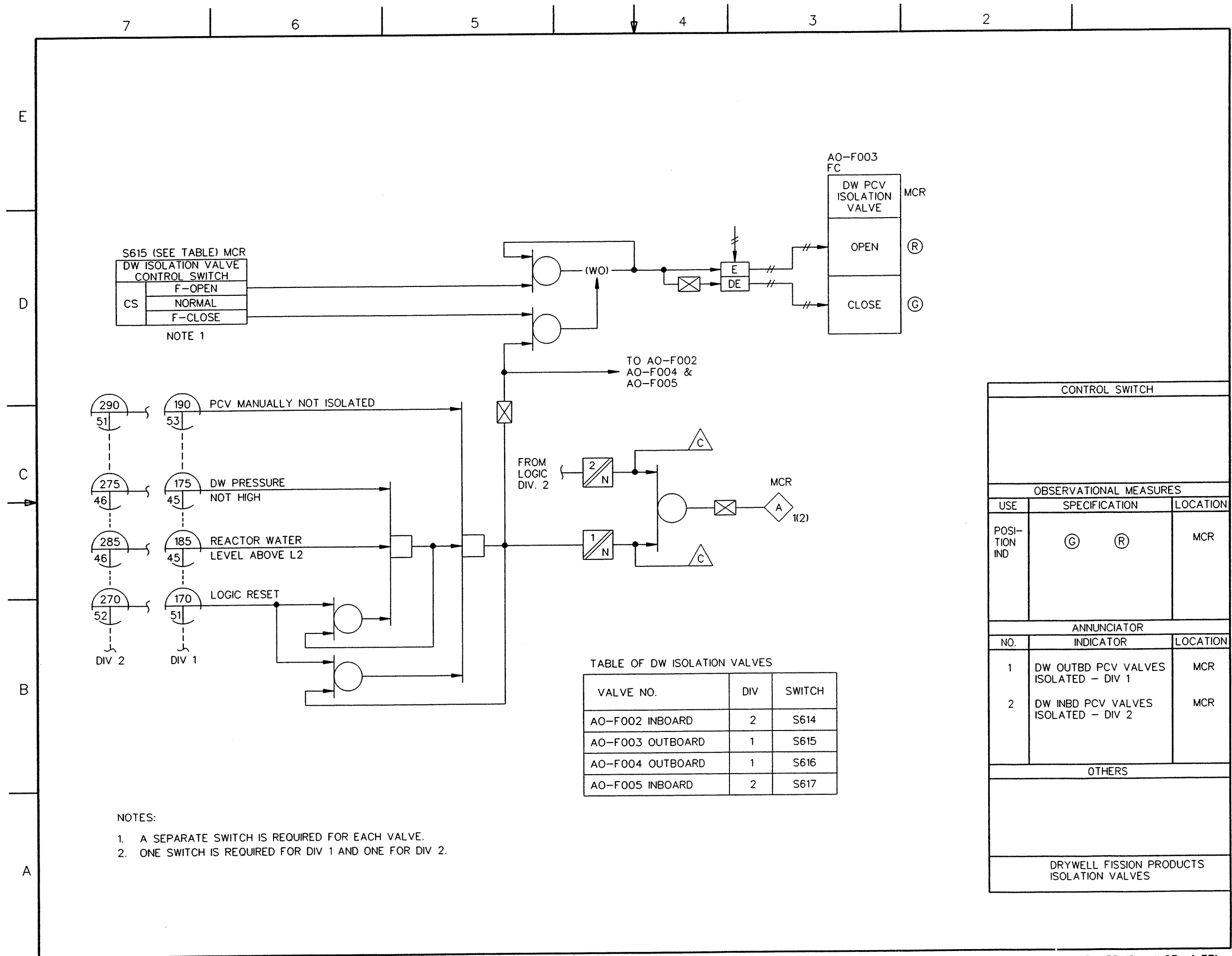


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 67 of 77)
ABWR DCD/Tier 2 Rev. 0 21-228.67

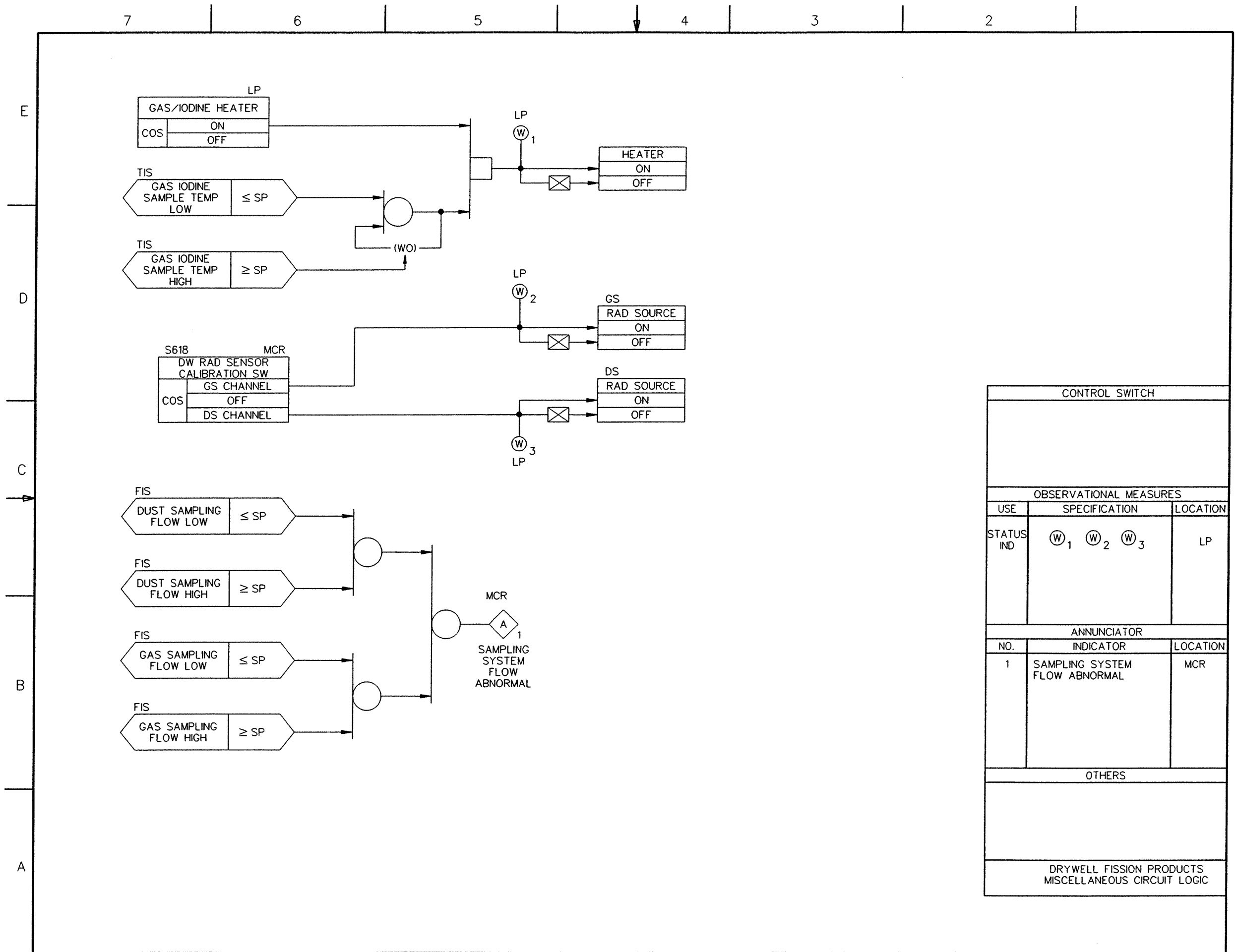
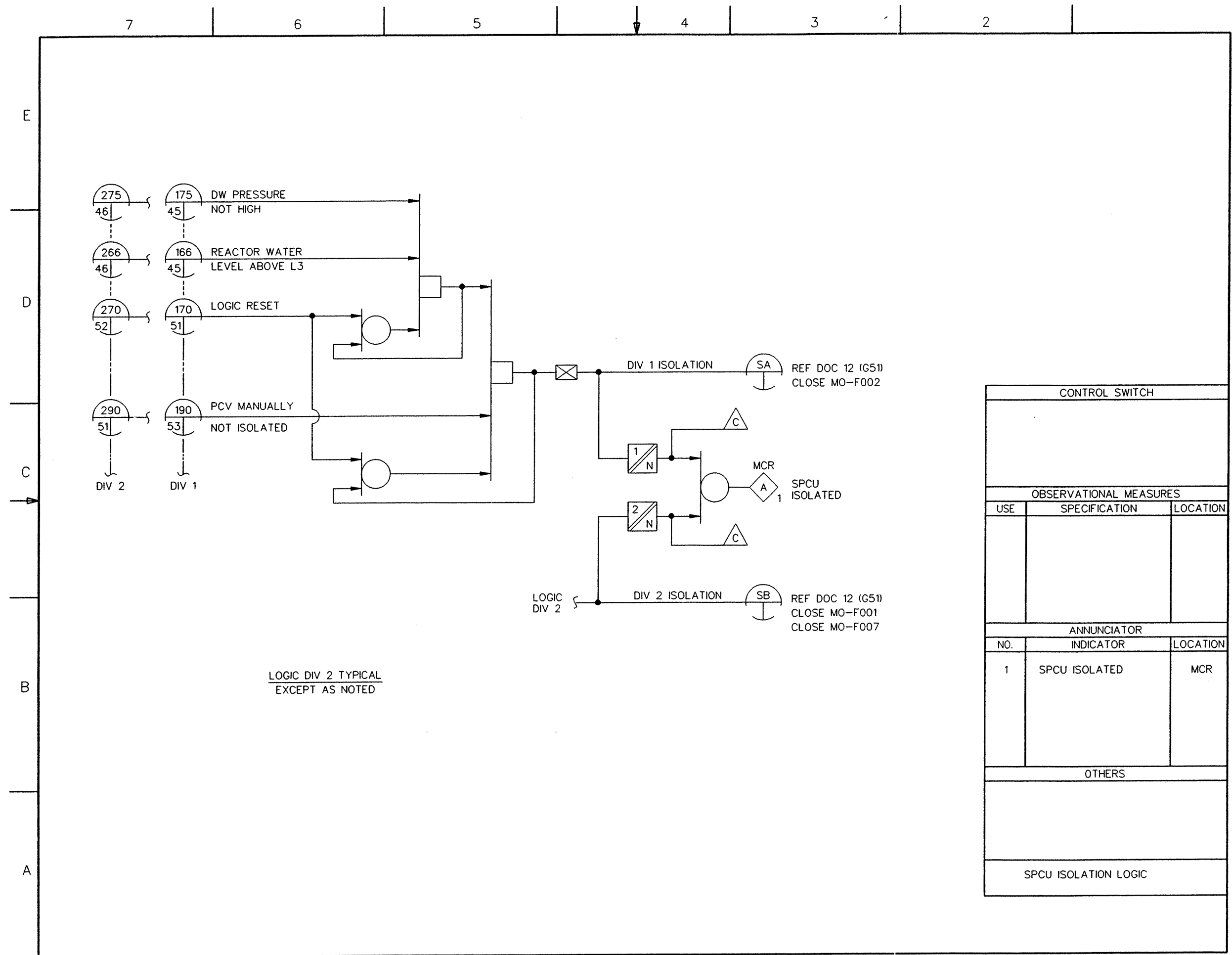


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 68 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.68



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	SPCU ISOLATED	MCR
OTHERS		
SPCU ISOLATION LOGIC		

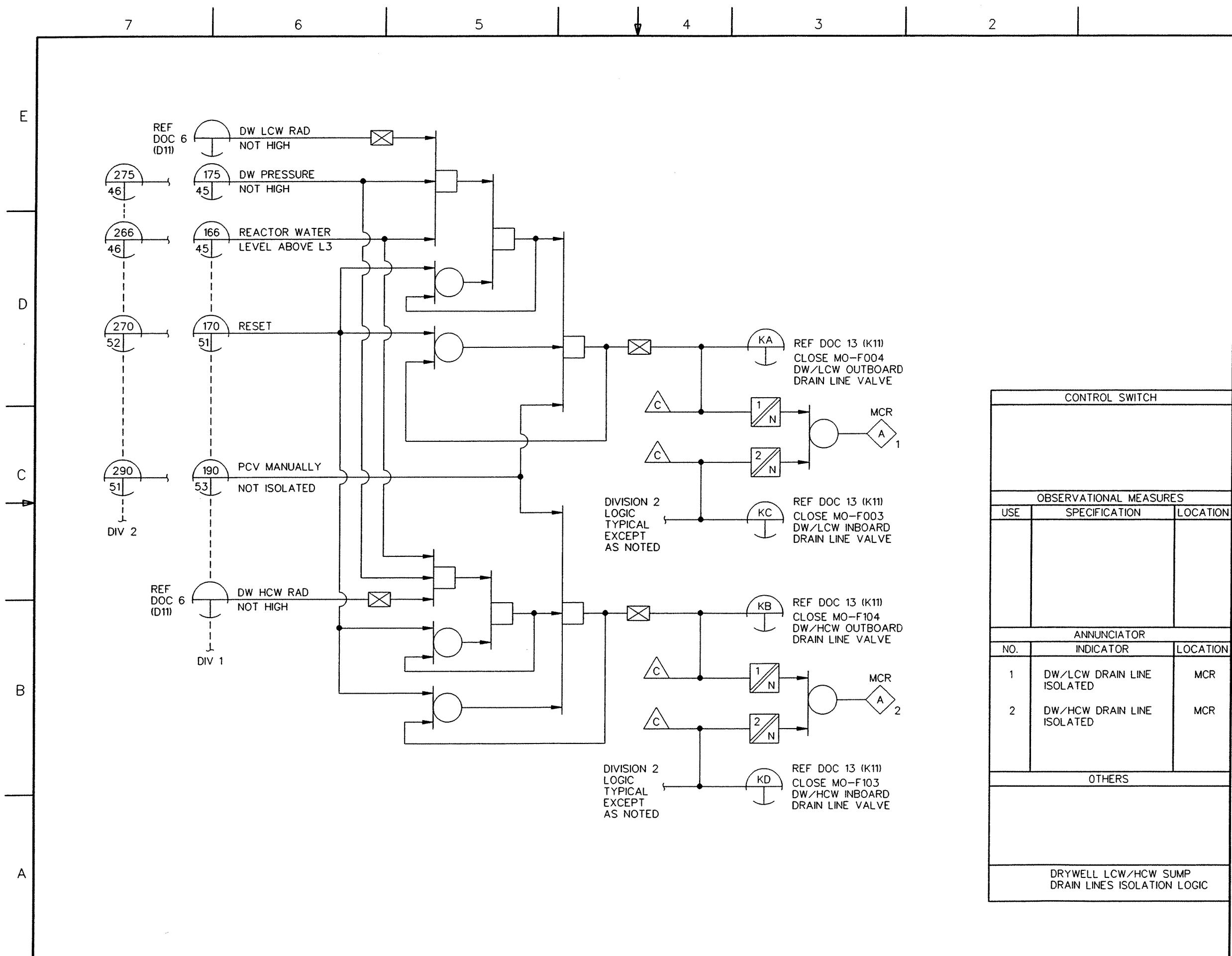
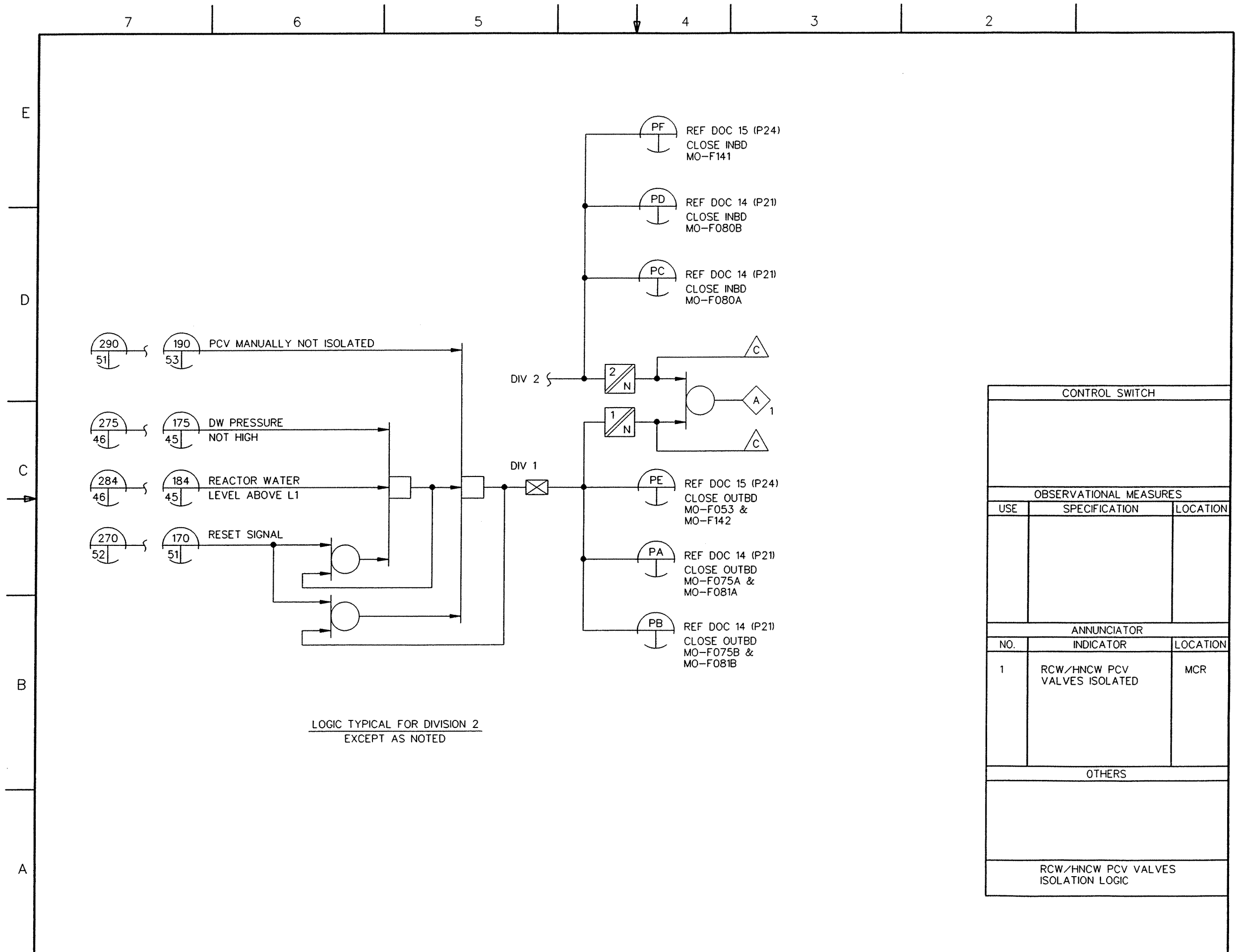


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 70 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.70



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	RCW/HNCW PCV VALVES ISOLATED	MCR
OTHERS		
RCW/HNCW PCV VALVES ISOLATION LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 71 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.71

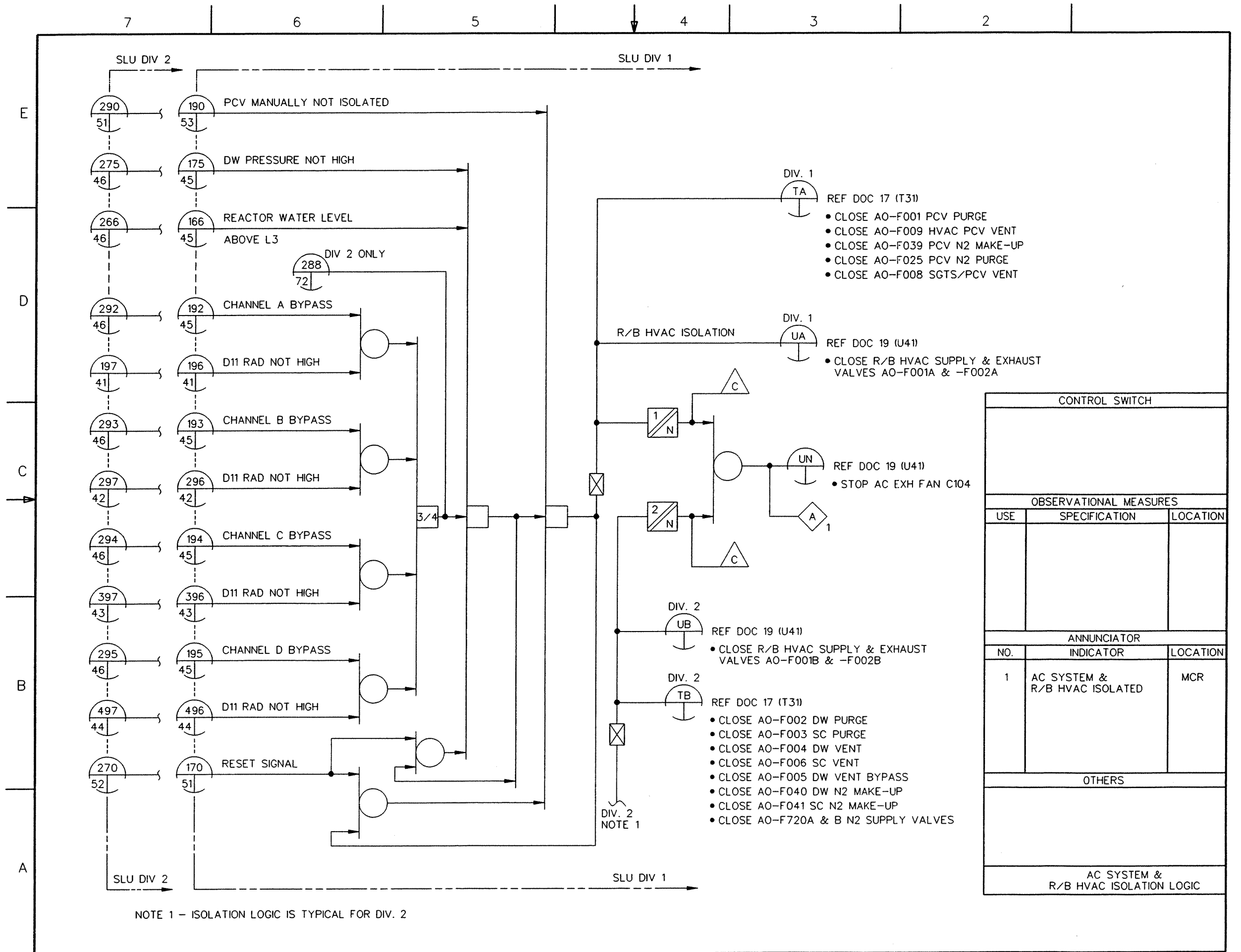


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 72 of 77)

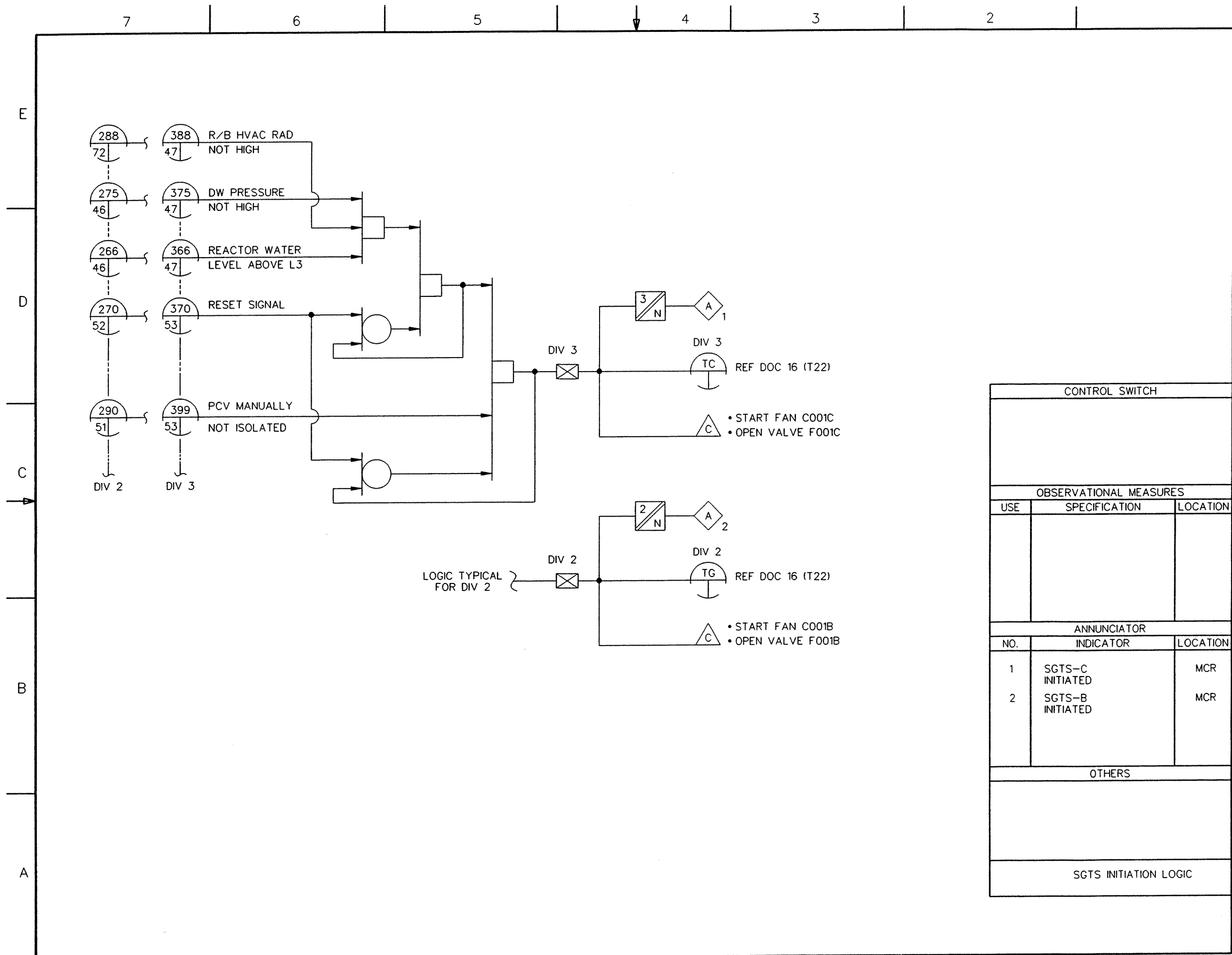
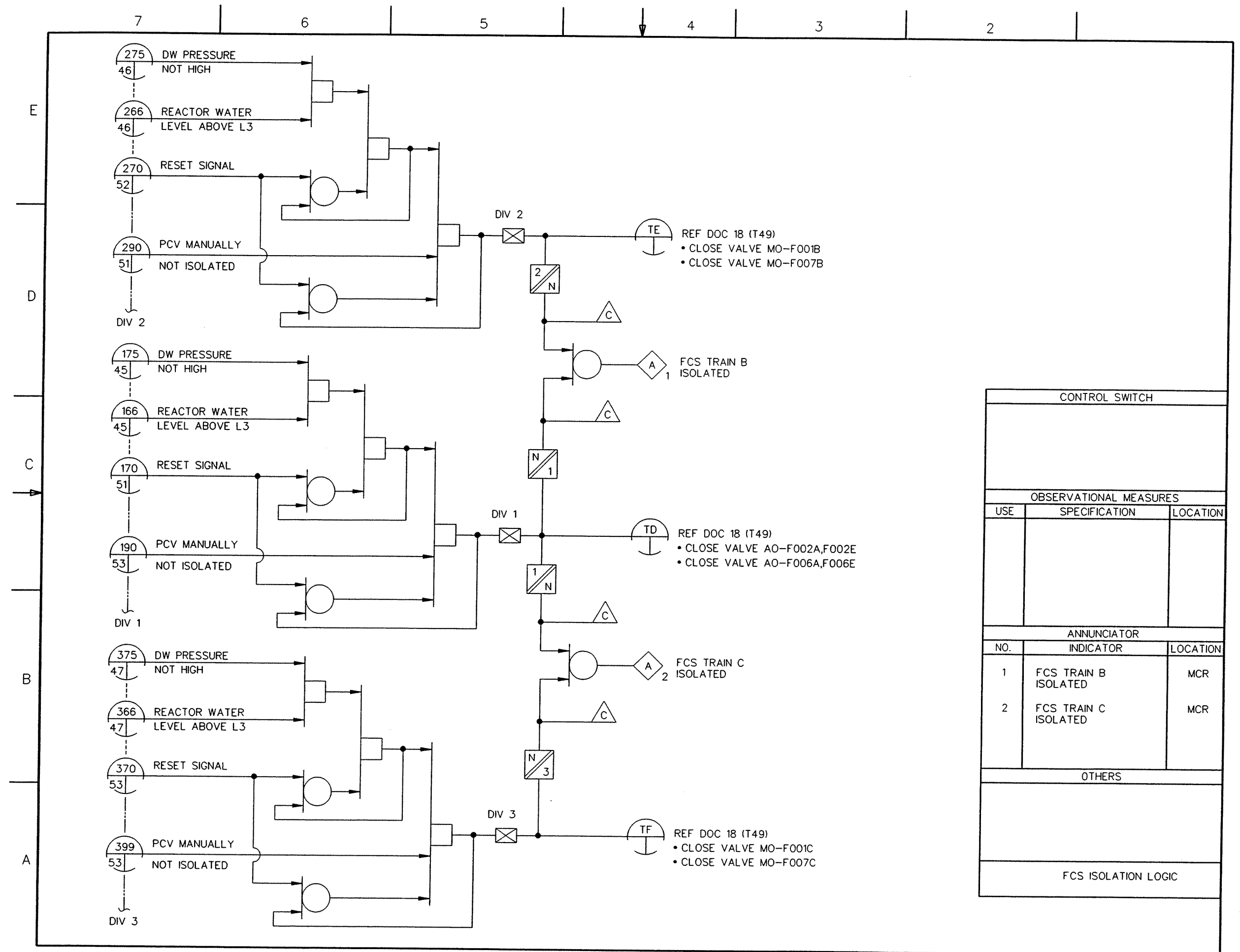


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 73 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.73



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
1	FCS TRAIN B ISOLATED	MCR
2	FCS TRAIN C ISOLATED	MCR
OTHERS		
FCS ISOLATION LOGIC		

FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 74 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.74

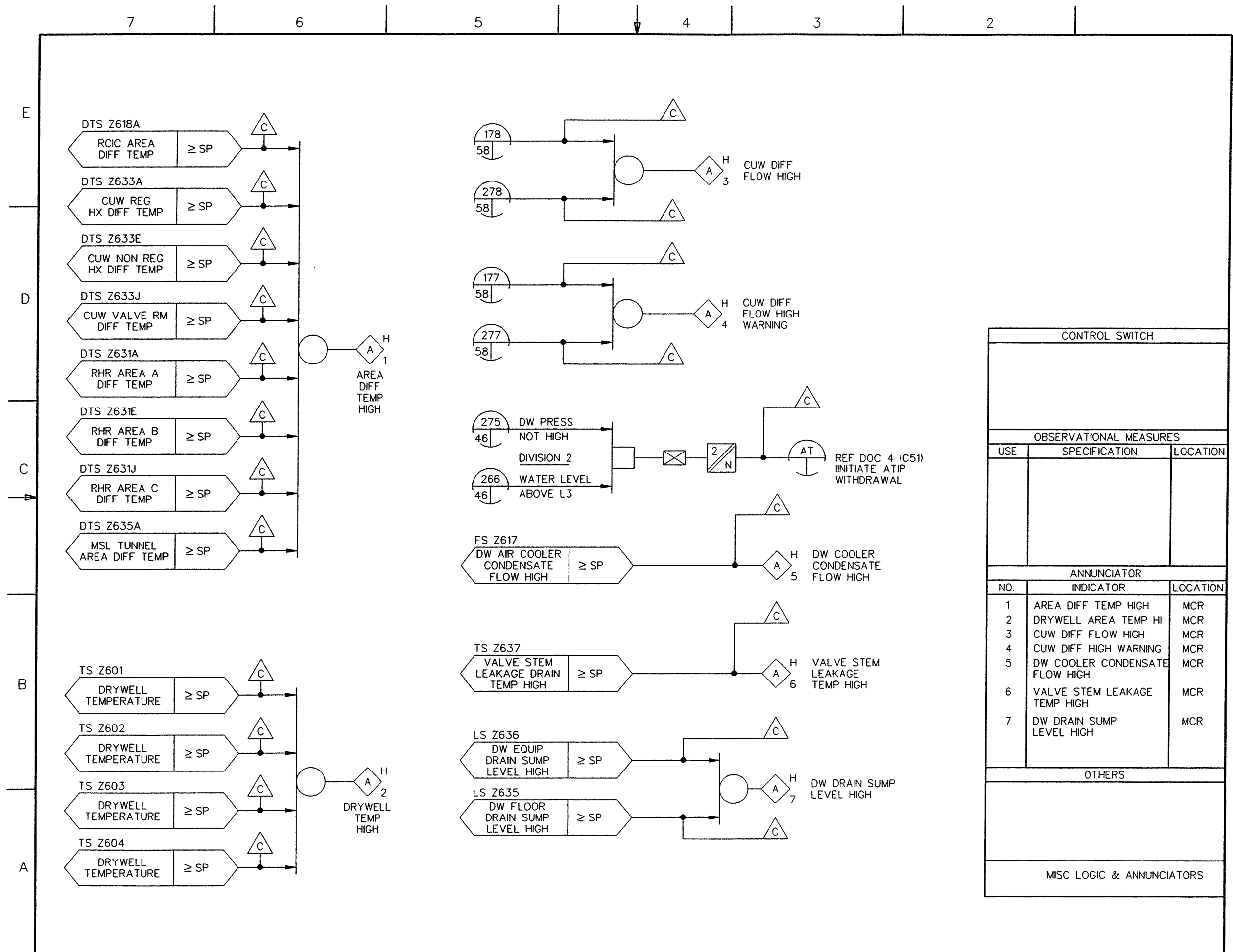


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 75 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.75

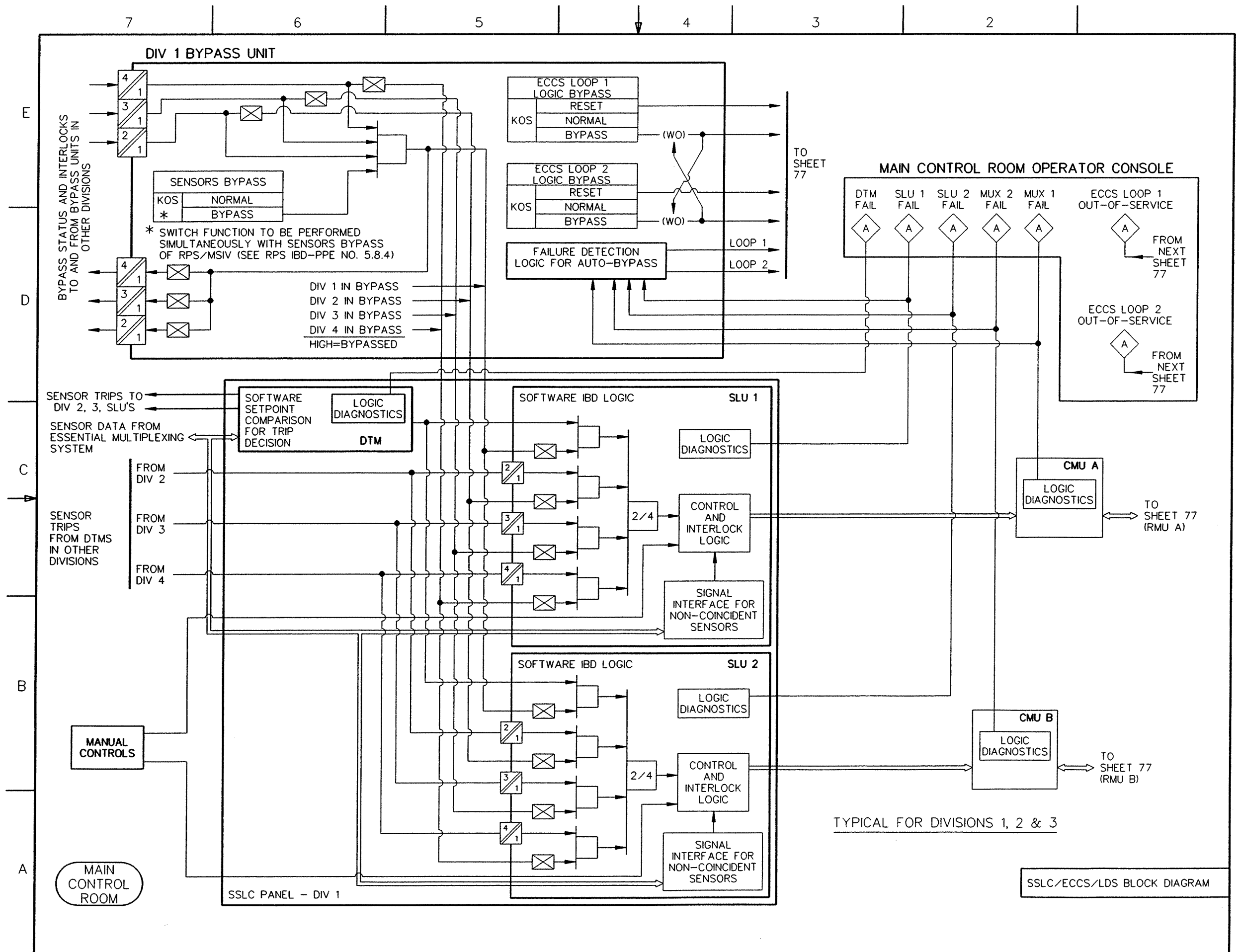


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 76 of 77)

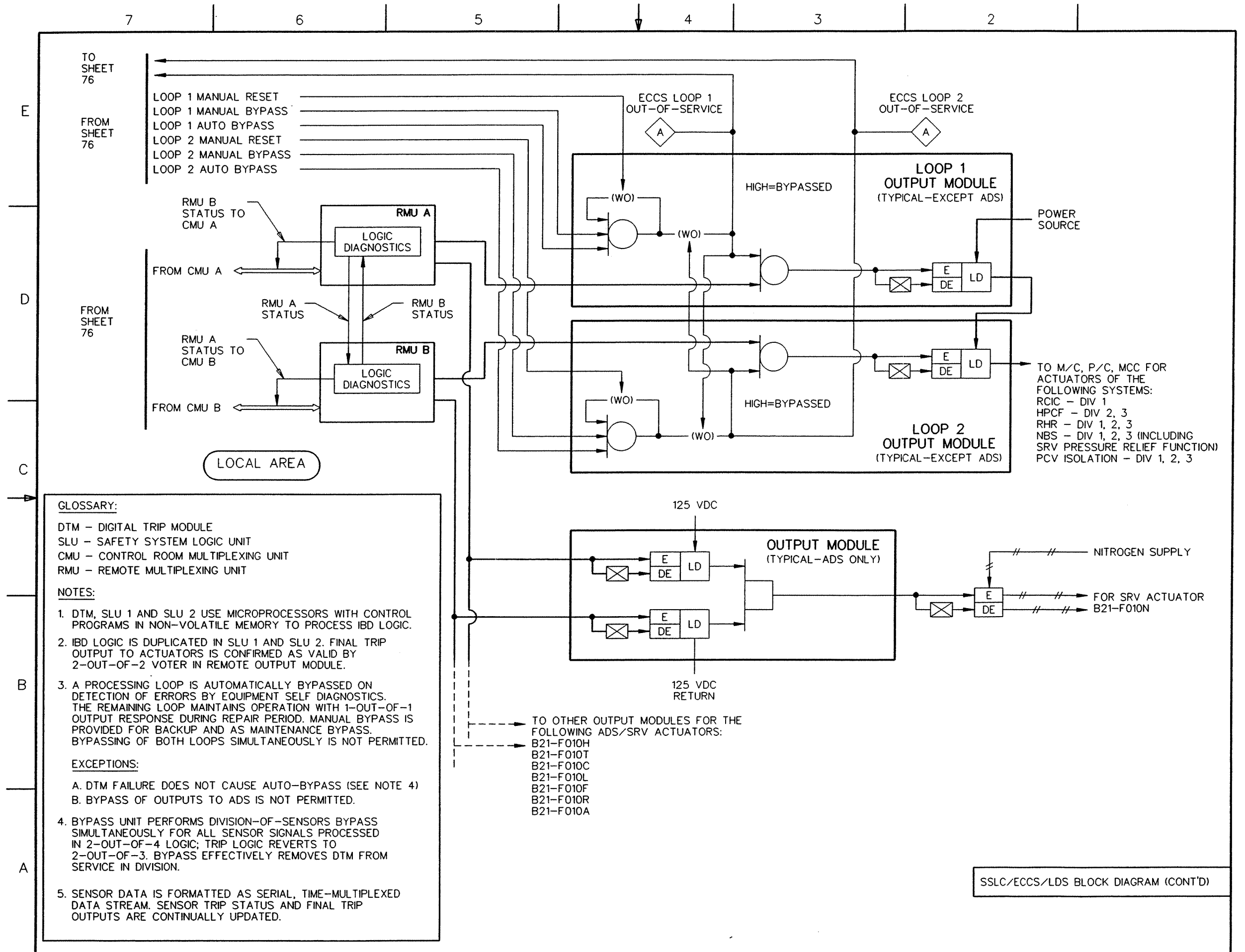
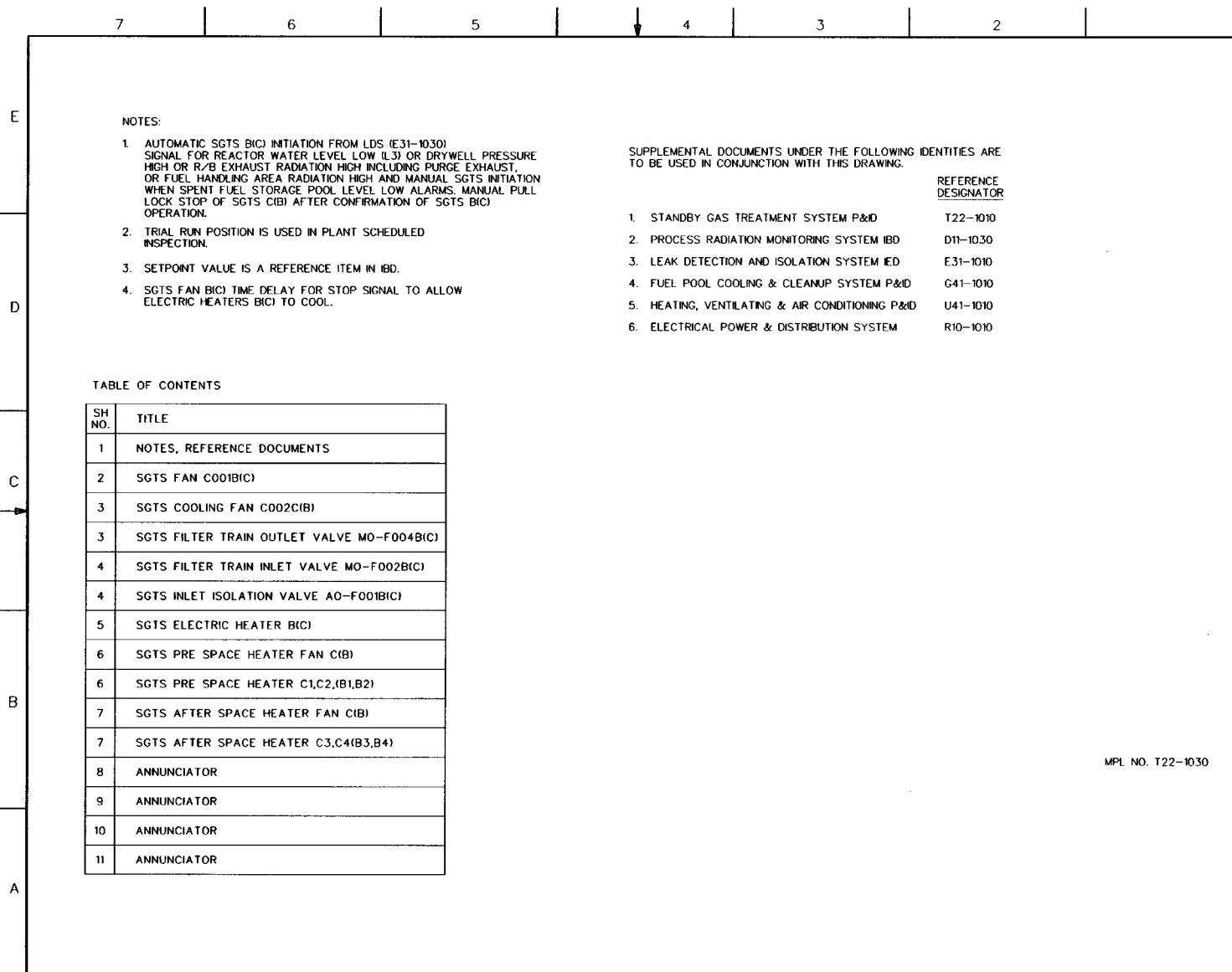


FIGURE 7.3-5 LEAK DETECTION AND ISOLATION SYSTEM IBD (Sheet 77 of 77)
 ABWR DCD/Tier 2 Rev. 0 21-228.77



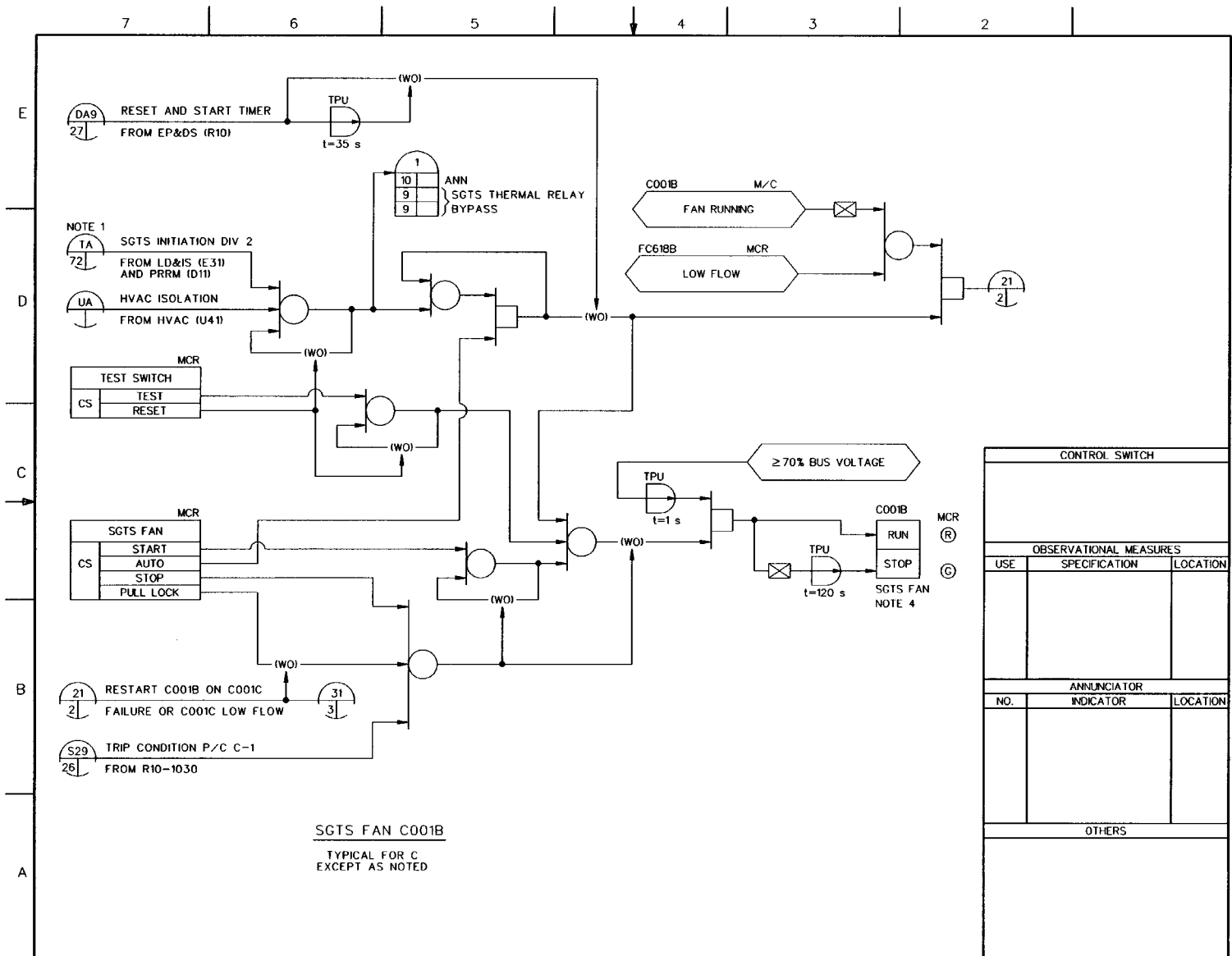


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 2 of 11)
ABWR DCD/Tier 2 Rev. 0 21-230

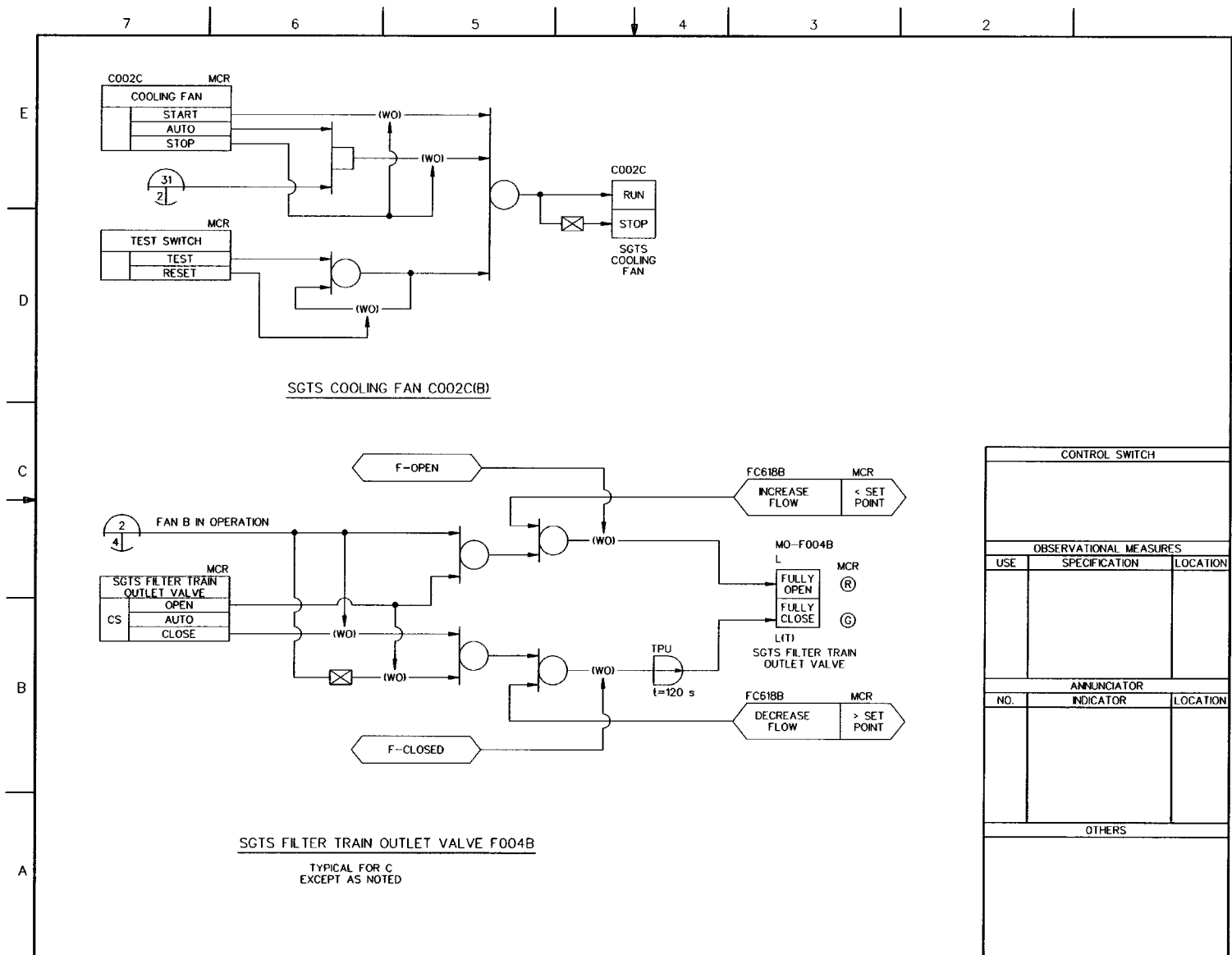


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 3 of 11)

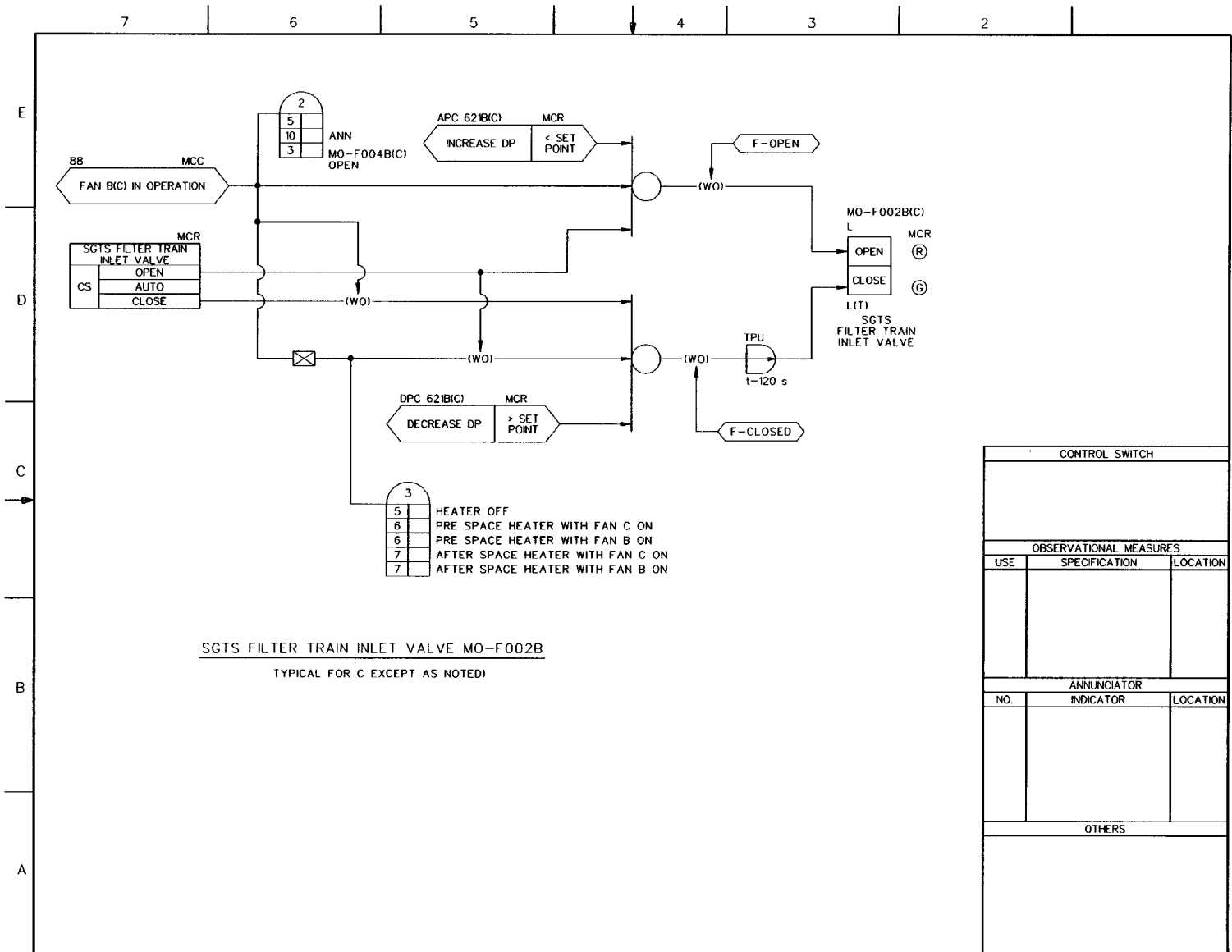


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 4 of 10)
ABWR DCD/Tier 2 Rev. 0 21-232

CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		

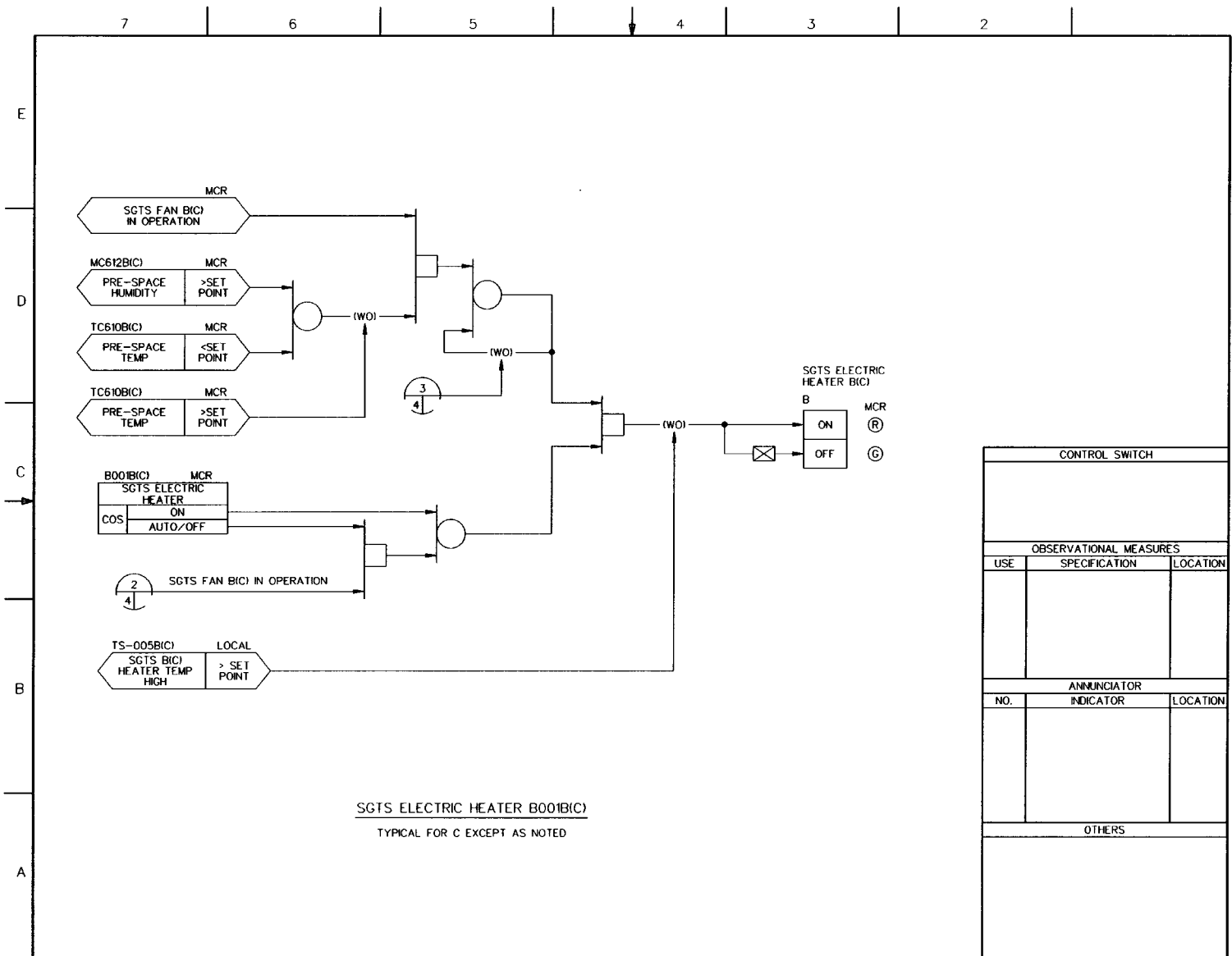


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 5 of 10)
ABWR DCD/Tier 2 Rev. 0 21-233

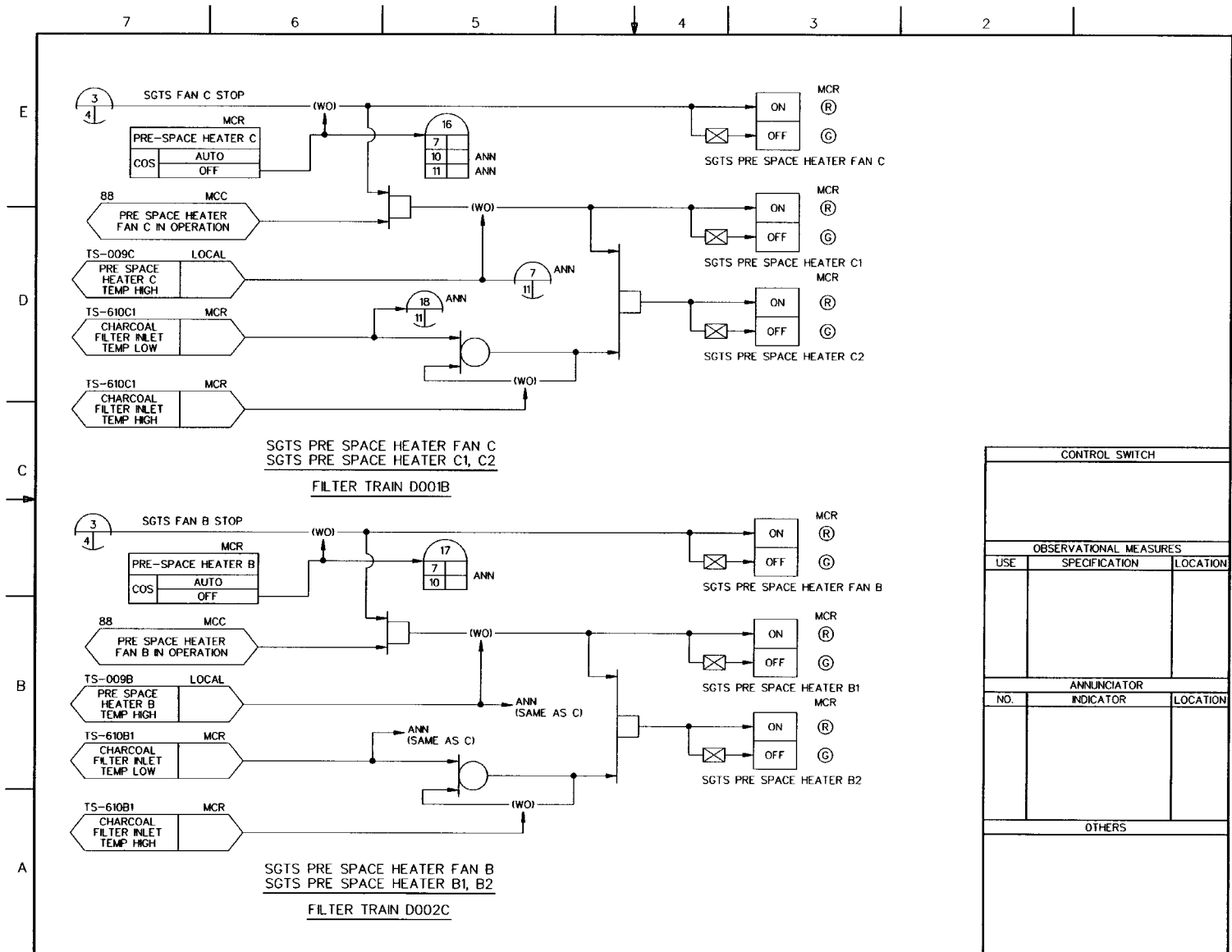


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 6 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-234

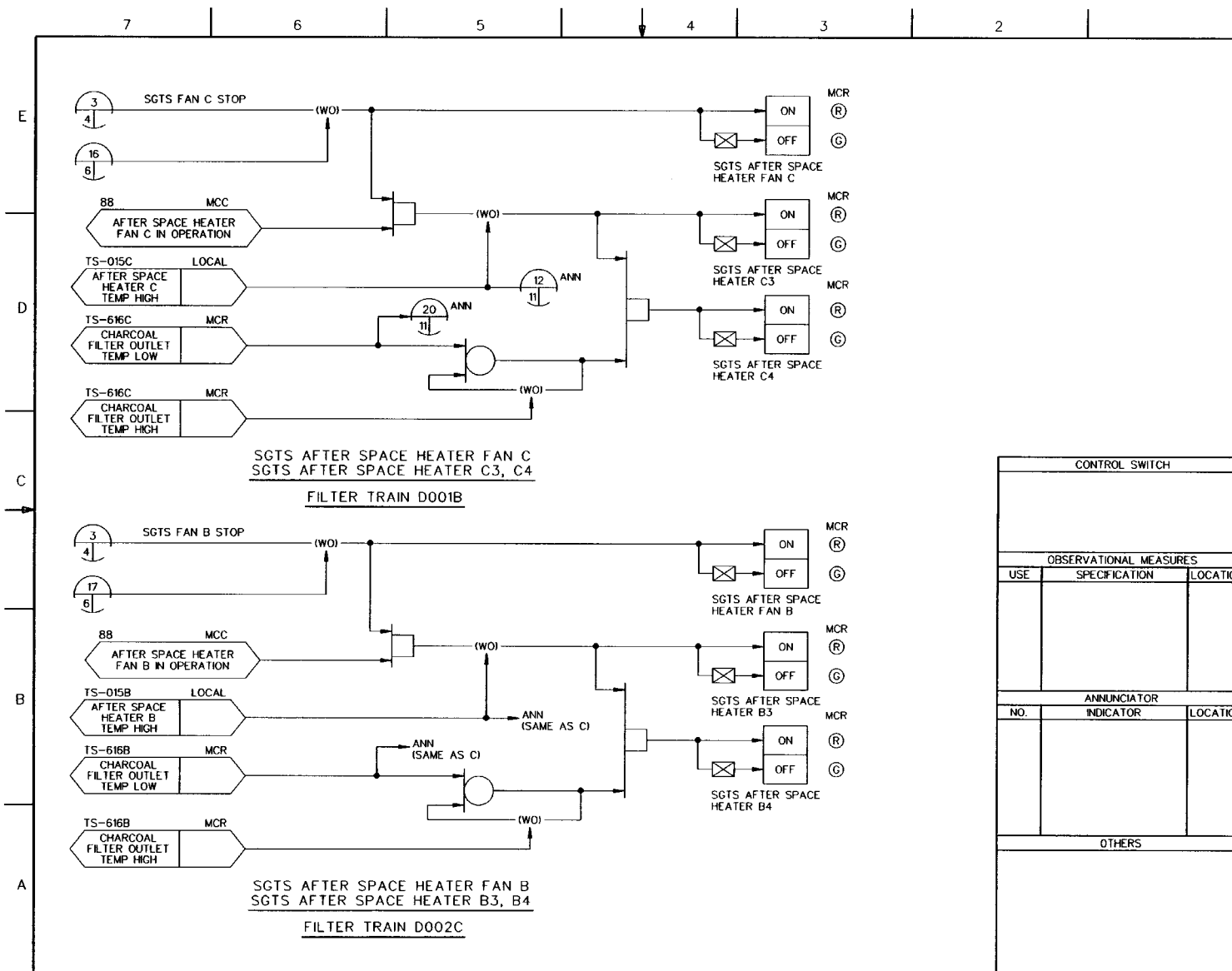


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 7 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-235

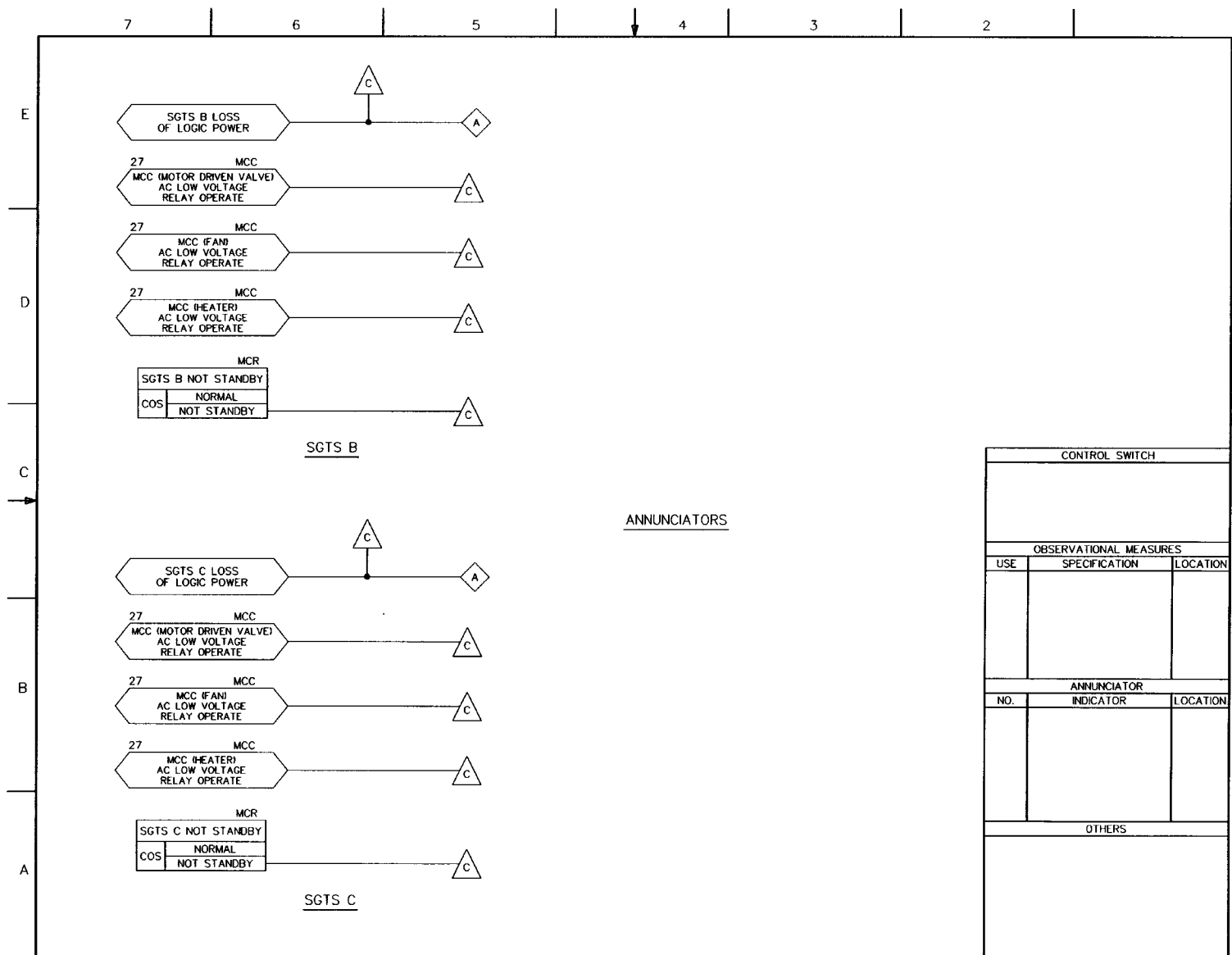


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 8 of 11)

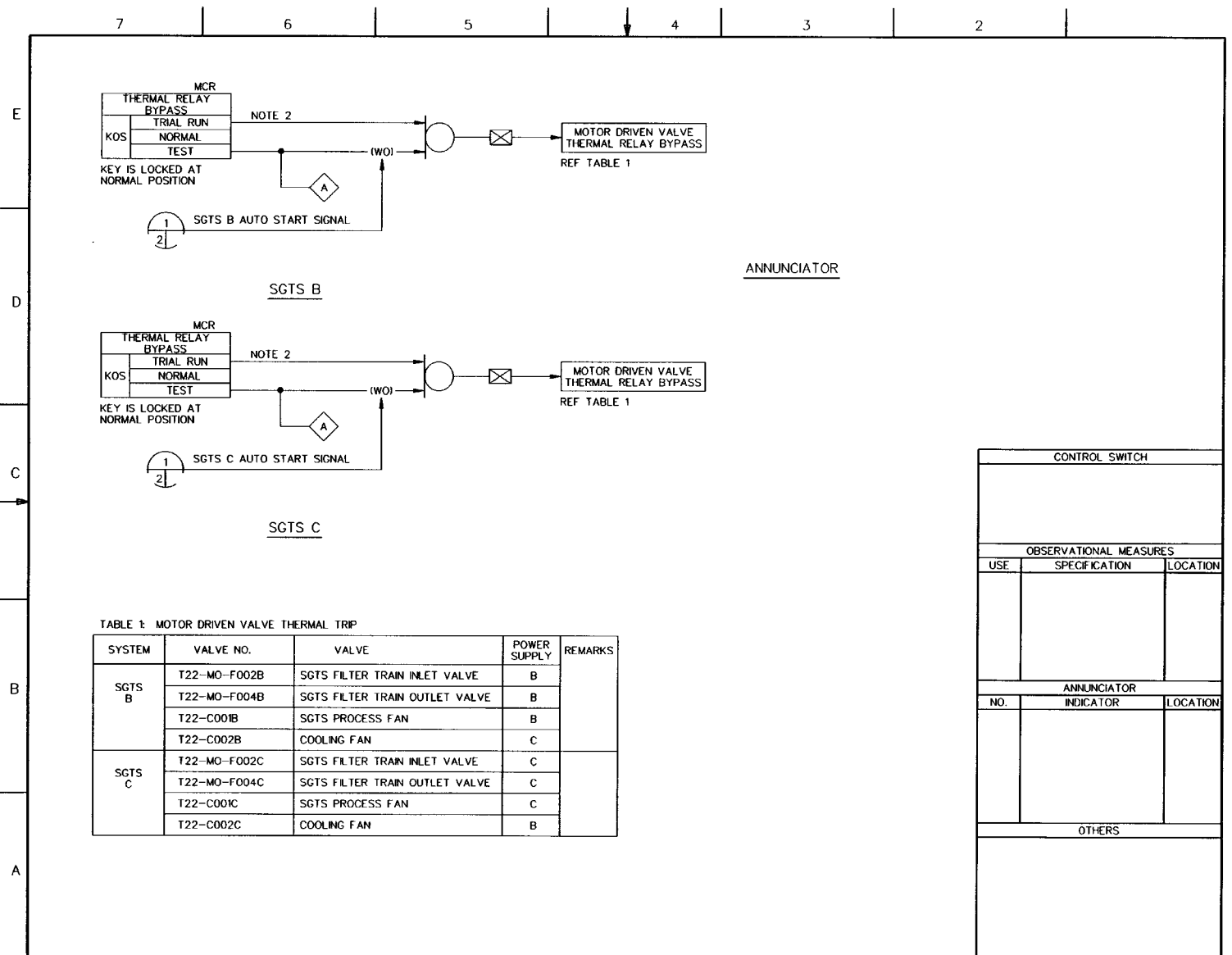


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 9 of 10)
ABWR DCD/Tier 2 Rev. 0 21-237

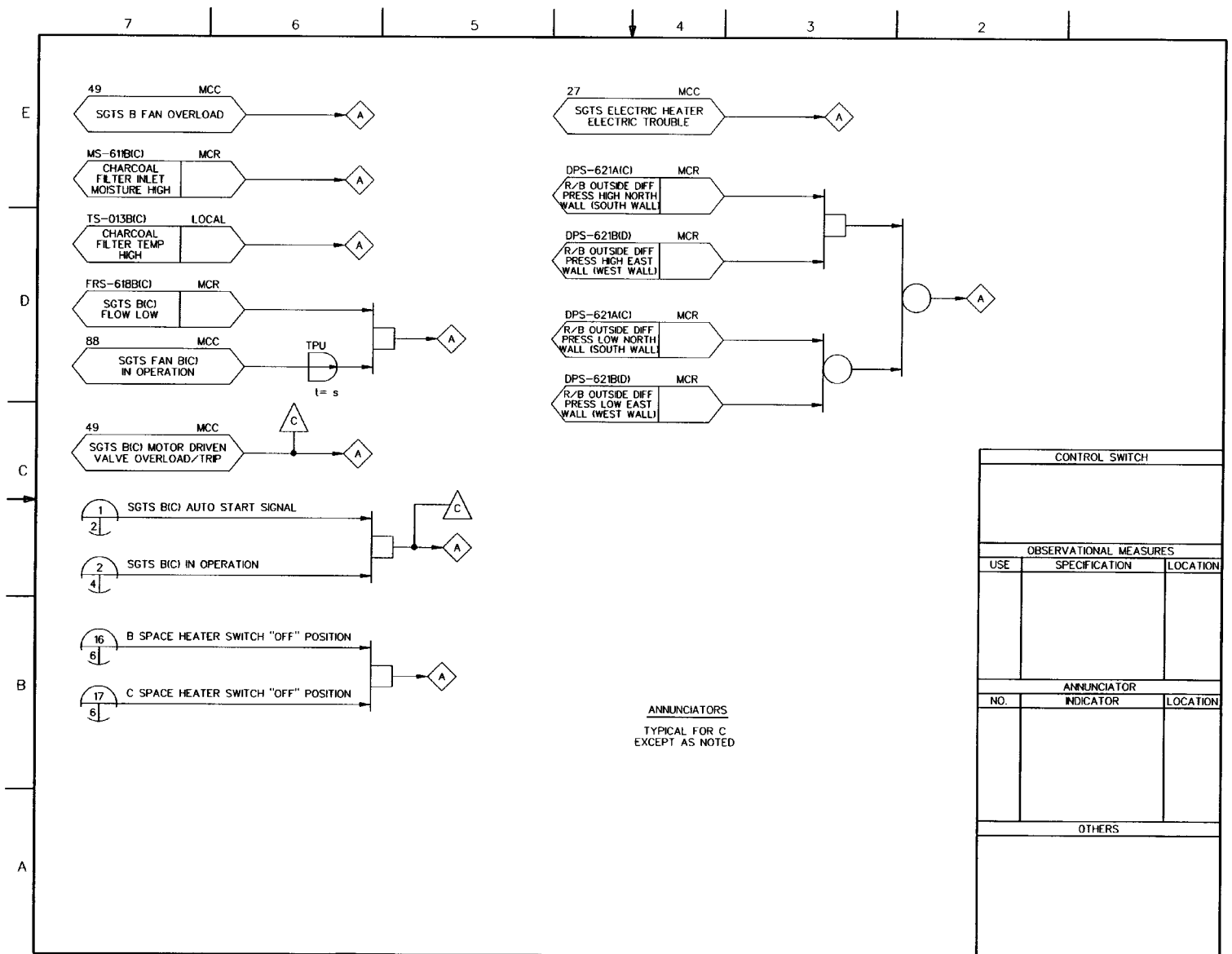


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 10 of 11)
ABWR DCD/Tier 2 Rev. 0 21-230

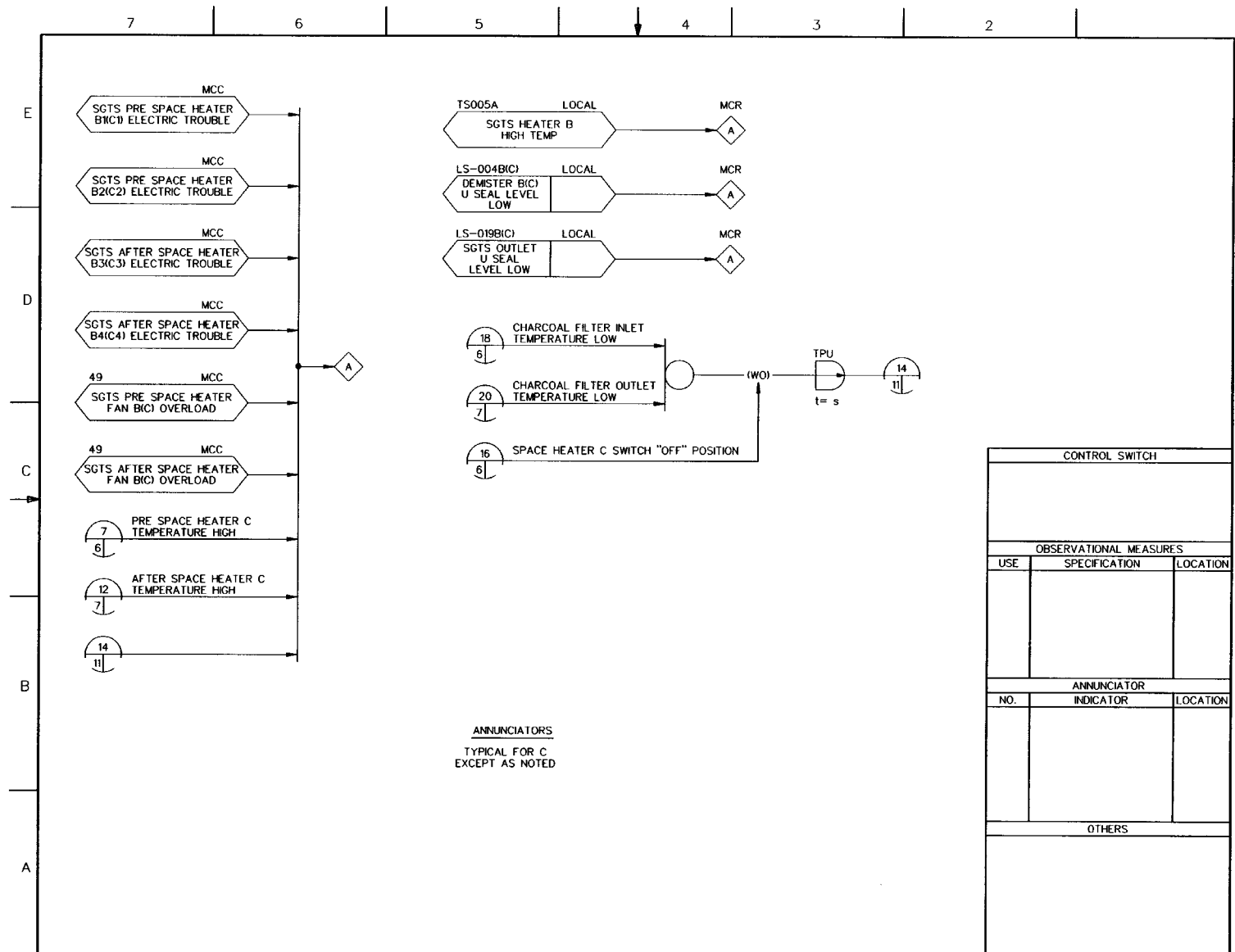
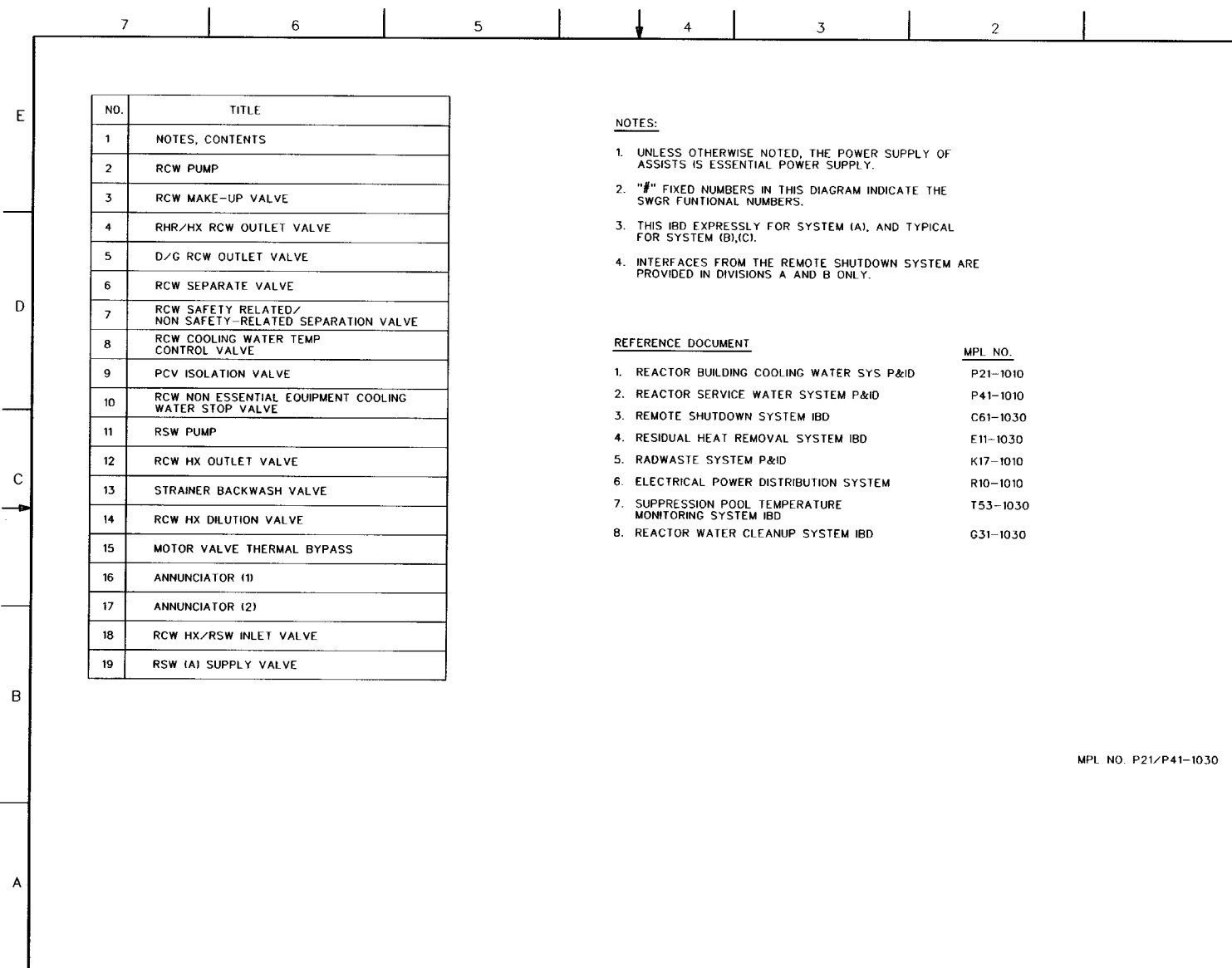


FIGURE 7.3-6 STANDBY GAS TREATMENT SYSTEM IBD (Sheet 11 of 11)
ABWR DCD/Tier 2 Rev. 0 21-239



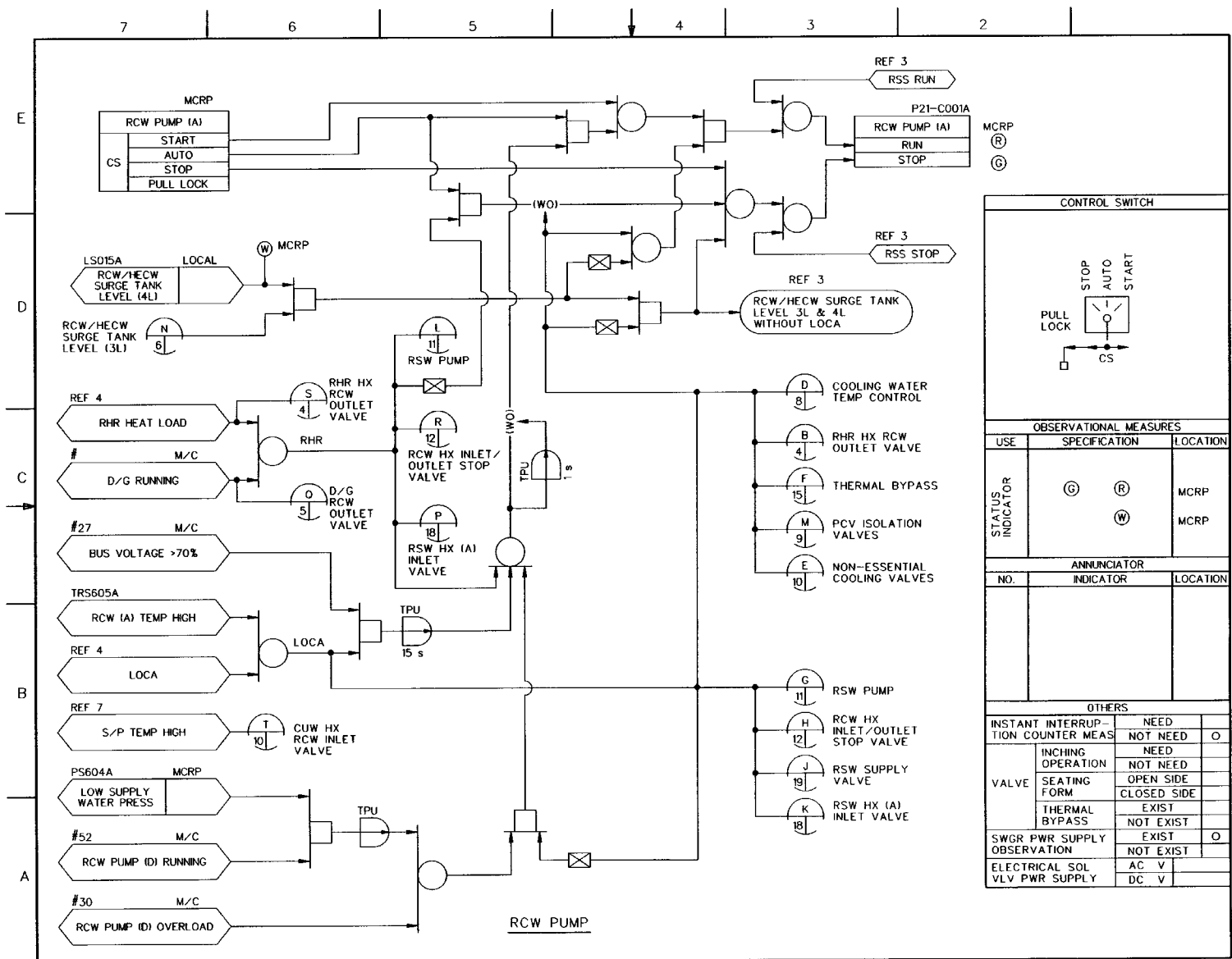


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 2 of 19)

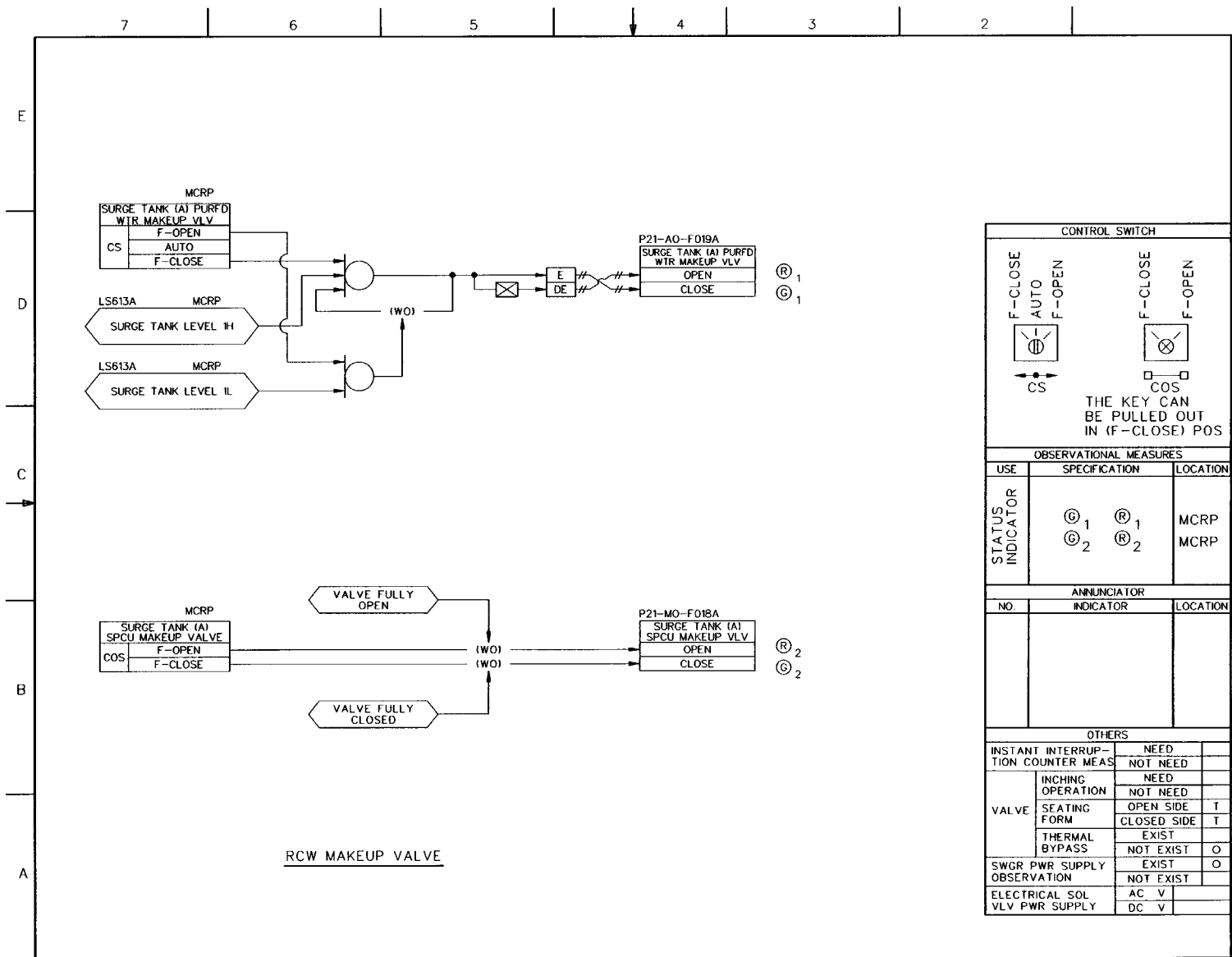


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 3 of 19)

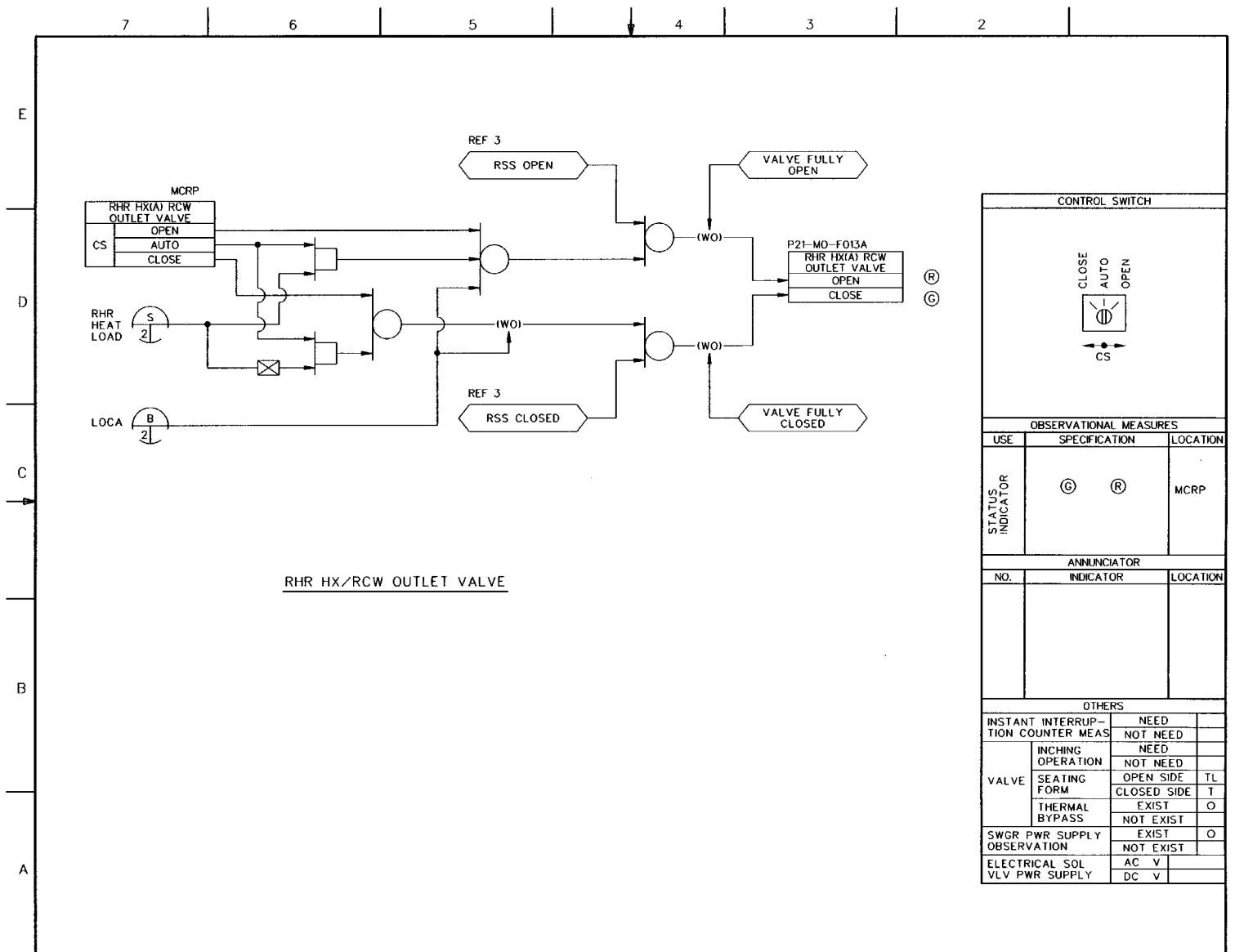


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 4 of 19)

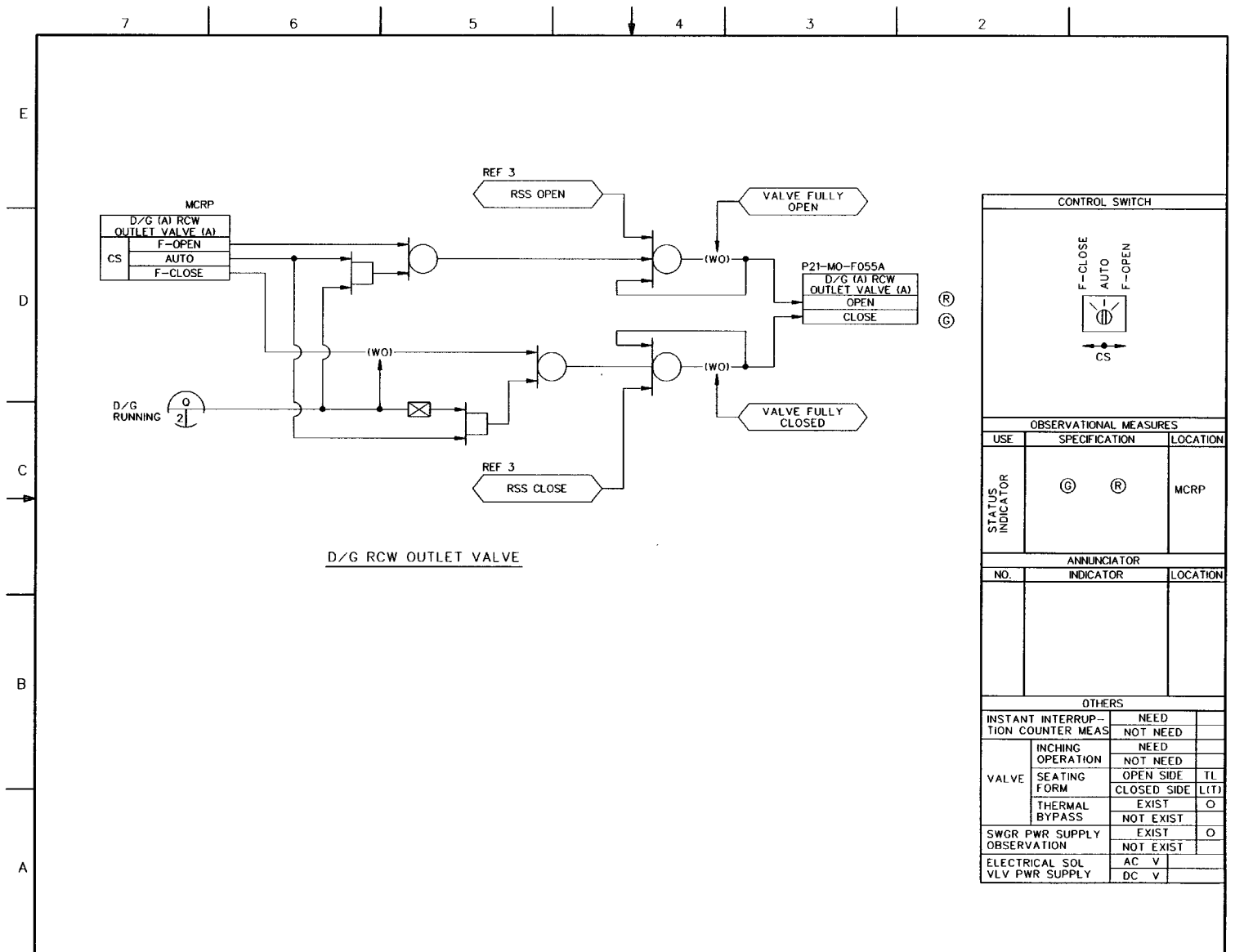


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 5 of 19)
ABWR DCD/Tier 2 Rev. 0 21-244

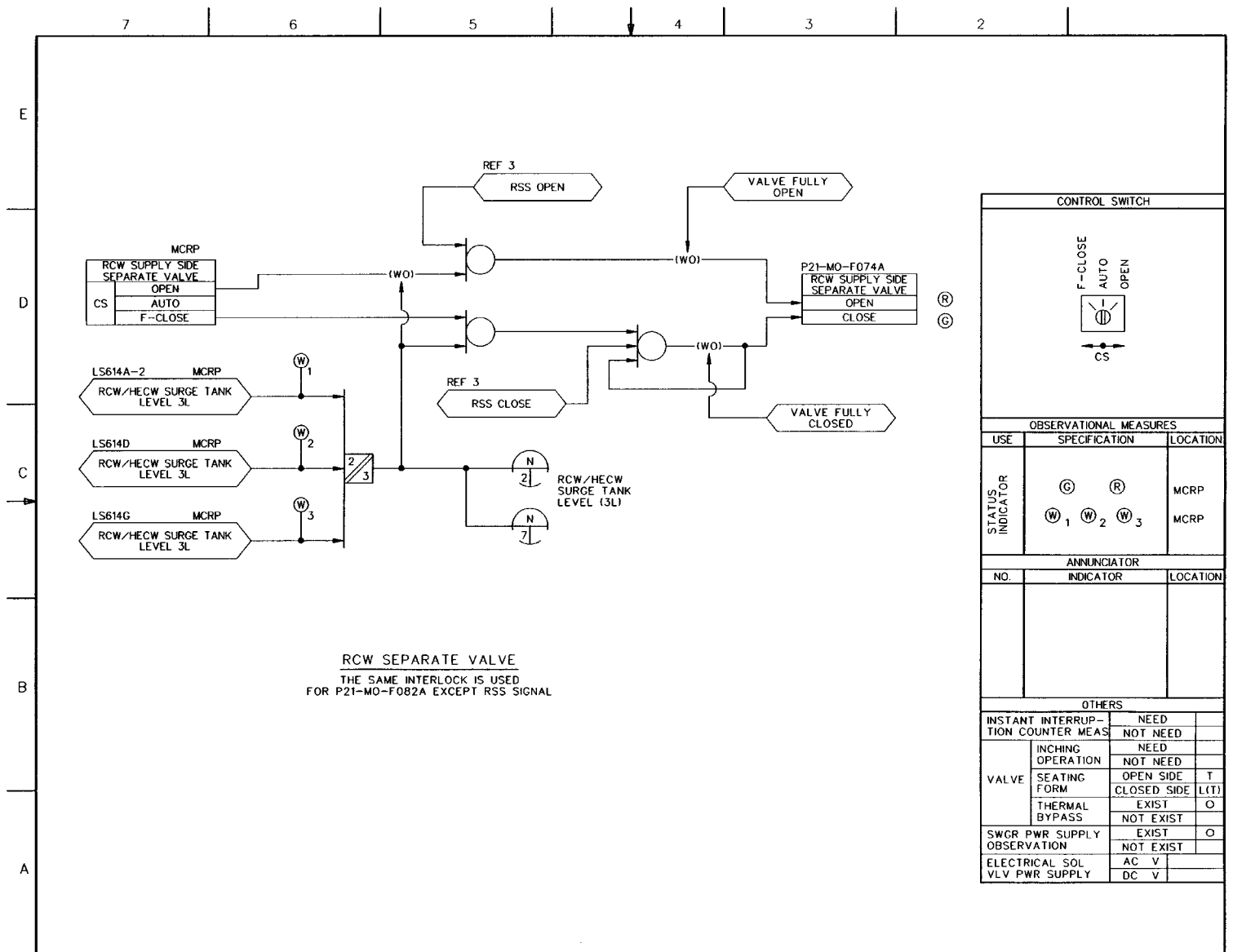


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 6 of 19)

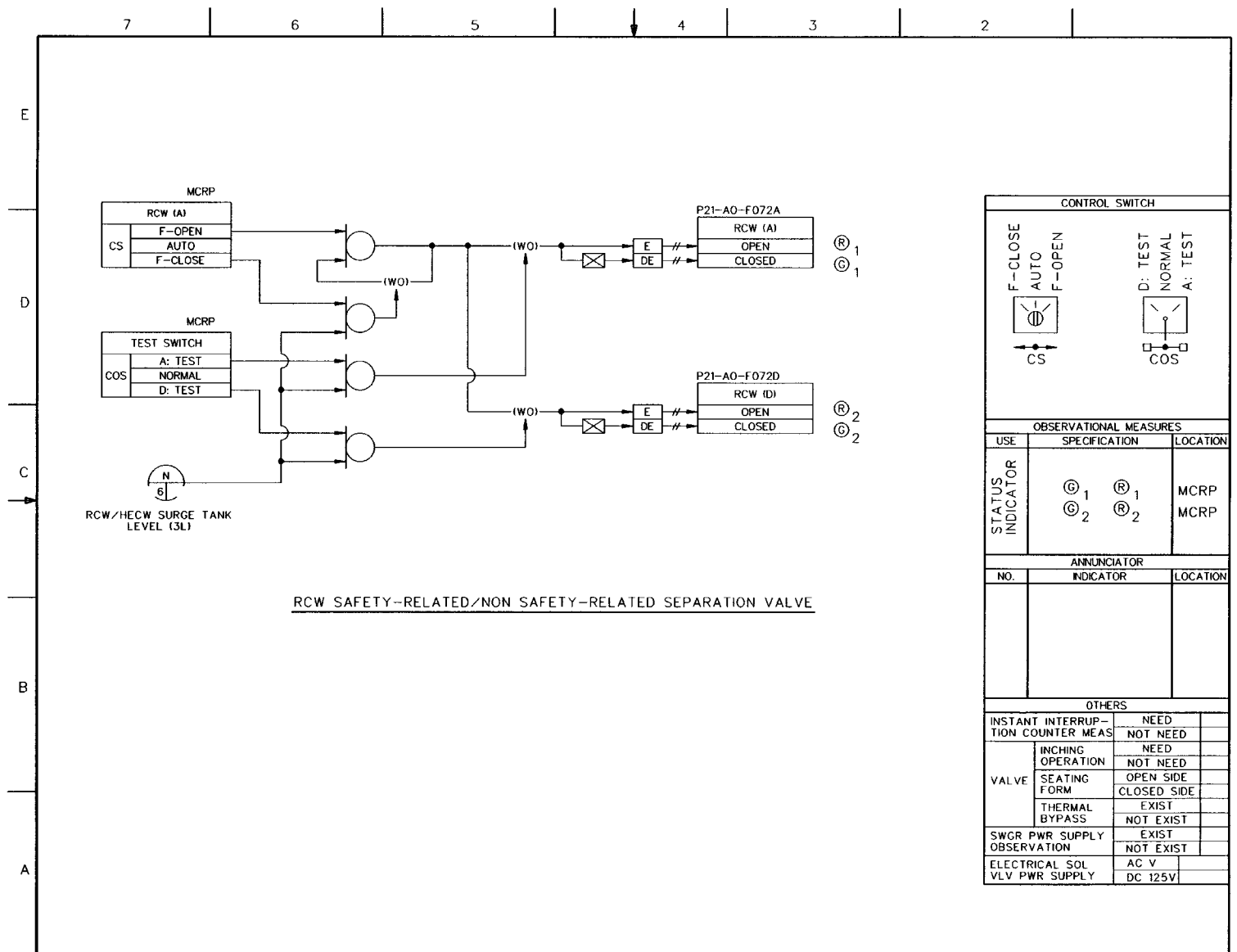


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 7 of 19)

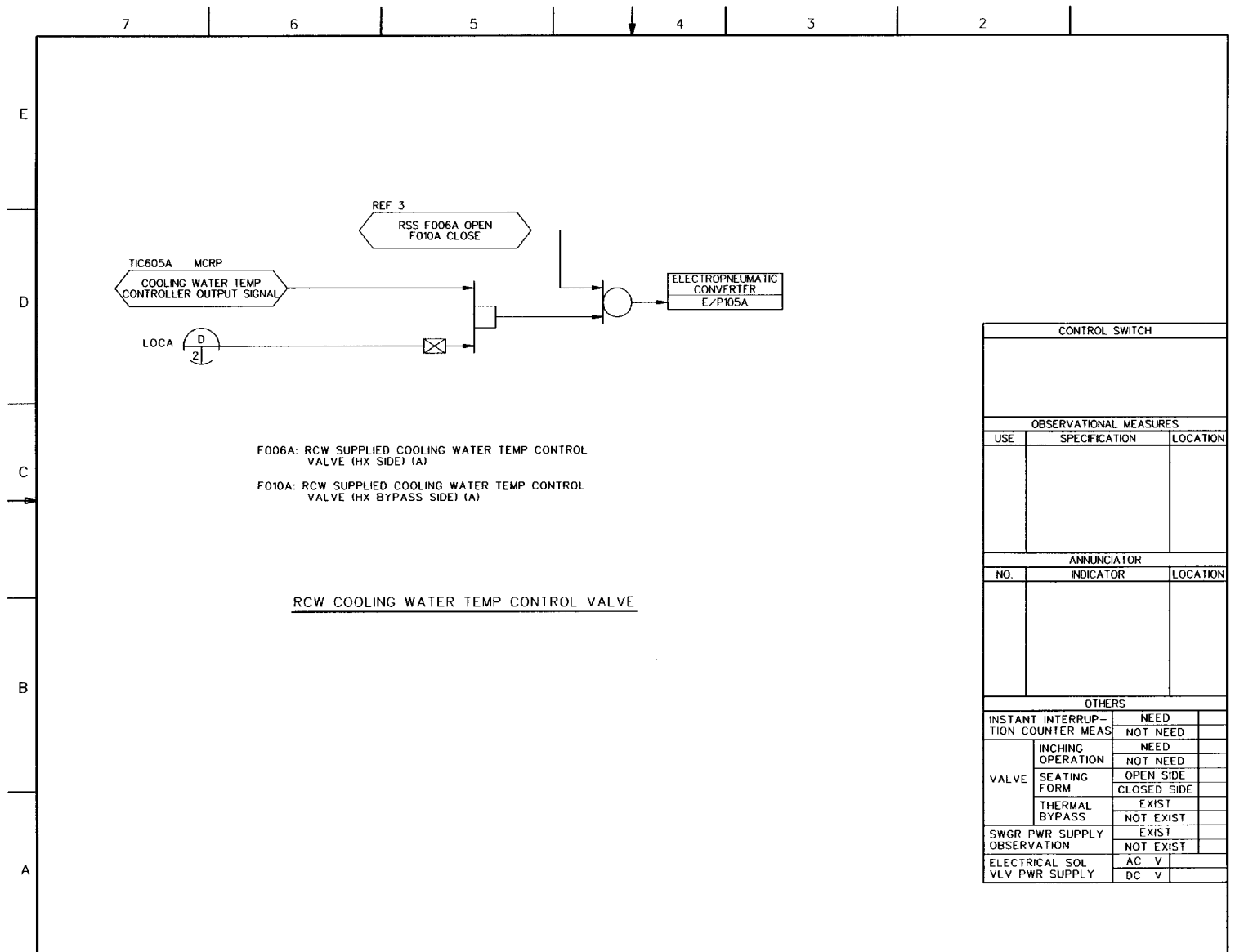


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 8 of 19)
ABWR DCD/Tier 2 Rev. 0 21-247

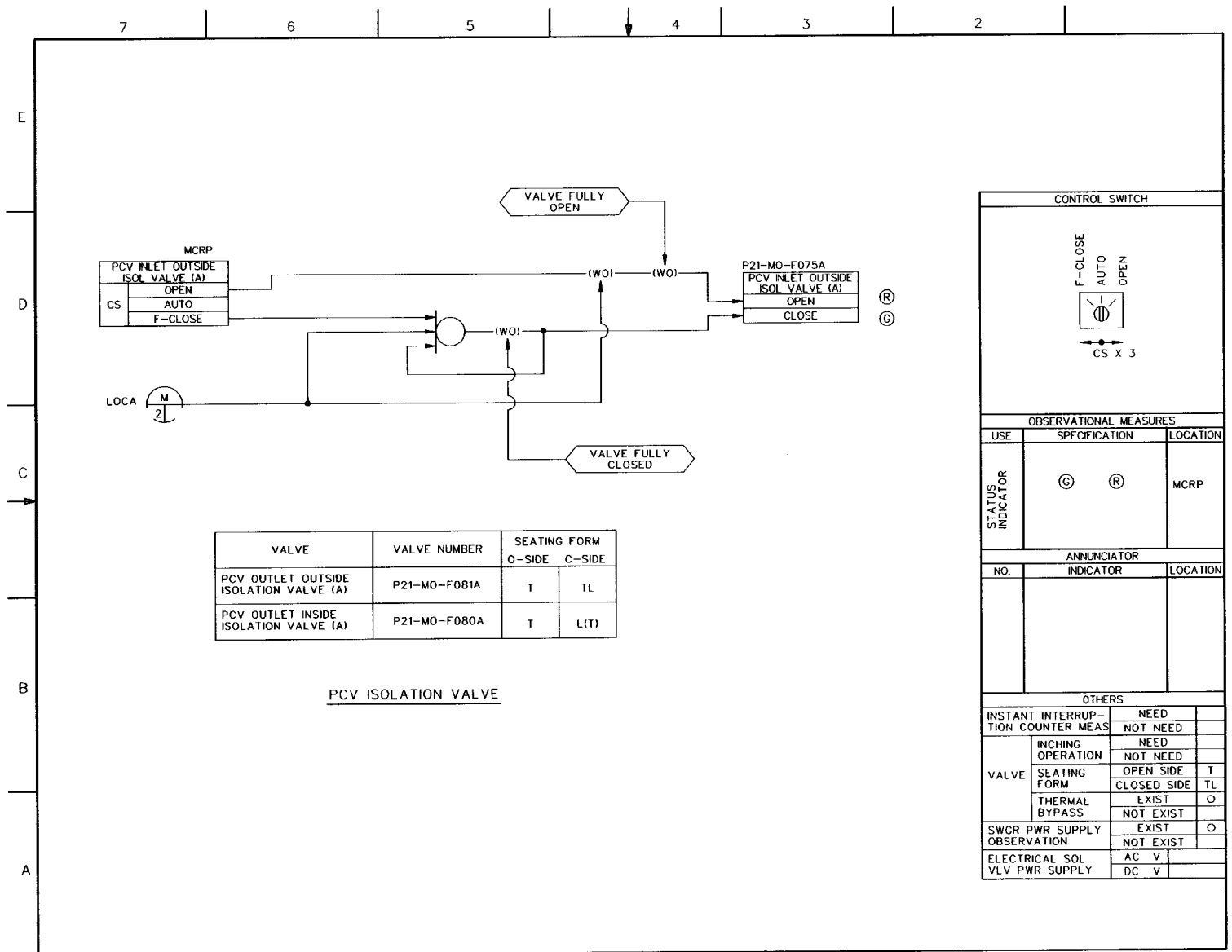


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 9 of 19)
ABWR DCD/Tier 2 Rev. 0 21-248

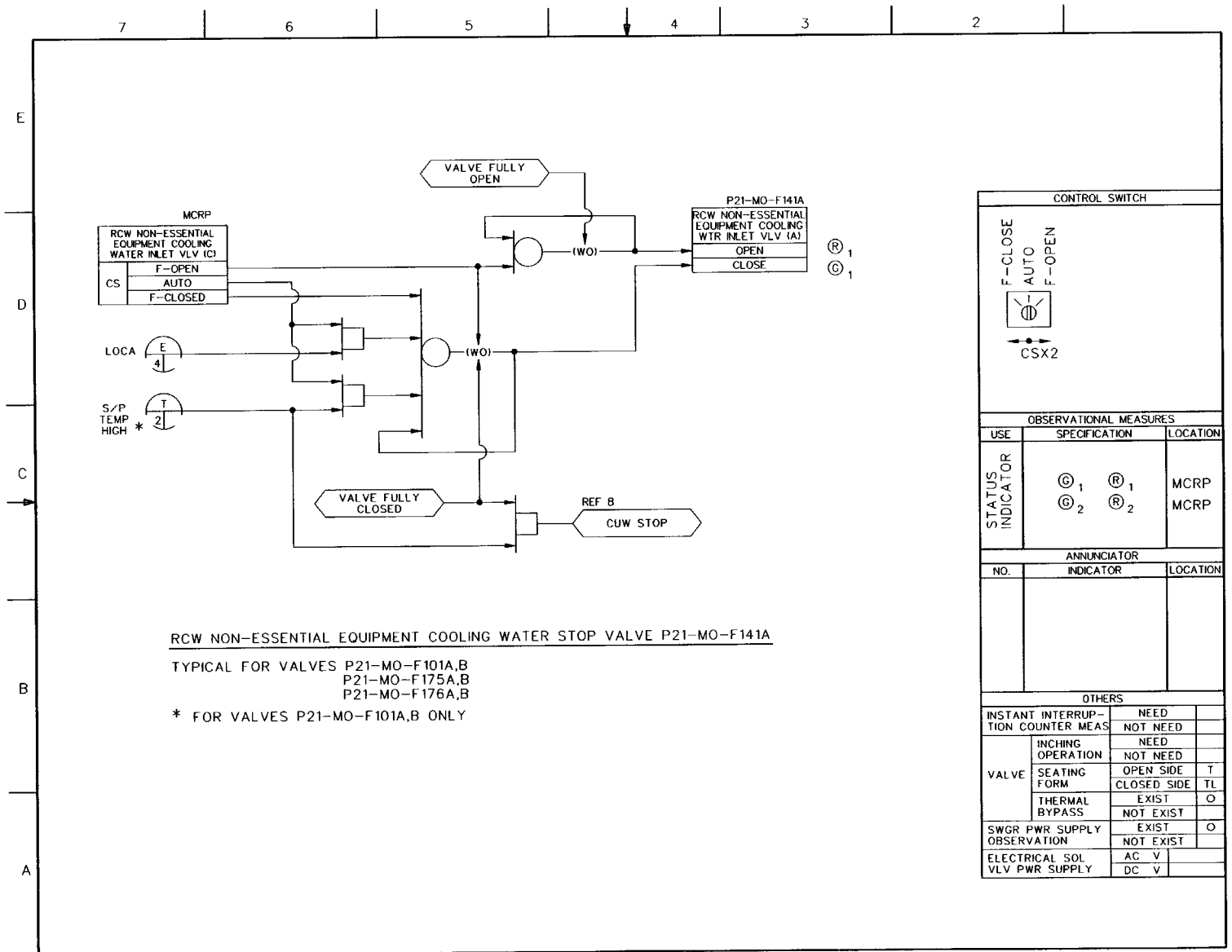


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 10 of 19)

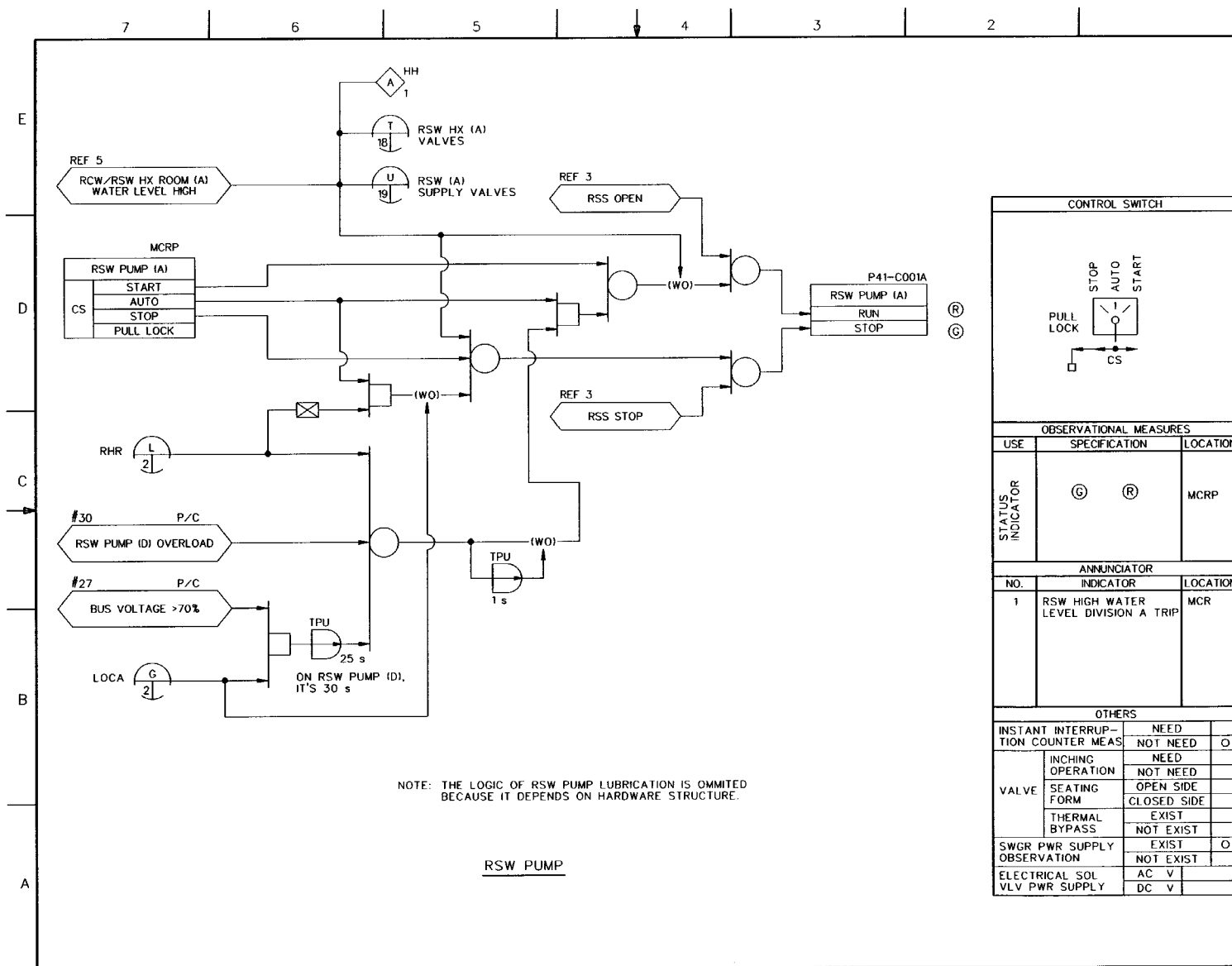


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 11 of 19)
ABWR DCD/Tier 2 Rev. 0 21-250

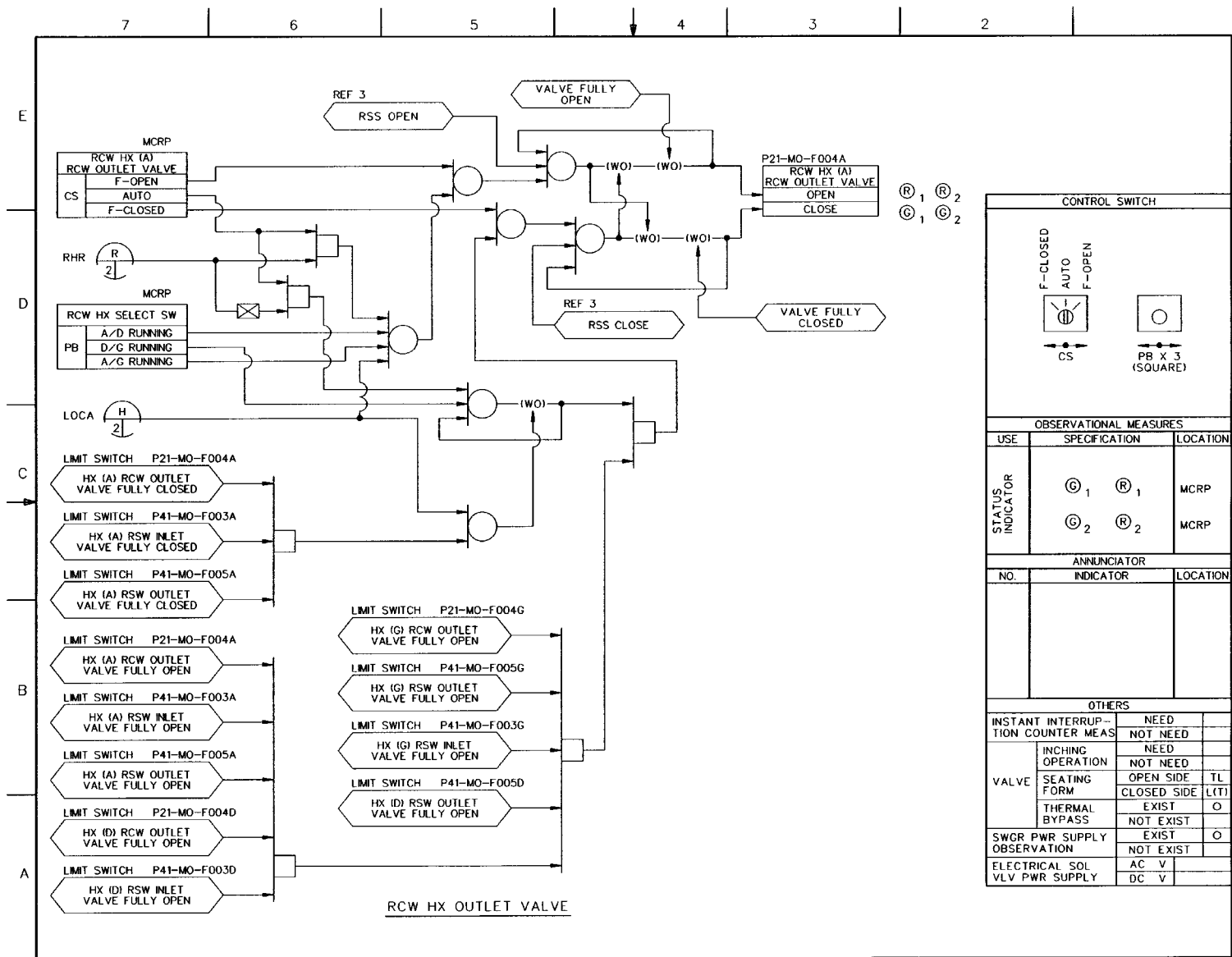


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 12 of 19)
ABWR DCD/Tier 2 Rev. 0 21-251

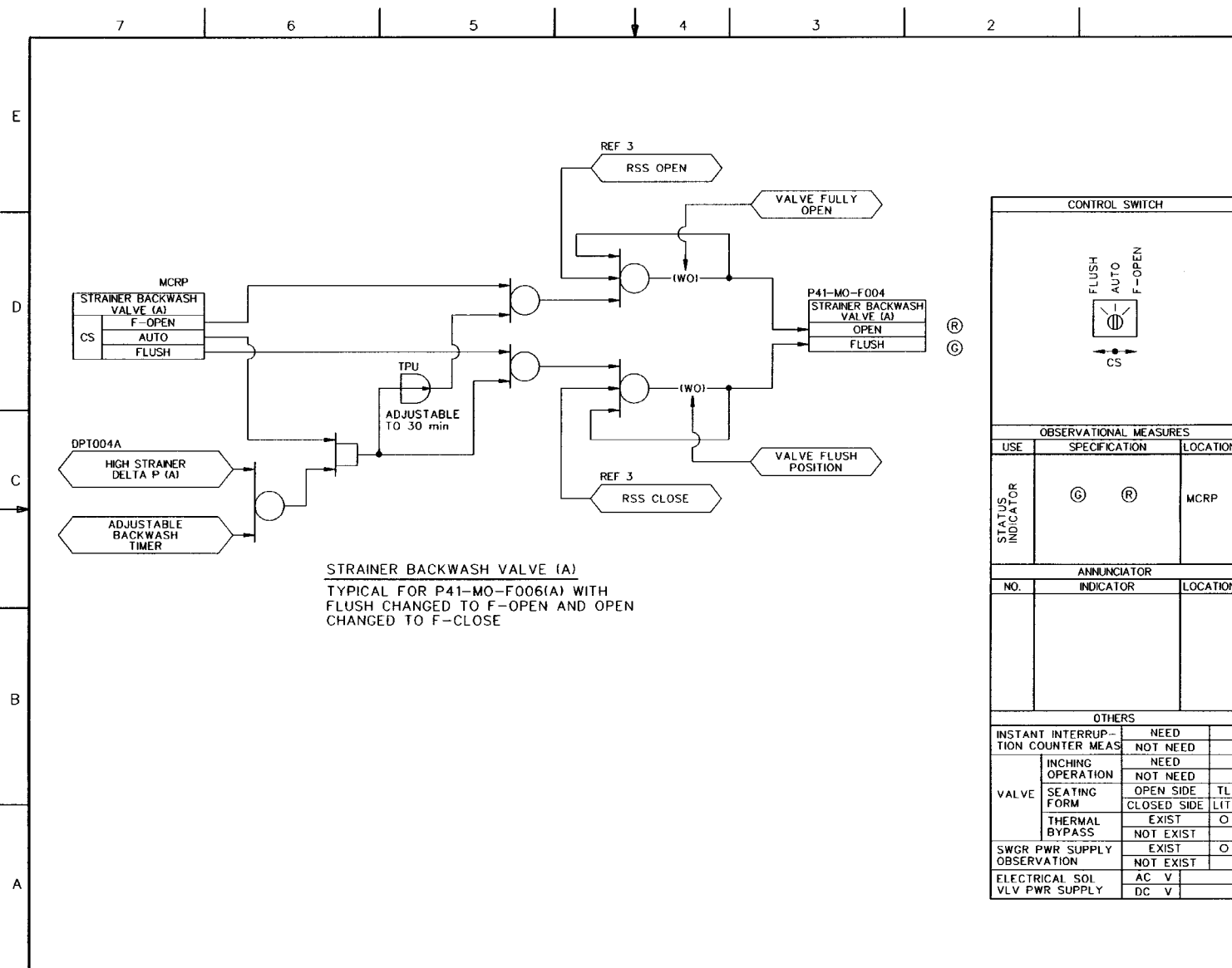


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 13 of 19)
ABWR DCD/Tier 2 Rev. 0 21-252

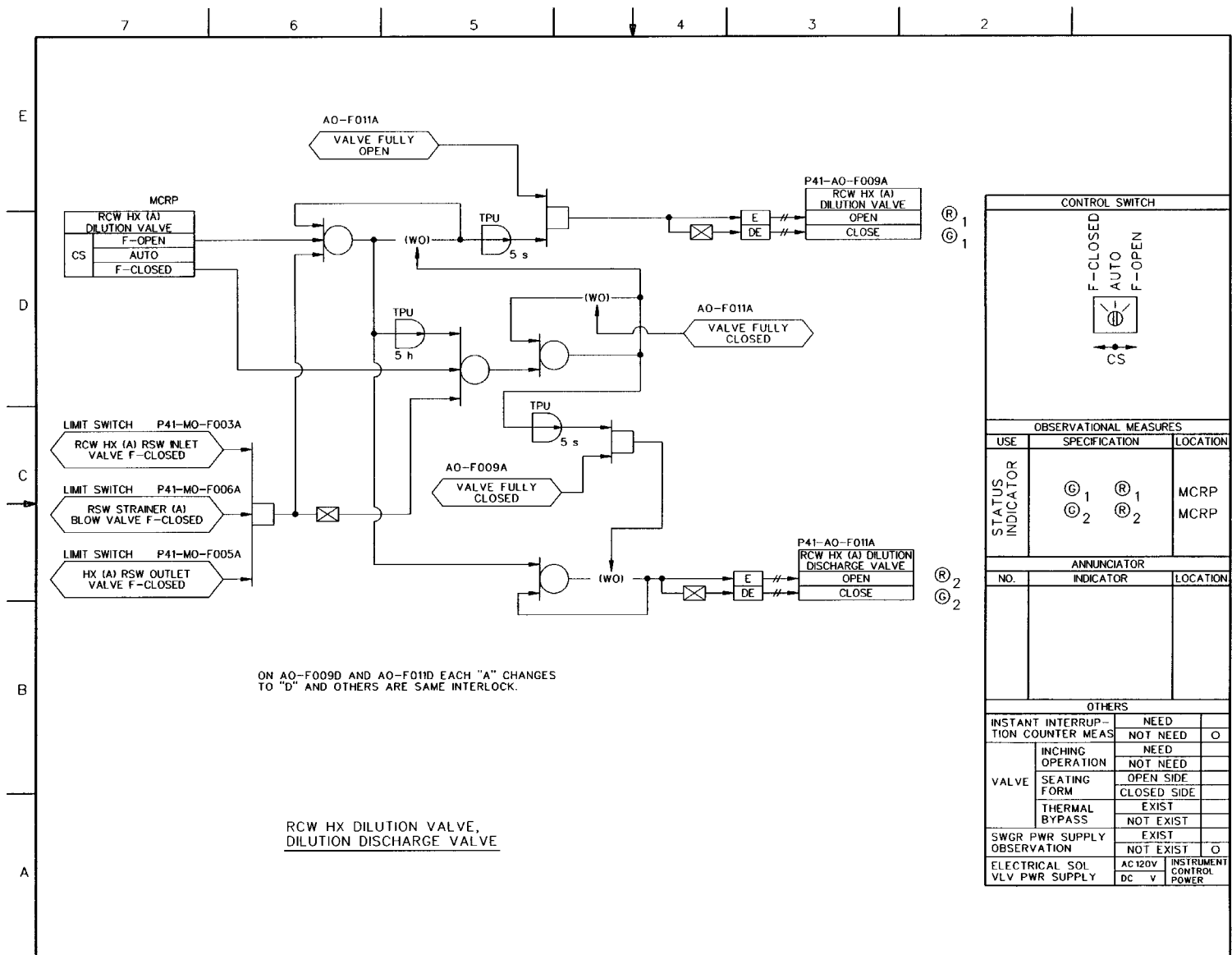


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 14 of 19)
 ABWR DCD/Tier 2 Rev. 0 21-253

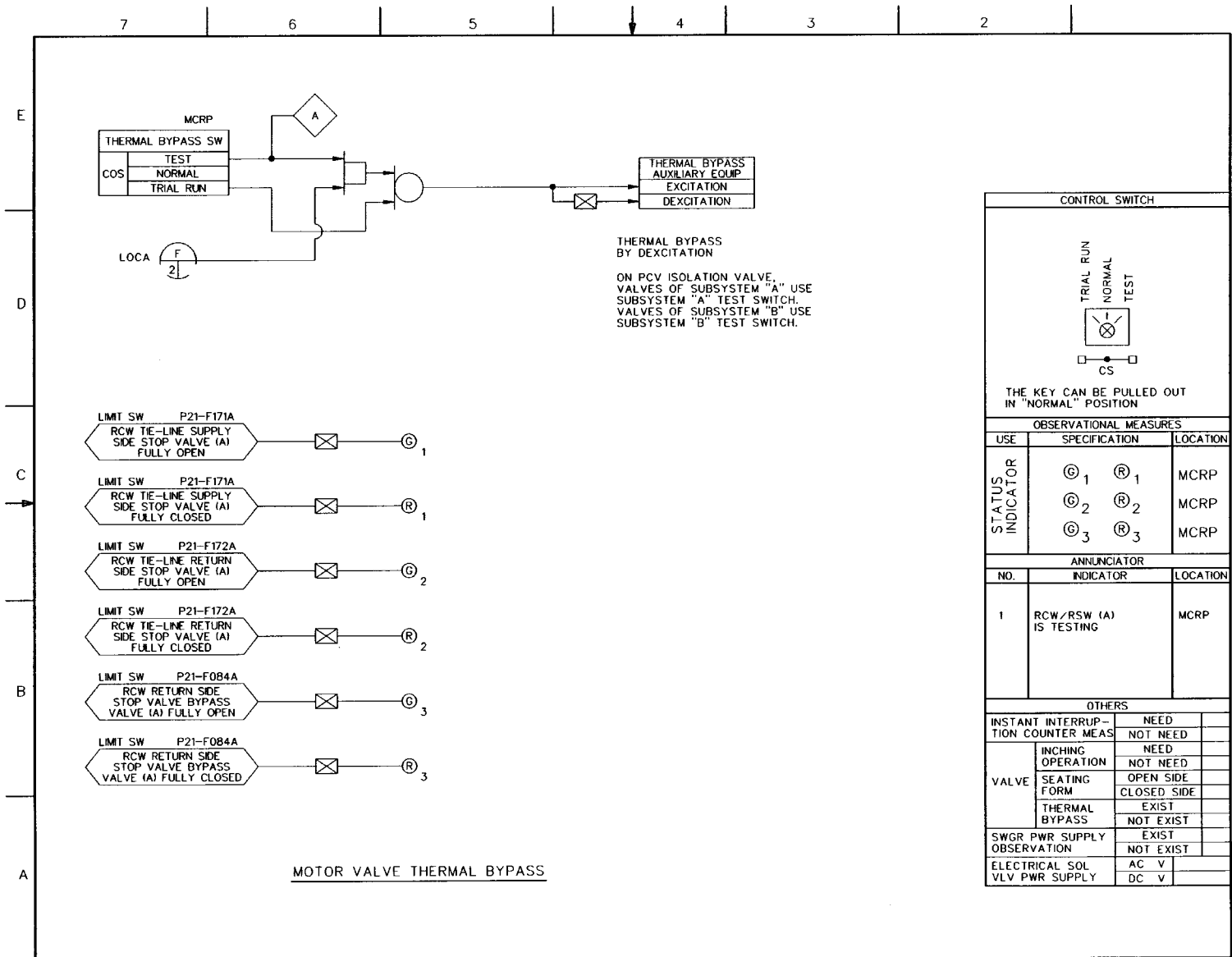


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 15 of 19)
ABWR DCD/Tier 2 Rev. 0 21-254

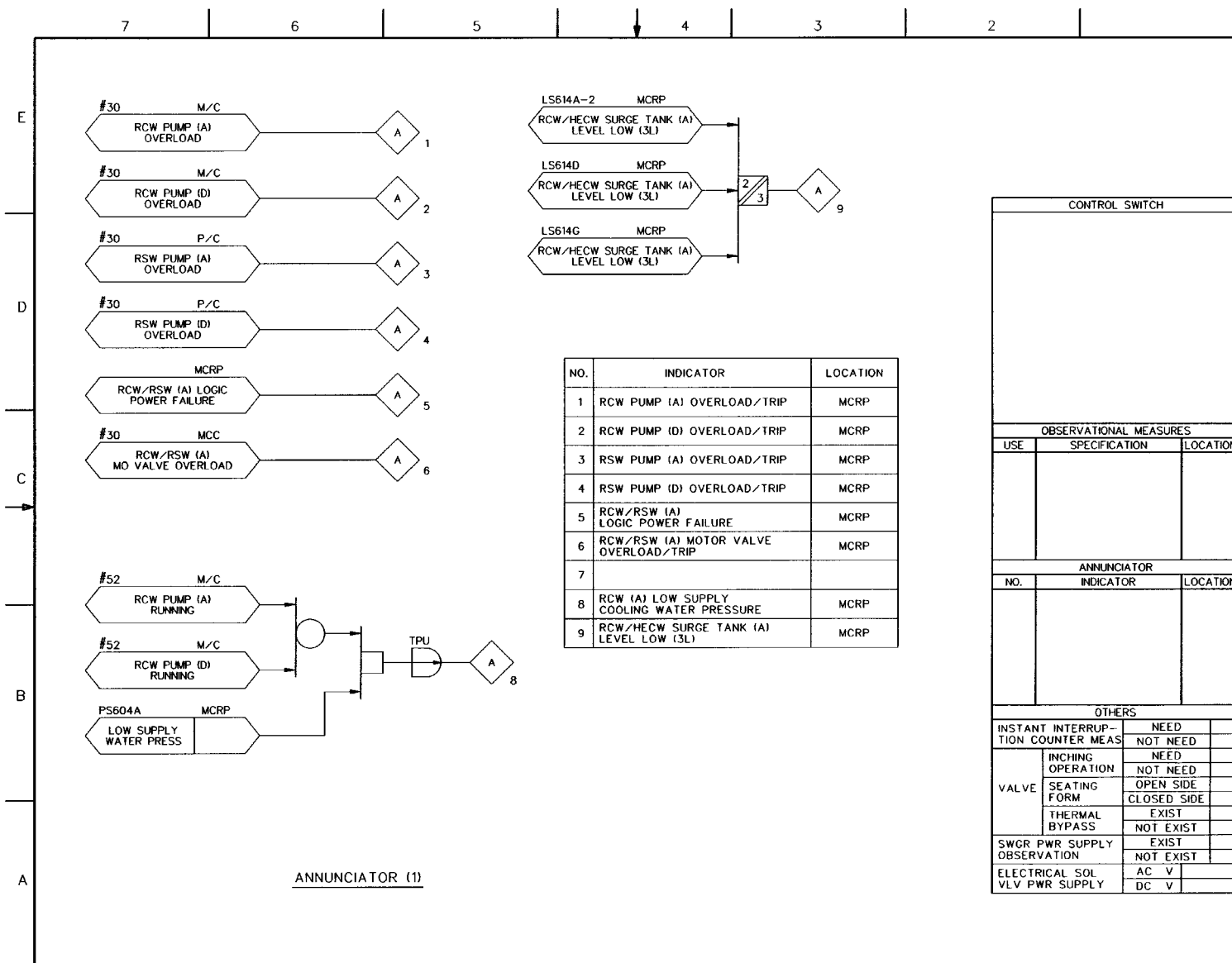


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 16 of 19)
ABWR DCD/Tier 2 Rev. 0 21-255

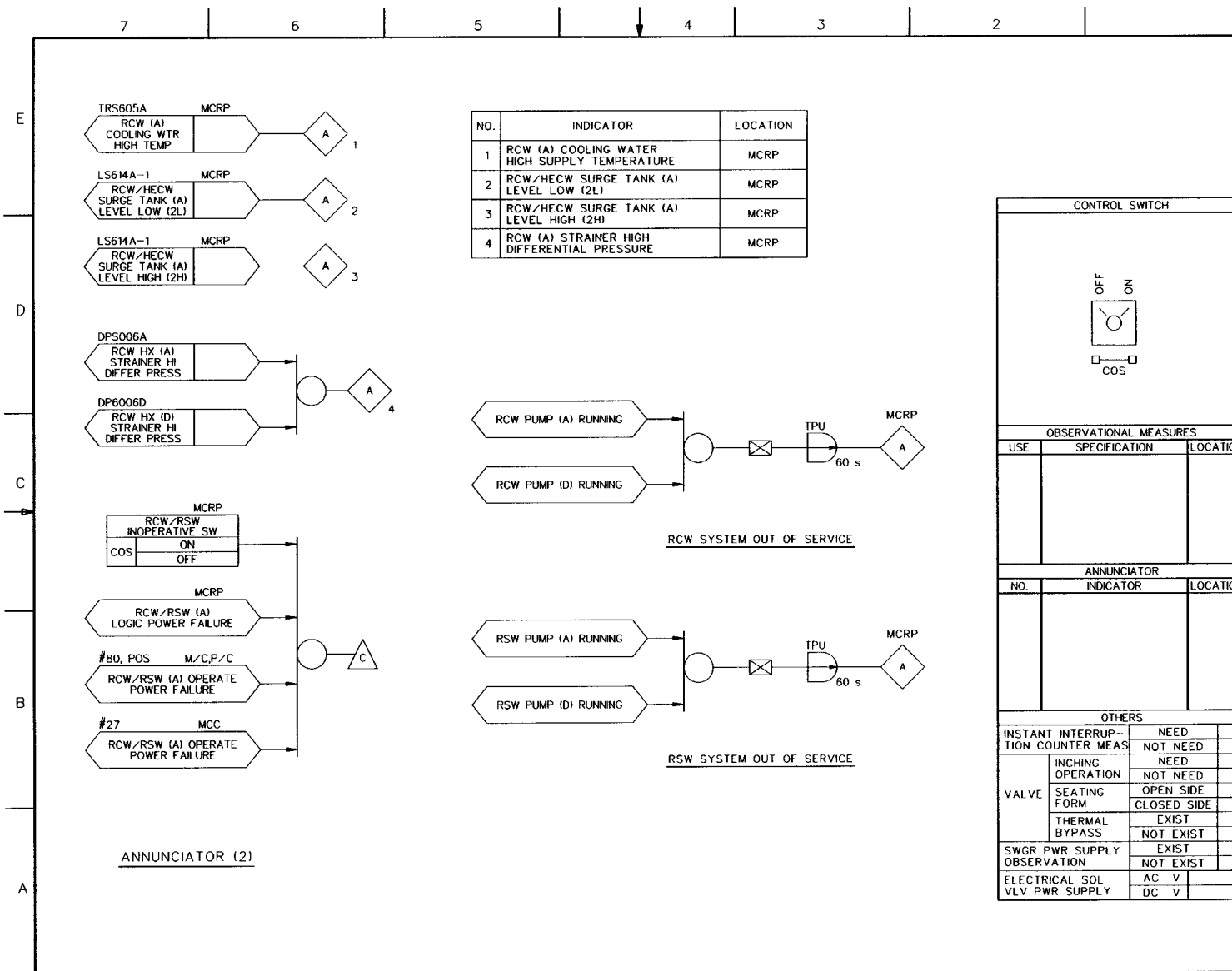


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 17 of 19)

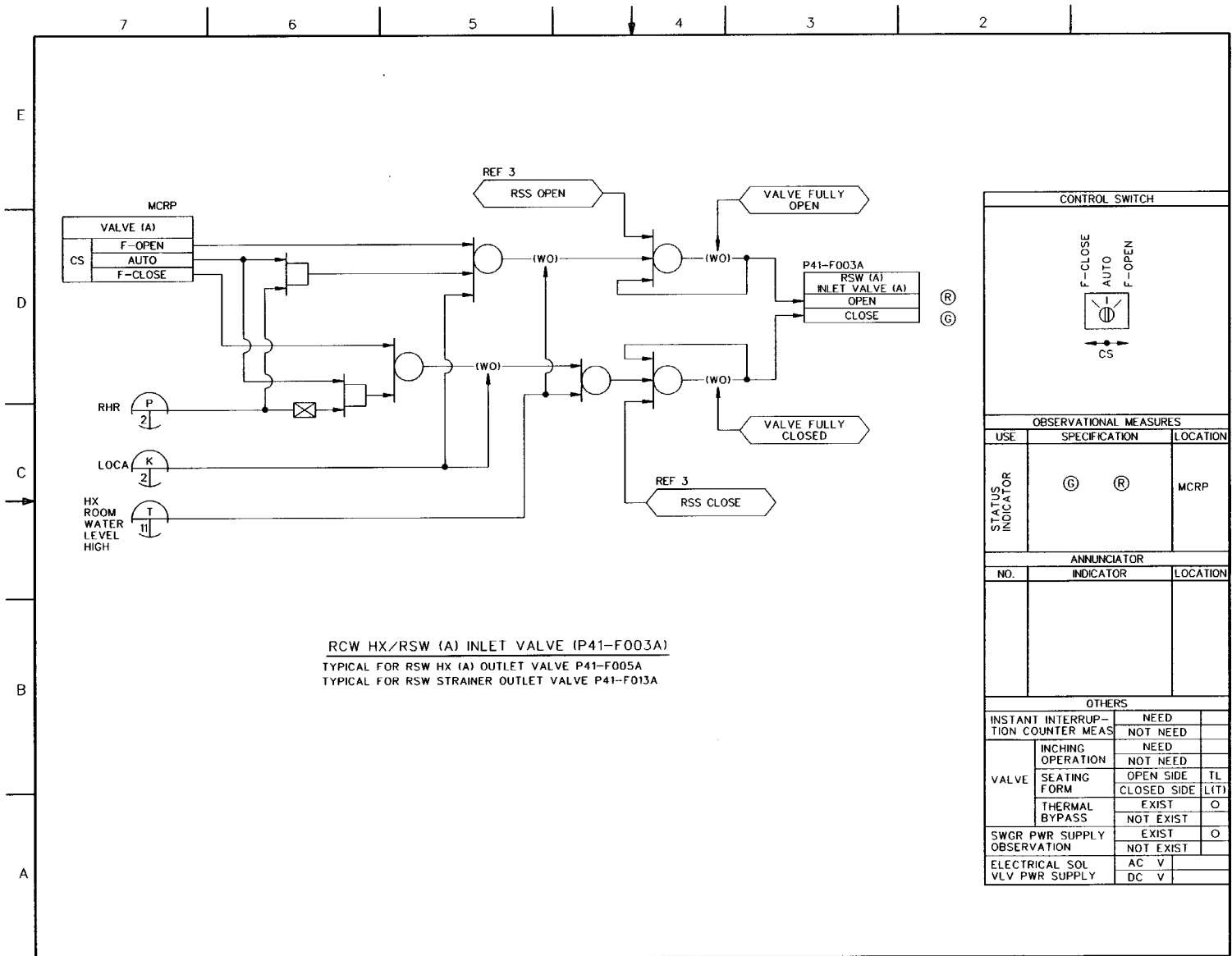
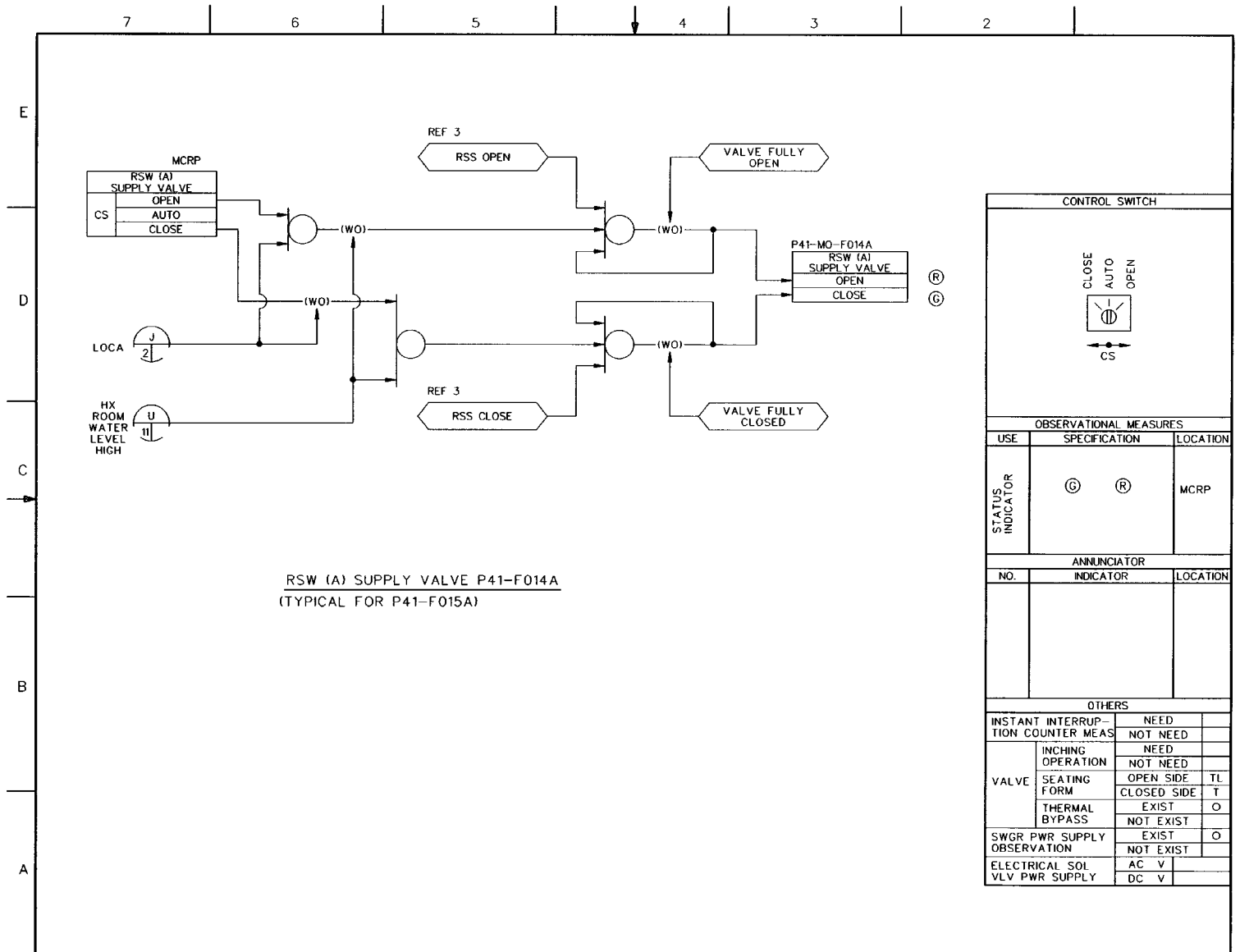
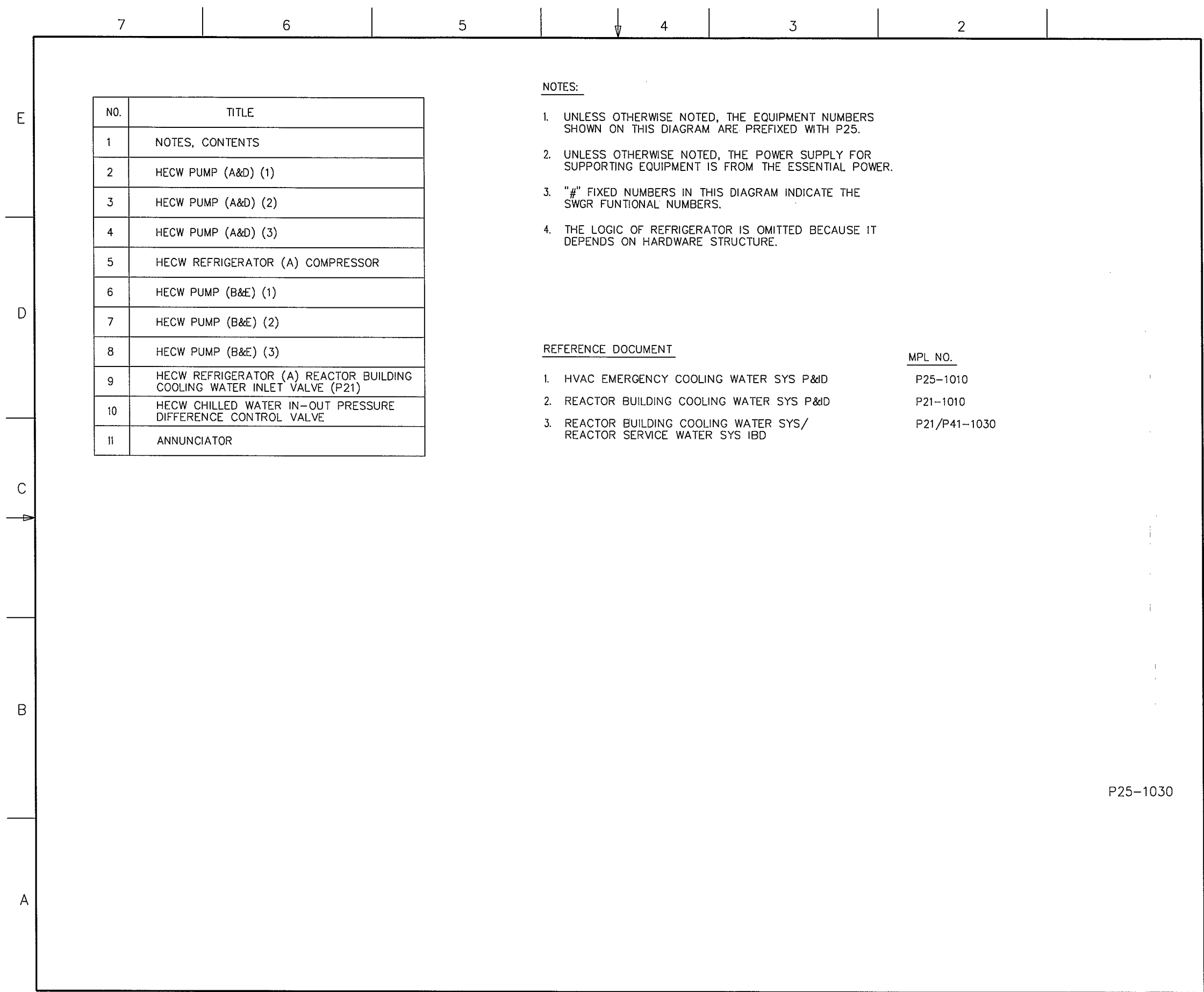
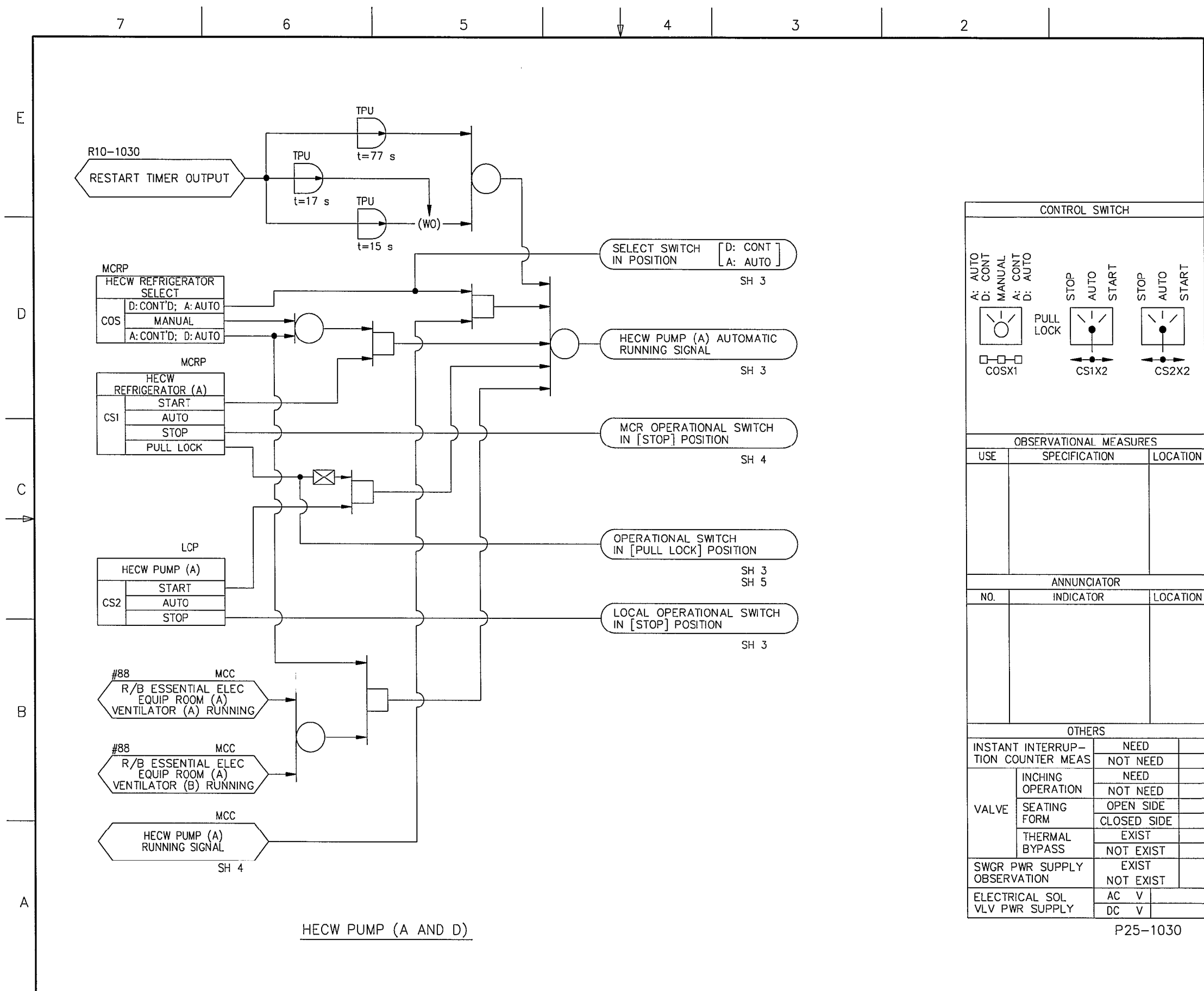


FIGURE 7.3-7 REACTOR BUILDING COOLING WATER/REACTOR SERVICE WATER SYSTEM IBD (Sheet 18 of 19)
ABWR DCD/Tier 2 Rev. 0 21-257



CONTROL SWITCH		
CLOSE AUTO OPEN CS		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
STATUS INDICATOR	Ⓒ Ⓓ	MCRP
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
INSTANT INTERRUPT- TION COUNTER MEAS	NEED NOT NEED	
INCHING OPERATION	NEED NOT NEED	
VALVE SEATING FORM	OPEN SIDE CLOSED SIDE	TL T
THERMAL BYPASS	EXIST NOT EXIST	O
SWGR PWR SUPPLY OBSERVATION	EXIST NOT EXIST	O
ELECTRICAL SOL	AC V	
VLV PWR SUPPLY	DC V	





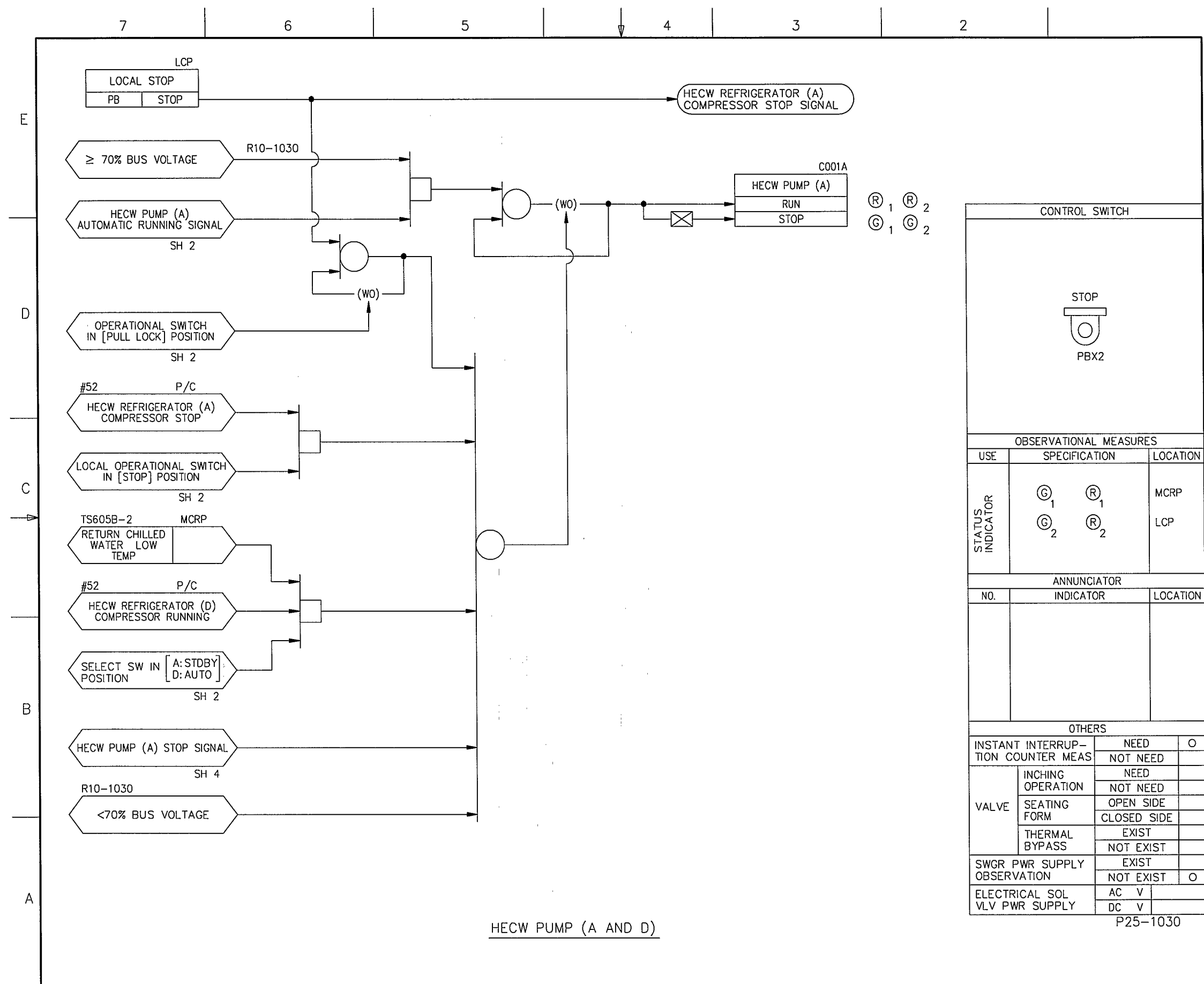
137C9445\B075002

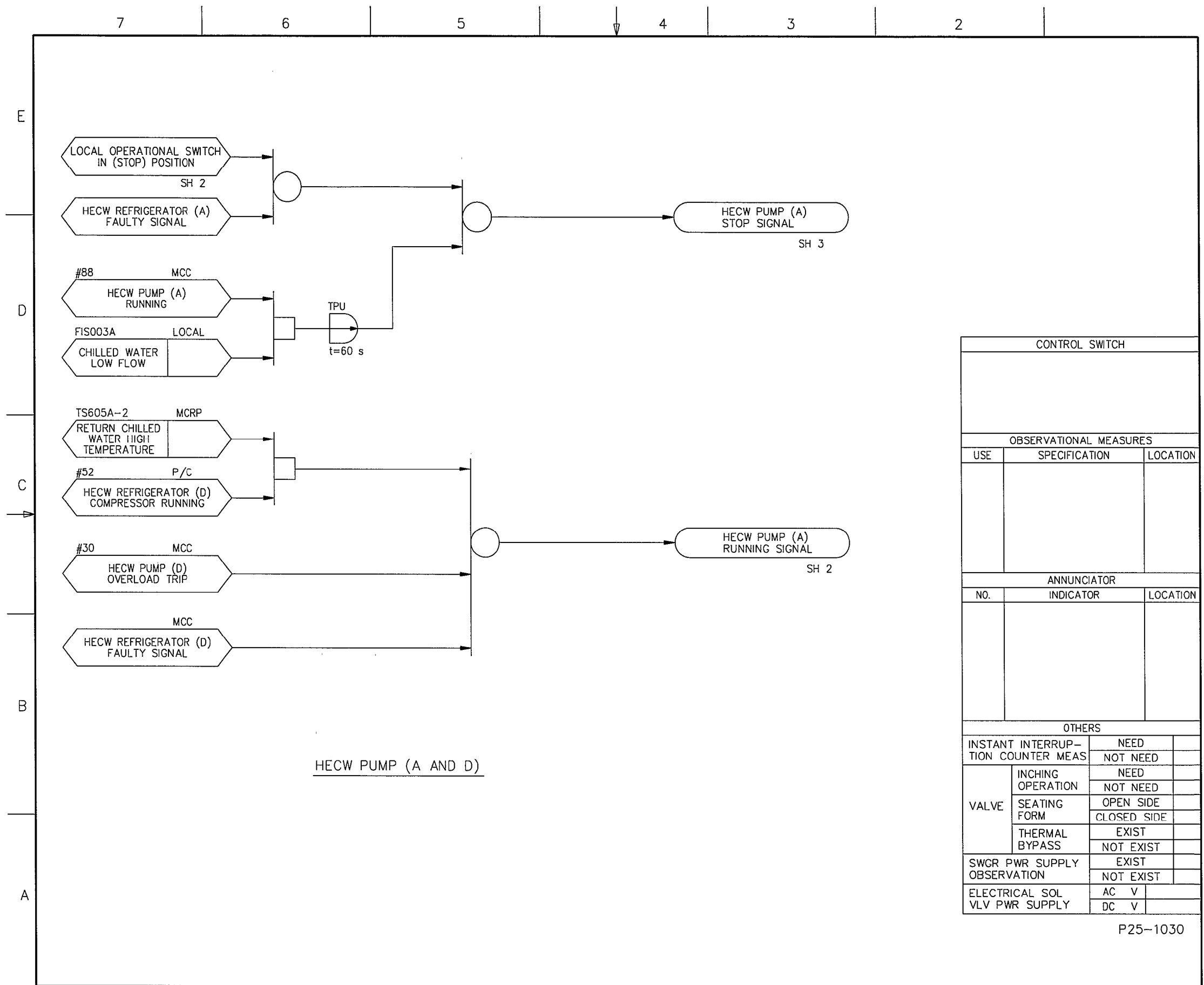
FIGURE 7.3-9 HVAC EMERGENCY COOLING WATER SYSTEM IBD (Sheet 2 of 11)

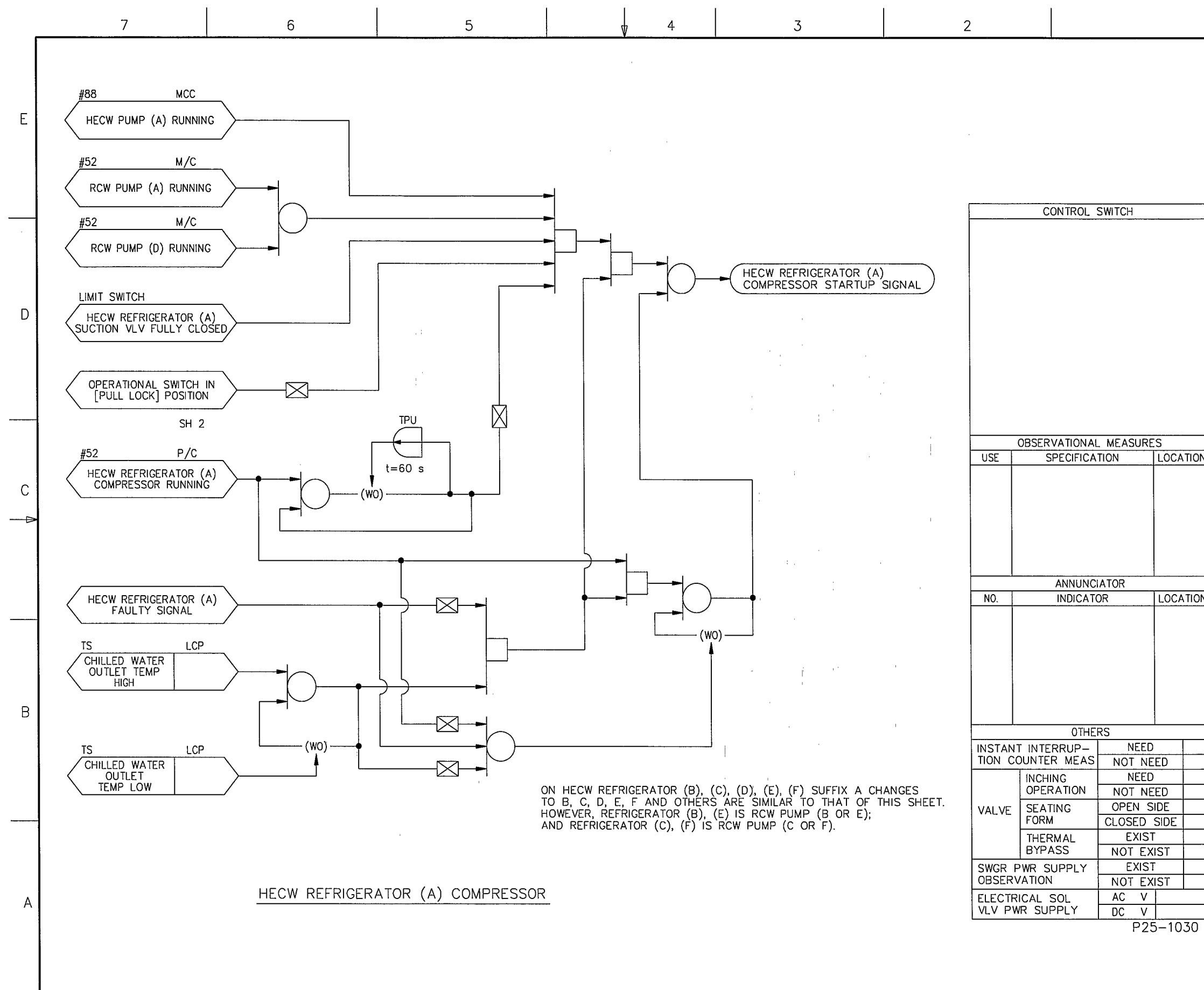
ABWR DCD/Tier 2

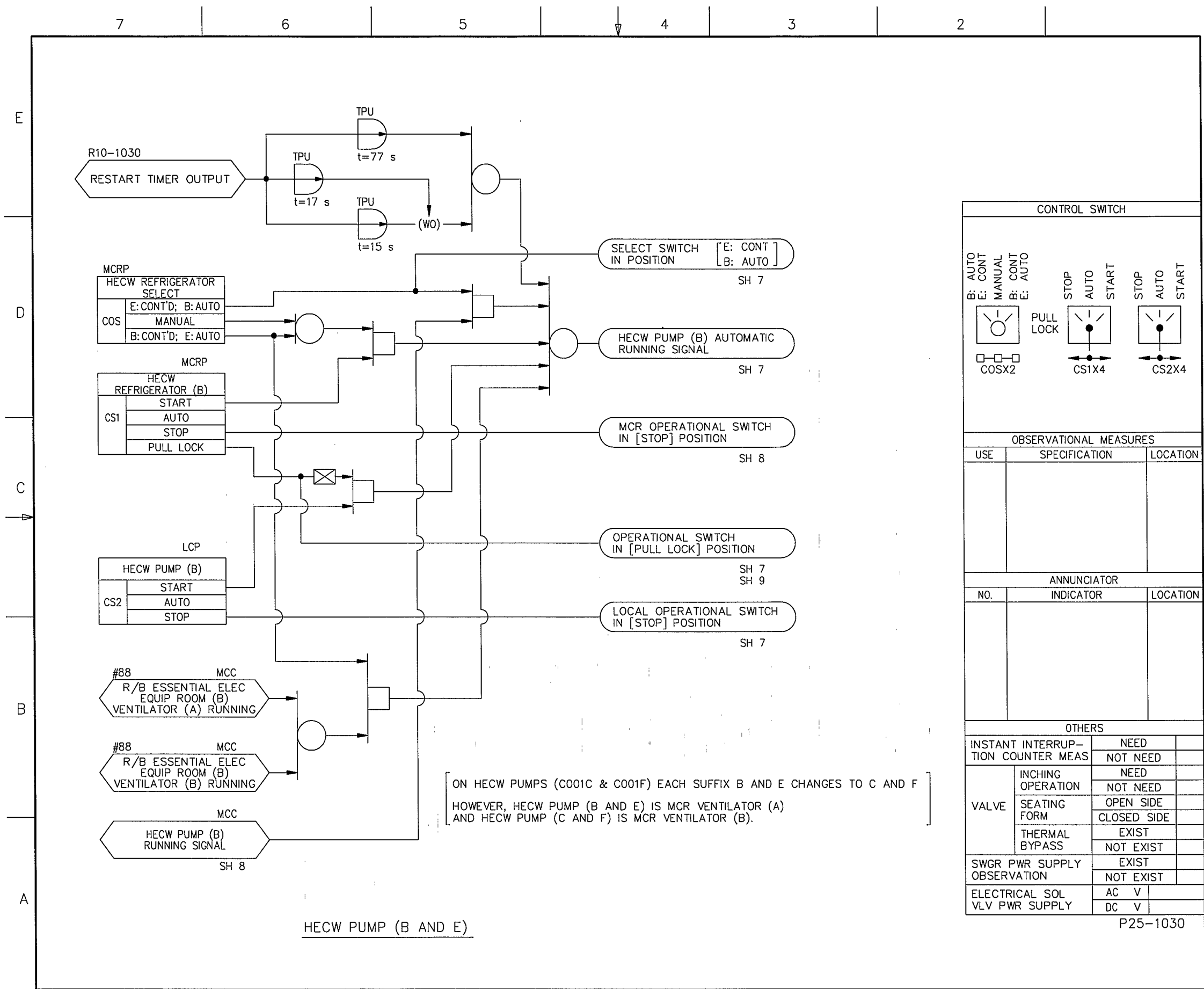
Rev 3

21-260



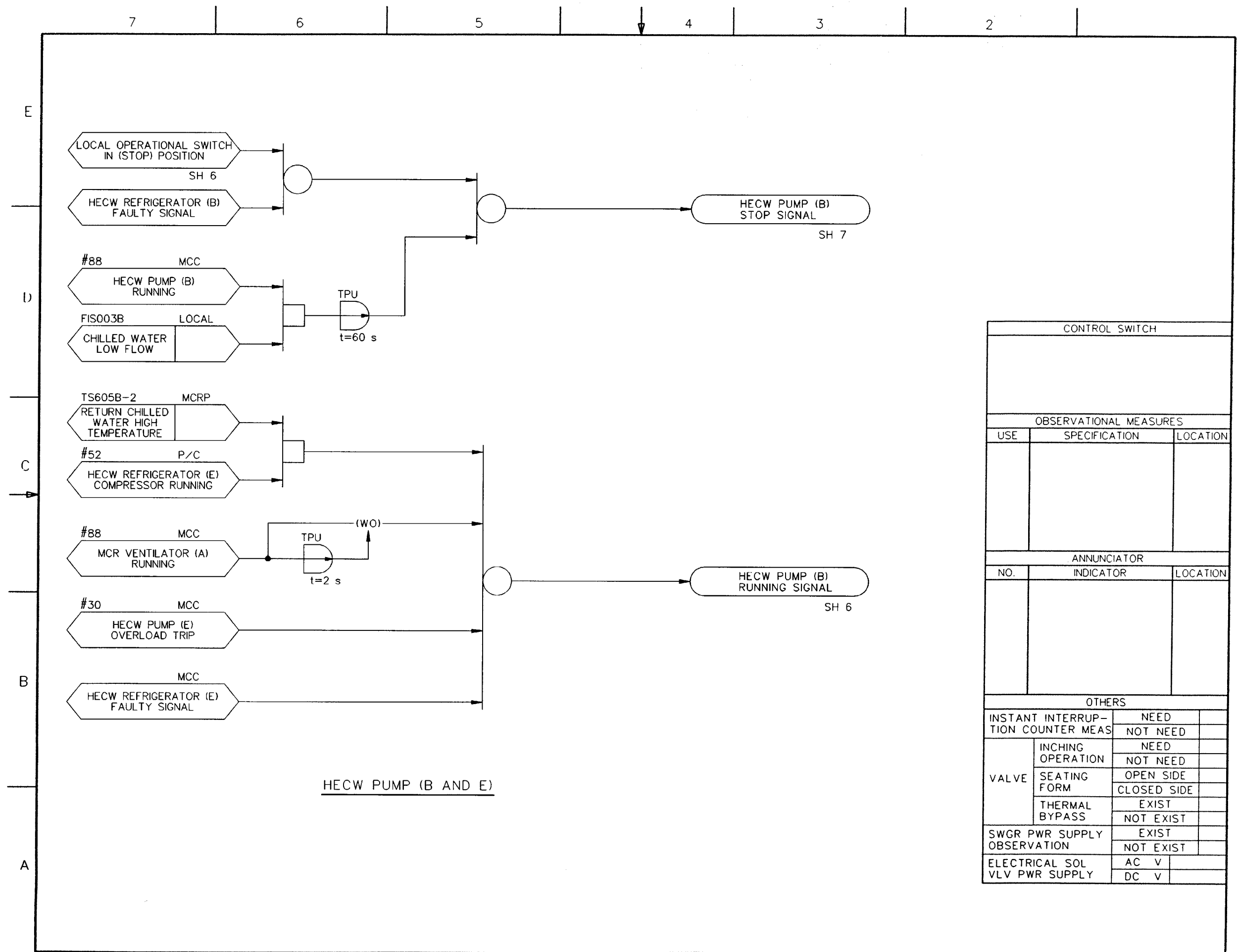






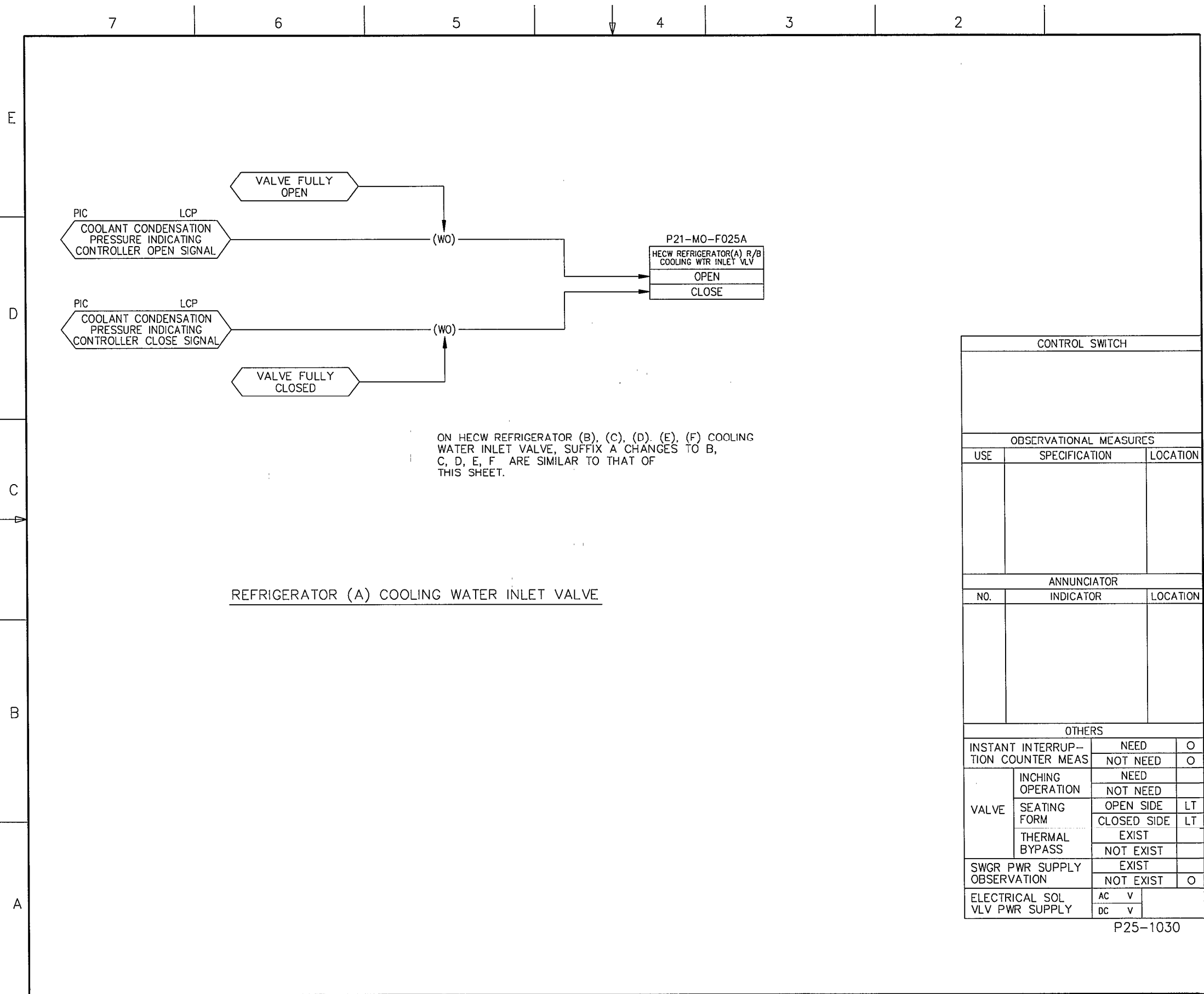
CONTROL SWITCH		
B: AUTO E: CONT MANUAL B: CONT E: AUTO	STOP PULL LOCK AUTO START	STOP PULL LOCK AUTO START
CSX2	CS1X4	CS2X4
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		
INSTANT INTERRUPT- TION COUNTER MEAS	NEED NOT NEED	
INCHING OPERATION	NEED NOT NEED	
SEATING FORM	OPEN SIDE CLOSED SIDE	
THERMAL BYPASS	EXIST NOT EXIST	
SWGR PWR SUPPLY OBSERVATION	EXIST NOT EXIST	
ELECTRICAL SOL VLV PWR SUPPLY	AC V DC V	

P25-1030



CONTROL SWITCH			
OBSERVATIONAL MEASURES			
USE	SPECIFICATION	LOCATION	
ANNUNCIATOR			
NO.	INDICATOR	LOCATION	
OTHERS			
INSTANT INTERRUPT- TION COUNTER MEAS		NEED	
		NOT NEED	
VALVE	INCHING OPERATION	NEED	
		NOT NEED	
	SEATING FORM	OPEN SIDE	
		CLOSED SIDE	
	THERMAL BYPASS	EXIST	
		NOT EXIST	
SWGR PWR SUPPLY OBSERVATION		EXIST	
		NOT EXIST	
ELECTRICAL SOL VLV PWR SUPPLY		AC V	
		DC V	

FIGURE 7.3-9 HVAC EMERGENCY COOLING WATER SYSTEM IBD (Sheet 8 of 11)
 ABWR DCD/Tier 2 Rev. 0 21-266



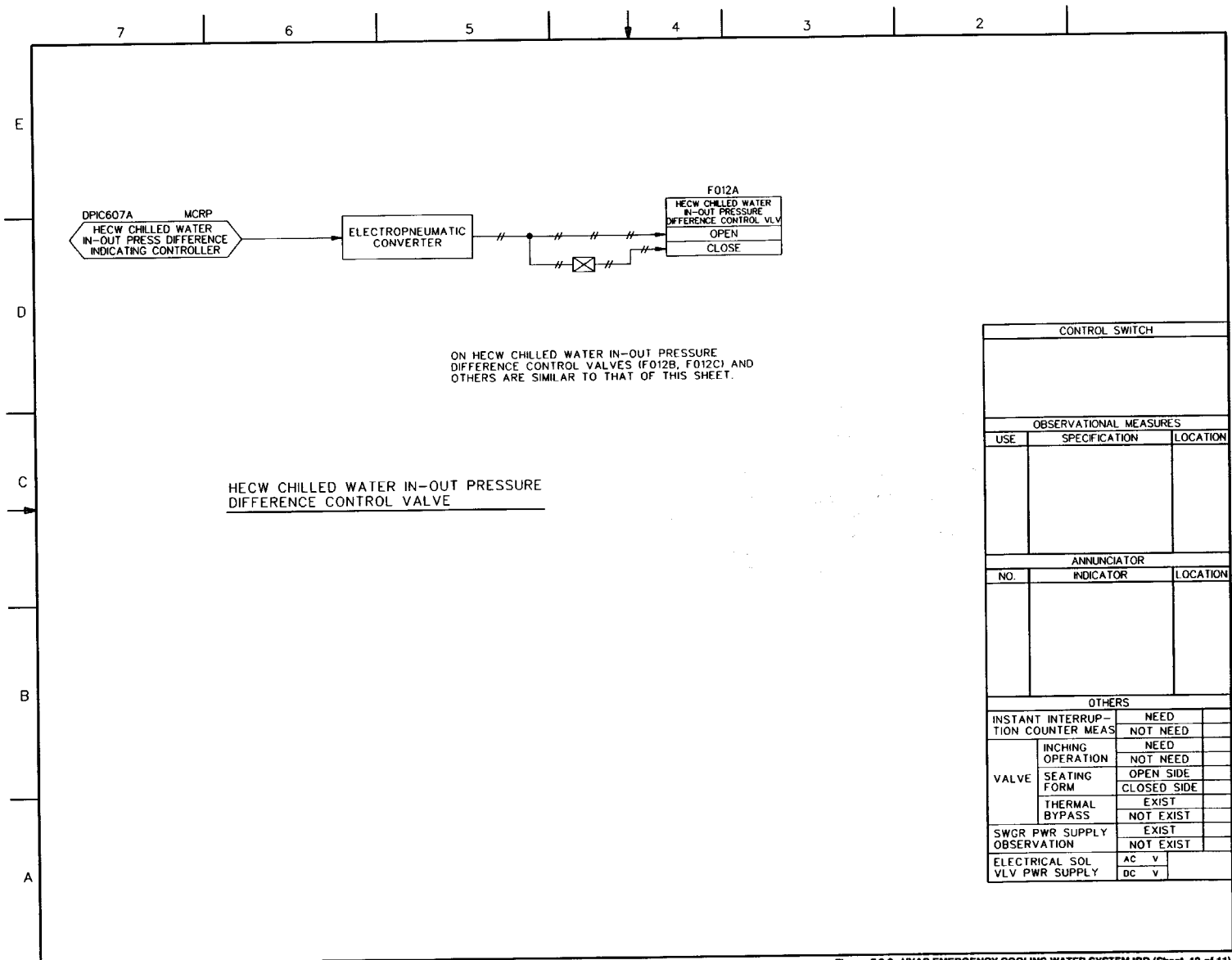


Figure 7.3-9 HVAC EMERGENCY COOLING WATER SYSTEM IBD (Sheet 10 of 11)
ABWR DCD/Tier 2 Rev. 0 21-268

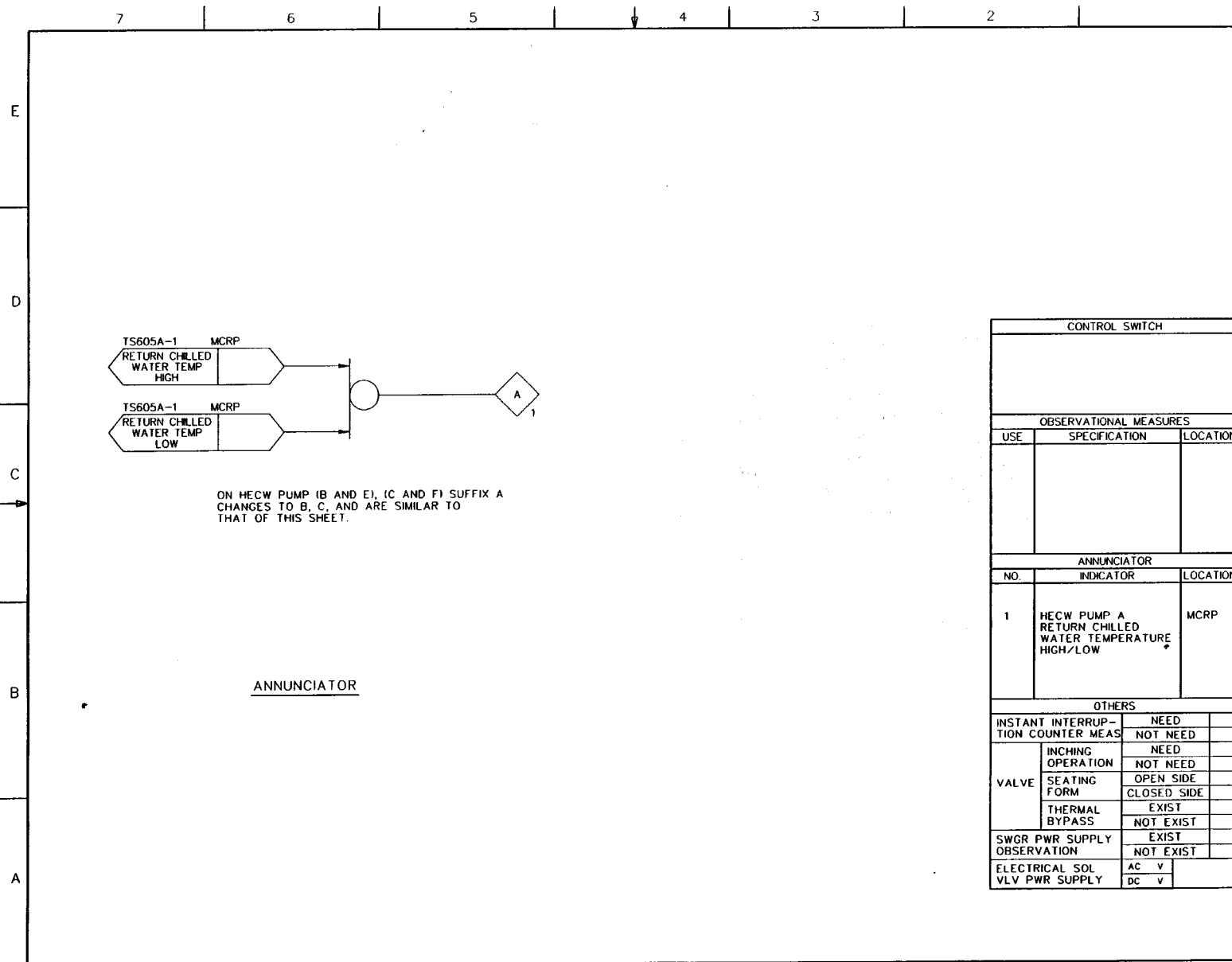
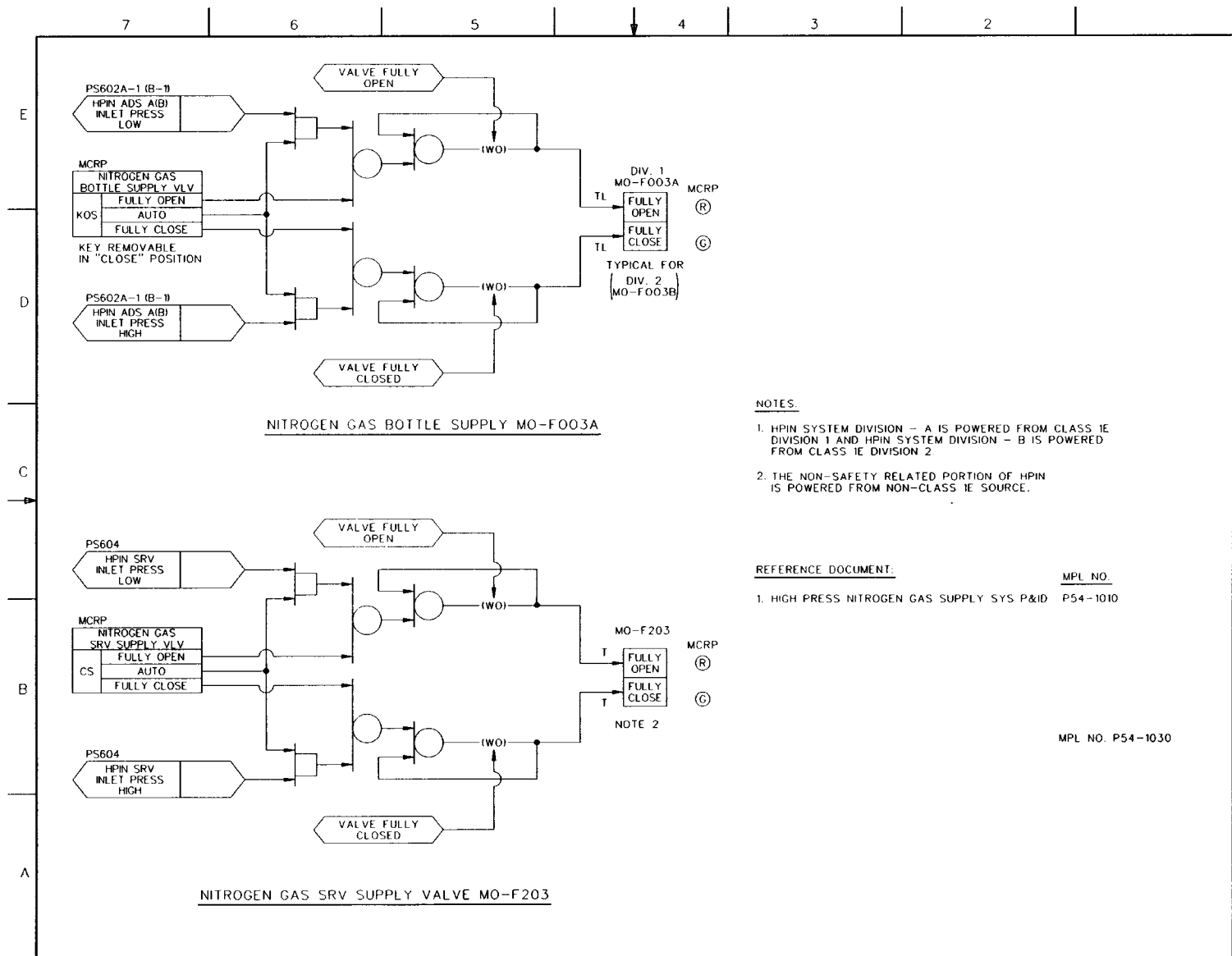


FIGURE 7.3-9 HVAC EMERGENCY COOLING WATER SYSTEM IBD (Sheet 11 of 11)
ABWR DCD/Tier 2 Rev. 0 21-269



NOTES:

1. HPIN SYSTEM DIVISION - A IS POWERED FROM CLASS 1E DIVISION 1 AND HPIN SYSTEM DIVISION - B IS POWERED FROM CLASS 1E DIVISION 2
2. THE NON-SAFETY RELATED PORTION OF HPIN IS POWERED FROM NON-CLASS 1E SOURCE.

REFERENCE DOCUMENT:

1. HIGH PRESS NITROGEN GAS SUPPLY SYS P&ID MPL NO. P54-1010

MPL NO. P54-1030

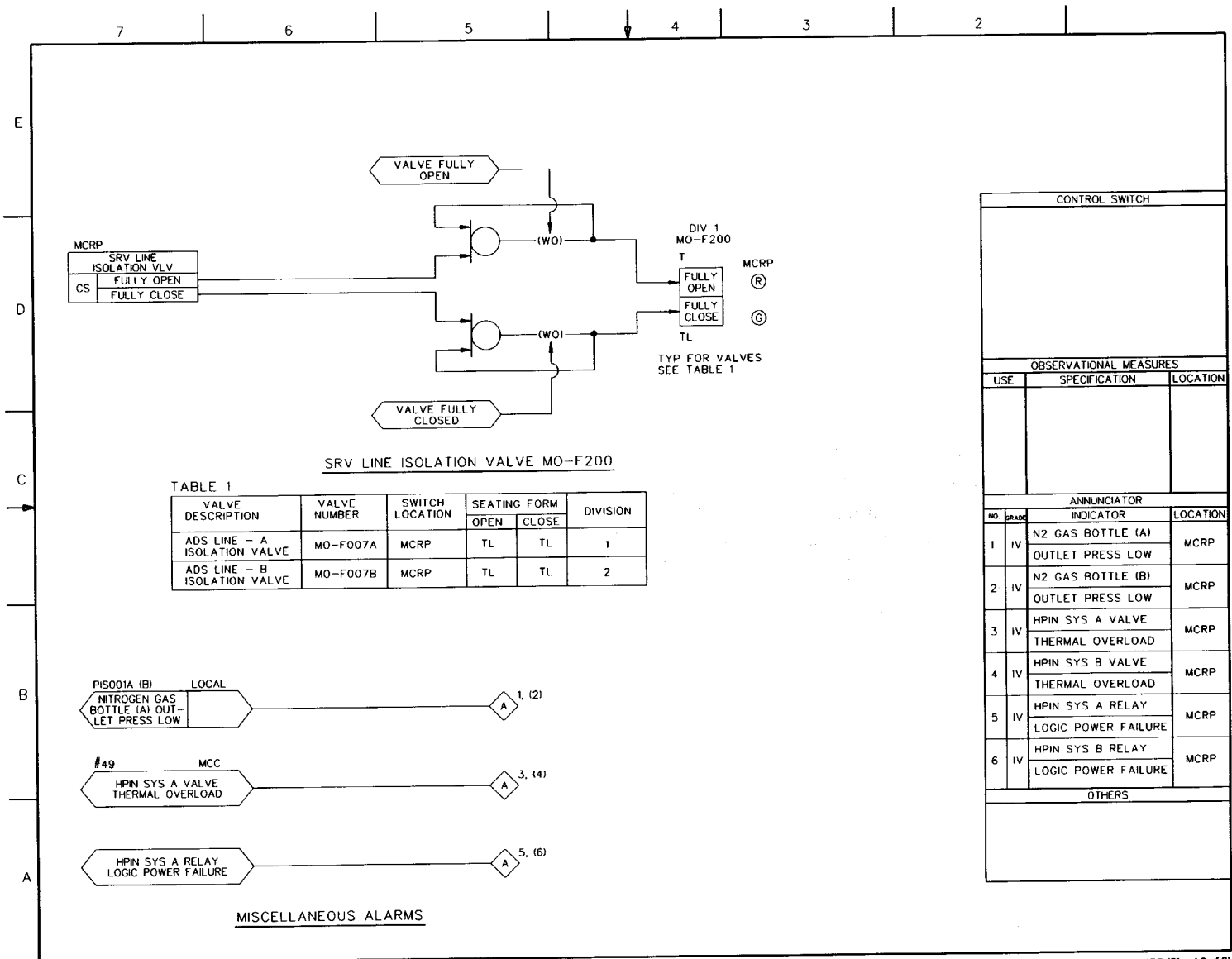


Figure 7.3-10 HIGH PRESSURE NITROGEN GAS SYSTEM IBD (Sheet 2 of 3)
ABWR DCD/Tier 2 Rev. 0 21-271

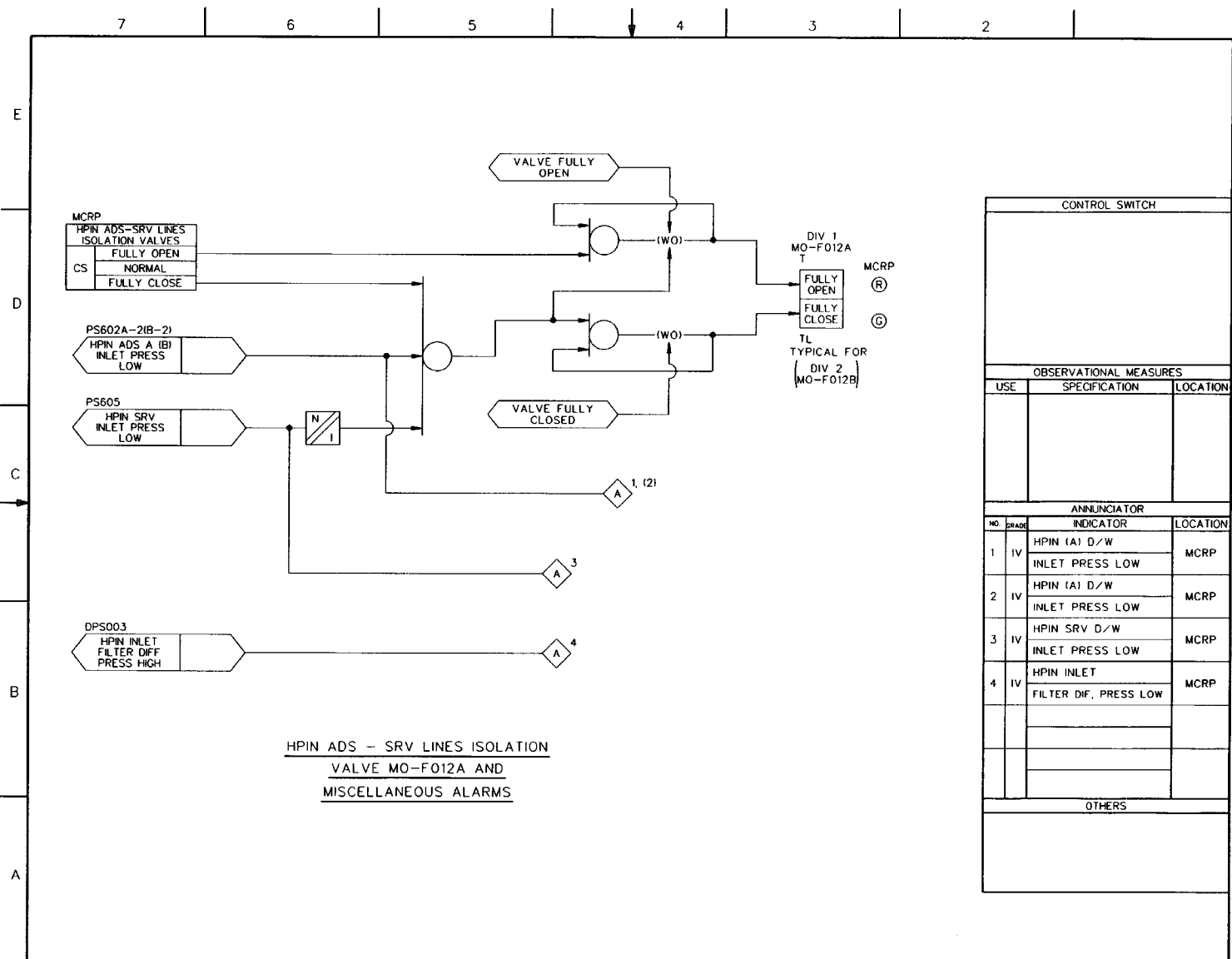
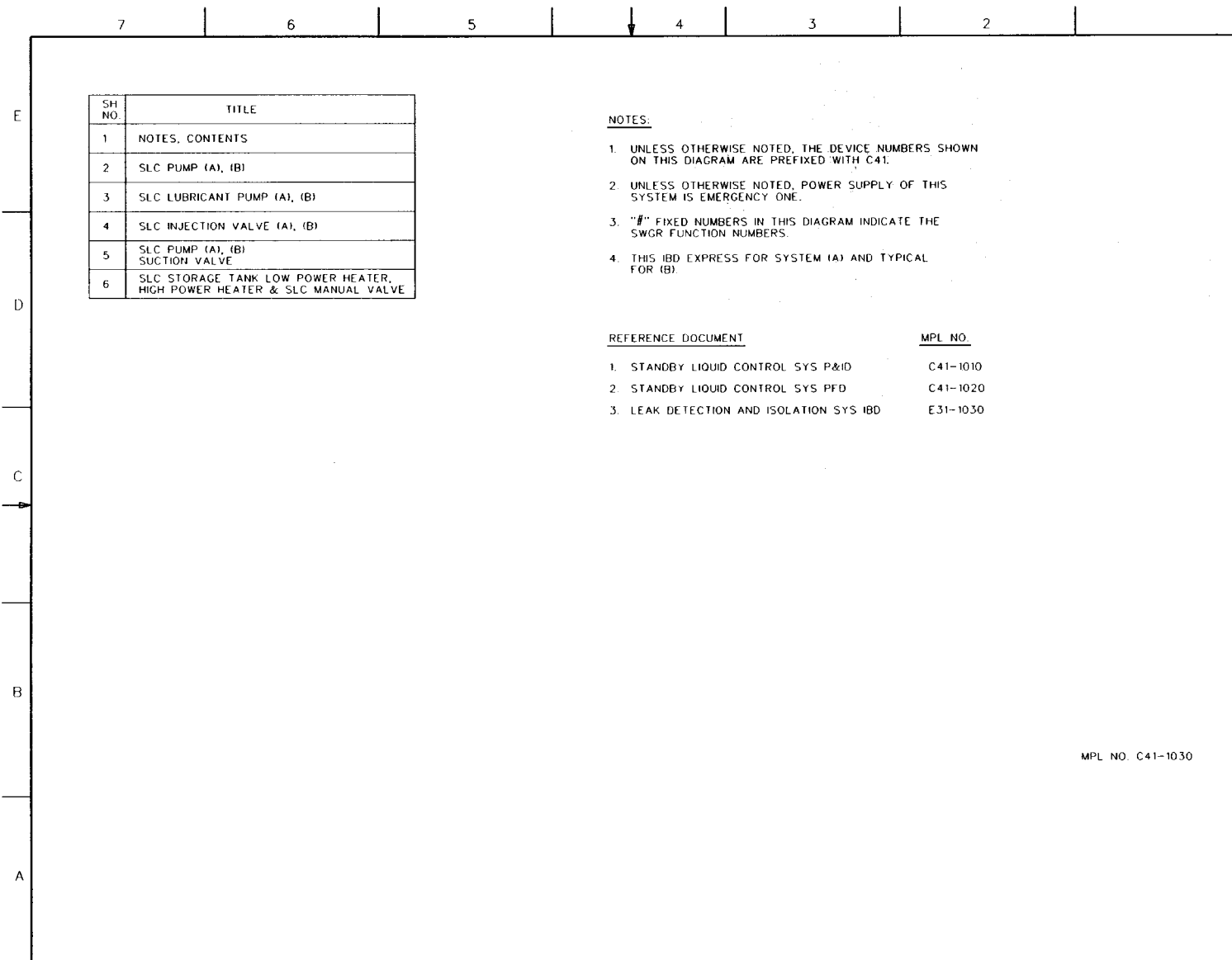


Figure 7.3-10 HIGH PRESSURE NITROGEN GAS SYSTEM IBD (Sheet 3 of 3)
ABWR DCD/Tier 2 Rev. 0 21-272



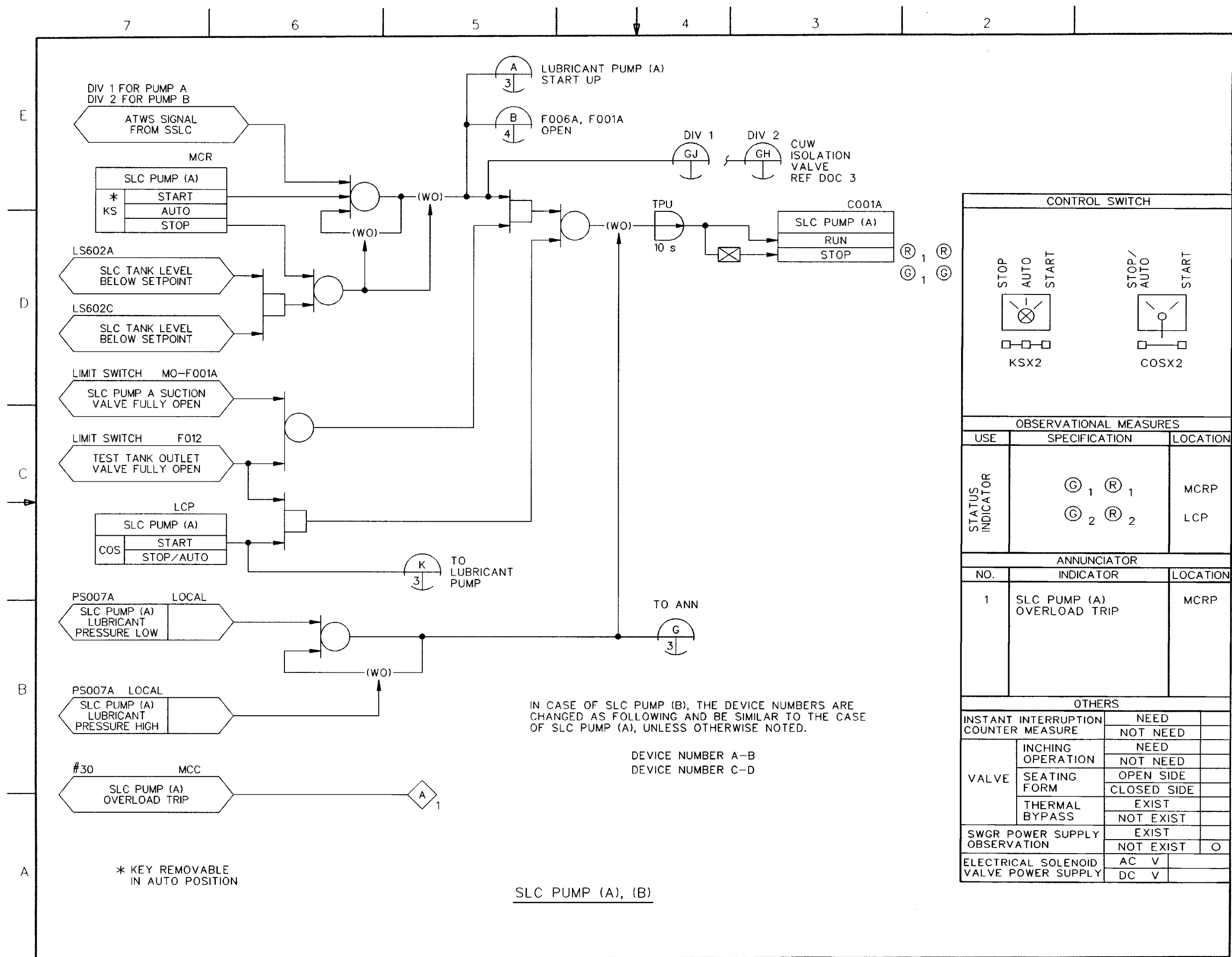


FIGURE 7.4-1 STANDBY LIQUID CONTROL SYSTEM IBD (Sheet 2 of 6)
 ABWR DCD/Tier 2 Rev. 0 21-274

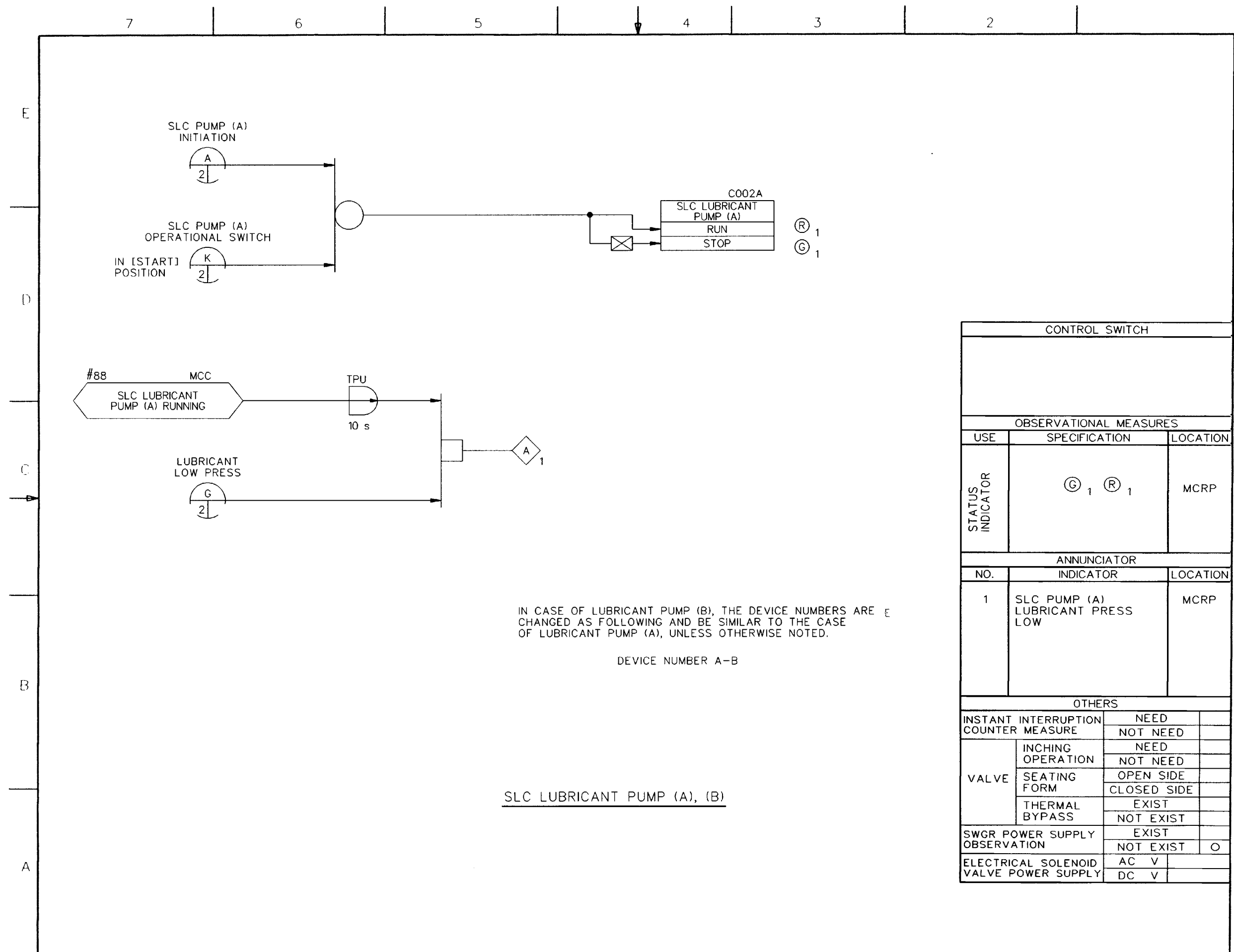


FIGURE 7.4-1 STANDBY LIQUID CONTROL SYSTEM IBD (Sheet 3 of 6)
ABWR DCD/Tier 2 Rev. 0 21-275

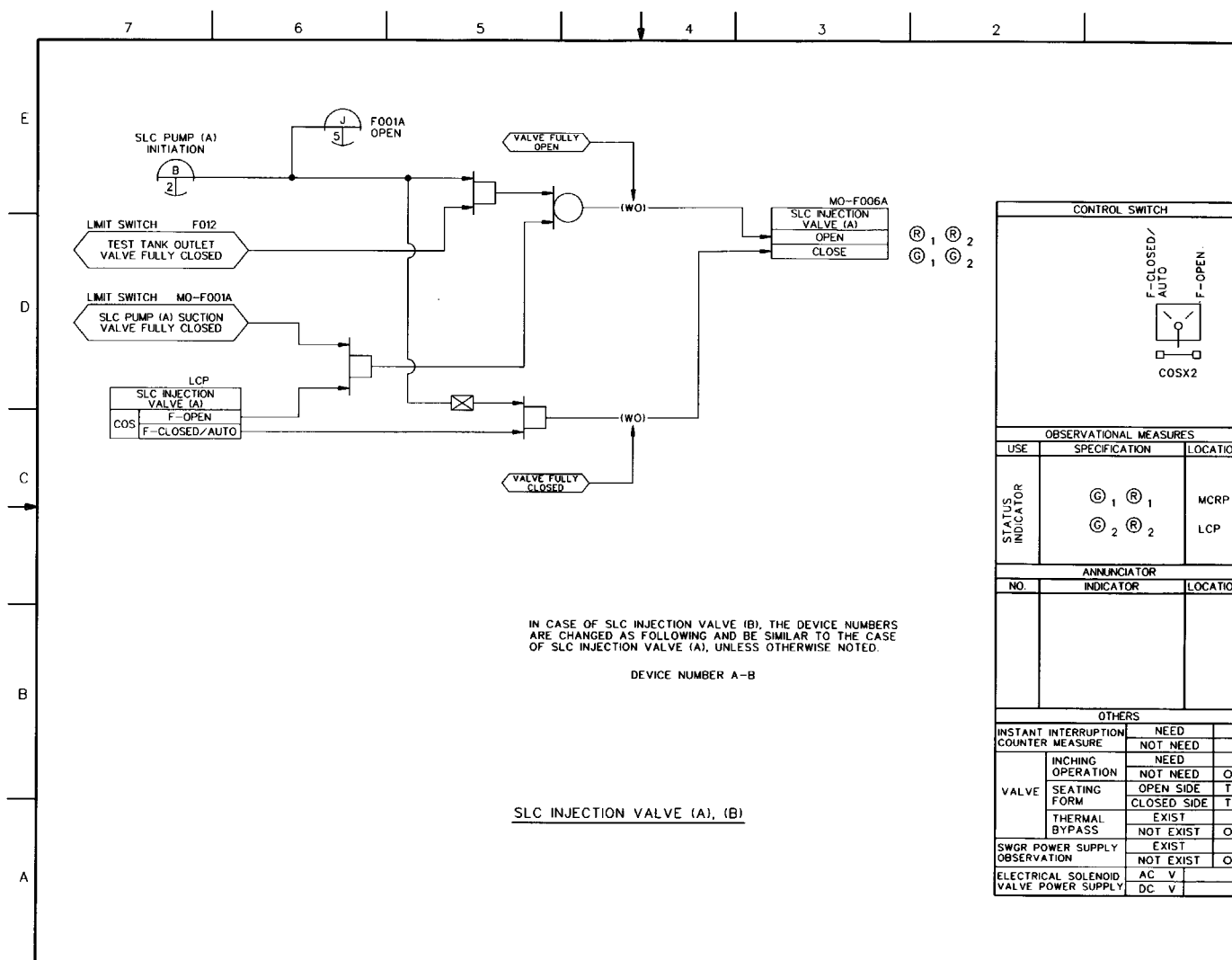


Figure 7.4-1 STANDBY LIQUID CONTROL SYSTEM IBD (Sheet 4 of 6)
ABWR DCD/Tier 2 Rev. 0 21-276

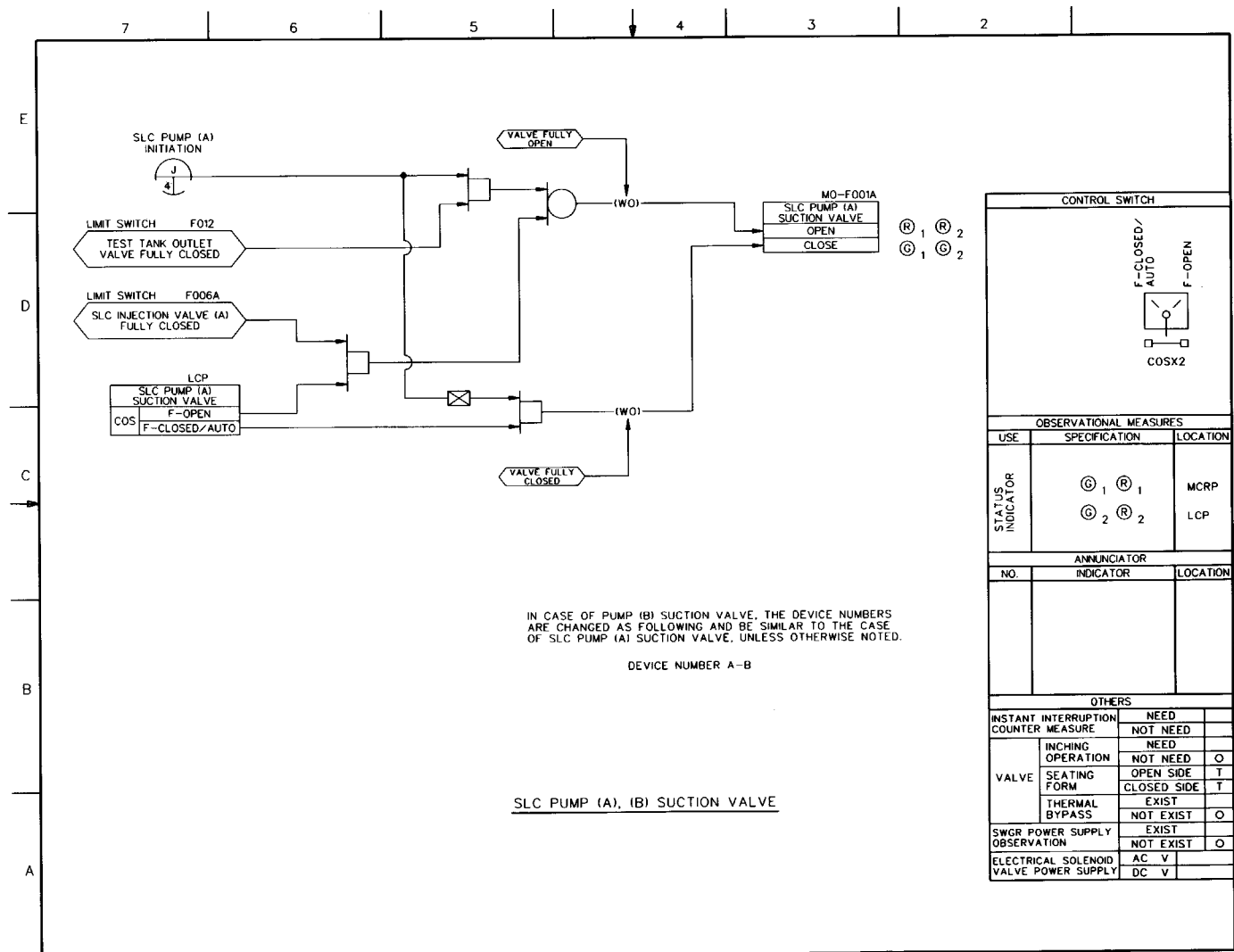


Figure 7.4-1 STANDBY LIQUID CONTROL SYSTEM IBD (Sheet 5 of 6)
ABWR DCD/Tier 2 Rev. 0 21-277

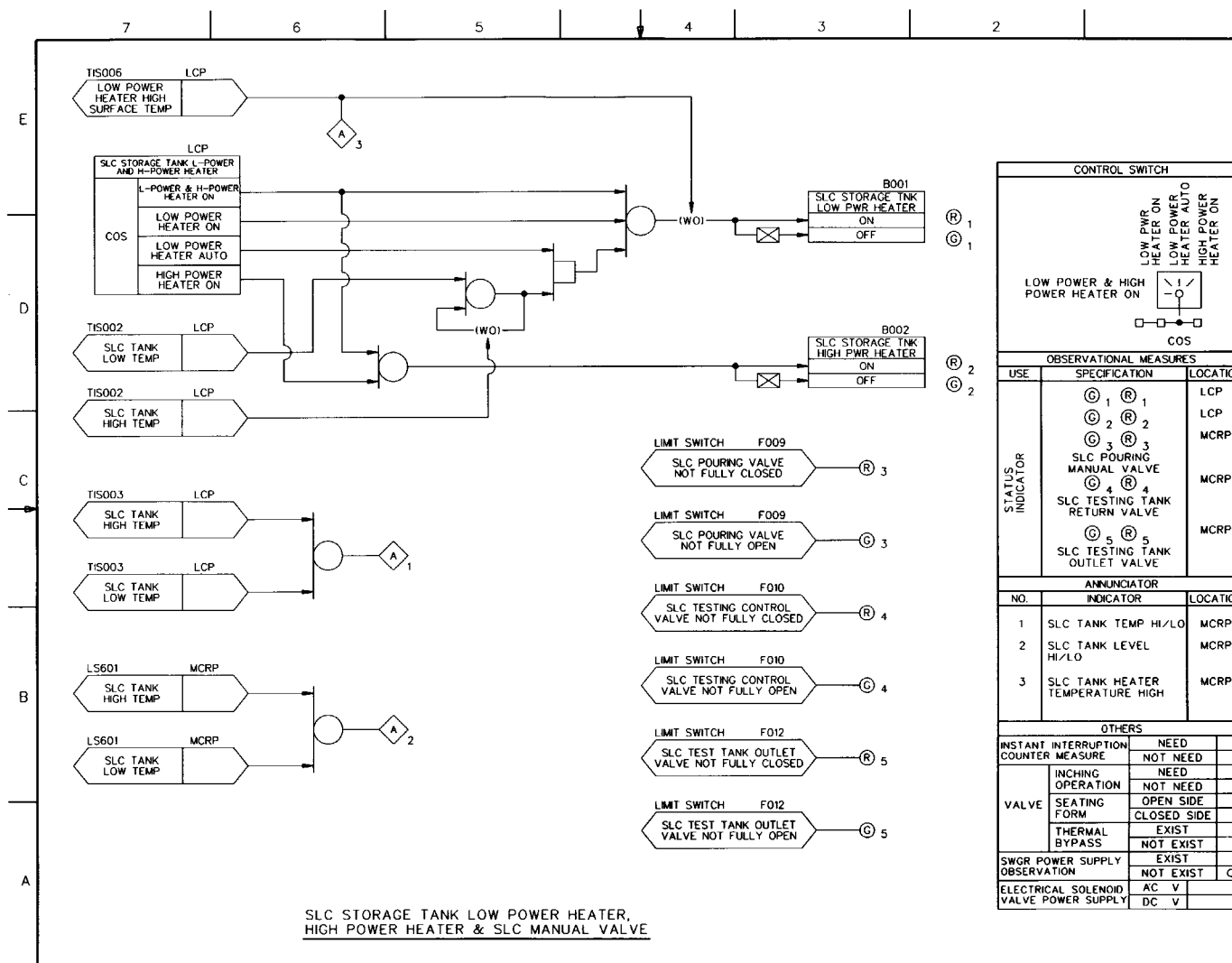


Figure 7.4-1 STANDBY LIQUID CONTROL SYSTEM IBD (Sheet 6 of 6)
ABWR DCD/Tier 2 Rev. 0 21-278

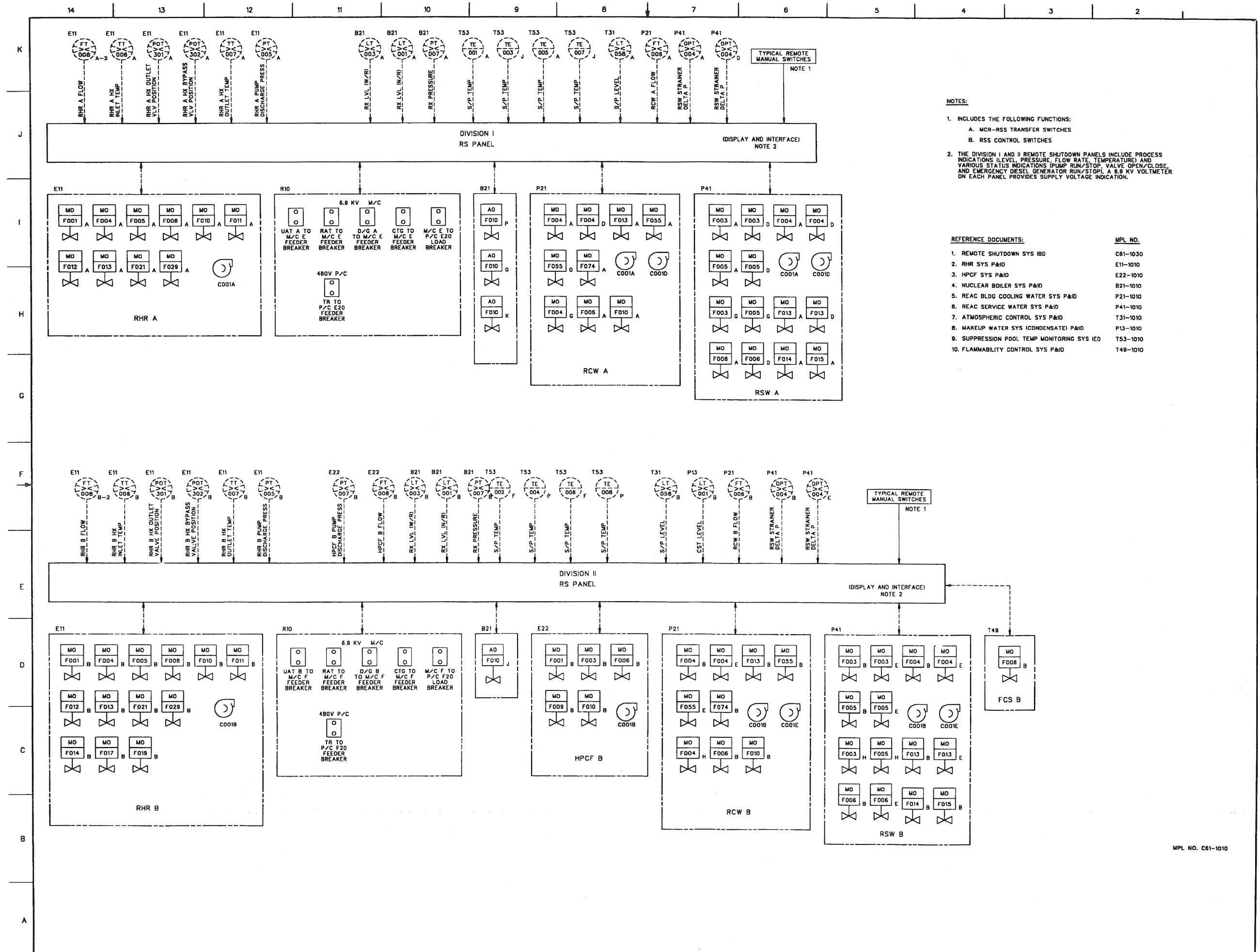


Figure 7.4-2 REMOTE SHUTDOWN SYSTEM IED
ABWR DCD/Tier 2
Rev. 0 21-279

	7	6	5	4	3	2	
E	<u>NOTES:</u> 1. MOTOR OPERATED VALVE CONTROL LOGIC ON SHEETS 6 AND 8 APPLIES TO MANY VALVES AS TABULATED ON SHEETS 2 AND 3. INTERFACE INFORMATION IS AVAILABLE FROM APPLICABLE REFERENCE DOCUMENTS LISTED ON SHEET 1.						
	<u>REFERENCE DOCUMENT</u> 1. REMOTE SHUTDOWN SYS IBD C61-1010 2. RESIDUAL HEAT REMOVAL SYS IBD E11-1030 3. HIGH PRESS CORE FLOODER SYS IBD E22-1030 4. REAC BLDG COOLING WATER SYS/ REACTOR SERVICE WATER SYS IBD P21/P41-1030 5. NUCLEAR BOILER SYS IBD B21-1030 6. FLAMMABILITY CONTROL SYSTEM P&ID T49-1010 7. ELECTRICAL POWER DISTRIBUTION ONE LINE DIAG R10-1010 8. MAKEUP WATER SYSTEM (CONDENSATE) P&ID P13-1010 9. ATMOSPHERIC CONTROL SYSTEM P&ID T31-1010 10. RESIDUAL HEAT REMOVAL SYSTEM P&ID E11-1010 11. HIGH PRESSURE CORE FLOODER SYSTEM P&ID E22-1010 12. NUCLEAR BOILER SYSTEM P&ID B21-1010 13. INTERLOCK BLOCK DIAGRAM (IBD) STANDARDS A10-3070						
D	<u>LEGEND:</u> MCR = MAIN CONTROL ROOM RSP = REMOTE SHUTDOWN CONTROL PANEL RSTS = REMOTE SHUTDOWN TRANSFER SWITCH RSCS = REMOTE SHUTDOWN CONTROL SWITCH						
C							
B							MPL NO. C61-1030
A							

		7	6	5	4	3	2	
E	RSW	RSW PUMP (A)	SH 5	P41-C001A	D/G	DIESEL GENERATOR A START-UP CIRCUIT	SH 11	-
		RSW PUMP (D)	SH 5	P41-C001D		DIESEL GENERATOR B START-UP CIRCUIT	SH 11	-
		RCW HX SERVICE WATER INLET VALVE (A)	SH 6	P41-F003A				
		RCW HX SERVICE WATER INLET VALVE (D)	SH 6	P41-F003D				
		RSW STRAINER INLET VALVE (A)	SH 26	P41-F004A	FCS	FCS COOLING WATER INLET VALVE B	SH 8	T49-F008B
		RSW STRAINER INLET VALVE (D)	SH 26	P41-F004D				
		RCW HX SERVICE WATER OUTLET VALVE (A)	SH 6	P41-F005A				
		RCW HX SERVICE WATER OUTLET VALVE (D)	SH 6	P41-F005D				
		RCW HX SERVICE WATER INLET VALVE (G)	SH 6	P41-F003G				
D		RCW HX SERVICE WATER OUTLET VALVE (G)	SH 6	P41-F005G				
		RSW STRAINER OUTLET VALVE (A)	SH 6	P41-F013A				
		RSW STRAINER OUTLET VALVE (D)	SH 6	P41-F013D				
		RSW STRAINER BACKWASH VALVE (A)	SH 6	P41-F006A				
		RSW STRAINER BACKWASH VALVE (D)	SH 6	P41-F006D				
		RSW (A) SUPPLY VALVE	SH 6	P41-F014A				
		RSW (A) RETURN VALVE	SH 6	P41-F015A				
C		RSW PUMP (B)	SH 5	P41-C001B	EPDS	6.9KV FEEDER BREAKER - UAT A TO M/C E	SH 10	-
		RSW PUMP (E)	SH 5	P41-C001E		6.9KV FEEDER BREAKER - RAT TO M/C E	SH 10	-
		RCW HX SERVICE WATER INLET VALVE (B)	SH 6	P41-F003B		6.9KV FEEDER BREAKER - D/G A TO M/C E	SH 10	-
		RCW HX SERVICE WATER INLET VALVE (E)	SH 6	P41-F003E		6.9KV FEEDER BREAKER - CTG TO M/C E	SH 10	-
		RSW STRAINER INLET VALVE (B)	SH 26	P41-F004B		6.9KV LOAD BREAKER - M/C E TO PC E20	SH 10	-
		RSW STRAINER INLET VALVE (E)	SH 26	P41-F004E		480V FEEDER BREAKER - TR TO P/C E20	SH 10	-
		RCW HX SERVICE WATER OUTLET VALVE (B)	SH 6	P41-F005B				
		RCW HX SERVICE WATER OUTLET VALVE (E)	SH 6	P41-F005E				
		RCW HX SERVICE WATER INLET VALVE (H)	SH 6	P41-F003H		6.9KV FEEDER BREAKER - UAT B TO M/C F	SH 10	-
		RCW HX SERVICE WATER OUTLET VALVE (H)	SH 6	P41-F005H		6.9KV FEEDER BREAKER - RAT TO M/C F	SH 10	-
		RSW STRAINER OUTLET VALVE (B)	SH 6	P41-F013B		6.9KV FEEDER BREAKER - D/G B TO M/C F	SH 10	-
		RSW STRAINER OUTLET VALVE (E)	SH 6	P41-F013E		6.9KV FEEDER BREAKER - CTG TO M/C F	SH 10	-
		RSW STRAINER BACKWASH VALVE (B)	SH 6	P41-F006B		6.9KV LOAD BREAKER - M/C F TO PC F20	SH 10	-
		RSW STRAINER BACKWASH VALVE (E)	SH 6	P41-F006E		480V FEEDER BREAKER - TR TO P/C E 20	SH 10	-
		RSW (B) SUPPLY VALVE	SH 6	P41-F014B				
		RSW (B) RETURN VALVE	SH 6	P41-F015B				
A								

FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 3 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-282

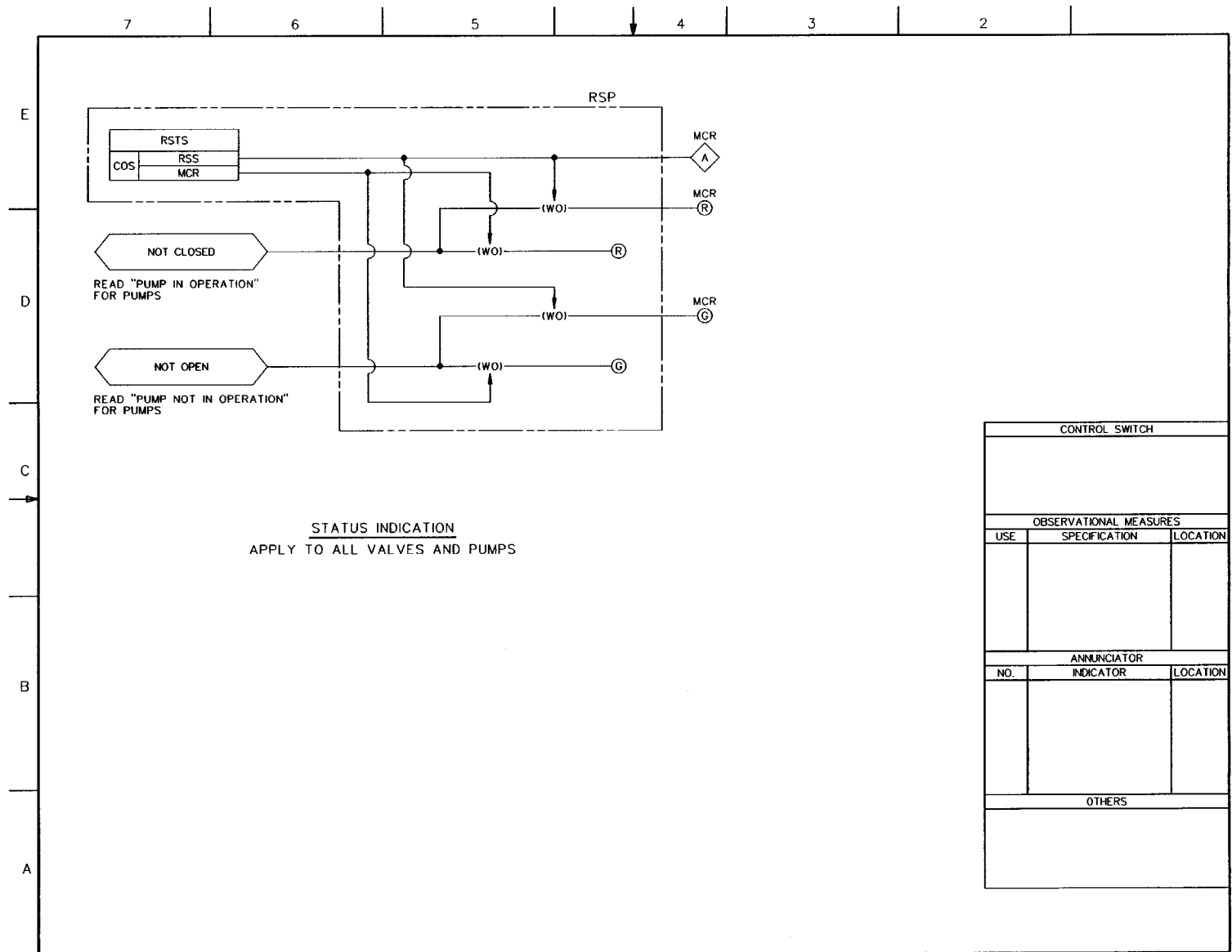


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 4 of 27)
ABWR DCD/Tier 2 Rev. 0 21-283

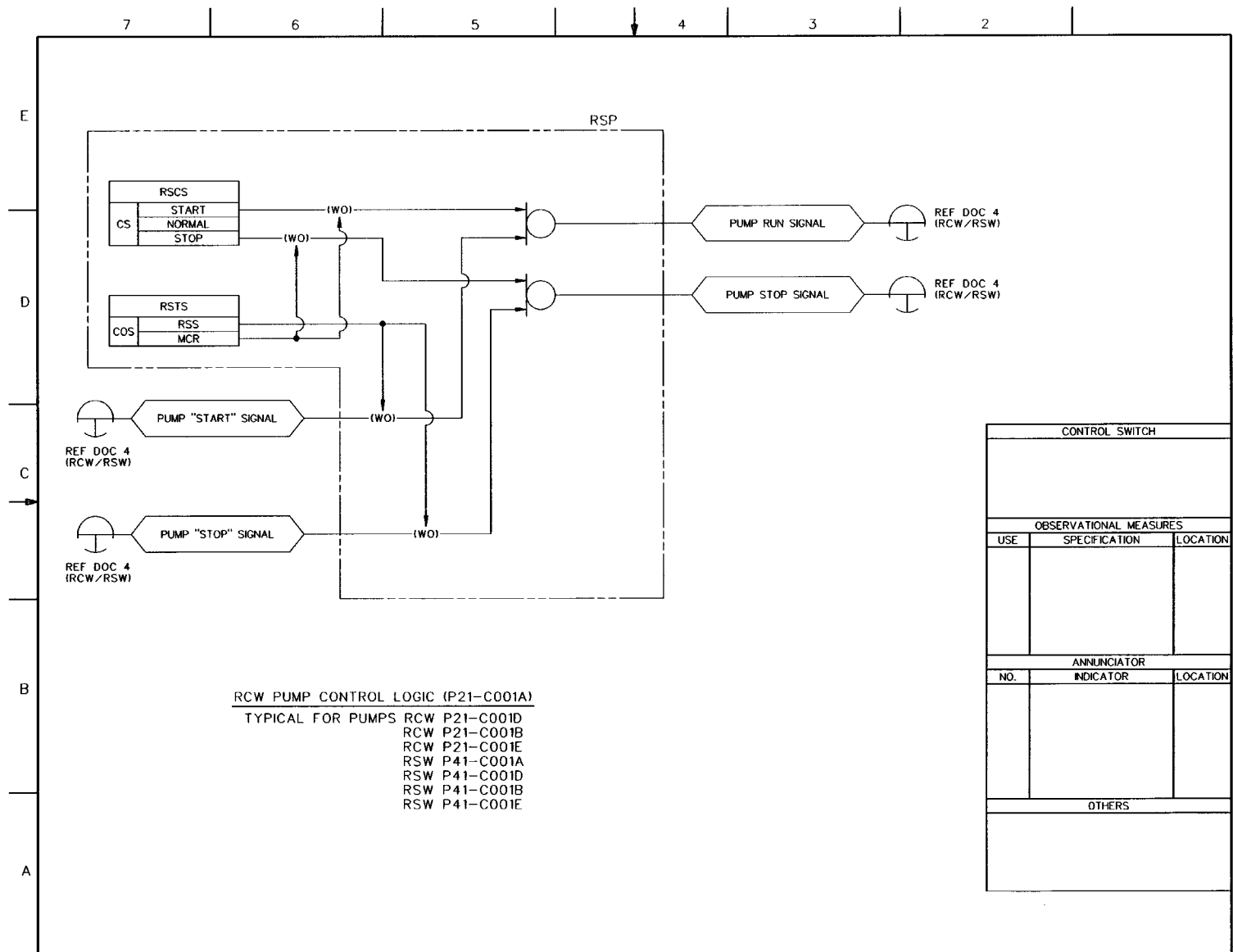
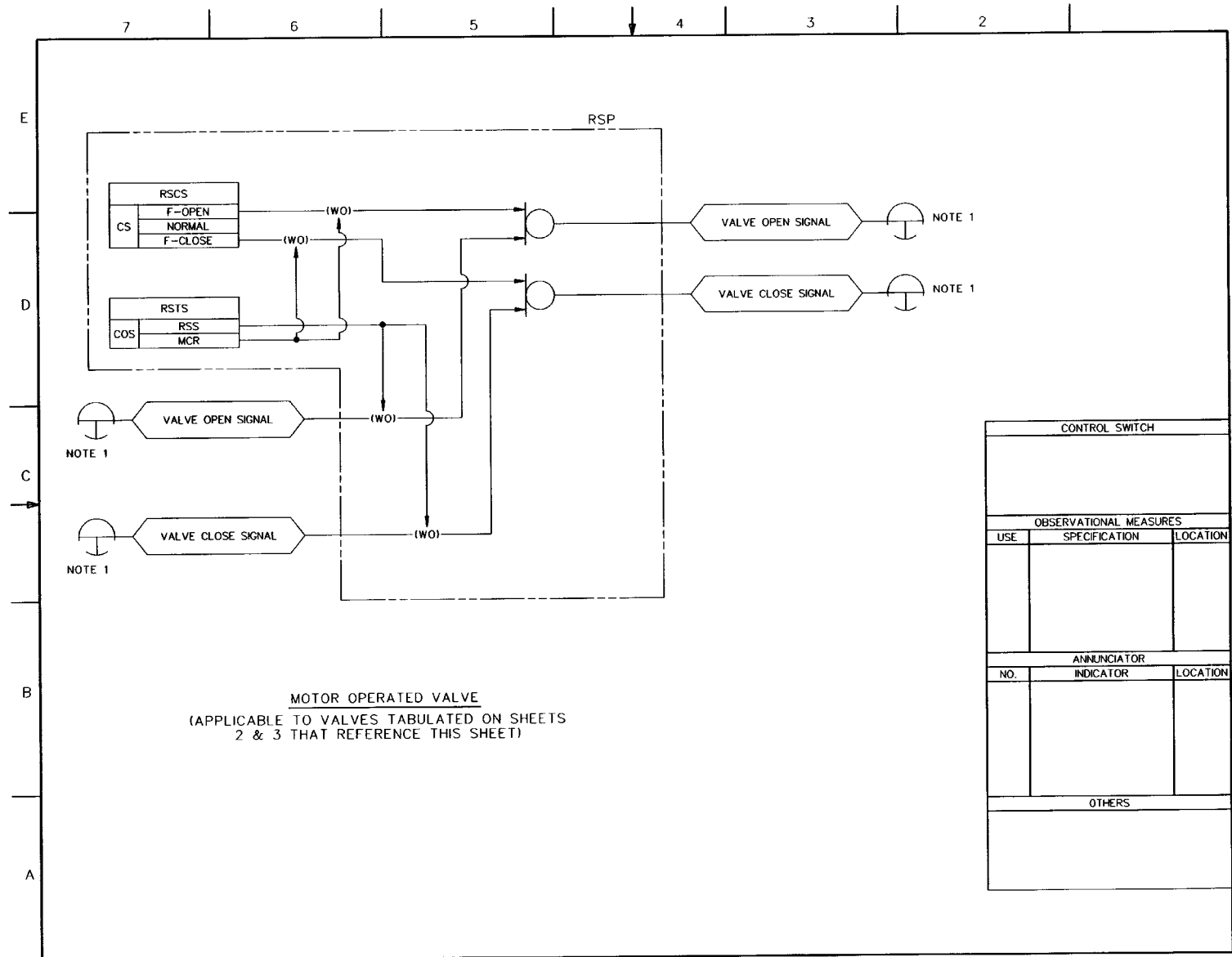


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 5 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-284



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		

FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 6 of 27)
ABWR DCD/Tier 2 Rev. 0 21-285

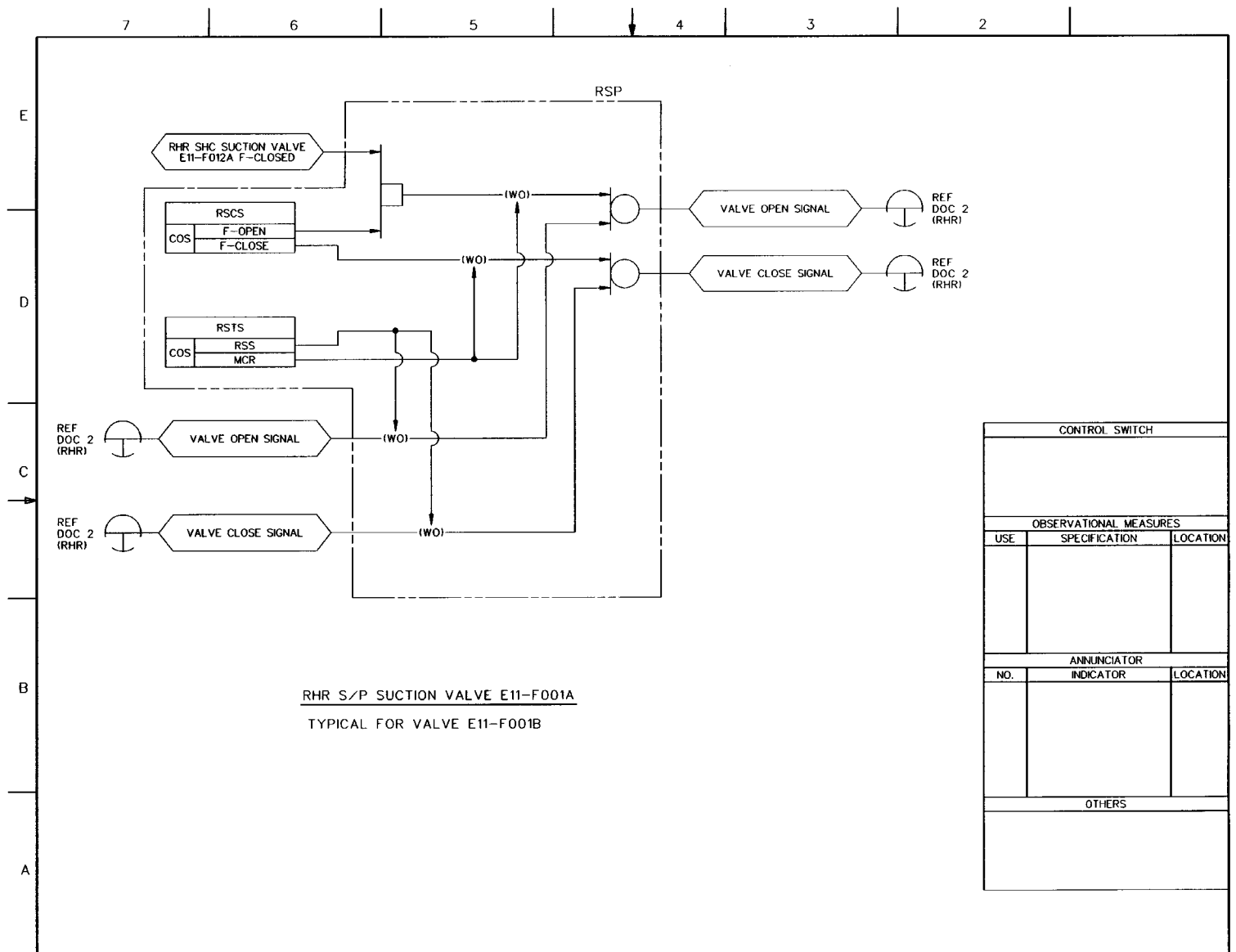
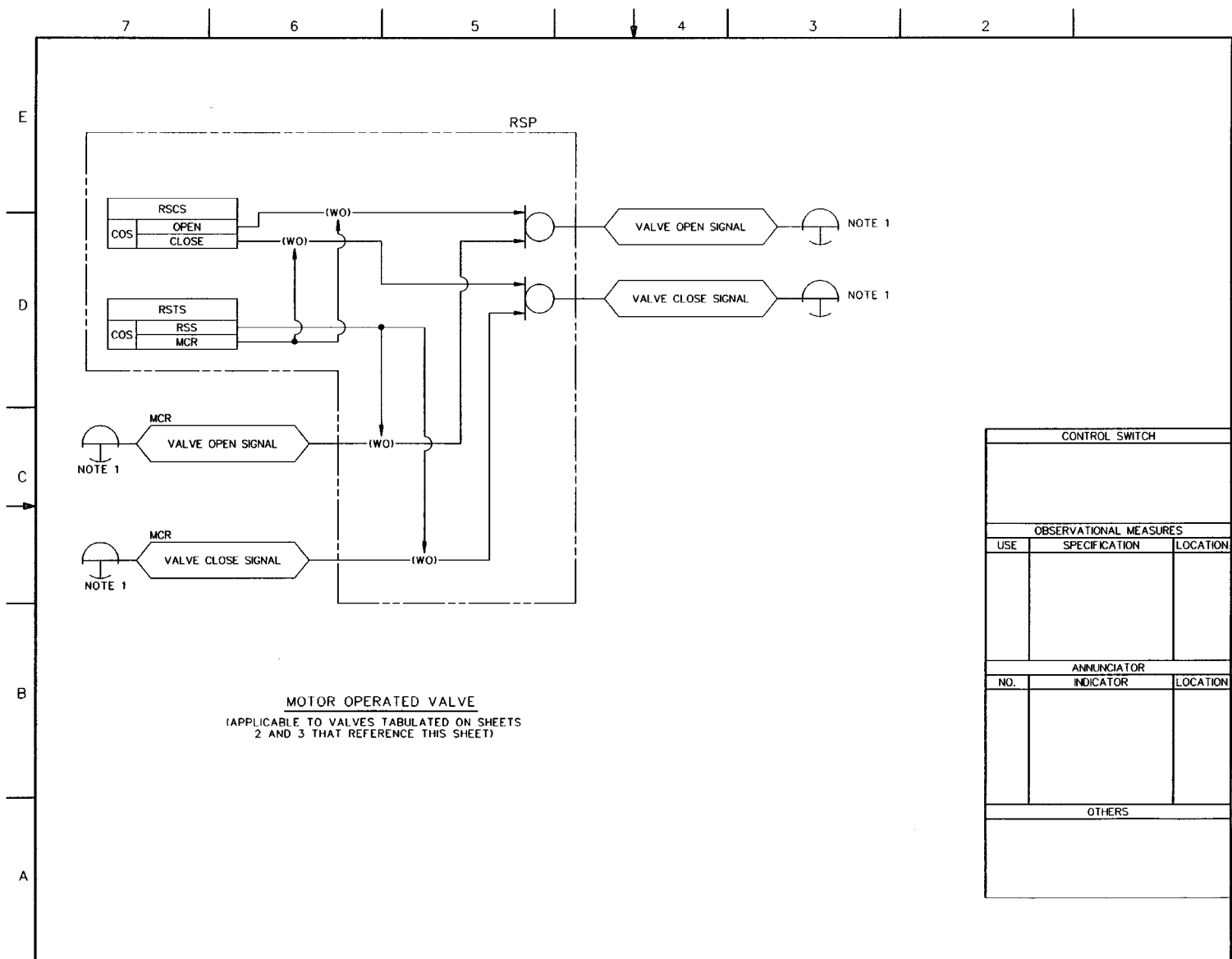


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 7 of 27)



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		

FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 8 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-287

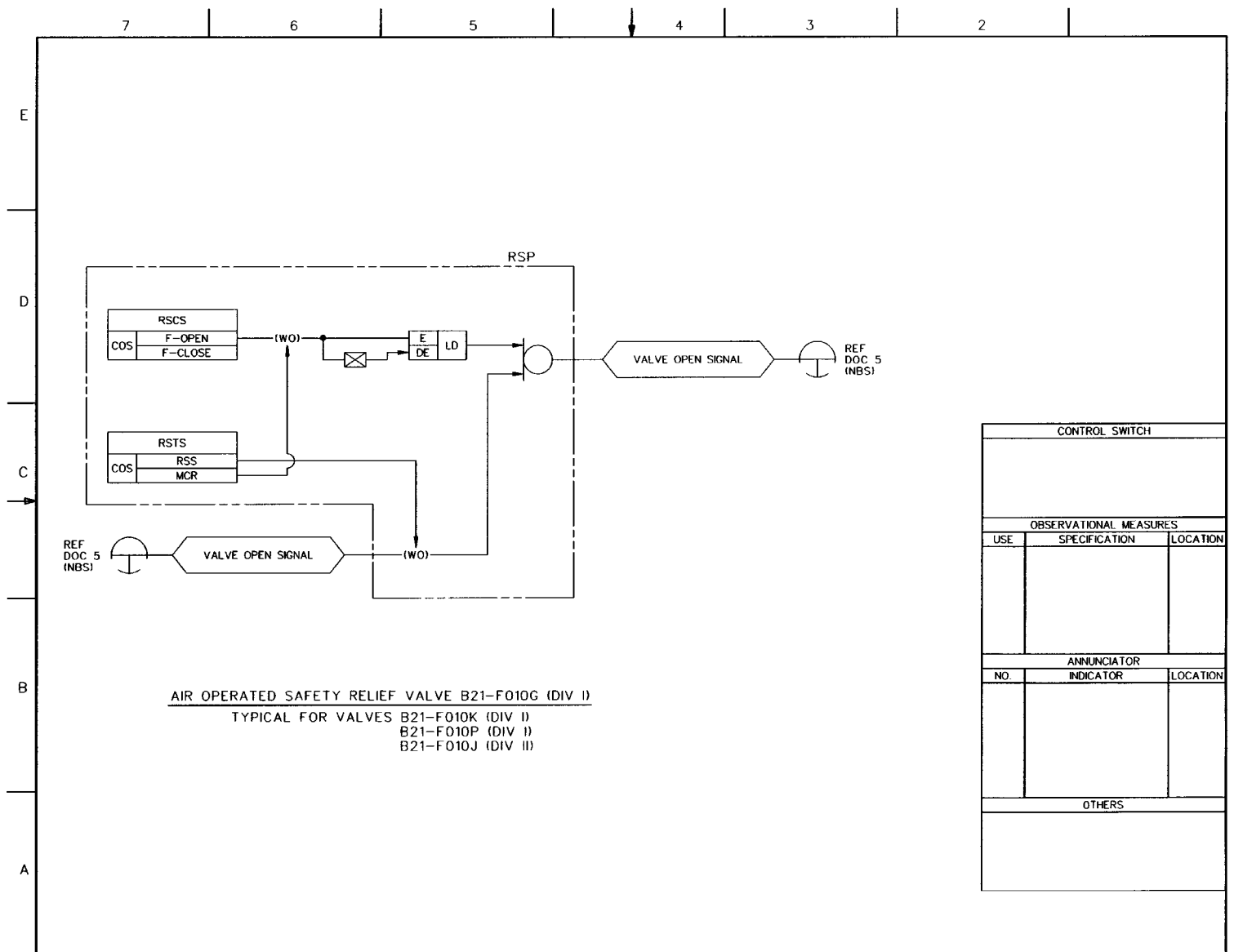
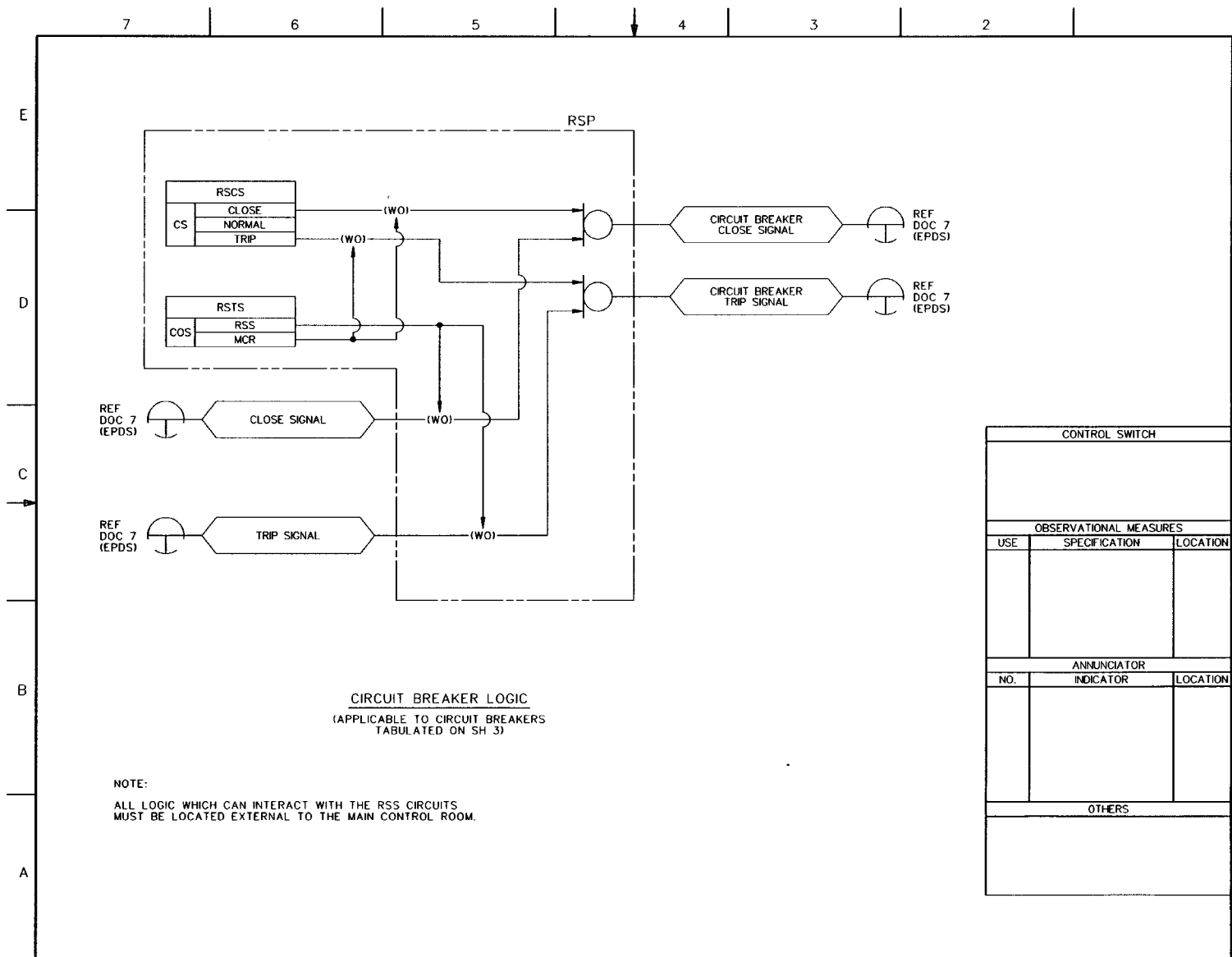
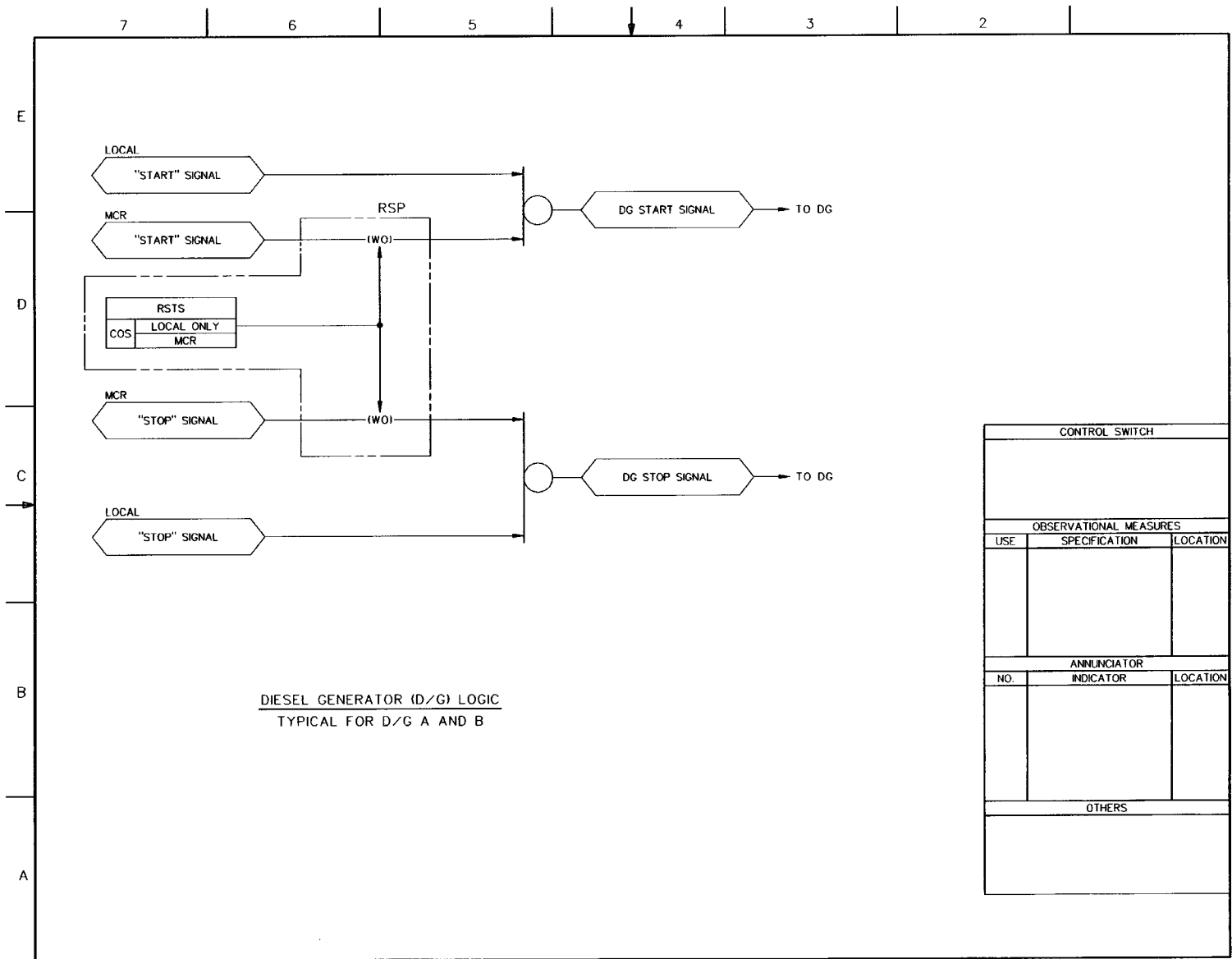


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 9 of 27)
ABWR DCD/Tier 2 Rev. 0 21-288





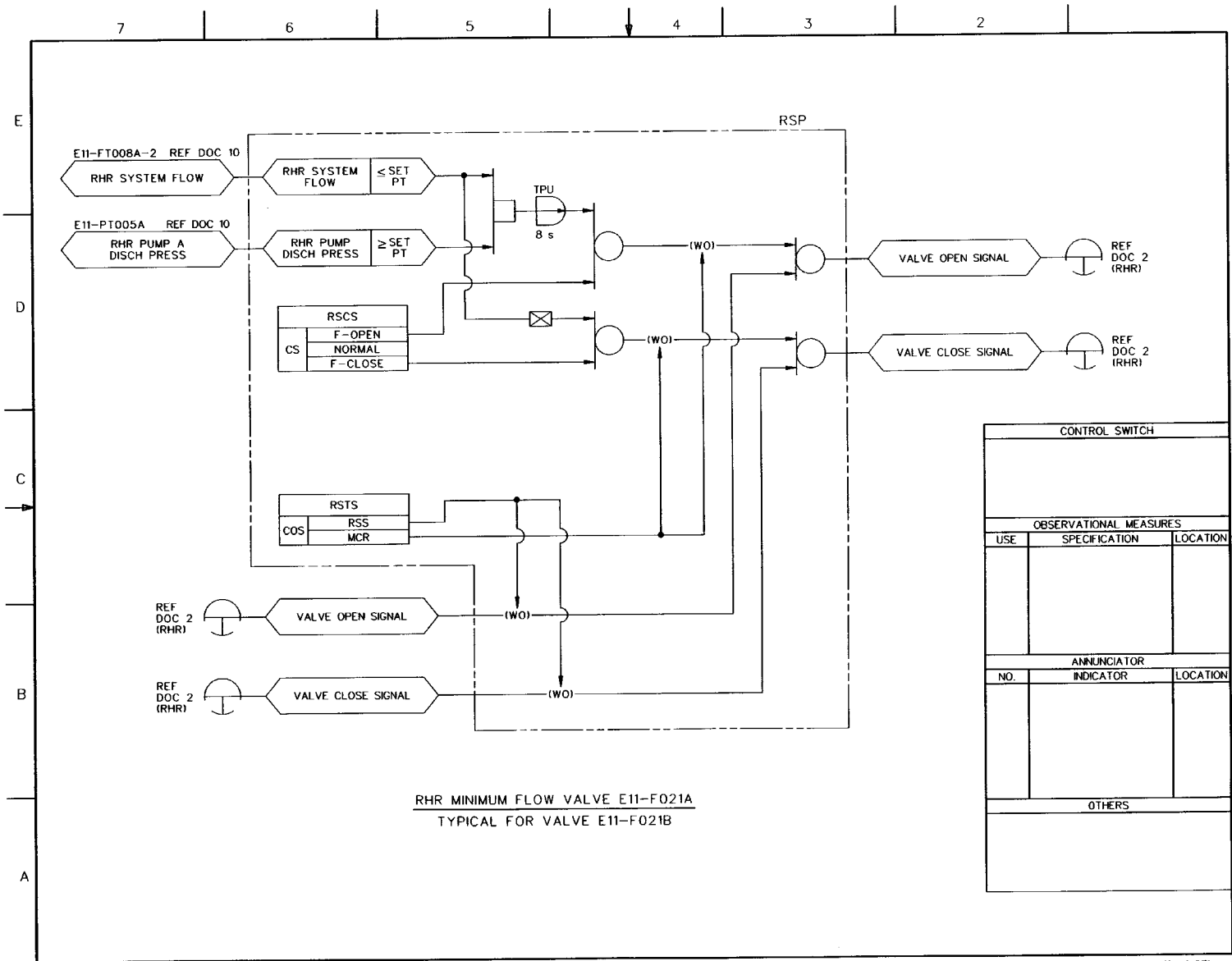


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 12 of 27)
ABWR DCD/Tier 2 Rev. 0 21-291

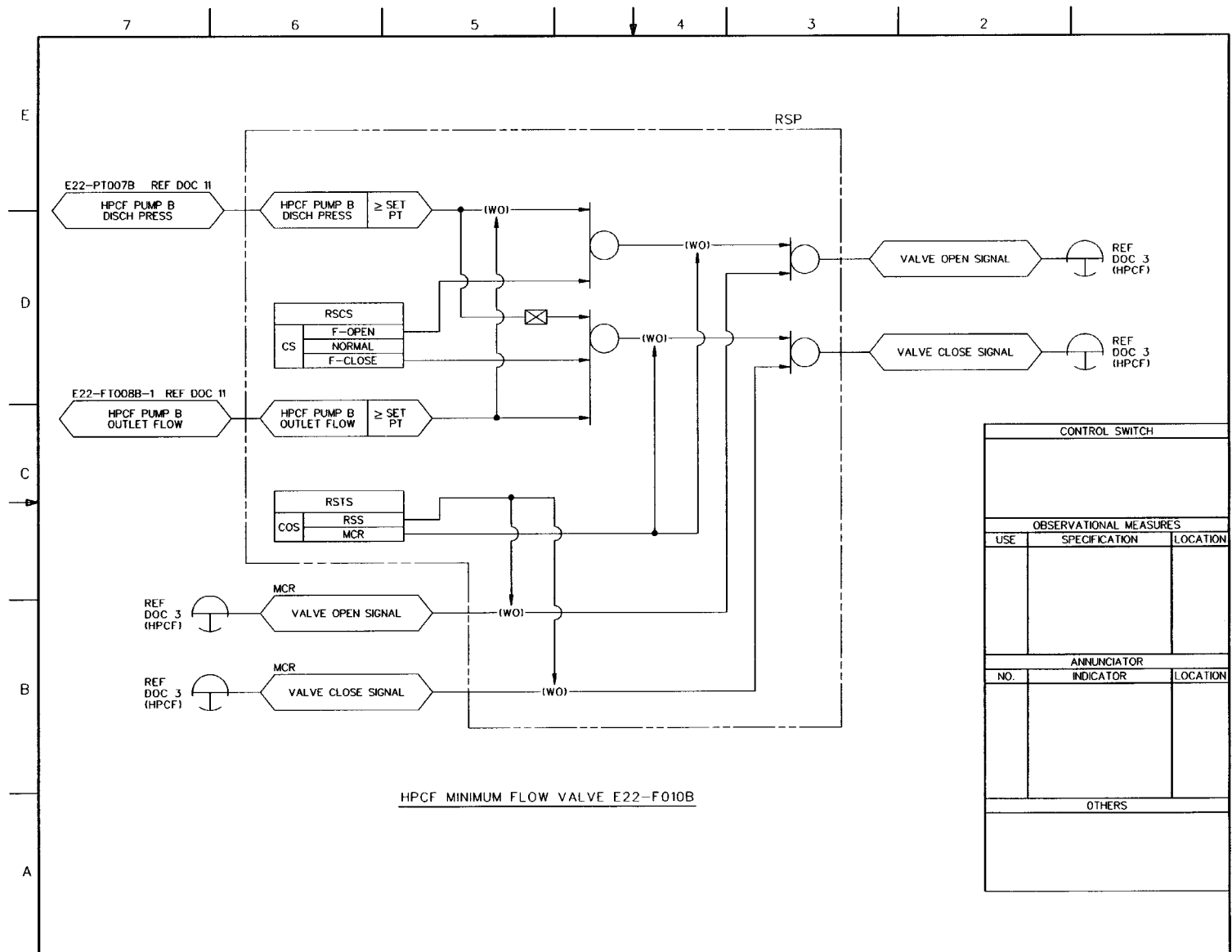
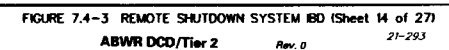


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 13 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-292



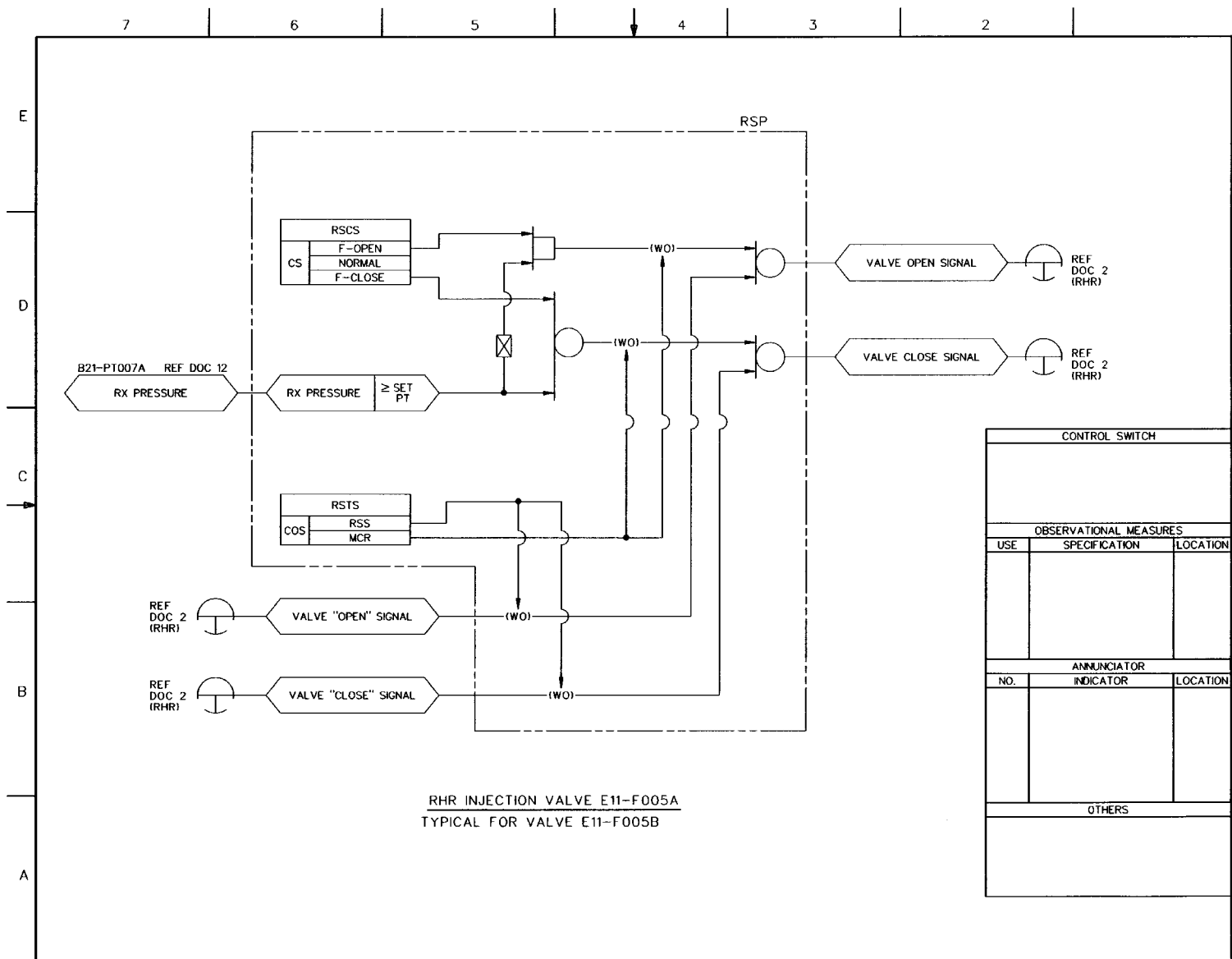


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 15 of 27)
ABWR DCD/Tier 2 Rev. 0 21-294

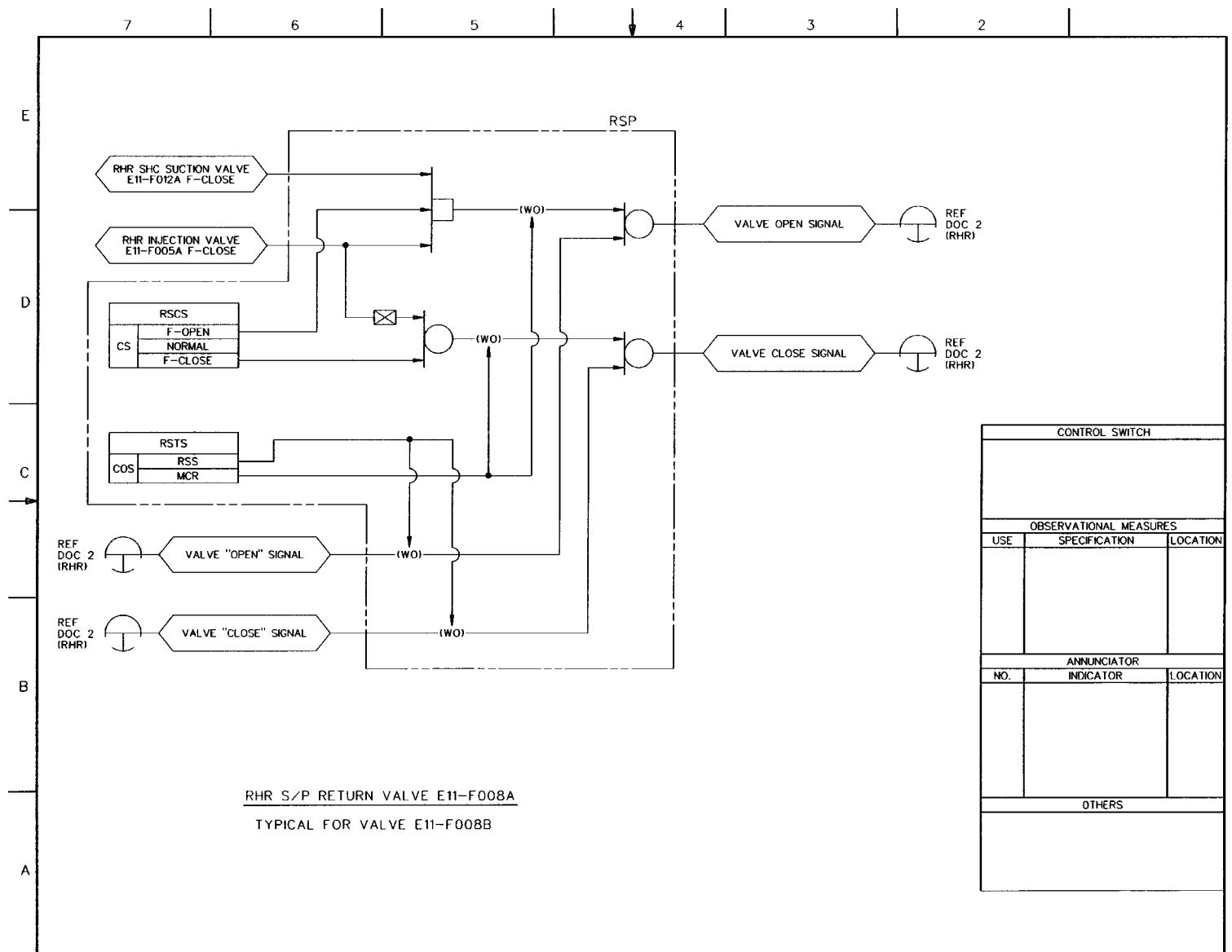


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 16 of 27)
ABWR DCD/Tier 2 Rev. 0 21-295

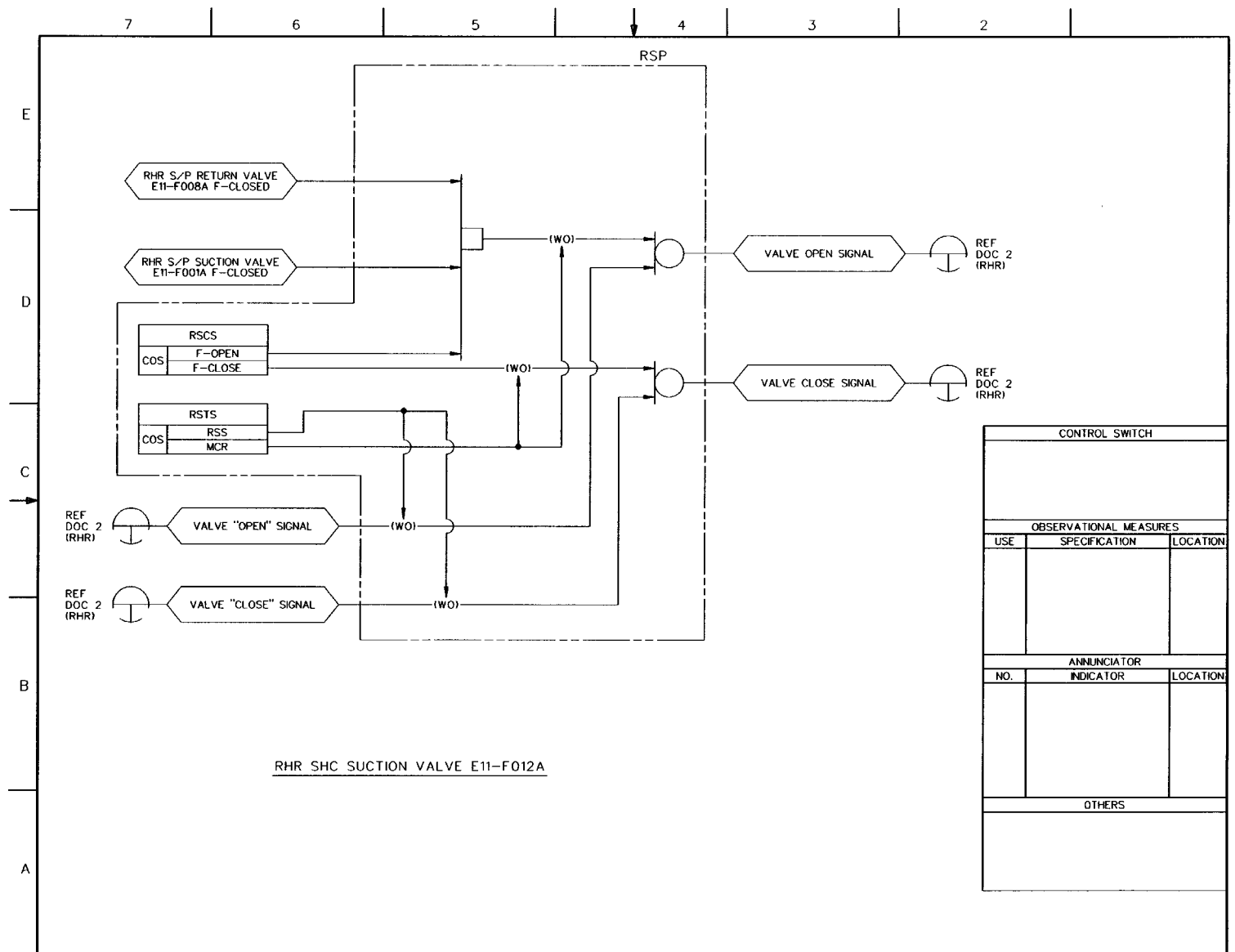


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM (Sheet 17 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-296

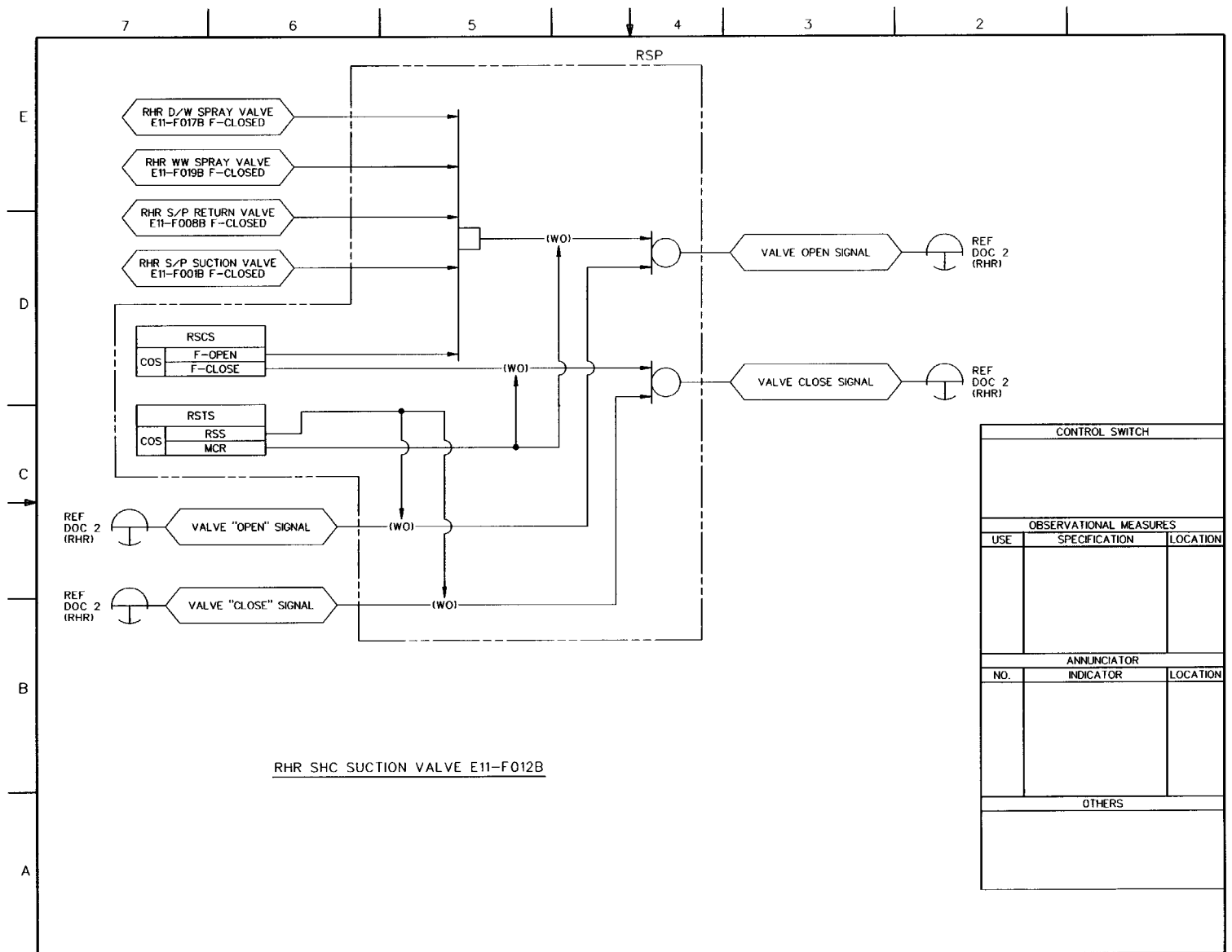


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 18 of 27)
ABWR DCD/Tier 2 Rev. 0 21-297

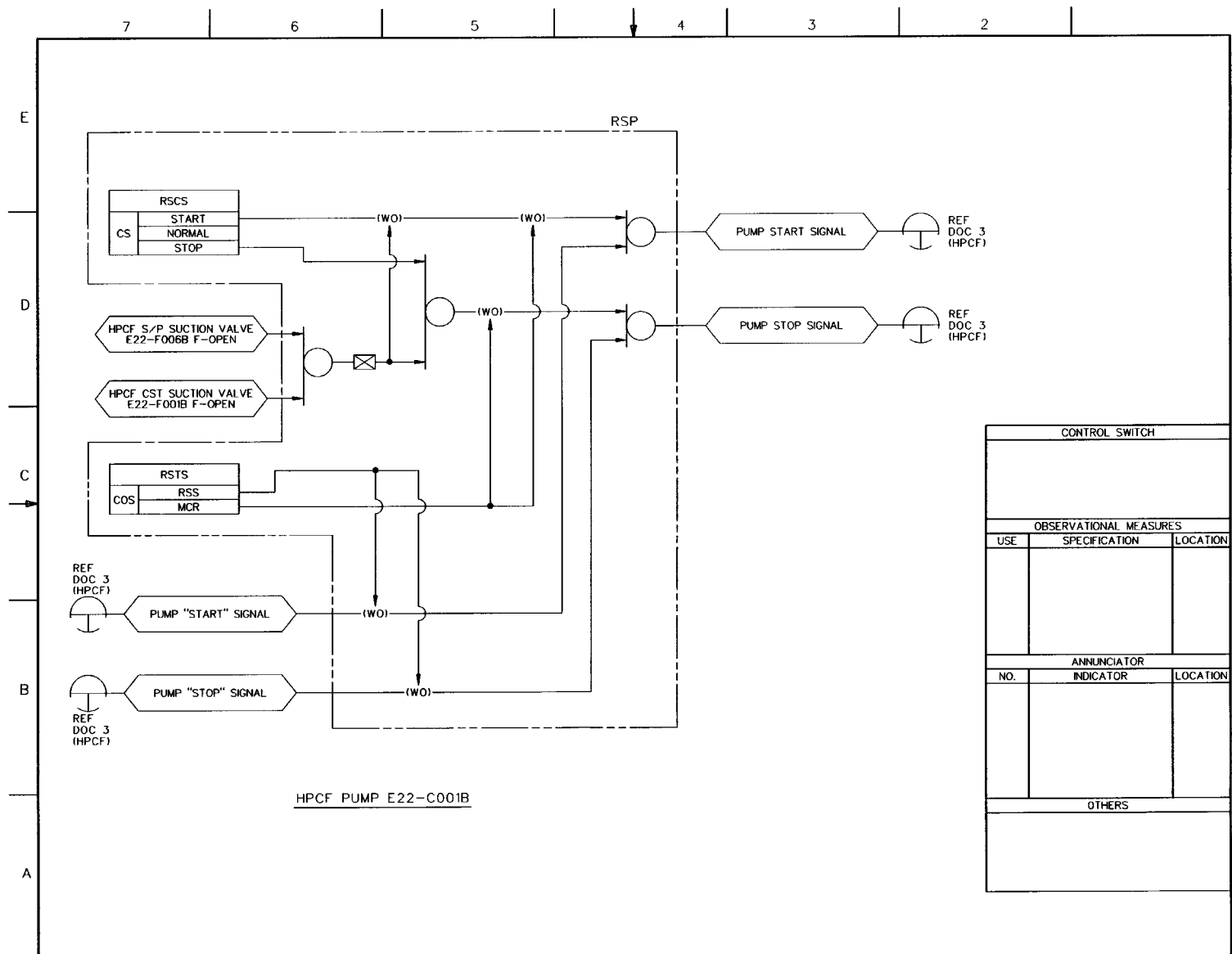


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 19 of 27)
ABWR DCD/Tier 2 Rev. 0 21-298

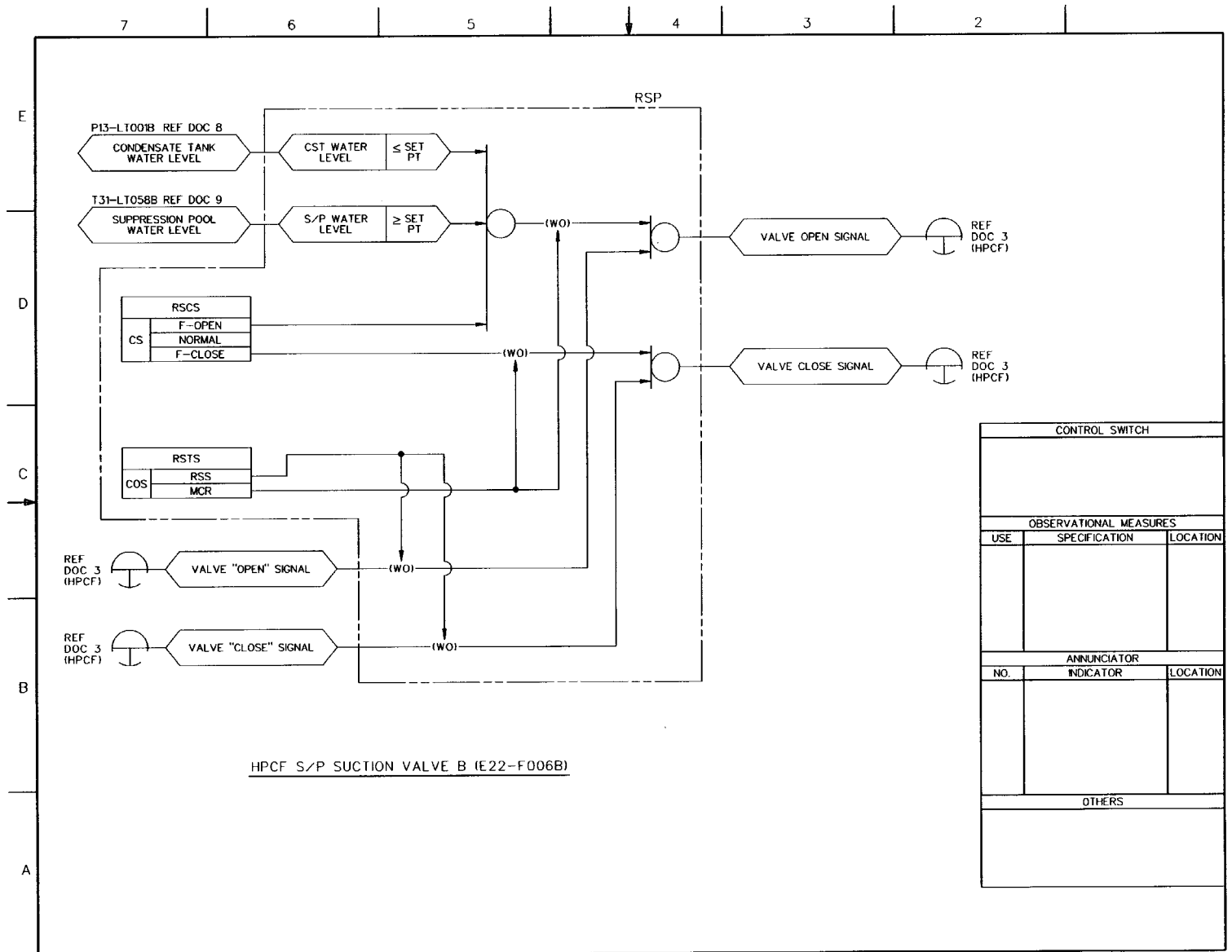


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 20 of 27)

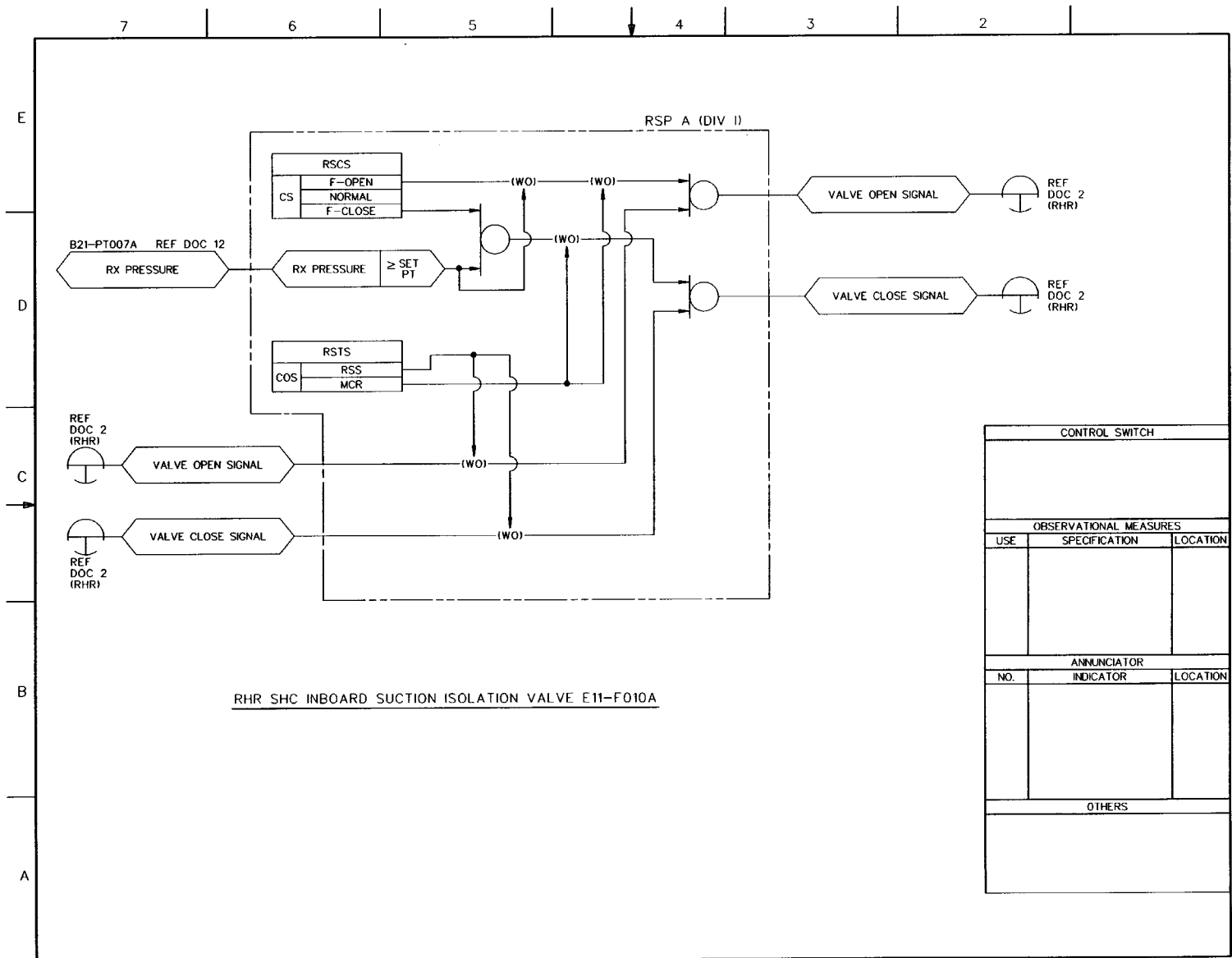


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 21 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-300

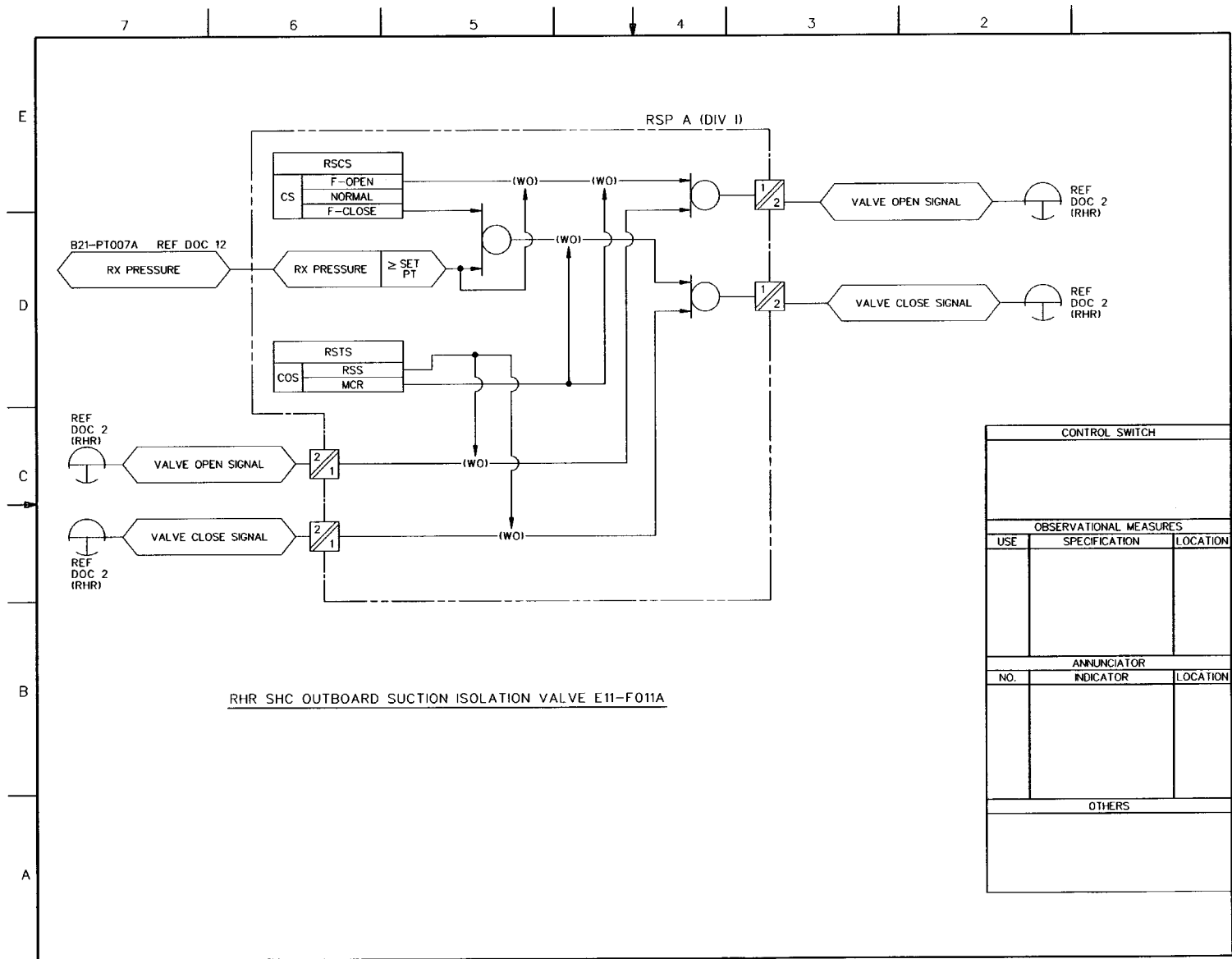


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 22 of 27)
ABWR DCD/Tier 2 Rev. 0 21-301

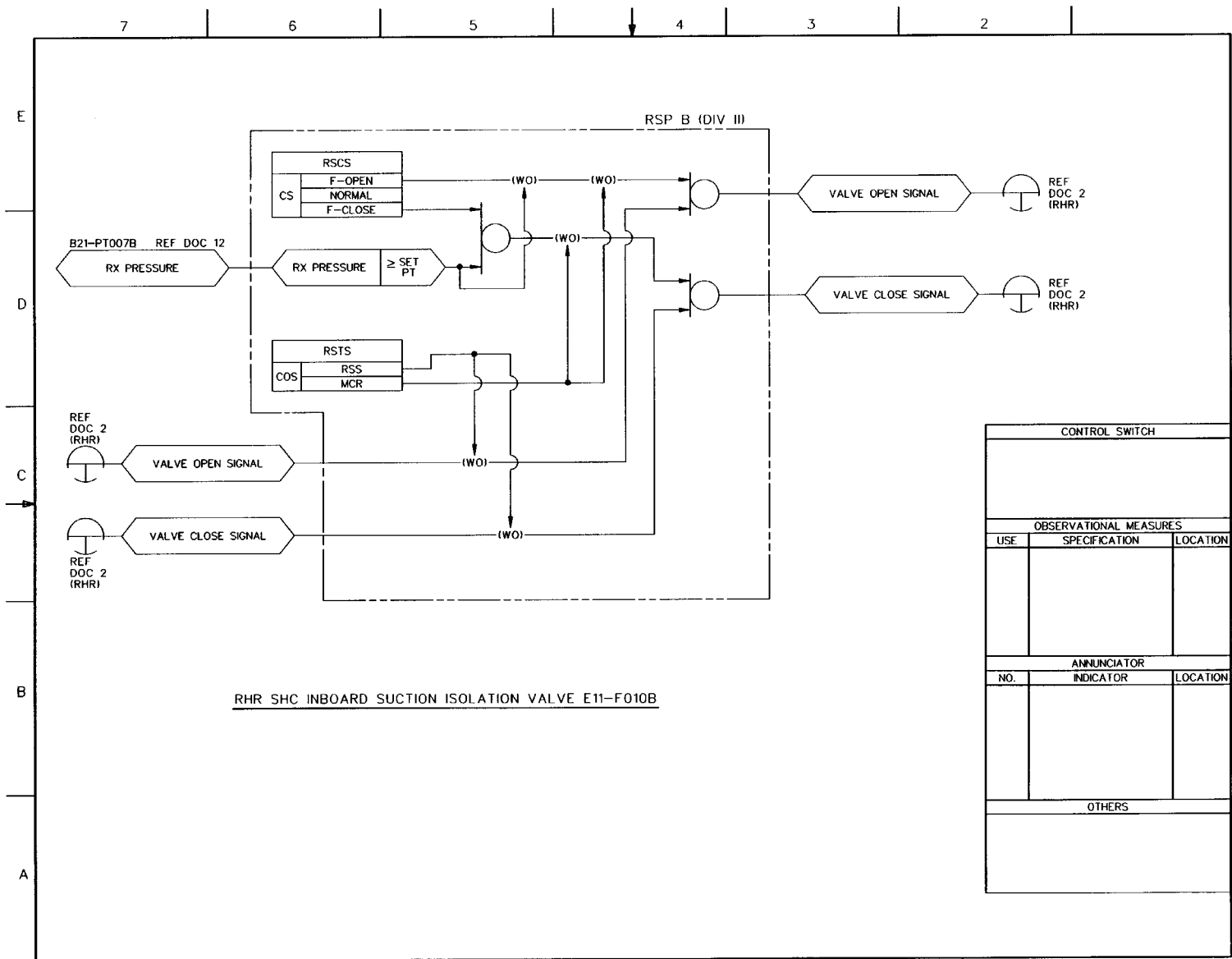


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 23 of 27)
 ABWR DCD/Tier 2 Rev. 0 21-302

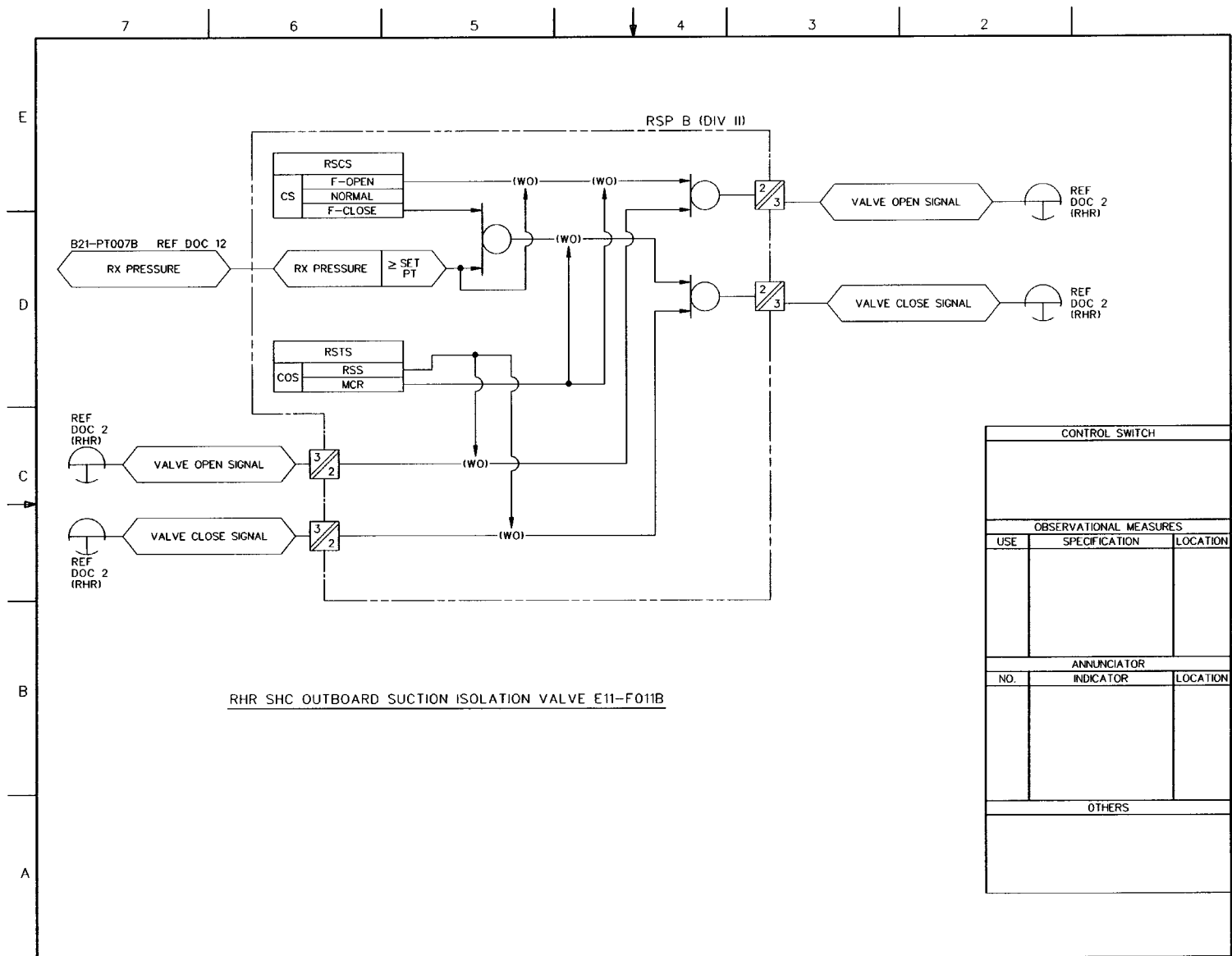


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 24 of 27)
ABWR DCD/Tier 2 Rev. 0 21-303

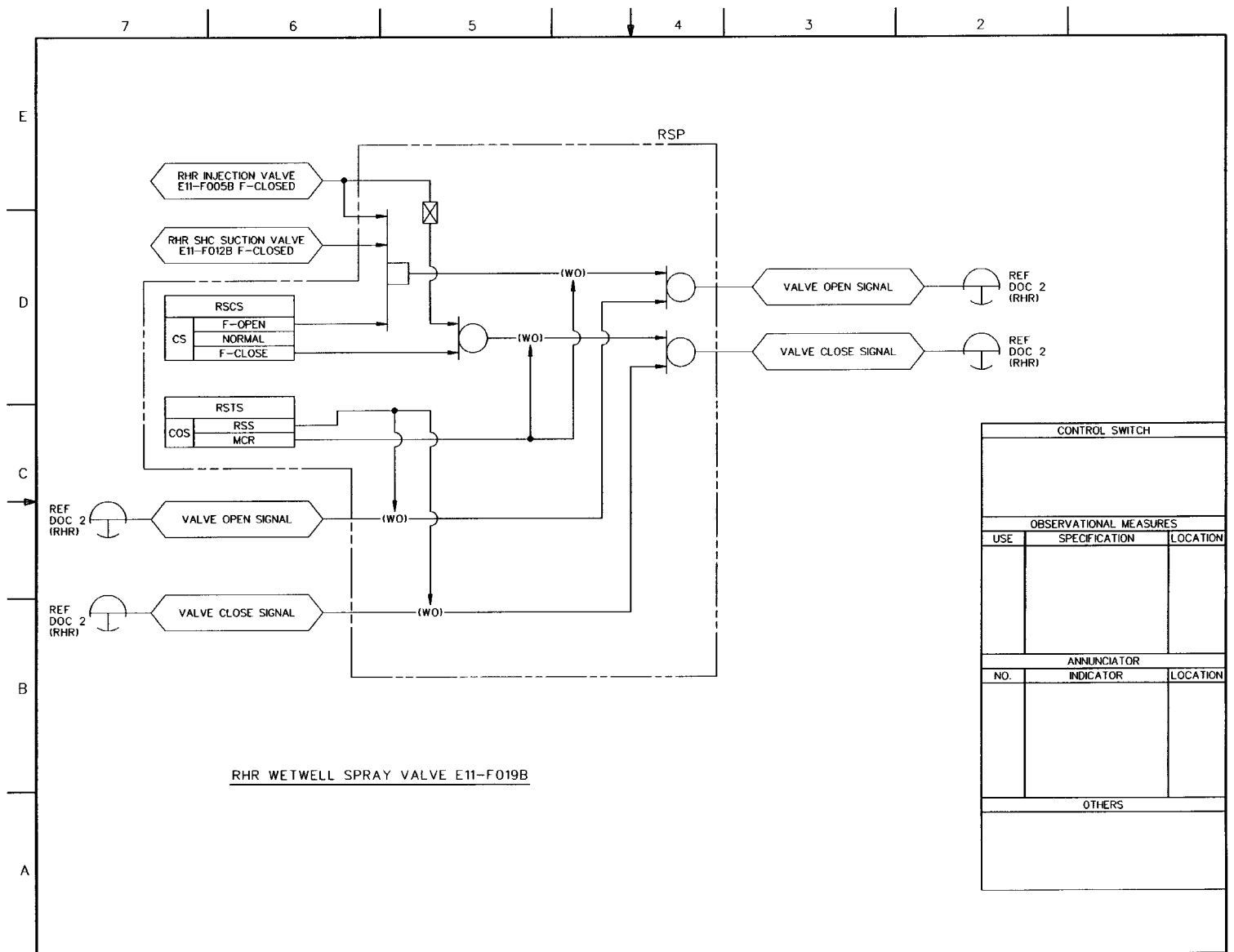


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM (Sheet 25 of 27)

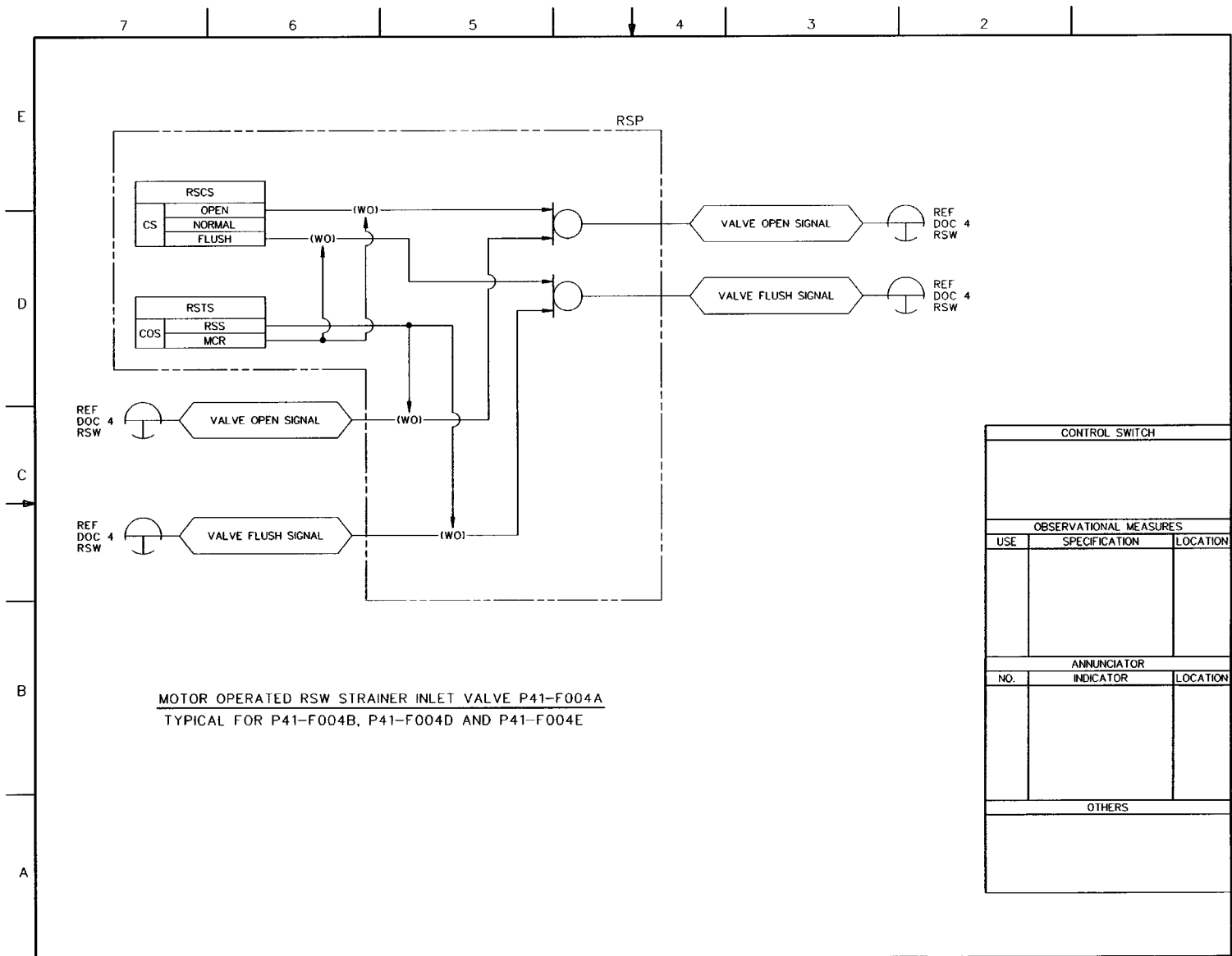
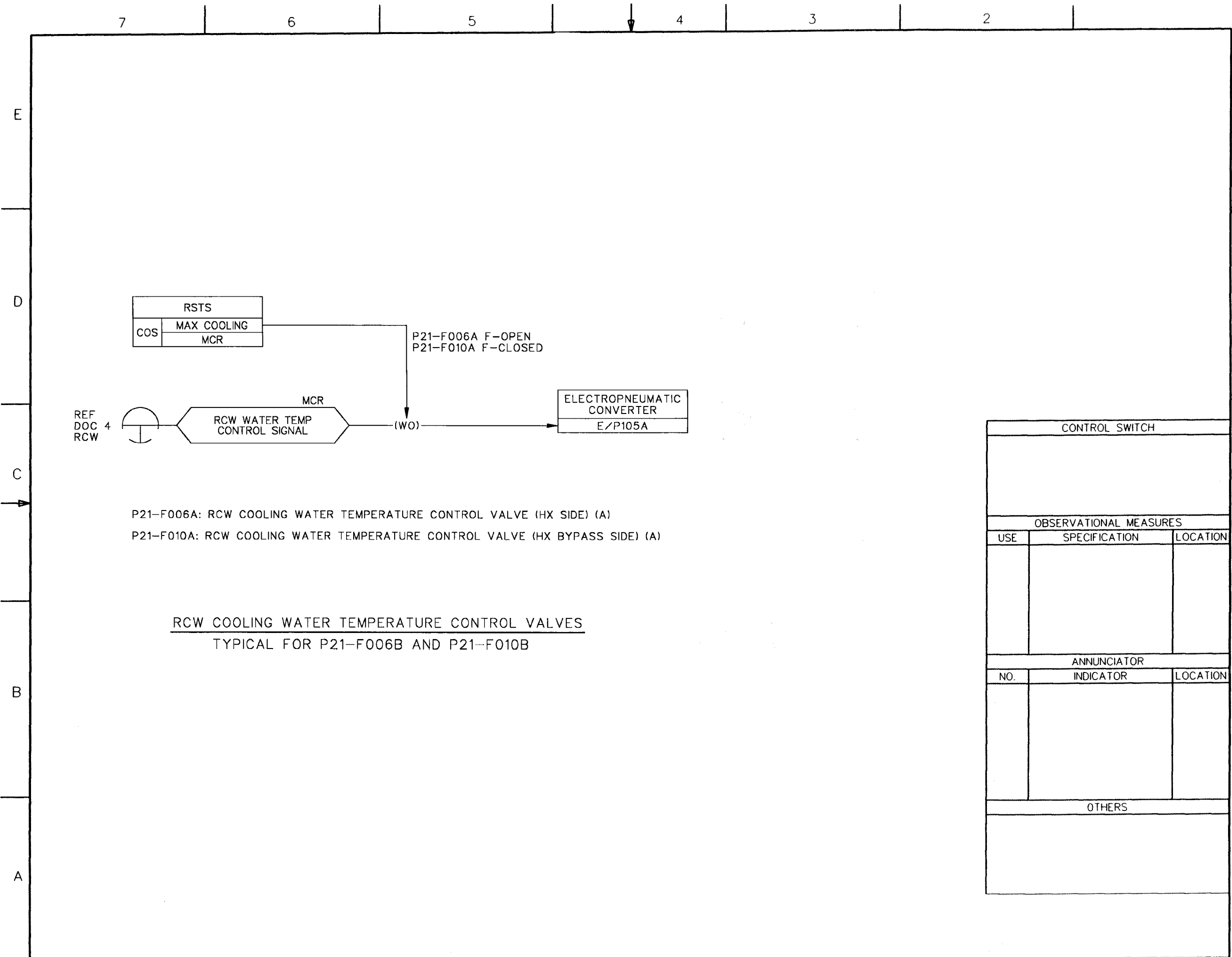


FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 26 of 27)
ABWR DCD/Tier 2 Rev. 0 21-305



CONTROL SWITCH		
OBSERVATIONAL MEASURES		
USE	SPECIFICATION	LOCATION
ANNUNCIATOR		
NO.	INDICATOR	LOCATION
OTHERS		

FIGURE 7.4-3 REMOTE SHUTDOWN SYSTEM IBD (Sheet 27 of 27)
ABWR DCD/Tier 2 Rev. 0 21-306